UNIVERSITY OF CANTERBURY

DOCTORAL THESIS

Devices and Applications of Square SnO_2 Nanotubes Grown by Mist Chemical Vapour Deposition

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Abstract

In this thesis, fabrication techniques have been developed to produce electronic devices, such as resistors, Schottky diodes, and field effect transistors using perfectly-square single-crystal nanotubes of tin oxide (SnO_2) grown by mist chemical vapour deposition (Mist-CVD). This involved optimizing the Mist-CVD growth conditions and the use of UV photolithography, scanning electron microscopy, RF magnetron sputtering, and electron beam evaporation to establish reliable ohmic and rectifying electrical contacts to individual and multiple square nanotubes. These devices were then electrically characterised and investigated for use in ultraviolet (UV) radiation sensing and neuromorphic applications. In the course of this work, valuable new insights were gained concerning the surface chemistry and surface electronic behaviour of these remarkable square nanostructures.

SnO_2 is a wide bandgap semiconductor and transparent conducting oxide, exhibiting the unusual combination of high optical transparency and strong electrical conductivity. In addition to these technologically useful properties, SnO_2 has a surface that is electronically sensitive to different gases and molecules. This is largely due to the presence of electron-donating surface hydroxyl groups that produce strong downward bending of the surface electronic bands and create a 2D surface electron accumulation layer. This surface sensitivity is advantageous in certain applications, such as gas and chemical sensing, but is a disadvantage in electronic devices that require stable operation. This surface sensitivity is particularly strong in the square SnO_2 nanotubes due to their high surface-to-volume ratios, and synchrotron X-ray spectroscopy techniques were combined with surface sulfur treatments to further understand and modify the degree of downward band bending.

Single-nanotube resistors were fabricated from individual undoped and 0.15% Sb-doped square SnO_2 nanotubes, with the Sb-doped nanotubes typically exhibiting 100 times higher conductances compared to their undoped counterparts. When illuminated with 200 μW cm^{-2} of 285 nm UV light, the current through an individual nanotube resistor (biased at + 2 V) increased by more than 20 times indicating a strong UV sensing ability. However, this was accompanied by an effect common in SnO_2 and other oxide semiconductors known as persistent photoconductivity (PPC), where the increased current through the nanotube decayed very slowly after removing the UV illumination. By systematically studying this PPC effect in different gas environments, it was found that both atmospheric oxygen and water vapour can play a significant co-operative role in minimising the PPC in SnO_2-based UV sensors - a significant new finding previously unreported in the literature.

Back-gated field-effect transistors (FETs) and Schottky diodes were also fabricated using individual square SnO_2 nanotubes. Transistors fabricated from undoped nanotubes exhibited strong n-type FET behaviour with an excellent on/off current ratio of $> 10^6$, however FETs fabricated using 0.15% Sb-doped nanotubes could not be completely pinched-off due to their high channel conductance. The electrical characterisation and modelling of the undoped FETs allowed an undoped nanotube carrier concentration of approximately 5×10^{16} cm^{-3} to be determined. Schottky diodes fabricated using IrO_x/Ir Schottky contacts on both undoped and 0.15% Sb-doped SnO_2 nanotubes showed excellent rectifying properties with forward/reverse bias current ratios exceeding 10^7, large effective barrier heights of up to 0.9 eV, and ideality factors as low as 1.04 indicating excellent lateral Schottky contact homogeneity. A Sb-doped SnO_2 nanotube Schottky diode back-illuminated with 360 μW cm^{-2} of 275 nm UV light exhibited an extremely high photocurrent response of 10^7 and a significantly faster PPC recovery compared the SnO_2 single-nanotube resistors. Surprisingly, both the FETs and Schottky diodes exhibited a significant (but reversible) decrease in performance with time when left exposed to air, which was attributed to atmospheric adsorbates such as oxygen and water vapour electronically interacting with the SnO_2 surface.

The low device yield and labour-intensive fabrication methods associated with

producing electronic devices based on individual square SnO_2 nanotubes were solved by the design of ionic polymer-gated transistors that use multiple SnO_2 nanotubes as the conducting channel between interdigitated source and drain contacts. These devices also displayed strong n-type transistor behaviour, with an on/off current ratio > 10^4. The $LiClO_4$/polyethylene oxide ionic gate used in this device had a strong time-dependent response which was used to produce several synaptic behaviours such as excitatory postsynaptic current and paired-pulse facilitation. This demonstrated that ionic-gated square SnO_2 nanotube transistors are promising candidates for use as artificial synapses in neuromorphic computing applications.

Synchrotron X-ray photoelectron spectroscopy (XPS) was used to further investigate the relationship between the surface chemistry and strong downward band bending of the square SnO_2 nanotubes. In-situ heating at 600 °C in vacuum was used to remove most of the electron-donating surface hydroxyl groups, but this also led to the creation of electron-donating oxygen vacancies and as a result the downward band bending was largely unchanged. A full recovery of the surface hydroxyl termination and repair of the oxygen vacancies could not be achieved by the use of in-situ exposure to water vapour and oxygen (even up to 8×10^8 and 1.35×10^4 Langmuirs, respectively), with a full recovery requiring multiple hours of exposure to atmosphere. Synchrotron XPS and near-edge X-ray absorption fine structure spectroscopy were employed to investigate the use of sulfur treatment to reduce the downward band bending of the nanotubes. Surface treatment using an aqueous Na_2S solution produced only small upward shifts in band bending, however these upward shifts significantly increased with increasing X-ray doses due to the chemical reduction of oxidized sulfur species on the nanotube surfaces and were sufficient to produce almost-flat electronic bands.

In summary, the practicality of semiconducting square SnO_2 nanotubes grown by the highly-scalable mist-CVD technique has been significantly extended by the fabrication and characterisation of a range of electronic devices using individual and multiple undoped and Sb-doped nanotubes. The potential use of these devices in UV sensing and neuromorphic applications was practically demonstrated.

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A doctoral thesis is no small feat to produce and accomplish, and this would have almost certainly been an impossible task without the support of dozens of other individuals who have each contributed in their own way. I'm thankful that so many people have been around to guide, assist and support me to make this thesis what it is today.

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Abbreviations

Notation	Description
2DEG	2-Dimensional Electron Gas
CVD	Chemical Vapour Deposition
DC	Direct Current
EDL	Electric Double Layer
EDX	Energy-Dispersive X-ray Analysis
EPSC	Excitatory Postsynaptic Current
EtOH	Ethanol
FET	Field-Effect Transistor
FWHM	Full Width at Half Maximum
GGA	Generalised Gradient Approximation
HRS	High Resistance State
IMFP	Inelastic Mean Free Path
IPA	Isopropanol
IV	Current-Voltage
LED	Light-Emitting Diode
LRS	Low Resistance State
MOS	Metal Oxide Semiconductor
Mist-CVD	Mist Chemical Vapour Deposition
NEXAFS	Near-Edge X-ray Absorption Fine-Structure
NT	Nanotube
NW	Nanowire

Notation	Description
OH	Hydroxyl
PEO	Polyethylene Oxide
PMMA	Polymethylmethacrylate
PPC	Persistent Photoconductivity
PPF	Paired-Pulse Facilitation
PVD	Physical Vapour Deposition
RF	Radio-Frequency
RMS	Root-Mean-Square
SEM	Scanning Electron Microscopy
SRDP	Spike-Rate-Dependant-Plasticity
STM	Scanning Tunnelling Microscopy
STP	Short-Term Plasticity
SnO_2	Tin (IV) Oxide, Tin Dioxide
TCO	Transparent Conducting Oxide
TDS	Temperature Desorption Spectroscopy
TEM	Transmission Electron Microscopy
TF	Thin-Film
UV	Ultraviolet
VB	Valence Band
VLS	Vapour-Liquid-Solid
XPS	X-ray Photoelectron Spectroscopy

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J.I. Scott, R.L. Adams, R.F. Martinez-Gazoni, L.R. Carroll, A.J. Downard, T.D. Veal, R.J. Reeves, and M.W. Allen. "Looking Outside the Square: The Growth, Structure, and Resilient Two-Dimensional Surface Electron Gas of Square SnO_2 Nanotubes". In: *Small* 19.28 (2023), p.2300520

R.L. Adams, M. Le Ster, H.N. Tran, J.G. Partridge, N. Gaston, T.D. Veal, R.J. Reeves, and M.W. Allen. "Combined role of H_2O and O_2 adsorbates on the persistent UV photoconductivity of perfectly-square SnO_2 nanotubes." In: *Nanoscale* 17 (2025), 23492-23502

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R.L. Adams, L.R. Carroll, R.J. Reeves, and M.W. Allen. "Tuning the Band Bending of Square Tin Oxide Nanotubes via Hydroxyl Coverage Modulation and Chemical Modification." The 5th International Conference on Emerging Advanced Nanomaterials (ICEAN 2024), 4-8 November 2024, Newcastle, Australia. Oral Presentation.

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Chapter 1

Introduction

1.1 Metal Oxide Semiconductors

Metal oxide semiconductors (MOSs) are materials which have received extensive research over the last fifty years, owing to their wide variety of properties such as high thermal and chemical stabilities, wide bandgaps, intrinsic electrical conductivity and electronically sensitive surfaces. These properties have led MOSs to be used in a range of applications, including in gas sensing[1–3], ultraviolet (UV) photodetectors[4–6], catalysis[7–9], biosensing[10–12] and energy harvesting[13–15]. Furthermore, almost all metals will readily oxidise when heated in a source of oxygen, and so many metal oxide materials are easily produced by a wide range of techniques. This has resulted in an explosion of research into these materials over the last 25 years. The interest in these materials can be seen by the number of scientific reports published on the topic each year, shown in Figure 1.1.

Transparent conductive oxides (TCOs) are MOSs that combine two usually conflicting properties: optical transparency and electrical conductivity. Generally, optically transparent materials have a wide bandgap, where photons with energies within the visible spectrum have insufficient energy to excite electrons from the material's valence band to the conduction band. As a result, these photons are able to pass through the material with little to no absorption. This wide bandgap, however, typically inhibits the formation of free charge carriers at room temperature due to large charge carrier ionisation energies and the Fermi level lying within the material bandgap. This leads to many transparent oxides being electrically insulating, such

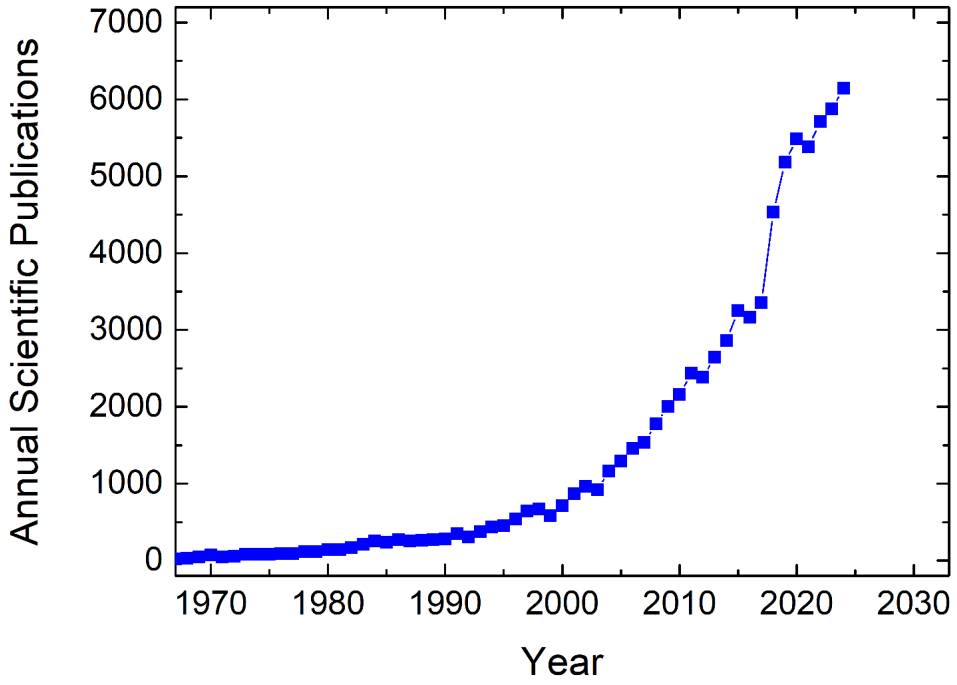


Figure 1.1: The annual number of Scopus-indexed scientific publications for the search term "Metal Oxide Semiconductors".

as Al_2O_3, SiO_2, ZrO_2 and HfO_2. Conducting materials, on the other hand, have small charge carrier ionisation energies that allow free carriers to be thermally excited to electronic bands that allow electronic conductivity. This, however, typically arises with materials that have a small or no bandgap, and this results in a strong absorption of photons within the visible spectrum. However, a select range of oxides have relatively small carrier ionisation energies while maintaining a wide bandgap, which leads to the unique combination of optical transparency and electrical conductivity. Some examples of TCOs include SnO_2, ZnO, Ga_2O_3, In_2O_3, CdO and NiO.

TCOs have found uses in a range of applications that benefit from the combination of optical transparency and electrical conductivity. Some examples of these include electron transport layers in third-generation solar cells[16, 17], transparent electronics for flat panel displays[18–20], and solar-blind UV photodetectors[21–23]. There are, however, additional properties of these materials that are useful in sensing and catalytic applications. These properties include an electrical sensitivity to different gases, catalytic surface sites, and the ability to functionalise the material surface for the detection of specific biomolecules. All of these properties result from effects that

take place at the surface of these materials, so for many applications it is desirable to have the largest possible surface area. In addition, changes in the surface electrical properties can often be relatively small to detect if there is a large amount of unchanging bulk material. For these reasons, significant effort has been invested in maximising the surface areas and surface-to-volume ratios of TCOs, which has led to the breakout of nanomaterials as an area of modern research.

1.2 Nanostructures

The desire to maximise surface areas and surface-to-volume ratios to exploit the surface properties of different materials has resulted in the development of material structures with dimensions that extend into the nanoscale, which are known as nanostructures. Nanostructures are often classified into different families depending on the number of spatial dimensions the nanostructure spans. Some examples of these families are 1-dimensional and 0-dimensional nanostructures.

The family of 1-dimensional nanostructures include nanowires, nanorods, nanofibres and nanotubes. The single-dimensional nature of these nanostructures means they offer increased surface-to-volume ratios while often maintaining good crystalinity and electrical properties. The range of different structures within this family of materials is due to the different methods used to produce them, which include thermal evaporation, electrospinning, hydrothermal methods and chemical vapour deposition. Often, the one-dimensional structure of these materials arises from the different surface energies of a single crystal material. Analogous to how surface tension minimises a water droplet's surface area, the growth of one-dimensional nanostructures arises from the minimising of the surface energy of the overall structure, often by maximising the surface area of the lowest energy crystal faces given a certain volume of material. This results in one axis that has a significantly increased growth rate compared to other growth directions, which gives rise to the one-dimensional structure. This minimisation of the surface energy means most nanostructures typically exhibit circular cross sections, as larger amounts of energy are required to form well defined edges and corners, making square and rectangular cross-sections relatively rare. A large downside of these structures, however, is that

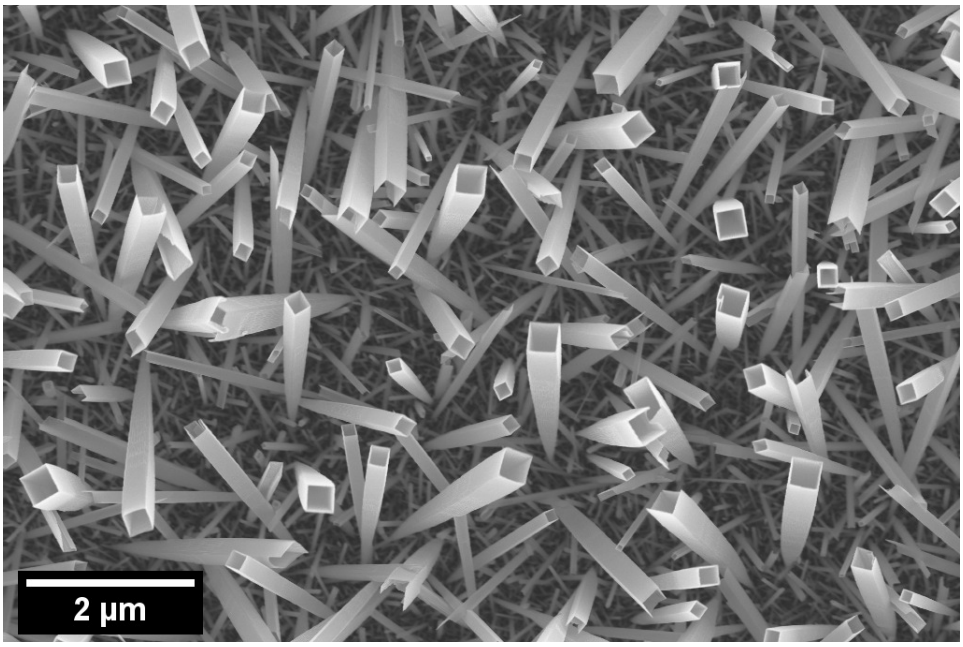
they are more difficult to incorporate into electronic devices compared to thin films, which can limit the commercial availability of these devices due to poor fabrication scaling.

The family of nanostructured materials known as 0-dimensional nanostructures primarily include nanoparticles such as nanospheres and nanoprisms. Perhaps the most famous use of nanoparticles is in quantum dots, where the small, highly controlled size of the nanoparticles causes strong photonic effects that have been applied in next generation solar cells[24, 25], light-emitting diodes[26, 27], and are responsible for many of the vibrant colours observed in stained-glass windows[28]. Nanoparticles exhibit the greatest surface-to-volume ratios of any nanoscaled material, however a clustering of the nanoparticles is required to achieve practical electrical conduction in devices that incorporate them, and this clustering results in many grain boundaries that negatively affect the electrical properties of the material. As a result, one-dimensional materials find themselves in an optimal point between maximising the surface-to-volume ratios of the material while still maintaining high crystalinity and good electrical properties.

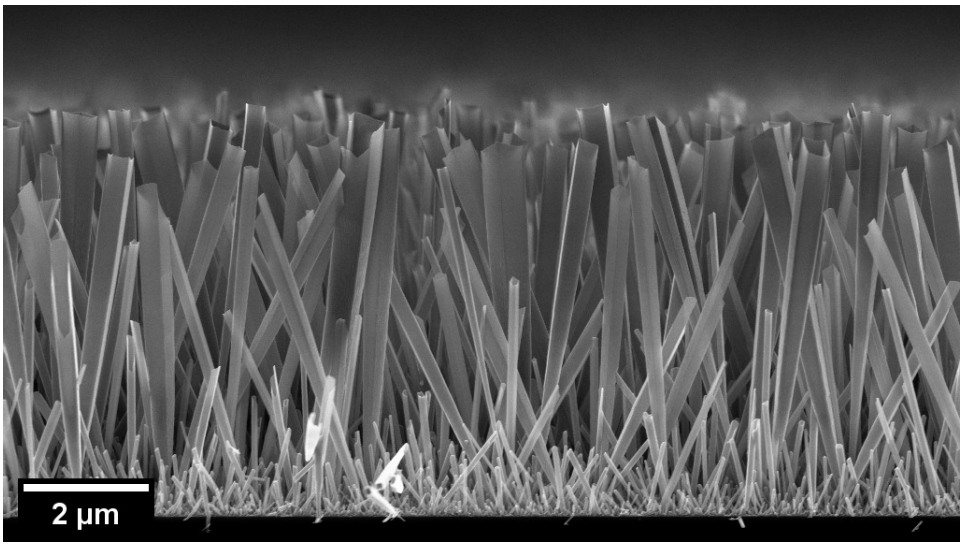
1.3 Square Nanotubes of Tin Oxide and Previous Work

Previously, research undertaken by Dr Jonty Scott resulted in the discovery of a process that produced hollow nanotubes of tin (IV) oxide (SnO_2) with perfectly square cross sections[29] — a nanostructure rarely reported in the literature. Scanning electron microscopy images of these square SnO_2 nanotubes can be seen in Figure 1.2. These structures were produced by a technique known as mist chemical vapour deposition (Mist-CVD), which is characterised as being a non-vacuum, solution-based technique that results in a low-energy, highly scalable process that can deposit TCO materials of a high crystal quality. Importantly, these structures exhibit incredibly large surface-to-volume ratios combined with a high crystal quality and highly unique geometry, properties that should be useful in a range of different applications.

The previous investigations of these nanostructures performed by Scott largely focused on their fundamental characteristics. Significant work was performed on



(a)



(b)

Figure 1.2: Scanning electron microscopy images of square SnO_2 nanotubes grown on a Si substrate by Mist-CVD. (a) Top-down view and (b) side-on view.

understanding and optimising their growth conditions to allow repeatable growths across a range of different substrates. The fundamental structural and chemical properties of the nanotubes have also been characterised, and the reader is invited to read Scott's doctoral thesis and published material for full details[29, 30]. However, the important findings from Scott's work are covered later in Chapter 2. While some investigations were performed on the electrical properties of the SnO_2 nanotubes, the results from this were preliminary in nature. The work performed by Scott has

laid a strong foundation for further investigations into the electrical properties of the SnO_2 nanotubes, their incorporation into different electrical devices and their use in different applications, which is the focus of this thesis.

1.4 Thesis Aims and Overview

This thesis aims to further expand the work of Scott and to investigate different device structures and applications for the square SnO_2 nanotubes. When these investigations revealed a range of unique phenomena that affects SnO_2 nanotube-based devices, further investigations were performed that focus on the fundamental properties of the nanotubes that contribute to these effects and potential processes to further tune these phenomena. Firstly, Chapter 2 presents the relevant background material for the rest of the thesis, and includes the fundamental properties of SnO_2 and its nanostructures, a review of different device structures and applications, and a detailed coverage of the previous work performed on the square SnO_2 nanotubes. Following this, Chapter 3 details the methodologies used for fabrication and characterisation throughout this thesis.

The remaining chapters present the body of work undertaken throughout this thesis. Chapter 4 focuses on optimising the contact method used to apply voltages and measure currents to individual SnO_2 nanotubes, before characterising their electrical properties and investigating this device structure for UV sensing. Chapter 5 presents a range of more advanced devices that incorporate the SnO_2 nanotubes, such as field-effect transistors, Schottky diodes and ionic-gated transistors that are investigated for applications such as UV sensing and neuromorphic computing. Chapter 6 investigates the fundamental properties of the SnO_2 nanotubes that contribute towards the interesting phenomena observed in the previous chapters, and details attempts to control and tune these fundamental properties for more stable devices. Finally, further growth optimisations and fundamental characteristics are presented as additional studies in Chapter 7, while the conclusions for this thesis are given in Chapter 8.

Chapter 2

Background

2.1 Introduction

Tin (IV) oxide, herein referred to as SnO_2, is a wide bandgap semiconductor and transparent conducting oxide that has received significant attention due to its wide range of unique properties. Such properties include controllable electrical characteristics, optical transparency, an electron-rich surface and a sensitivity to a wide range of different gases. In addition to these properties, SnO_2 has an atmospherically stable stoichiometry and a hardness of 7 on the Mohs scale, resulting in a physically robust material. Like many other metal oxides, SnO_2 has a number of surface-related properties that are more pronounced at larger surface-to-volume ratios, leading to the creation of SnO_2 nanostructures to exploit these properties for different applications.

In this chapter, the fundamental properties of SnO_2 are discussed with a focus on its structural, optical, electrical and surface properties. Following this, the various types of SnO_2 nanostructures present in the literature are introduced, alongside their comparative benefits and disadvantages. The applications of SnO_2 nanostructures are discussed after this, illustrating the existing work and challenges in maximising the potential of this material. Finally, the previous work involving the SnO_2 nanotubes is summarised, with the limitations of these findings identified.

2.2 The Fundamental Properties of Tin Oxide

2.2.1 Structural Properties

The most stable phase of SnO_2 crystallises in a rutile structure with a tetragonal unit cell in the $P4_2/mnm$ group, and can be seen in Figure 2.1. The unit cell of the bulk crystal has lattice spacings of a = b = 4.737 Åand c = 3.185 Å. Each Sn atom is six-fold coordinated with neighbouring O atoms and is in the +4 oxidation state, while each O atom is three-fold coordinated to neighbouring Sn atoms with an oxidation state of -2.

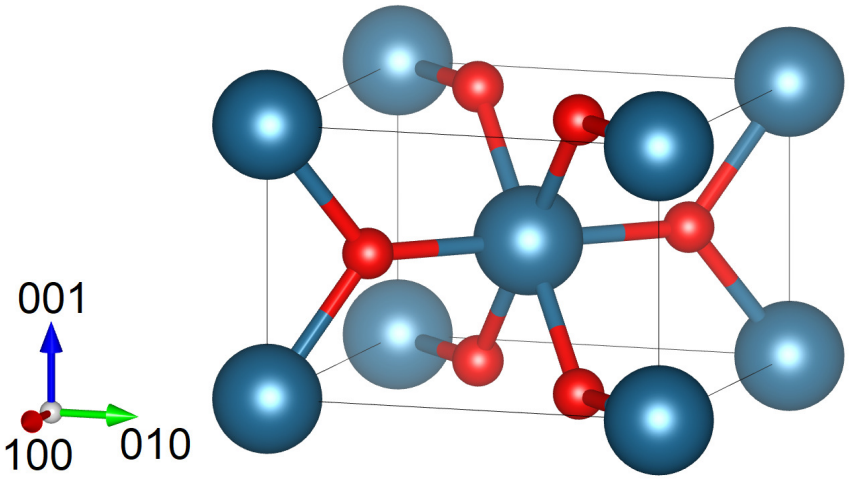
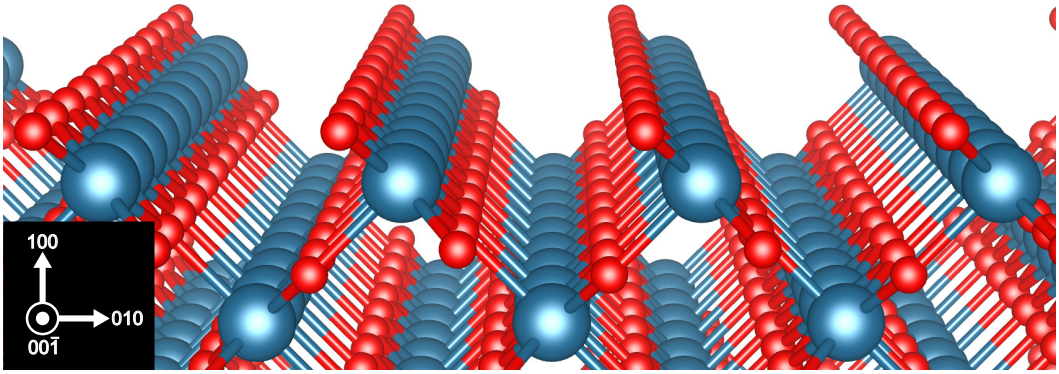


Figure 2.1: The tetragonal unit cell of rutile SnO_2, consisting of Sn^{4+} (blue) and O^{2-} (red) atoms.

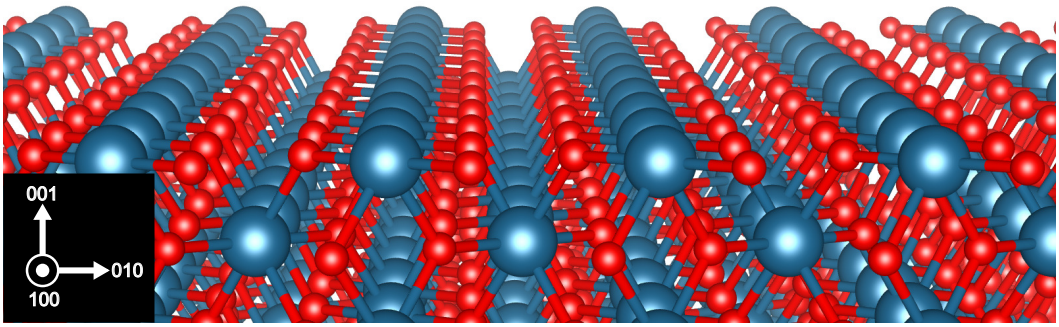
There are other, less stable phases of tin oxide that are known to exist. The next most reported form of tin oxide is the monoxide, SnO, with the Sn atoms in a +2 oxidation state. This form of tin oxide is studied comparatively less than rutile SnO_2, primarily because SnO is a metastable phase known to oxidise to SnO_2 when left exposed to air[31, 32]. Despite this main drawback of material oxidation, there continues to be an interest in SnO as it is a rare p-type conducting oxide[31, 33]. Other intermediate tin oxide phases have been reported which contain a mixture of Sn^{2+} and Sn^{4+}, and are generally described as Sn_3O_4[34–36] or Sn_2O_3[37–39]. These phases, however, have limited reproducibility and are rarely reported, with little agreement on their stoichiometry or crystal structure.

The tetragonal unit cell of rutile SnO_2 results in four low-index surface terminations, which correspond to the (100), (001), (110) and (101) lattice orientations[40]. Each of these surface terminations can be seen in Figure 2.2. As SnO_2 is a material with a highly sensitive surface, each surface termination is important due to the differing density and availability of surface sites. Each stoichiometric surface of SnO_2 consists of surface Sn and O atoms that are under-coordinated compared to the bulk crystal structure. In the (100), (101) and (110) surface terminations the surface Sn atoms are 5-fold coordinated with one dangling bond, which is often considered the active site for gas adsorption onto the SnO_2 surface[41–43]. The Sn atoms on the (001) surface however are 4-fold coordinated, which results in two dangling bonds per Sn. Each of these surfaces also contains surface O atoms which are 2-fold bonded to other Sn atoms, which are referred to as bridging oxygen. These bridging oxygen atoms also have important roles on the surface chemistry of SnO_2, as they can assist in chemical reactions and can change the valency of the surface Sn atoms from +4 to +2 if they are removed from the surface.

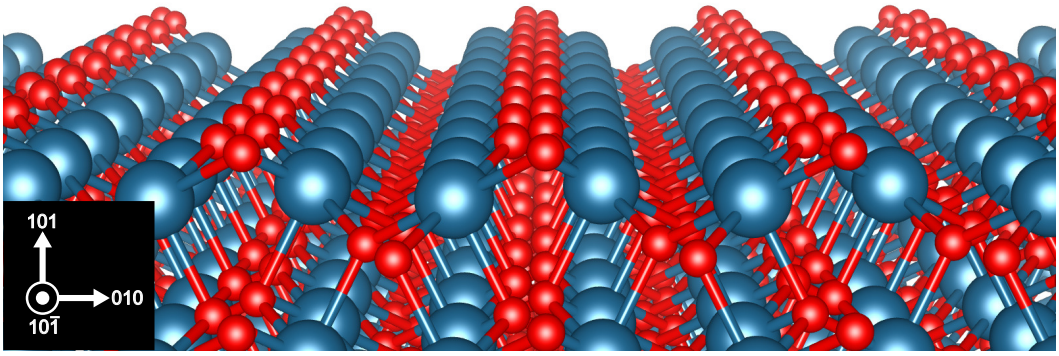
As mentioned in Section 1.2, the different surface energies of a material are highly important when considering different nanostructures that the material may form. Several groups have calculated the surface energies of the different SnO_2 surfaces by different methods, primarily by the generalised gradient approximation (GGA)[44, 45] or by the shell model[46, 47]. A table summarising the surface energies of the four low-index surfaces can be seen in Table 2.1. Despite variations within the calculated energies between the different calculation models, all reports agree that the lowest energy surface of SnO_2 is the (110) face, and the (001) face has a significantly higher surface energy compared to the other surface terminations. This is illustrated in Figure 2.3, which shows a vapour-phase transport grown single crystal of SnO_2 annotated with its different surface terminations. The (110) face has the largest surface area, dominating the (100) face which is significantly smaller. The (001) face is not present due to its high surface energy, with the crystal preferring the (101) face to terminate its top surface.



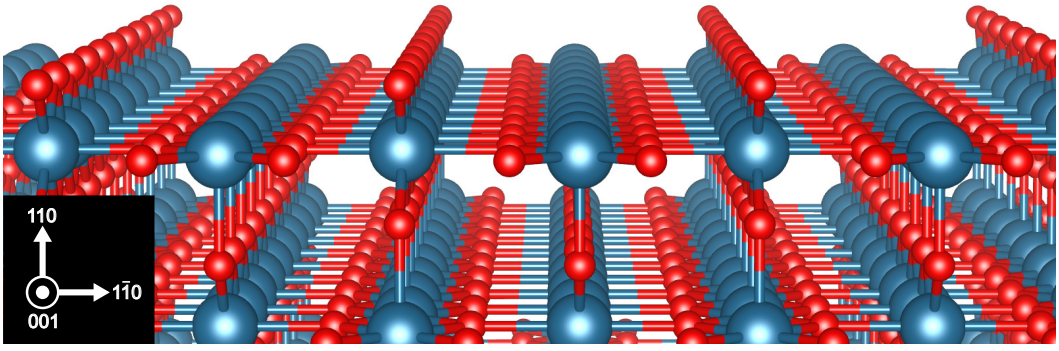
(a)



(b)



(c)



(d)

Figure 2.2: Stoichiometric terminations of the (a)(100), (b)(001), (c)(101) and (d)(110) surfaces of SnO_2. Sn atoms are shown in blue while O atoms are shown in red.

Table 2.1: Calculated values of the surface energies of stoichiometric SnO_2 surfaces in Jm^{-2} by different calculation models.

SnO_2 Surface	GGA[44]	GGA[45]	Shell[46]	Shell[47]
(100)	1.14	1.27	1.648	1.648
(001)	1.72	1.84	2.366	2.363
(101)	1.33	1.43	1.554	1.554
(110)	1.04	1.20	1.380	1.401

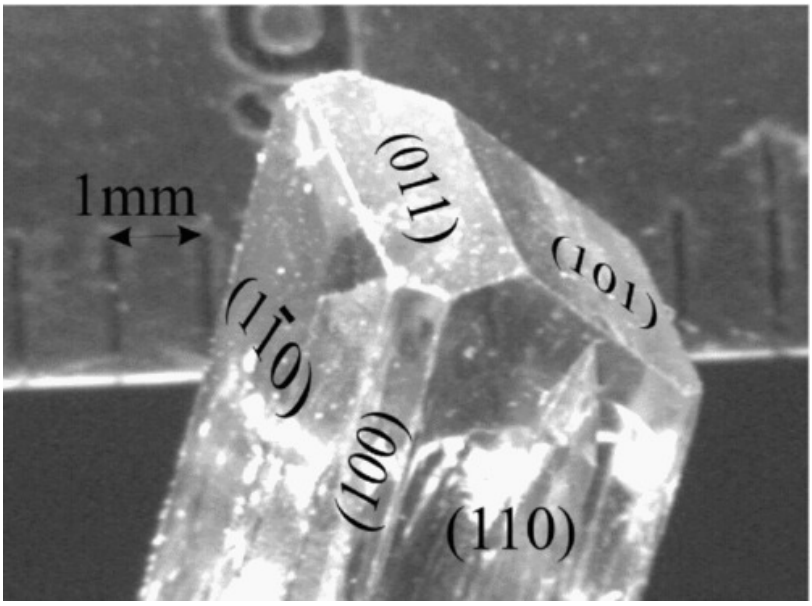


Figure 2.3: A vapour-phase transport grown crystal of SnO_2 with annotated surface terminations. Note that the (1$\bar{1}$0) and (011) faces are equivalent to the (110) and (101) faces, respectively. Reproduced with permission from [48, 49]

2.2.2 Optical Properties

SnO_2 is a wide-bandgap semiconductor with a well-accepted fundamental bandgap of ~3.6 eV[50–52]. However, this fundamental bandgap is dipole forbidden[53], which means electrons cannot simply be excited from the valence band maximum to the conduction band minimum by photons with the same energy as the bandgap. Higher energy photons are required for this process to occur. When combined with other effects such as Burstein-Moss shifts and variations in defects, stoichiometry and crystal quality, this leads to the various reports of direct optical bandgaps of SnO_2 that range from 3.6 to 4.3 eV[53–58].

The wide bandgap of SnO_2 leads to a high transparency in the optical region of light, with reports demonstrating up to 97% transparency in this region for 1 µm

thick films[59, 60]. SnO_2 also has a refractive index of ~2.0[54, 61]. These properties explain why SnO_2 is a highly suitable material for transparent electronics and solar cell applications.

2.2.3 Electrical Properties

As a TCO, SnO_2 possesses the unusual combination of electrical conductivity alongside its optical transparency. Undoped SnO_2 is insulating when the crystal is highly stoichiometric[62, 63], however, defects that occur within the crystal lattice are believed to act as electron donors which results in a commonly occurring n-type conductivity in the undoped material[59, 64, 65]. SnO_2 has a wide range of controllable electrical properties, with carrier concentrations ranging from insulating to 10^{21} cm^{-3}[66–68], a maximum observed electron mobility of 260 $cm^2V^{-1}s^{-1}$[69], and resistivities from > 10^3 to 10^{-4} Ωcm[66–68].

Although it is commonly accepted that defects within the SnO_2 lattice are the cause of the intrinsic n-type conductivity within the material, the exact nature of the responsible defects is still under debate. The most common theory is that oxygen vacancies are responsible for much of the n-type conductivity by the creation of shallow electron donor levels that can be thermally ionised. The support of this theory largely stems from a range of deposition and annealing studies where an increasing oxygen concentration results in lower conductivities and carrier concentrations in the resulting SnO_2 samples[65, 70–74]. However, two problems generally arise with this theory[59]. Firstly, it has been noted that atomic vacancies tend to create deep electron donor levels within the bandgaps of semiconducting materials (such as Si and GaAs), which has little overall effect on the material conductivity. Secondly, undoped SnO_2 has been reported with intrinsic carrier concentrations up to 10^{20} cm^{-3}[75], which would in turn require a large density of defects that are typically unfeasable in most binary oxides.

Several alternative defects have been proposed to contribute to the intrinsic n-type conductivity present in SnO_2. A computational study by Kilic and Zunger reported that tin interstitials are also a likely contributor to the n-type conductivity due to a small calculated defect formation energy and a resulting donor level ~200 meV

above the conduction band minimum[59]. Singh et al. disagree with this finding however, and reported that both oxygen vacancies and tin interstitials are unlikely to contribute to conductivity in SnO_2, due to large calculated defect formation energies and ionisation energies, respectively [76]. They instead propose that interstitial and substitutional hydrogen are responsible for n-type conductivity in SnO_2 due to small calculated values for the defect formation energies and ionisation energies. Further complicating matters, Godinho et al. reported that oxygen vacancies were the most likely intrinsic defect to be responsible for n-type conductivity in undoped SnO_2, according to their own computational results[77].

While the exact defects responsible for the intrinsic n-type conductivity in SnO_2 are still debated, the doping of extrinsic impurities to increase its carrier concentration is better understood. The most common added impurities include Sb[66–68], which substitutes on a Sn site, and F[78–80], which substitutes on an O site. Both of these impurities act as electron donors, adding an additional shallow donor level for each substituted atom that increases the electron carrier concentration and n-type conductivity in SnO_2. Other donor impurities such as Ta[81, 82] and Cl[83, 84] have been investigated, although these studies occur less frequently in the literature.

Attempts have been made to dope SnO_2 with acceptor atoms to achieve p-type conductivity. Commonly used acceptor atoms include In[85, 86], Ga[87, 88], Al[89, 90] and Mg[91, 92], which all substitute on a Sn lattice site. The results from these studies are disappointing however, as the addition of acceptor dopants tends to just produce an increase in material resistivity and low free electron concentrations. Some studies do report p-type conductivity with holes as the majority carrier, however these moderate hole carrier concentrations are accompanied by low hole mobilities that still result in a high material resistivity.

There are two main reasons why p-type conductivity is difficult to achieve in SnO_2. Firstly, the position of the Fermi level within the material bandgap has a direct effect on the intrinsic defect formation energies. Figure 2.4 shows calculated intrinsic defect formation energies in SnO_2 by Kilic and Zunger[59], plotted alongside the position of the Fermi level. This shows that when SnO_2 is p-type (with the Fermi level close to the valence band maximum) there is a negative formation energy

for electron-donating defects like oxygen vacancies and tin interstitials. This means these defects are very likely to occur in p-type material and oppose the formation of high hole carrier concentrations. Secondly, many TCOs exhibit a wide bandgap due to a low valence band maximum associated with O 2p states[93]. In materials with a high oxygen stoichiometry, this tends to result in the formation of localised hole polarons that causes large hole ionisation energies and low hole mobilities. This means that even in heavily acceptor doped SnO_2, low hole carrier concentrations and hole mobilities give high material resistivities.

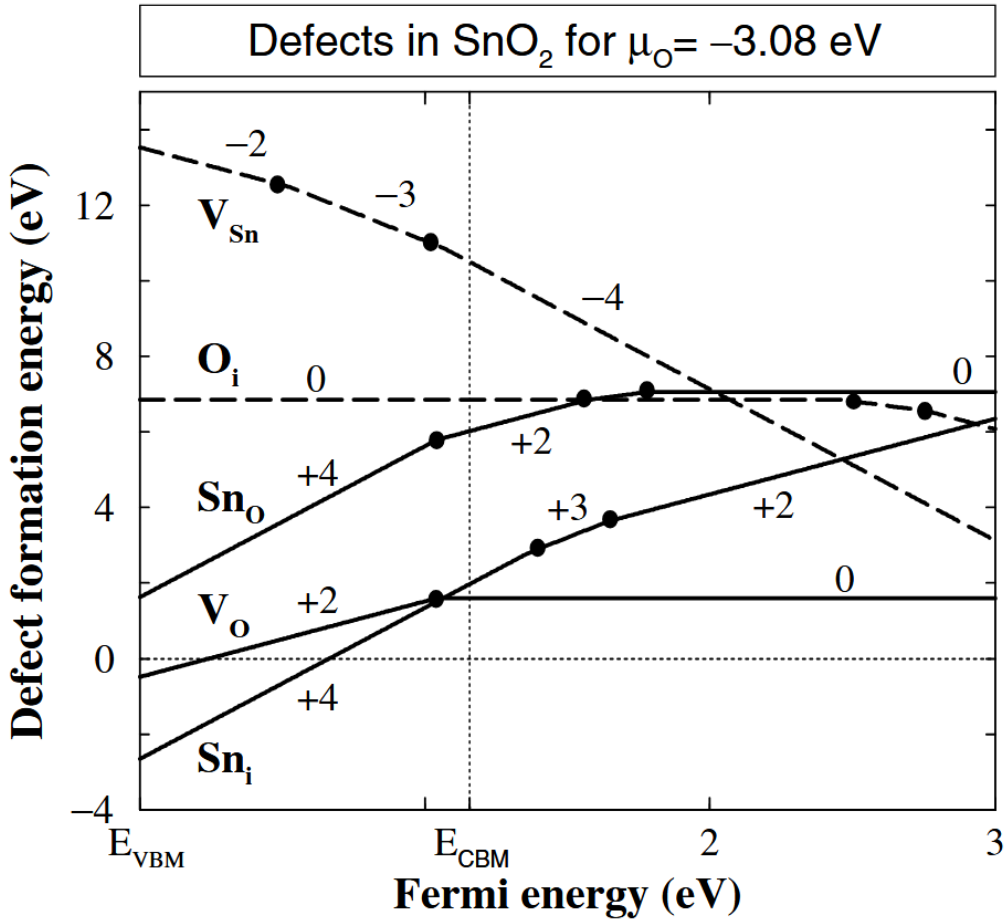


Figure 2.4: Calculated intrinsic defect formation energies in SnO_2 against the position of the Fermi level. Reused with permission from [59].

2.2.4 Band Bending at the SnO_2 Surface

In a crystalline material, the repeating crystal structure gives rise to many of the bulk electrical and optical properties. However, the crystal eventually has to terminate at a surface, which can often adopt a range of different structures that give rise

to different properties compared to the bulk material. One such phenomenon that can occur is known as band bending, where differences in the electrical properties between the surface and the bulk result in the electronic bands bending upwards or downwards relative to the Fermi level. In the case of undoped SnO_2, the material tends to experience downward band bending[94], caused by the presence of electron-donating groups at the surface. This often causes the conduction band minimum to bend below the Fermi level, which results in a significant surface electron accumulation layer known as a 2-dimensional electron gas (2DEG)[94]. An illustrative diagram showing this effect can be seen in Figure 2.5. The sensitivity to different gasses and the catalytic properties of SnO_2 are generally attributed to this 2DEG[49], as the difference in charge density at the surface attracts electrophillic molecules that

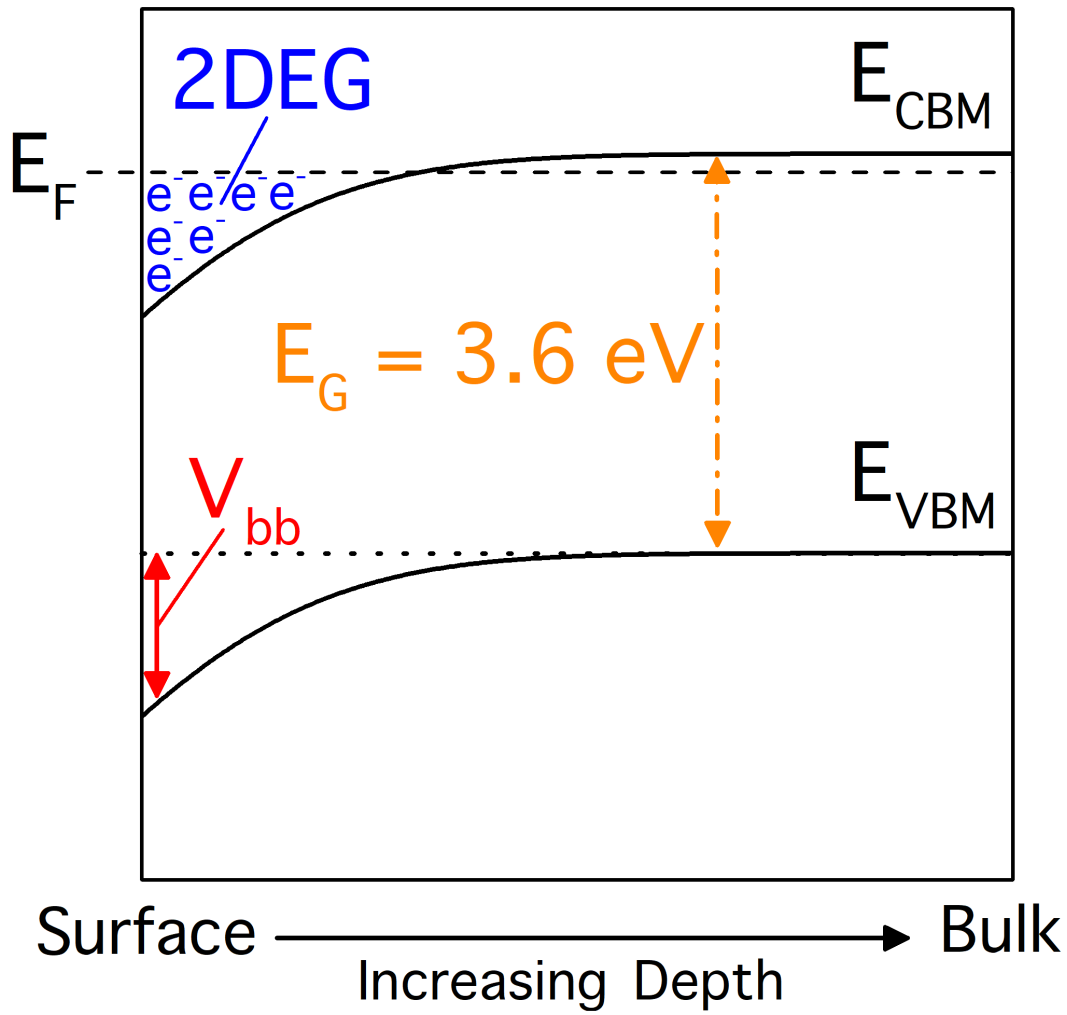


Figure 2.5: An illustrative diagram showing downward band bending at the SnO_2 surface, which results in a 2D electron gas at the surface as the conduction band bends below the Fermi level.

can further interact with the surface.

Several surface species have been associated with the downward band bending present on undoped and moderately doped SnO_2 surfaces. Unlike the stoichiometric surfaces shown in Section 2.2.1, SnO_2 surfaces exposed to air have at least a partial coverage of terminating surface hydroxyl (-OH) groups that arise from the dissociation of water molecules on the surface[95, 96]. These groups have been reported to be electron-donating[30, 70, 97], and are thought to be the main contributor to the downward band bending on SnO_2 surfaces that have been exposed to the atmosphere. Oxygen vacancies on the reduced SnO_2 surface provide another electron-donating defect, which can be produced by annealing at high temperatures in vacuum that causes increased downward band bending[71].

2.3 SnO_2 Nanostructures

Nanostructures of SnO_2 can generally be grouped into one of two categories. The first is nanostructures which are polycrystalline or amorphous, which are typically porous and contain large numbers of grain boundaries. The second category is nanostructures which are single crystal, with an ordered, aligned lattice that remains oriented throughout the nanostructure. These different categories of nanostructures each have their own advantages and disadvantages in various applications.

2.3.1 Polycrystalline and Amorphous SnO_2 Nanostructures

Polycrystalline and amorphous SnO_2 nanostructures have unique structures, which allow for many advantages in specific applications. Many of these structures are constructed out of a large collection of SnO_2 grains, which results in a rough surface and a high surface-to-volume ratio. Furthermore, these structures are also designed to be hollow and porous, which further increases the available surface area. Despite these advantages, however, there are two significant disadvantages that these structures possess. The first is that the high number of grain boundaries causes scattering of the electrical carriers, decreasing carrier mobility and increasing electrical resistivity. The second disadvantage is also due to the large number of grains in these structures, as the bonding between grains is much weaker than in a single crystal

material. This causes these nanostructures to be more fragile than their single crystal counterparts, which complicates the fabrication of devices for various applications.

Polycrystalline and amorphous SnO_2 nanostructures have been created using a number of different techniques, which have resulted in a wide range of nanostructure morphologies. One of the most common methods used to produce these structures are solution-based techniques such as hydrothermal, solvothermal or reflux methods. Many of these nanostructures are created using a templating technique, where a sacrificial nanostructure of another material serves as a template on which a layer of SnO_2 is deposited from the precursor solution. This offers a distinct advantage as a wide range of nanostructured shapes are able to be created depending on the initial shape of the template, many of which do not crystallise naturally in SnO_2.

An example of polycrystalline SnO_2 nanotubes formed by this templated deposition process can be seen in Figure 2.6. Zhang et al. reported the creation of these structures through the use of a carbon nanofibre as a template for hydrothermal SnO_2 deposition[98]. A thermal annealing step removes the carbon nanofibre template, leaving a hollow SnO_2 nanotube with a polycrystalline surface. The surface of these nanotubes was then decorated with a thin carbon layer and used as a Li-ion battery anode with a high cycling stability. Several other types of structures have been deposited using this templating method, such as hollow SnO_2 nanospheres templated from polystyrene beads[99], and hollow SnO_2 nanoboxes constructed from Cu_2O nanocube templates[100].

While there are many unique nanostructure shapes that have been created using a template with hydrothermal, solvothermal and reflux techniques, there are also many examples of nanostructures that form by template-free self-assembly within the precursor solutions. Different precursor chemicals used in hydrothermal, solvothermal and reflux techniques produce different effects which promote the self assembly of nanostructures from precursors dissolved in solution. Some examples of this include the formation of hollow SnO_2 nanospheres by the Oswald ripening method[101], SnO_2 nanotubes formed by the Kirkendall effect[102], SnO_2 octahedra formed by 2D aggregation[103], and SnO_2 nanowires formed by linear polymerisation and van der Waals interactions[104].

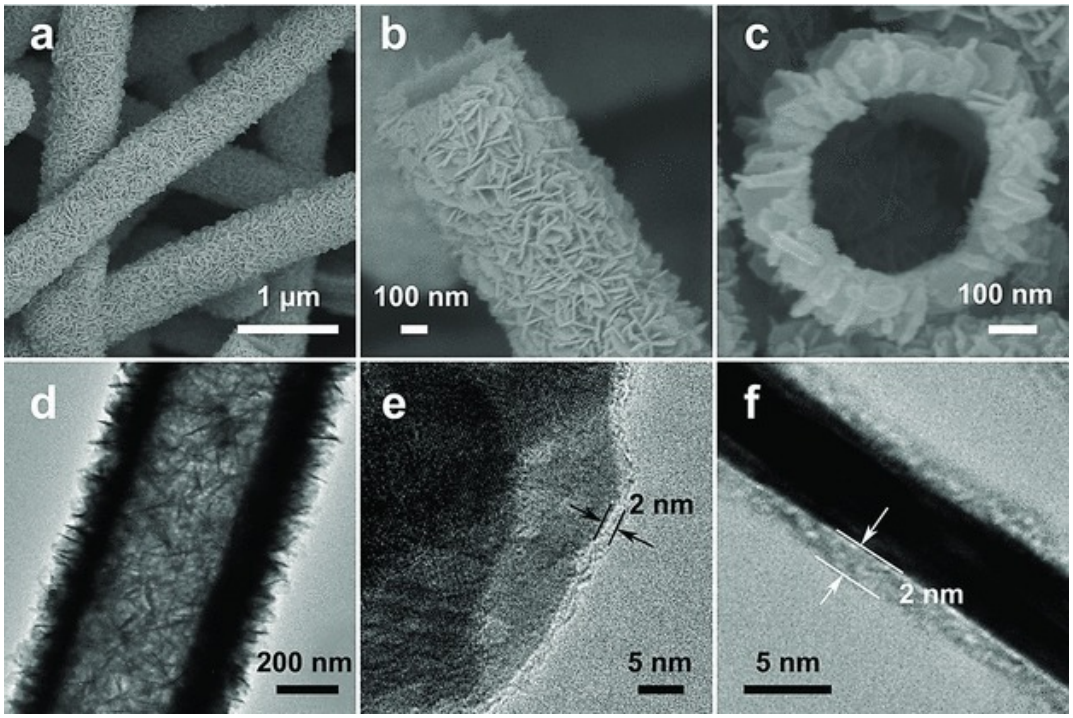


Figure 2.6: (a-c) Scanning electron microscopy and (d-f) transmission electron microscopy images of carbon-coated polycrystalline SnO_2 nanotubes formed by a carbon nanotube templating. Retrieved with permission from [98].

Electrospinning is another technique used to create nanostructured materials, specifically, long and thin nanofibres that are typically amorphous or polycrystalline. In this technique, a precursor solution is pumped out of a spinning tip while exposed to a strong electric field. This strong electric field causes the surface of the fluid to accumulate an electrical charge which opposes the surface tension of the fluid. This causes the fluid to be drawn into a thin fibre as it is spun, resulting in the formation of long and thin nanofibres[105]. These nanofibres can then be annealed to crystallise the fibres. This technique has been used to create nanofibres of SnO_2 for a range of applications[106–108].

2.3.2 Single Crystal SnO_2 Nanostructures

Unlike polycrystalline and amorphous SnO_2 nanostructures, single crystal SnO_2 nanostructures have smooth faces and a consistent crystal lattice throughout the nanostructure, resulting in the absence of grain boundaries. This has benefits for a variety of properties, such as increased structural stability and an increase in carrier mobility when compared to polycrystalline or amorphous nanostructures. Single

crystal nanostructures also typically have facets with a single surface termination, which can be advantageous for catalytic applications as different surface terminations can possess different catalytic properties[109].

There is a wide range of single crystal nanostructure morphologies which have been observed in the literature. The most common types of morphologies are 1-dimensional nanostructures such as nanobelts, nanowires, nanorods and nanotubes. These nanostructure morphologies all have large surface-to-volume ratios and are very surface sensitive while maintaining good electrical properties due to their single crystalinity, so they are popular for sensing, catalytic and energy storage applications.

Nanobelts were one of the first 1-dimensional nanostructures of SnO_2 to be fabricated. These structures are typically fabricated using a thermal evaporation method (also known as vapor-phase deposition). In this technique, powders of SnO_2 or Sn are evaporated at elevated temperatures (800-1350°C) and the vapor is collected on a substrate. Vapor-solid growth then results in the formation of SnO_2 nanobelts, such as in Figure 2.7.a. These nanobelts typically grow along the (101) lattice direction[110–113], however, growth along other lattice directions such as (100) has been observed[114]. Variations in the growth parameters and gas flow have also been observed to produce nanobelts in a zig-zag pattern[110, 115].

SnO_2 nanowires are the most common single crystal SnO_2 nanostructures in the literature, and an example of these structures can be seen in Figure 2.7.b. One possible reason for their popularity over other SnO_2 nanostructures is due to the wide variety of techniques that can produce them. SnO_2 nanowires have been synthesised by thermal evaporation[116–119], different varieties of chemical vapour deposition (CVD)[120–123], hydrothermal methods[124, 125], electrodeposition[126] and pulsed laser deposition[127, 128]. Unlike SnO_2 nanobelts, which grow using a catalyst-free vapour-solid mechanism, many reports on SnO_2 nanowires utilise a catalyst to promote nanowire growth. The most common catalyst material is Au, where a thin layer (2-50 nm) is usually deposited on a substrate such as Si or sapphire before the nanowire growth. At the beginning of nanowire growth, the thin Au

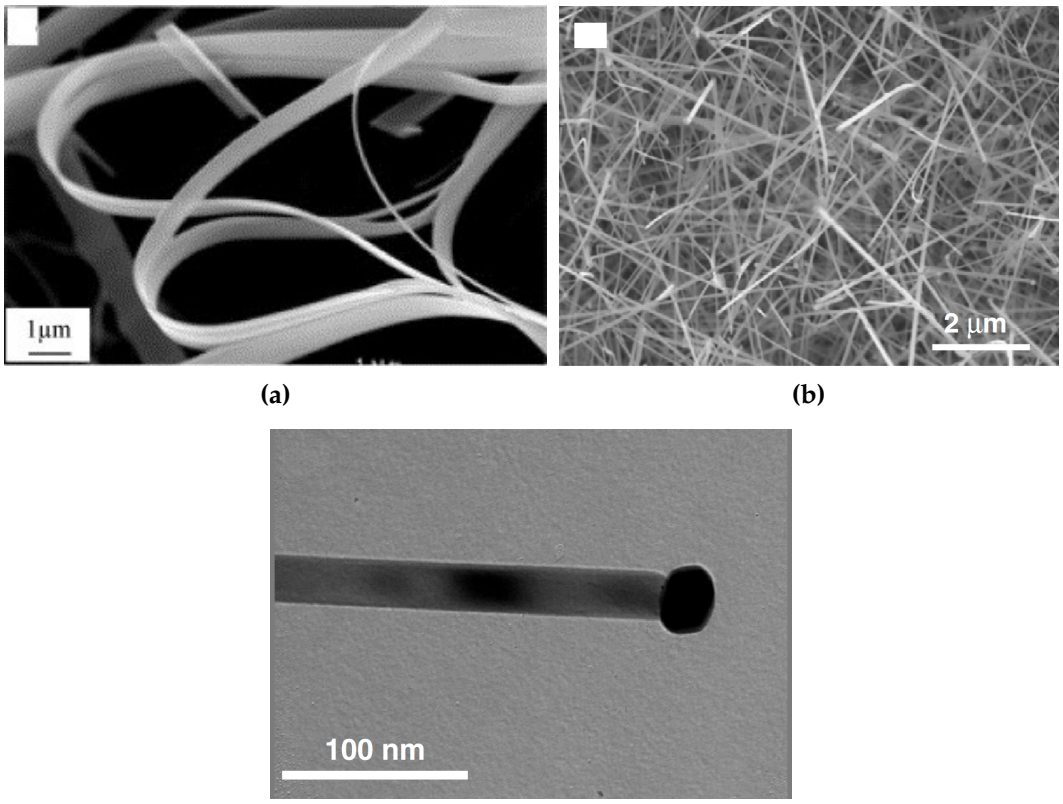
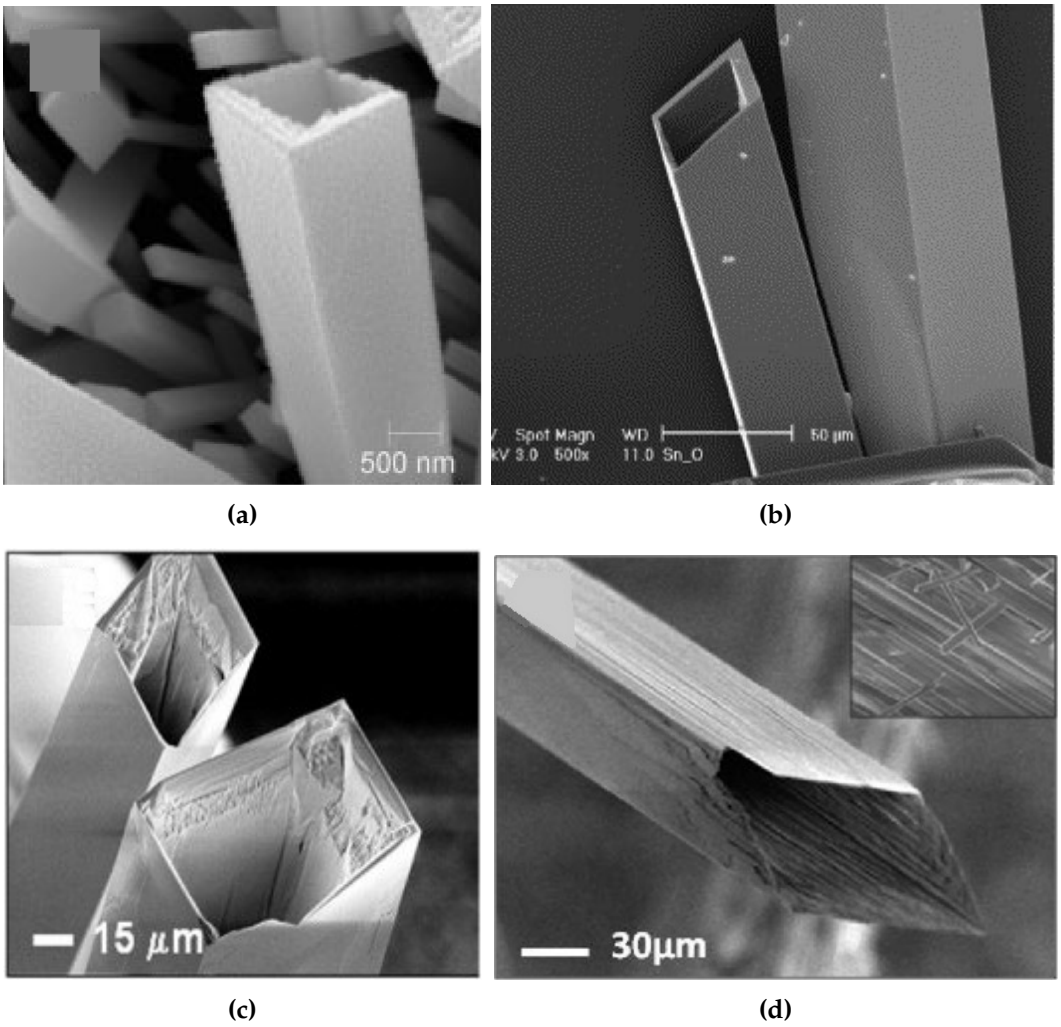


Figure 2.7: Scanning electron microscopy images of (a) SnO_2 nanobelts and (b,c) SnO_2 nanowires. (c) shows the Au nanoparticle at the tip of a nanowire which has grown by the VLS mechanism. Retrieved with permission from (a)[111] and (b,c)[116].

layer dewets into Au nanoparticles, which absorb SnO_2 vapour until the nanoparticle becomes supersaturated with SnO_2. SnO_2 is then deposited underneath the Au nanoparticle by precipitation, resulting in nanowire growth (Figure 2.7.c). This mechanism for nanowire growth is known as the vapour-liquid-solid (VLS) mechanism[129]. Although Au is the most common metal catalyst, several other metals have been observed to follow the VLS mechanism, including Ag[130] and Sn[131]. The wide variety of growth techniques used to create SnO_2 nanowires has resulted in a variety of crystal growth orientations. SnO_2 nanowires have been grown along the (001), (101), (110) and (100) directions[118, 120, 123, 125, 127].

There are only a limited number of reports in the literature which cover hollow single crystal SnO_2 nanostructures such as nanotubes. The first hollow-tubular nanostructures were reported by Liu et al., who reported the synthesis of well aligned "Nano-box beams" of SnO_2 (Figure 2.8.a)[132, 133]. These nanostructures were 0.5-2

µm in width and up to 7 µm in length, with rectangular and square cross-sections. The four square faces of the nanostructure were determined to be the (110) surface termination, indicating that the nano-box-beams grew along the (001) direction. These nanostructures were formed using combustion CVD, in which an organic solvent with a dissolved tin precursor was atomised and combusted using a flame at a system temperature of 950-1150°C. The resulting vapours were collected on quartz substrates, on which the nano-box-beams formed.



(a) **(b)**

(c) **(d)**

Figure 2.8: Scanning electron microscopy images of various single crystal SnO_2 square (a) nanotubes and (b-d) microtubes. Retrieved with permission from (a) [132], (b) [134], (c) [135] and (d) [136].

Duan et al. have also reported square tubular structures of SnO_2, in the form of square microtubes (Figure 2.8.b)[134]. These microtubes were considerably larger than those presented by Liu et al., with microtube widths and lengths of 20-80 µm and 1 mm respectively. Similarly, however, the side faces of the microtubes were

(110) terminated, with the microtube growing along the (001) lattice direction. The growth of the microtubes was performed in an optical float zone system operating at 1550°C, with a stick of sintered SnO_2 acting as both the SnO_2 source and the substrate.

SnO_2 microtubes have also been reported by Maestre et al., who sintered compacted SnO_2 powders at 1400°C under Ar gas flow (Figure 2.8.c)[135]. These microtubes had widths on the order of 50-150 µm, with the length unreported. Interestingly, the exterior faces of these microtubes were (010) and $(10\bar{1})$ terminated, with the microtube growing along the (101) direction.

The most recent report in the literature on SnO_2 hollow single crystal microtubes is from García-Tecedor et al. in 2016, who also sintered compacted SnO_2 powders at 1400°C (Figure 2.8.d)[136]. These microtubes had widths of tens of micrometres and lengths of hundreds of micrometres. More interestingly, despite very similar growth procedures to Maestre et al., the exterior faces were all (110) terminated with microtube growth along the (001) direction.

2.4 Applications of SnO_2 Nanostructures

Many of the nanostructures covered in Section 2.3 have been utilised in a wide variety of applications that benefit from their enhanced surface-to-volume ratios. This section covers the different device structures that have been created using SnO_2 nanostructures and a range of different applications of the SnO_2 nanostructures present in the literature.

2.4.1 Electronic Devices of SnO_2 Nanostructures

Many applications require the electrical characteristics of SnO_2 to be monitored in order to detect changes in the local environment, such as the presence of gas or UV light. This requires electrical devices which incorporate the SnO_2 nanostructures to be fabricated. This presents a new challenge, as electrical contacts need to be made to the nanostructured material. Typically, this involves placing a one-dimensional nanostructure such as a nanowire on an insulating substrate, before performing

lithographic patterning and the deposition of metal layers to create the electrical contacts. By using and expanding upon this method, a range of simple and more advanced electrical devices can be made that incorporate SnO_2 nanostructures, such as resistive sensors, transistors and Schottky diodes. All of these electrical devices each have their own advantages in different applications.

Resistors

The simplest electrical devices made using SnO_2 nanostructures involve a nanostructure that has an ohmic electrical contact on each end, allowing current to freely flow through the nanostructure under an applied voltage. This forms a resistor using the nanostructure, with the electrical resistance of the nanostructure dominating the electrical properties of the device. As the resistance of the SnO_2 nanostructure can change in the presence of external stimuli, such as various gasses and UV light, this electrical device can detect these stimuli and act as a sensor. The largest downside of this device is the comparatively high currents that occur when there are no external stimuli. This often results in a low sensor sensitivity as the increases or decreases in the nanostructure resistance are relatively small. Despite this disadvantage, there are numerous reports in the literature that utilise this type of device for gas sensing[116, 123, 137, 138] and UV sensing[120, 139].

Transistors

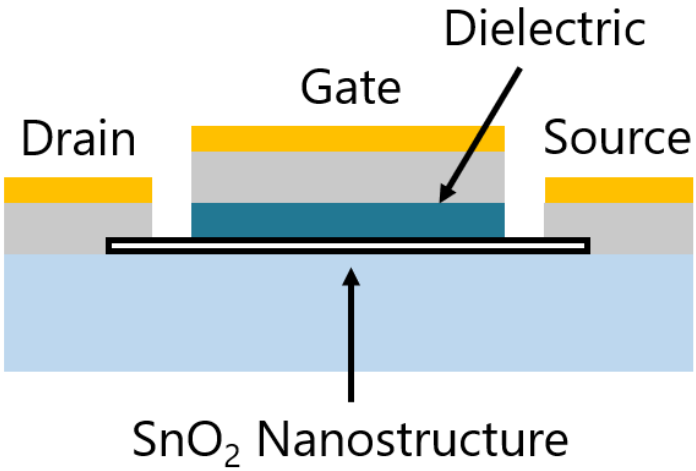
Field-effect transistors (FETs) are another device that can be created using SnO_2 nanostructures. Like the resistor-based device, this device has two ohmic contacts on each end of a SnO_2 nanostructure, known as the source and drain electrodes. However, there is a third contact present in the device, known as the gate. The gate is electrically isolated from the SnO_2 nanostructure by a thin layer of insulating dielectric material, such that when a voltage is applied to the gate an electric field is produced across the SnO_2 nanostructure. This electric field causes a field-effect, which is able to deplete or enhance the SnO_2 nanostructure of carriers depending on the magnitude and polarity of the charge on the gate. If a voltage is applied across the drain and source, the resulting current through the nanostructure is able to be modulated by changes in the applied gate voltage. The ability to control the current through the nanostructure is the basis for sensing applications, as an optimal operating current

can be chosen that maximises the sensor response. For example, FET-based sensors often operate with the gate voltage near the device threshold voltage, as electrical interactions to the nanostructure produce large changes in the drain-source current. FETs constructed from SnO_2 nanostructures are widely present in the literature[140–146].

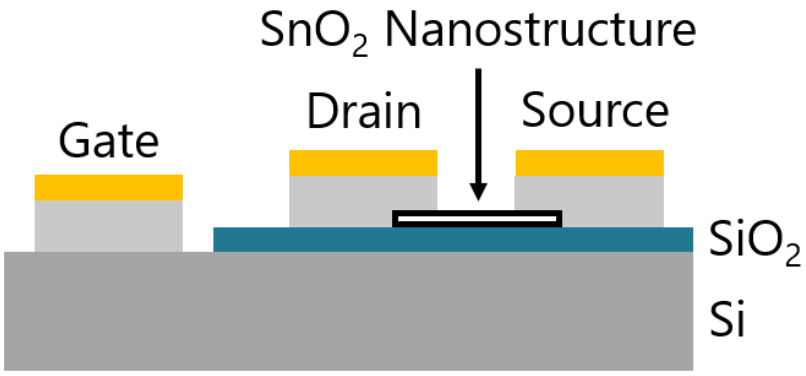
There are a number of different device structures that can be used to create field-effect transistors. A commonly used device structure is known as a top-gated FET, where the insulating dielectric material and metal gate contact is deposited on top of the SnO_2 nanostructure. A diagram of this device structure can be seen in Figure 2.9.a. The deposition of the dielectric on top of the nanostructure generally results in a more stable device as gaseous molecules cannot interact with the SnO_2 surface. This does mean that this device structure cannot generally be used for gas sensing or biosensing applications, but is better employed in transparent electronics[147, 148].

A simpler FET structure that is employed in the literature is known as a back-gated FET. In this device structure a conducting substrate is used to produce the electric field in the SnO_2 nanostructure through a dielectric layer, as seen in Figure 2.9.b. A conducting Si substrate is typically employed as the conducting layer as the surface of the substrate can be easily thermally oxidised to SiO_2, which acts as an excellent insulating dielectric. This removes the need for the deposition of the dielectric layer, which can often be difficult to produce with a sufficiently high quality. Furthermore, the surface of the SnO_2 nanostructure is exposed in this device structure, which allows the device to be used in gas sensing and biosensing applications[140, 149].

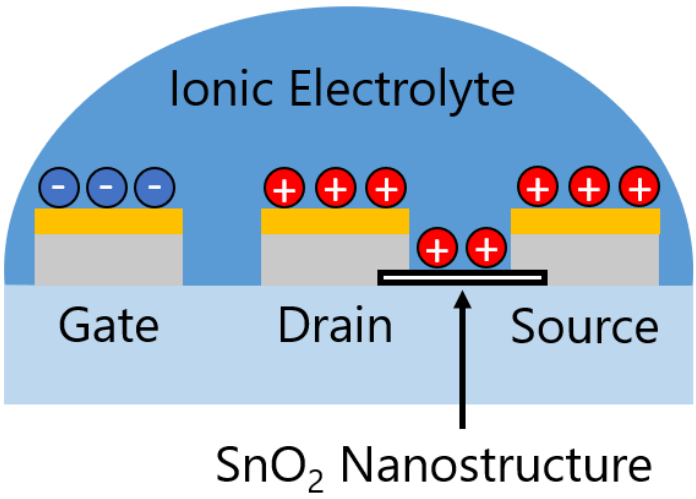
A third transistor structure that is growing in popularity among researchers is known as an ionic-gated transistor. This device structure operates differently to that of a typical FET, in that an insulating dielectric is not used between the gate and the conducting channel. Instead, an ionic electrolyte is used in direct contact with both the gate and the conducting channel. When a voltage is applied to the device gate, the electric field created causes the migration of ions in the electrolyte that opposes the electric field. This causes an accumulation of ions of one polarity at the device gate and an accumulation of ions of the opposite polarity at the SnO_2 channel, as seen in Figure 2.9.c. This accumulation of ions at the gate and channel



(a) Top-gated FET



(b) Back-gated FET



(c) Ionic-gated Transistor

Figure 2.9: The different transistor structures that utilise SnO_2 nanostructures.

is known as an electric double layer (EDL)[150]. In the SnO_2 channel, electronic charges accumulate due to the large electrostatic potential created by the ions at the surface of the channel. In this way, the number of charge carriers in the SnO_2 channel can be depleted or enhanced by applying gate voltages that cause different accumulations of ions at the SnO_2 channel. A significant advantage of this device structure is that the EDL is not separated from the channel by a dielectric, and so large electric fields can be produced even with applied gate voltages much smaller than those used in conventional FET structures. This leads ionic-gated transistors to have a low overall power consumption, with potential applications in low-power devices, neuromorphic applications and biosensing[150–153].

Schottky Diodes

Another type of electrical device that can be fabricated using SnO_2 is a Schottky diode. In this device, the junction between an n-type semiconductor and a metal with a sufficiently large work function forms a rectifying junction known as a Schottky barrier, where the surface electronic bands of the semiconductor bend upward at the interface between the two materials (Figure 2.10). The height of the Schottky barrier, Φ_B, is typically modelled by the Mott-Schottky relationship, where the barrier height is equal to the difference between the size of the metal work function and the semiconductor electron affinity[154]. However, deviations from this relationship are often observed as surface effects such as Fermi level pinning and image-force lowering tend to have a large effect on the effective barrier height[155, 156]. The upward band bending produced at the Schottky junction forms a depletion region in the semiconductor that is devoid of charge carriers, which can then be modulated by applying a voltage bias across the metal-semiconductor junction. Under a sufficiently large forward bias, the potential of the semiconductor with respect to the metal is increased and the effective barrier height decreases, allowing current to flow across the junction. Under a reverse bias, however, the potential of the semiconductor with respect to the metal is decreased and the effective barrier height increases, widening the depletion region and preventing the flow of current. Effectively, a Schottky diode only allows current to flow in one direction through the device, with the exception of a small amount of leakage current when the device is

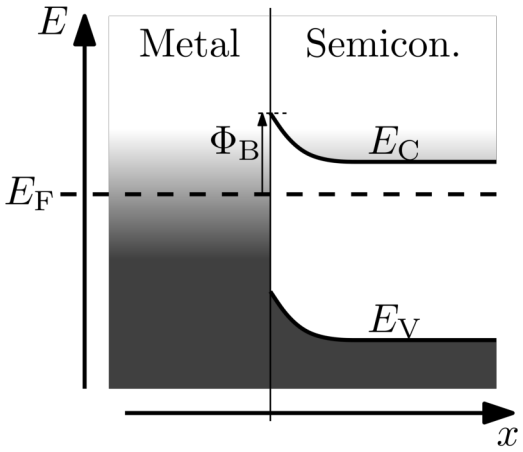


Figure 2.10: A band diagram of a Schottky barrier formed by a metal and an n-type semiconductor, such as SnO_2. Retrieved with permission from [157].

reverse biased.

Schottky diodes that utilise SnO_2 are not that common in the literature, with only one report that utilises SnO_2 nanostructures. Chen et al. fabricated a Schottky diode using an SnO_2 nanowire with a Pt Schottky contact[158]. This device had a rectification ratio (the ratio of forward to reverse bias current) of ~10^3, indicating that a reasonable Schottky barrier had been created. More reports have been published for Schottky diodes fabricated using thin films of SnO_2, with Schottky contact metals that include Au[159, 160], Pt[160, 161], Pd[160] and Ni[162]. These reports, however, typically involve Schottky diodes with less-than-ideal properties, which is indicative of Schottky contacts that contain a relatively large amount of defects and inhomogeneities. It is known to be particularly difficult to form high quality Schottky contacts to SnO_2, as the intrinsic downward band bending present at the material surface opposes the upward band bending created by the Schottky contact[94]. In spite of this, Dang et al. reported high quality Schottky contacts to SnO_2 thin films that were significantly better performing than other reports in the literature[163]. These devices utilised an oxidised AgO_x layer as the Schottky contact. Oxidised noble metal Schottky contacts such as AgO_x, PtO_x, PdO_x and IrO_x have been previously reported to create high quality Schottky contacts to other metal oxides such as β-Ga_2O_3[164, 165] and ZnO[166, 167] due to their high work functions and their ability to reduce the formation of oxygen vacancies at the Schottky junction. This

suggests that these materials could be used to produce high quality Schottky contacts to SnO_2 nanostructures.

2.4.2 Gas Sensing

Semiconducting metal oxides are some of the most commonly used gas sensing materials. This is due to a variety of factors, such as low cost, high sensitivity and controllable electrical characteristics. SnO_2 is one of the most popular metal oxides used in gas sensing, due to its intrinsic n-type conductivity and high thermal and chemical stability[168, 169]. The fundamental gas sensing mechanism of SnO_2 is based on the chemical and subsequent electrical interaction between a gas molecule and the SnO_2 surface. As mentioned in Section 2.2.4, defects and dissociated water molecules on the SnO_2 surface contribute to downward band bending and surface electron accumulation. In air, atmospheric oxygen is attracted towards this surface and is adsorbed, capturing an electron from the surface and forming an ionosorbed superoxide (O_2^-) ion[41, 170]. Reducing gases (such as H_2, CO or H_2S) can react with this adsorbed O_2^- ion, removing it from the surface and releasing the trapped electron which increases the surface conductivity[169]. Oxidising gasses (such as NO, NO_2 and O_3) operate in the opposite manner, adsorbing onto the surface of SnO_2 and trapping further electrons. This results in a decrease in the surface conductivity of the SnO_2 gas sensor. As the changes in electrical conductivity are occurring at the surface of SnO_2, nanostructured devices with large surface-to-volume ratios exhibit greater conductivity changes and subsequently have higher sensitivities.

In addition to adsorbed oxygen, water vapour is also able to adsorb onto the SnO_2 surface which causes changes in the surface conductivity[40]. As the day-to-day humidity often changes, this can affect the baseline conductivity of gas sensors without the presence of other gases. To avoid this problem, SnO_2-based gas sensors are generally operated at elevated temperatures between 200-400 °C, which prevents the adsorption of water vapour but still allows oxidising and reducing gases to interact with the SnO_2 surface.

The first reports of 1D SnO_2 nanostructures used in gas sensors utilised nanobelts of SnO_2. Law et al. used nanobelts synthesised by thermal evaporation to detect NO_2

gas at concentrations as low as 2 parts-per-million (ppm) and achieved response ratios over 8 with 100 ppm NO_2[171]. Similarly, Comini et al. utilised nanobelts to detect various concentrations of CO, NO_2 and ethanol. Their device had a response ratio of 2 for 500 ppm CO, 41.6 for 250 ppm ethanol, and 15.5 for 0.5 ppm NO_2[138].

Recent reviews in the literature demonstrate the advances made in gas sensors that utilise SnO_2 nanostructures. SnO_2 gas sensors have been used to detect a wide range of dangerous gases, such as acetone[172], H_2[173], CO[174], CO_2[175], H_2S[176] and other flammable and explosive compounds[169]. However, this wide range of detectable gases is also one of the leading disadvantages of SnO_2-based gas sensors. As SnO_2 can detect a wide range of gases, this results in a low selectivity which can prevent the immediate identification of the absorbed gas and the gas concentration. Different approaches have been taken to enhance the selectivity and sensitivity of SnO_2 gas sensors, such as through the doping of SnO_2 with different elements[177, 178], and by the creation of a range of heterostructures with other materials[179–181]. The sensitivity and selectivity of SnO_2 nanostructure gas sensors can also be improved by the decoration of the SnO_2 surfaces with noble metal nanoparticles, as recently reviewed by Zhu et al.[182].

2.4.3 UV Sensing

A common application for SnO_2 nanostructures is in UV photodetectors. As SnO_2 has a wide bandgap of ~3.6 eV, incident photons with a wavelength longer than ~345 nm have only small effects on the conductivity of SnO_2, primarily due to defects that create electronic states within the bandgap. Photons with a wavelength shorter than 345 nm have a much larger effect on the conductivity of SnO_2, as these photons have energies larger than the bandgap. When an incident photon with energy larger than the bandgap enters SnO_2, an electron near the top of the valence band may absorb the photon and be promoted into the conduction band, leaving behind a hole in the valence band. This electron-hole pair created by the incident photon increases the conductivity of SnO_2, however, the electron-hole pair is also able to recombine, reducing the conductivity to its original state. This means that under UV illumination, the conductivity quickly increases as electron-hole pairs are generated. Eventually,

the increase in conductivity saturates as the recombination rate of electron-hole pairs matches the generation rate of electron-hole pairs. Finally, when the UV source is removed electron-hole pairs are no longer being generated, allowing the remaining electron-hole pairs to recombine and the conductivity to return to its initial state.

SnO_2 nanostructures have a particular advantage for UV photodetector applications, which is due to the extremely small sizes of the nanostructures. The intensity of light as it passes through an absorbing material can be described by the Beer-Lambert law, where the intensity of light decreases exponentially with the distance the light travels through the material[183]. This means that UV absorption occurs primarily near the surface of SnO_2. With large surface-to-volume ratios, SnO_2 nanostructures achieve higher sensitivities to UV light. This effect can be further amplified by incorporating the SnO_2 nanostructures in devices such as Schottky diodes, as the built-in potential of the Schottky barrier is able to separate the electron-hole pairs, preventing the competing recombination mechanism.

The most significant disadvantage of using SnO_2 in UV photodetectors arises from an effect known as persistent photoconductivity (PPC), which significantly increases the recovery time of the photodetector. In this effect, the photocurrent of an SnO_2 UV photodetector rises quickly when UV light is first illuminated on the device; a desired result. Although, the rise then becomes increasingly slow towards saturation, which is undesired. When the UV illumination stops, the induced photocurrent falls much more slowly compared to the initial photocurrent rise, such as in Figure 2.11, which is even more undesired. Because of this, integrating the photodetector signal leads to a large overestimation of the UV dose, preventing applications in UV dose monitoring. The prevailing theory in the literature attributes this effect to the UV-induced desorption of adsorbed oxygen molecules[184–186]. In this case, holes from photogenerated electron-hole pairs recombine with electrons trapped by the oxygen molecules (in the form of superoxide ions), releasing the oxygen molecules from the surface. This leaves free photogenerated electrons in the conduction band which increases the photodetector conductivity. The process of oxygen desorption occurs very quickly when the photodetector is first illuminated, leading to the fast initial photocurrent rise. When the UV illumination stops, the desorbed oxygen has

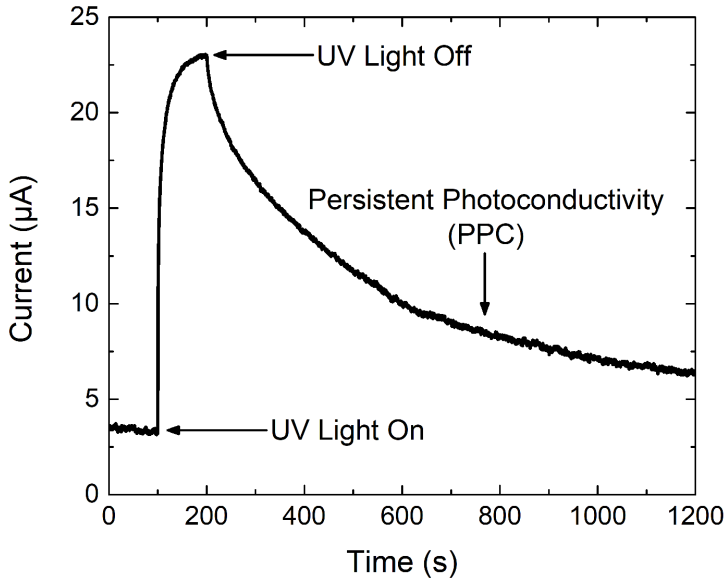


Figure 2.11: Persistent photoconductivity in an SnO_2-based UV photodetector, demonstrating the fast photocurrent rise under UV illumination and slow photocurrent decay afterwards.

to readsorb back onto the SnO_2 surface and recapture conduction band electrons to return the photodetector to its initial conductivity state. This readsorption process occurs on a much longer time scale compared to the desorption process, and persistent photoconductivity occurs as a result. While this is the prevailing theory on the cause of PPC in the literature, other origins have also been proposed, such as a potential role of water vapour[187] or contributions from mid-gap trapping states[188].

Persistent photoconductivity has been reported in tin oxide materials as early as 1958[189], and is currently one of the primary limitations preventing SnO_2 from widespread adoption in UV photodetector applications. As SnO_2 nanostructures have large exposed surface areas, UV photodetectors using SnO_2 nanostructures are severely affected by this process.

2.4.4 Neuromorphic Applications

In recent years, there has been a growing interest in neuromorphic computing to combat rising issues such as the end of Moore's Law[190] and the increased power consumption of generative and large-language model AI technologies[191, 192]. In

particular, spiking artificial neural networks are a promising future device architecture, as the sparse, event-driven processing of these devices is believed to enable lower power consumption and higher efficiency computation[193, 194]. However, biological systems are highly non-linear and are difficult to emulate in inorganic systems, and significant work is still required to create the fundamental devices for these systems that operate in a scalable and reliable way.

Artificial synapses are a fundamental component of neuromorphic systems, which connect neurons to other neurons creating a hugely interconnected neural network. These devices demonstrate a range of unusual behaviours such as short- and long-term plasticity, where the devices are capable of retaining elements of their previous electrical state in a time-dependant fashion. Artificial synapses are generally constructed in two different device structures: two-terminal memristors and three-terminal synaptic transistors.

Two-terminal memristors are a simple synaptic device typically constructed with two metal contacts to an active material layer. This device has two different resistance states: a high resistance state (HRS) and a low resistance state (LRS). The default state of this device is typically the HRS, where small applied voltages do not result in significant current through the device. Under a large enough voltage bias however, the device transitions from the HRS to the LRS, dramatically increasing the current through the device. Once the LRS has been set, this state persists even at the small applied biases that previously resulted in little current through the device. Transitioning back to the HRS typically requires a reset process, such as a negative applied bias. SnO_2 nanostructures have been incorporated into memristors as the active channel material. In particular, the resistive switching effect tends to occur when SnO_2 nanomaterials are contacted with metals that form back-to-back Schottky contacts[195–199]. In these devices, Fowler-Nordheim tunnelling occurs at certain bias voltages and leads to large amounts of current hysteresis that forms the HRS and LRS.

Synaptic transistors are the other neuromorphic devices that emulate synaptic behaviours. In these devices, the transistor gate experiences time-dependent hysteresis effects caused by phenomena such as charge accumulation, which can be used

to produce time-dependent neuromorphic behaviours such as an excitatory postsynaptic current and paired-pulse facilitation. These time-dependent gate hysteresis effects can be achieved through the use of ionic-gated transistors, as the diffusion and accumulation of ions at the channel is a time dependent process. Ionic-gated transistors that utilise SnO_2 nanostructures have been reported in the literature[200, 201]. Field-effect transistors can also be used to create synaptic transistors, as demonstrated by Gou et al.[153]. They created a back-gated FET with a chitosan dielectric layer and an SnO_2 nanowire channel that demonstrated several synaptic behaviours.

Persistent photoconductivity in SnO_2 has also been used in synaptic transistor-like devices, where the "gate" of the device is a UV source[202, 203]. In this case, the persistent photoconductivity encountered by UV photodetectors closely resembles an excitatory postsynaptic current, allowing neuromorphic behaviours such as paired-pulse facilitation to be emulated.

2.4.5 Li-Ion Batteries

Lithium-ion batteries are used in a large number of electronic devices, such as smartphones, laptops and electric vehicles. A significant amount of research has investigated various electrode materials for improving the performance of lithium-ion batteries. Two characteristics of lithium-ion batteries primarily determine the performance of the battery: charge capacity and cycling performance. Charge capacity determines the amount of energy that can be stored per unit mass of material, with higher charge capacity resulting in more energy and longer device runtime before needing to be recharged. Cycling performance analyses how the charge capacity of the battery degrades as it is charged and discharged over its lifetime. An ideal battery would have high charge capacity while suffering little capacity degradation over a large number of charge/discharge cycles.

SnO_2 has been investigated as an anode material for lithium-ion batteries. The reaction that occurs at the anode can be seen in the following equations[204]:

$$SnO_2 + 4Li^+ + 4e^- \rightarrow Sn + 2Li_2O \tag{2.1}$$

$$Sn + xLi^+ + xe^- \leftrightarrow Li_xSn \ (0 \leq x \leq 4.4) \tag{2.2}$$

Equation 2.1 shows an irreversible reaction that occurs during the first charge cycle, in which SnO_2 reacts with Li^+ ions and is reduced to Sn metal alongside the formation of Li_2O. Equation 2.2 shows the reversible reaction that occurs during the charge/discharge cycle, where Sn metal reacts with Li^+ to form an alloy of Li_xSn. This reversible reaction has a large specific charge capacity of up to 780 mAh g^{-1}[204–206]. Conventional graphite anodes used in commercial lithium-ion batteries have a lower specific charge capacity of 372 mAh g^{-1}[207, 208], so SnO_2 appears to be a promising material for lithium-ion battery anodes. A significant disadvantage of SnO_2 in lithium-ion batteries, however, is that the anode material expands up to 300% in volume during charging[204, 209, 210]. This causes large mechanical stresses, which results in degradation of the material and poor cycling performance. SnO_2 nanostructures have been proposed as a promising way to overcome this downside, as they are better able to withstand the increases in volume[206], improving the cycling performance compared to thin films of SnO_2 while maintaining the high specific charge capacity.

2.4.6 Catalysis

SnO_2 has wide ranging applications as a catalyst material, particularly as an electroreduction catalyst for the conversion of CO_2 to formic acid (HCOOH). Formic acid is an important chemical for various industrial and agricultural processes, and can also be used in formic acid fuel cells. Currently, formic acid is produced commercially from fossil fuels, resulting in a high amount of carbon emissions. The electrocatalytic reduction of CO_2 to formic acid is a promising carbon-negative technique that could replace current production techniques. SnO_2 nanostructures are promising electrocatalytic reduction materials due to the high surface area producing large numbers of catalytic sites. These catalytic sites reduce CO_2 to HCOOH in

the following manner[211]:

$$CO_2 + 2H^+ + 2e^- \rightarrow HCOOH \tag{2.3}$$

This is the standard reaction for producing formic acid, however, there are two other reactions that can also take place which produce undesirable byproducts. The first is the electrolysis of two protons, known as the hydrogen evolution reaction, which produces H_2 gas:

$$2H^+ + 2e^- \rightarrow H_2 \tag{2.4}$$

The second reaction reduces CO_2 to CO and H_2O:

$$CO_2 + 2H^+ + 2e^- \rightarrow CO + H_2O \tag{2.5}$$

The ability of an electrode to catalyse the formation of formic acid over CO and H_2 is called the selectivity of the electrode. SnO_2 has been shown to be a promising candidate for formic acid production, with multiple studies demonstrating a high selectivity towards formic acid production[212–215].

In addition to CO_2 reduction, SnO_2 has been used in a number of other catalytic applications. Examples in the literature include the photocatalytic degradation of organic dyes[216–218], the oxidation of CO[219–221] and the treatment of various chemicals in wastewater[222–225].

2.5 Previous Work on the Square SnO_2 Nanotubes

In recent years, a process to create square, hollow nanotubes of SnO_2 by Mist-CVD was discovered by Dr Jonty Scott[29]. These square nanotubes were produced by initially depositing a ~2 nm Au film on Si substrates by electron-beam evaporation. The substrate was then placed into a furnace tube at a growth temperature of 650°C. This resulted in the thin Au film dewetting into Au nanoparticles that covered the substrate surface, which acted as the seeding catalyst for nanotube growth. A precursor solution consisting of tin (II) chloride ($SnCl_2$) and hydrochloric acid (HCl) dissolved in deionised water was then ultrasonically atomised into a fine mist and

carried into the heated furnace tube by N_2 gas. This simple procedure resulted in the growth of square, hollow nanotubes of SnO_2, seen in Figure 2.12.

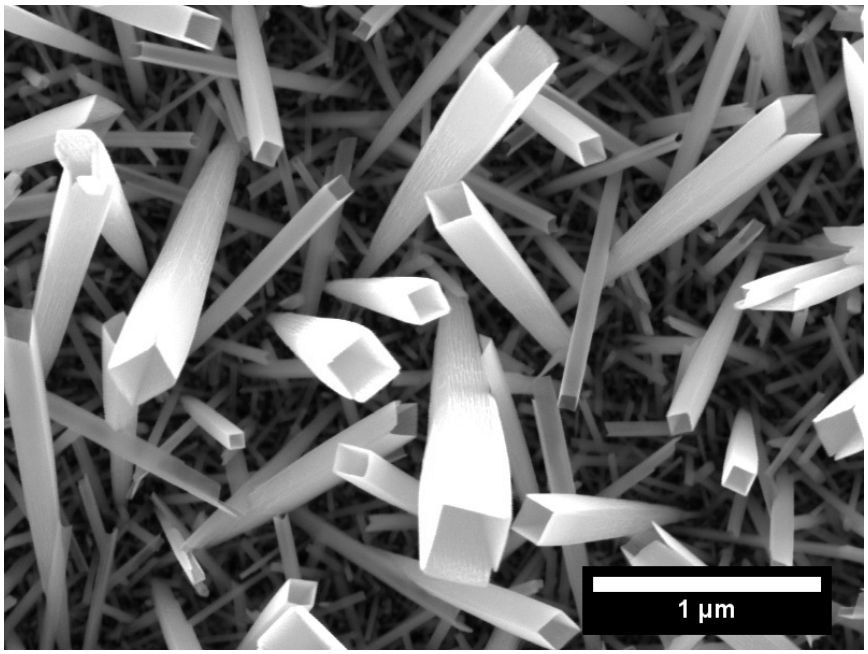


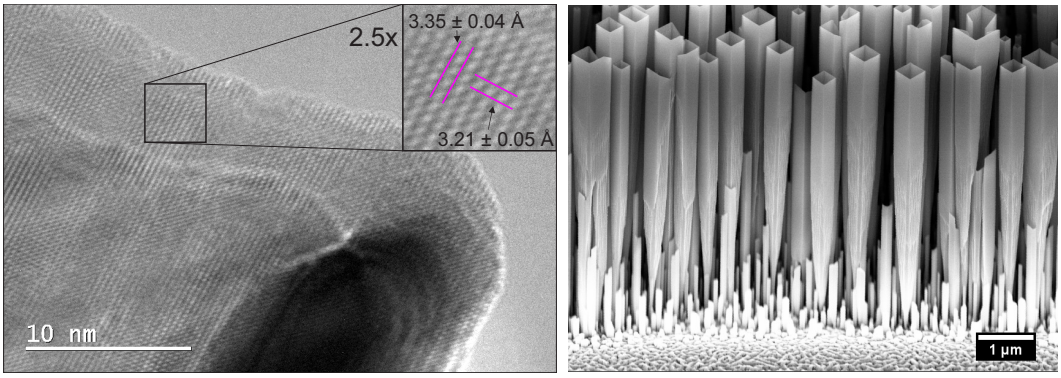
Figure 2.12: A scanning electron microscope image of square SnO_2 nanotubes grown on a (111) Si substrate for 80 minutes by Mist-CVD.[29]

Various structural, chemical and electrical studies were undertaken by Scott on the SnO_2 nanotubes, and the detailed findings of these investigations can be found in his PhD thesis[29]. The key results from that body of work are summarised here. Initial studies on the SnO_2 nanotubes focused on optimizing the growth parameters for nanotube formation. Growth parameters that were varied and optimized include the tin (II) chloride concentration in the precursor solution, the deposition temperature, the Au seeding layer thickness and the growth substrate. The optimised values for the growth parameters alongside the range of parameters that resulted in nanotube growth can be seen in Table 2.2.

Table 2.2: Optimised parameters for the growth of square SnO_2 nanotubes by Mist-CVD.

Growth Parameter	Optimised Value	Growth Range
$SnCl_2$ Concentration	0.0075 M	0.005-0.01 M
Growth Temperature	650°C	550-750°C
Au Thickness	2 nm	0.9-3.8 nm
Growth Substrate	Si, quartz, m-plane Sapphire	-

Structural studies were undertaken by several techniques in order to determine the crystal structure and orientation of individual nanotubes. Transmission electron microscopy (TEM) revealed that the square SnO_2 nanotubes consisted of single crystalline rutile SnO_2, which was further confirmed through X-ray diffraction measurements. Lattice spacings measured from TEM images revealed the nanotubes grew along the (001) lattice direction and had four equivalent exterior faces terminated with the (110) surface (Figure 2.13.a). M-plane sapphire had been previously shown to promote epitaxial growth along the (001) direction when used as a growth substrate for SnO_2 thin films[29] and was subsequently used as a substrate for the deposition of vertically aligned nanotubes (Figure 2.13.b).



(a) **(b)**

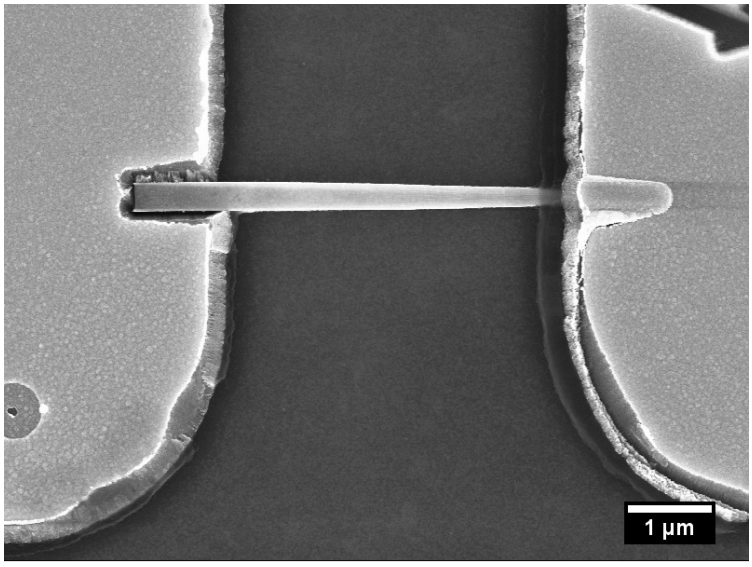
Figure 2.13: (a) Transmission electron microscope image of the head of an SnO_2 nanotube, with lattice spacings corresponding to the (110) surface and growth along the (001) axis. (b) Vertically aligned SnO_2 nanotubes grown epitaxially on m-plane sapphire.[29]

The surfaces of the SnO_2 nanotubes were studied by X-ray photoelectron spectroscopy (XPS) to reveal more information about their electrical and chemical properties. These investigations revealed that the surfaces of the SnO_2 nanotubes exhibit unusually strong downward band bending that results in the formation of an electron accumulation layer in the resulting potential well[30]. O 1s core level measurements showed there was a large density of terminating hydroxyl groups on the surface of the SnO_2 nanotubes. These are thought to act as the donors that are largely responsible for the strong downward band bending in the as-grown nanotubes. Surprisingly, annealing studies revealed that unlike other metal oxide semiconductors, the thermal desorption of these surface hydroxyls at temperatures up to 600 °C did

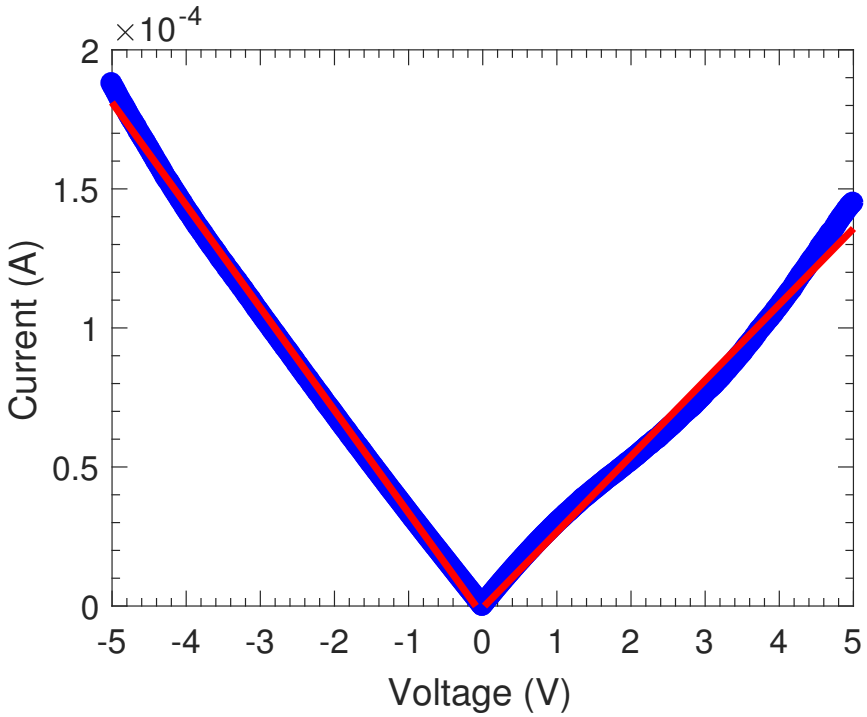
not significantly change the downward band bending. This is likely due to the formation of in-plane oxygen vacancies (which also act as donors) at higher temperatures which compensates for the removal of the surface hydroxyl groups[30, 71].

Sb-doping of the SnO_2 nanotubes to increase their conductivity was also achieved by Scott. Different amounts (0.1-1% Sb/Sn) of antimony (III) chloride ($SbCl_3$) were added to the precursor solution before Mist-CVD growth, resulting in Sb being incorporated into the SnO_2 nanotubes. The morphology of the nanotubes was unaffected up to doping concentrations of ~0.6%, above which the density of nanotubes on the substrate significantly decreased.

To measure the differences in electrical conductivity between the undoped and Sb-doped nanotubes, current-voltage (IV) measurements were undertaken. Individual undoped and 0.3% Sb-doped SnO_2 nanotubes were bridged between Ti/Au contacts by UV photolithography to allow the current through individual nanotubes to be measured. A contacted 0.3% Sb-doped nanotube can be seen in Figure 2.14.a. This nanotube exhibited linear IV characteristics, with an estimated resistivity of 3×10^{-2} Ωcm (Figure 2.14.b). This resistivity was similar to that of a 0.25% Sb-doped thin film grown by Mist-CVD[29]. One significant problem encountered during the device fabrication was the loss of the Ti/Au capping layer from the top of the nanotube, which resulted in a poor electrical contact. This was due to the poor adherence and/or an insufficient thickness of the Ti/Au layer, resulting in the top contact being removed during the photolithography liftoff step. These poor contacts resulted in very few measurable devices, with no undoped nanotubes that were able to be electrically measured. As such, the data on the electrical characteristics of the SnO_2 nanotubes and the effect of the Sb-doping remained limited and uncertain.



(a)



(b)

Figure 2.14: (a) Scanning electron microscopy image of a contacted 0.3% Sb-doped SnO_2 nanotube with the top cap removed. (b) The IV measurement through the nanotube (blue) and the linearly fitted resistances under positive and negative voltage biases (red).[29]

2.5.1 Band Bending Flattening of the SnO_2 Nanotubes

The downward band bending present at the surface of the SnO_2 nanotubes can be problematic for certain applications that require low environmental sensitivity or upward band bending (such as in Schottky diodes). In the thesis of Dr Liam Carroll[226], a range of chemical modifiers were covalently attached to the surfaces of the SnO_2 nanotubes and the resulting effect on the surface band bending was investigated using XPS. It was found that the surface band bending could be bi-directionally controlled depending on the choice of surface modifier. In particular, a sulfide surface modification method was able to almost completely remove the downward band bending under certain conditions.

Sulfur treatments are a method used commonly in the literature for the electrical passivation of III-V semiconductor surfaces. In these materials the native oxide that forms at the surface results in a large amount of surface defect states that pin the Fermi level, leading to band bending at the surface. For example, the oxide layer on GaAs pins the Fermi level near the middle of the bandgap[227, 228], causing upward band bending in n-type GaAs and downward band bending in p-type GaAs. Sulfur treatments for III-V semiconductors rely on the removal of the native oxide layer and their associated surface defect states, while the surface is passivated with a sulfur monolayer preventing oxidation of the top layers of material[229].

The majority of sulfur passivation on semiconductors is performed by wet chemical methods, typically by treating a sample in heated solutions of Na_2S or $(NH_4)_2S$. These methods of sulfur passivation have resulted in an upward shift of band bending in various materials. The group of McConville reported a range of band bending shifts on $(NH_4)_2S$ treated InN, reducing the amount of downward band bending at the material surface by 0.08-0.15 eV[230–232]. McNeill observed a significant upward shift in the band bending of Na_2S treated ZnO, with a 1.75 eV valence band shift between the untreated and treated samples[233].

There is limited research in the literature on the effect of sulfur treatments on the surface of SnO_2. Some research shows SnO_2 acts as a polysulfide trap in Li-S batteries. Liu et al. performed XPS measurements on nanopowdered SnO_2 treated with

a solution of Li_2S_4 in tetrahydrofuran[234]. S 2p spectra revealed the absorption of S_4^{2-} onto the SnO_2 surface. This absorption produced a -0.4 eV shift in the Sn 3d core level suggesting a corresponding upward shift in surface band bending. Similar shifts of the Sn 3d peak were observed by Shen et al.[235]. Zervos et al. converted the surface of SnO_2 nanowires to SnS_2 by annealing in H_2S at 300 °C, although they observed a decrease in resistance which was attributed to an increase in electron accumulation at the SnO_2/SnS_2 interface[236].

In the thesis of Dr Carroll, core level and valence band spectra of the SnO_2 nanotubes were measured by XPS before and after treatment in a 0.8 M Na_2S 1:1 EtOH: H_2O solution[226]. The S 2p spectrum revealed a complicated sulfur environment, with four different sulfur species present. These four sulfur species were attributed to SO_4^{2-}, SO_3^{2-}, S_x^{2-} and S^{2-}, indicating that significant sulfur oxidation had occurred. Such a degree of oxidation was unexpected, as no significant amounts of oxidised sulfur species were observed on Na_2S treated ZnO[233], GaAs[237, 238] or InP[239]. Valence band spectra showed only a small upward shift in the band bending following Na_2S treatment, however, extended exposure to the X-ray beam resulted in a significant upward shift in the surface band bending from -0.47 eV to -0.09 eV. Observations of the S 2p spectra under increasing X-ray exposure revealed the reduction of SO_4^{2-} to SO_3^{2-} and S^{2-}, indicating that the surface band bending change was caused by x-ray induced reduction of the sulfur species at the SnO_2 surface.

Chapter 3

Methodologies

3.1 Introduction

A selection of methodologies were used to fabricate and characterise a range of electronic devices that incorporated the SnO_2 nanotubes, assess their suitability for different applications, and further investigate their fundamental properties. These methodologies are discussed in this chapter, beginning with an overview of Mist-CVD and the process used to deposit the square SnO_2 nanotubes. Following this, the general fabrication methodologies are discussed, including UV photolithography, metal contact deposition and sulfur-based treatments of the SnO_2 nanotube's surface. Finally, the characterisation methodologies are introduced, which includes scanning electron microscopy, the electrical characterisation methods, X-ray spectroscopy techniques and scanning tunnelling microscopy.

3.2 Mist Chemical Vapour Deposition

Mist chemical vapour deposition (Mist-CVD) is an experimental method typically used to deposit thin films of different materials onto a substrate. It belongs to a family of deposition techniques known as chemical vapour deposition (CVD), where source precursors undergo different chemical reactions or decomposition to deposit a film on a substrate. This differentiates these techniques from physical vapour deposition (PVD) methods such as electron-beam evaporation and sputtering, in which the deposited film is very similar to that of the source material. The wide variety of

CVD methods and precursors allows the deposition of a wide variety of materials, such as metals, oxides, nitrides, carbides and a full range of other ceramics.

The defining characteristic of Mist-CVD compared to other forms of CVD is the use of ultrasonic transducers to atomise a source solution into a mist, which is then used as the deposition precursor. In other forms of CVD, such as metal-organic CVD and atomic layer deposition, volatile precursors are transported to the substrate in the gas phase, where they react with the substrate surface. In Mist-CVD, the source solution typically consists of a metal salt dissolved in a non-volatile solvent such as water or methanol. The atomised solution remains in the liquid phase in the form of micrometre-sized droplets, which are light enough to be carried by a carrier gas such as nitrogen or oxygen. The carrier gases deliver the mist droplets into a furnace tube where the material deposition occurs on a heated substrate.

There are multiple advantages that Mist-CVD exhibits over other deposition techniques. The use of soluble salts and non-volatile solvents as material precursors offers a high degree of safety, particularly when compared to conventional CVD techniques which often use volatile and hazardous precursors[240]. Additionally, ultrasonic atomisers produce droplets with a narrow size distribution, which increases the uniformity of the deposited film[241]. Mist-CVD also has low financial and environmental costs, as the equipment involved can be purchased at a comparatively low cost while the absence of any high-vacuum equipment considerably decreases the relative energy consumption of the technique. Finally, high-quality films can still be obtained with these advantages over other deposition techniques.

The primary disadvantage of Mist-CVD is shared with other techniques that are performed at atmospheric pressure. The temperature gradients created as the room-temperature carrier and dilution gases flow into the heated furnace tube result in convection currents. Furthermore, the dimensions of equipment such as the reaction chamber and substrate holder can result in eddy currents near the substrate surface. These convection and eddy currents can dramatically affect the film thickness and uniformity across the substrate. Mist-CVD, however, is more resilient to these effects compared to atmospheric deposition techniques that transport precursors in the gas phase. The inertia of the mist droplets provides some resistance to

sudden changes in velocity, leading to higher film uniformity. It should be noted however, that methods which utilise vacuum chambers suffer much less from these effects compared to both Mist-CVD and other atmospheric pressure techniques, and exhibit much higher film uniformity and thickness consistency. Another disadvantage of Mist-CVD is that continuous flow of gases through the heated reaction vessel results in higher heat losses, requiring more energy to maintain a defined deposition temperature. Significant care is also needed to maintain the cleanliness of the Mist-CVD equipment, as upstream impurities can be carried by the gas flow into the reaction vessel and incorporated into the deposited film.

3.2.1 History of the Technique

The formation of thin films by the decomposition of a precursor aerosol dates back to the late 1940s - early 1950s, with patents describing the formation of metal oxide films by the spraying of a precursor solution onto heated substrates[242, 243]. An in-depth methodology for the production of thin films by spray deposition was first published by Chamberlin and Skarman in 1966, who deposited a range of sulphides and selenides after dissolving the precursors in a single sprayed solution[244]. While these methods describe a technique very similar to that of Mist-CVD, it should be noted that the solutions were sprayed using a simple air-based atomiser, which tends to produce droplets on the order of 50 - 200 µm in diameter[245]. The use of ultrasonic transducers to produce sub- 10 µm droplets for the purpose of spray deposition was first reported in 1972 in a French patent by Spitz and Viguie[246]. This method was called the pyrosol process, and is functionally identical to Mist-CVD[241].

The simplicity of Mist-CVD and its relative advantages over conventional spray deposition techniques has led to the convergent evolution of techniques that have different names, yet are functionally very similar to Mist-CVD. Alternative names for Mist-CVD in literature include the pyrosol process[241, 247, 248], aerosol assisted CVD[249–251], ultrasonic spray pyrolysis[252–254], ultrasonic spray CVD[255–257] and liquid source CVD[258, 259]. The name of Mist-CVD arose in the early 2000s[260], and was popularised by groups such as Fujita *et al*[261–263] and Kaur *et al*[264–266].

The Mist-CVD system used in this work is the same used by Scott[29], itself an extension of the work performed by Okuno et al.[267] and Dang et al.[163]. For continuity purposes, Mist-CVD is the name adopted for this technique in this work.

3.2.2 Growth Mechanisms of Mist-CVD

There are two competing growth models that are used to explain the deposition of material by Mist-CVD, which primarily differ about the point at which the mist droplets evaporate.

The first mechanism is known as the Leidenfrost model, where mist droplets remain in the liquid phase the entirety of the way to the heated substrate. Once a mist droplet reaches the surface of the heated substrate, heat conduction from the substrate quickly vaporises the droplet at the droplet-substrate interface, where the vapour pressure created by this reaction is strong enough to suspend the mist droplet above the substrate surface. This effect is known as the Leidenfrost effect, and results in a material-heat exchange between the droplet and substrate. The vaporised material from the droplet deposits on the substrate surface, while the increased vapour pressure at the interface allows the transfer of heat from the substrate to the droplet, promoting more vaporisation. Furthermore, the Leidenfrost effect allows the droplet to continue moving with the net flow of gas across the substrate, resulting in an even material deposition across the substrate.

The second proposed mechanism of material deposition by Mist-CVD is known as the vaporisation model. In this model, the mist droplets evaporate before reaching the substrate due to the combination of radiative heat transfer alongside conductive and convectional heating as the carrier gas and mist are heated by the walls of the furnace tube. This vaporised source material then deposits on the heated substrate, resulting in film formation.

Several papers support the Leidenfrost model as an effective explanation for the deposition of material by Mist-CVD. Ha et al. performed a variety of experimental studies and numerical simulations to investigate several factors that influence material deposition by Mist-CVD[268]. Critically, a high speed camera was used to directly observe the Leidenfrost effect occurring on a tilted substrate surface, as

seen in Figure 3.1. Kawaharamura found that the central column of air flowing at 3 ms^{-1} over a 700K surface remained at low temperatures, while the air flowing near the surface heated almost immediately[269]. This supports the theory that much of the mist can remain unevaporated until it reaches the substrate surface, where it is deposited by the Leidenfrost effect.

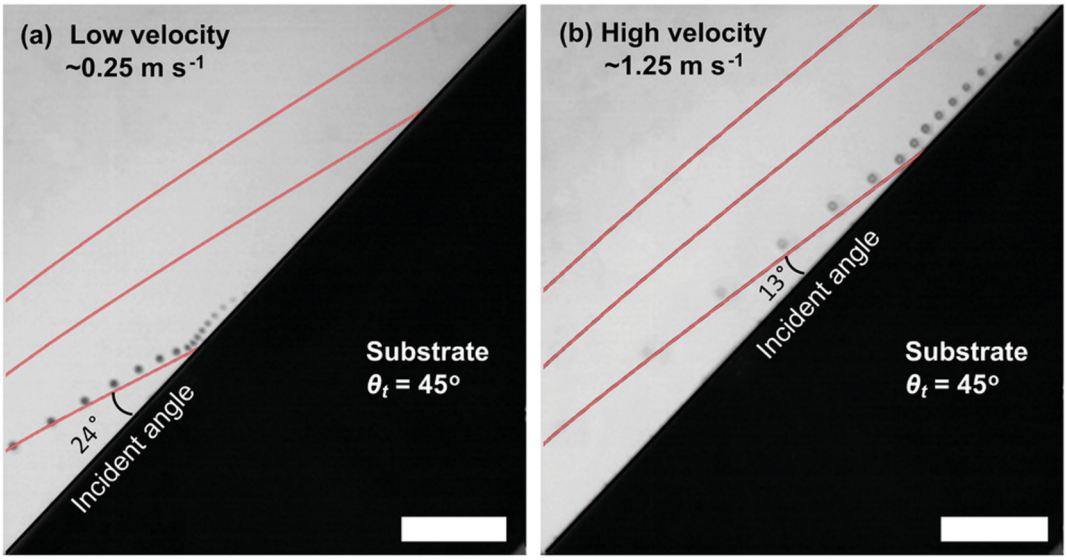


Figure 3.1: The Leidenfrost effect occurring on a 45° tilted substrate at 300°C, captured by a high-speed camera at (a) low and (b) high mist stream velocities. Retrieved with permission from [268]

The vaporisation model also has support in the literature. A significant study was performed by Uno et al., who investigated the deposition of ZnO and ZnS using a dual-source Mist-CVD system[270]. They performed depositions of ZnO with one source containing Zn and O^{16} precursors and the other source containing a mixture of O^{16} and O^{18} precursors. The resultant film contained a ratio of O^{16} and O^{18} equivalent to that expected if the two solutions had been combined, suggesting the precursors had vaporised and mixed together before deposition on the substrate. A similar result was obtained using separate sources containing Zn and S precursors for the deposition of ZnS films. Okamura et al. performed a similar study with the deposition of Cu_2SnS_3 films by dual source Mist-CVD, utilising different sources for the metal and S precursors[271]. Whilst they obtained films containing precursors from both sources, the film stoichiometry was found to vary widely with changes in the carrier gas flow rate and the position of the substrate in the furnace, suggesting

a more complicated interaction between the source precursors and growth mechanism. An apparent counterexample to the vaporisation model are the various papers that report spherical nanopowders of material deposited by Mist-CVD, which argue that individual spheres are formed by the complete evaporation of individual droplets[272–274].

There is also the possibility that both processes can occur concurrently, with a variety of parameters that can influence the dominant deposition mechanism such as the gas flow rate, furnace tube temperature, precursor composition and substrate position inside the reaction vessel. Viguié and Spitz proposed a range of four deposition processes that occur at increasing deposition temperatures, seen in Figure 3.2[255]. It is unlikely that the transition between each of these proposed processes is discrete, and it is quite possible that two of the proposed processes can occur alongside one another. Kuddus et al. found that the average diameter of precursor mist particles for the deposition of TiO_2 decreased from 200 nm to 50 nm after flowing through a 300 mm section of tube furnace at 300°C[275]. This decrease in droplet diameter represents a total volume decrease of 98.4% assuming the droplets are perfectly spherical, indicating that much of the source material has been converted to

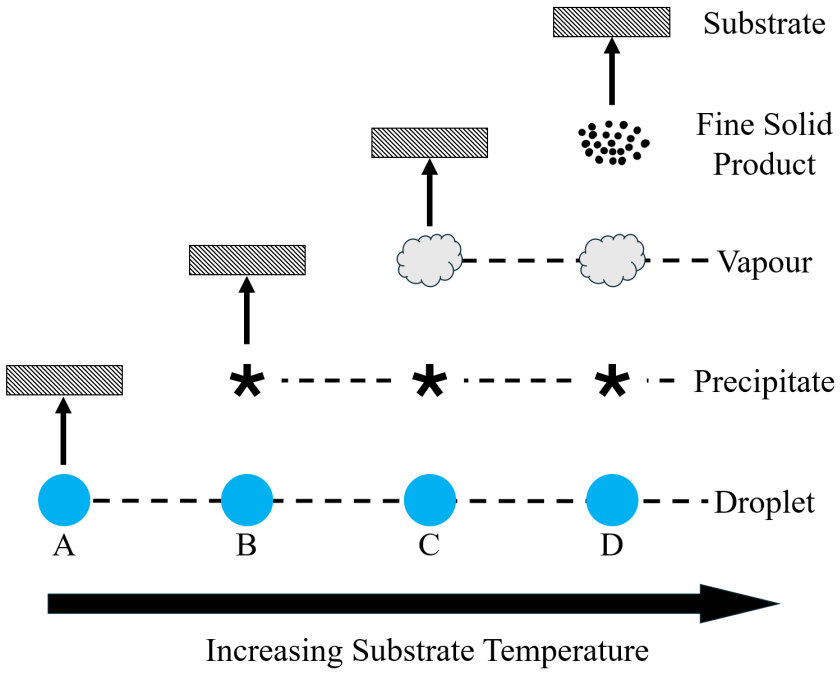


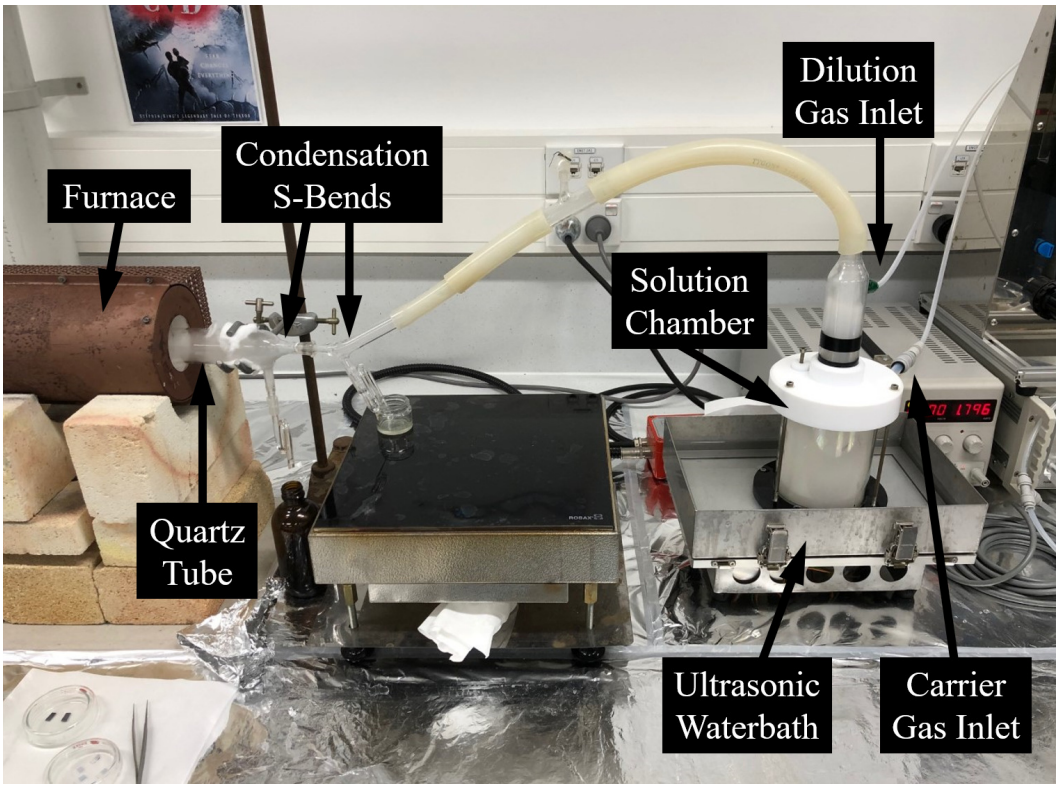
Figure 3.2: Four processes, A, B, C and D, which show film deposition occurring by different final precursor states as a function of substrate temperature[255].

the vapour phase. It is, however, difficult to predict if the vaporised material is largely solvent, leaving a droplet of highly concentrated metal precursor to deposit via the Leidenfrost effect, or whether the metal precursor is also vaporised resulting in majority deposition by the vaporisation model. Regardless of the proposed growth mechanism, the process required for optimal deposition of thin films is likely to vary widely based on the material system deposited and the precursors utilised for deposition.

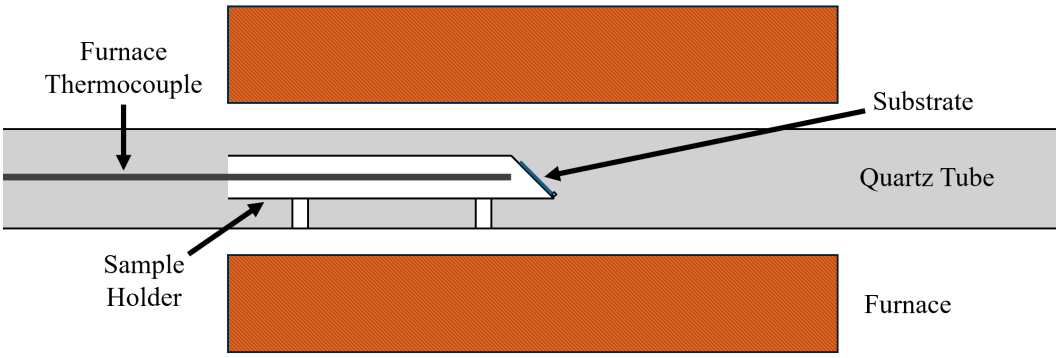
3.2.3 Mist-CVD in this Thesis

Mist-CVD was exclusively used throughout this work to prepare samples of SnO_2 nanotubes for further investigation and use in different applications. An in-house designed hot-wall type Mist-CVD system was used for all depositions, seen in Figure 3.3.

Precursor atomisation was achieved through the use of an ultrasonic waterbath and a solution chamber incorporating a plastic diaphragm bottom. The ultrasonic waterbath used three 2.4 MHz piezoelectric transducers driven by a RF oscillator circuit, which produced mist droplets of approximately 2.3 µm in diameter according to a capillary wave model[276]. The waterbath was nominally filled with 1.1 L of deionised water, which acoustically coupled the precursor solution in the solution chamber with the ultrasonic transducers. The solution chamber was constructed from glass and Teflon, with an interchangeable plastic diaphragm which was usually replaced after each day of Mist-CVD growth. Nitrogen carrier and dilution gases were used to carry the atomised precursor through tubing to the reaction vessel, which consisted of a quartz tube enclosed by a cylindrical furnace tube. Nitrogen was produced in-house by the evaporation of liquid nitrogen from a liquid nitrogen dewar. Carrier and dilution gas flow rates of 1.6 and 4.5 $Lmin^{-1}$ were used for all depositions, controlled by Showa PM-1170 control valves. Solution condensation in the transfer tubing and at the intake to the reaction vessel was collected by s-bends, which prevented this from travelling into the reaction vessel and evaporating. Depositions of SnO_2 nanotubes took place on an inserted substrate, which was held at



(a)



(b)

Figure 3.3: (a) The front end of the Mist-CVD system used for the deposition of square SnO_2 nanotubes. (b) The interior of the deposition furnace and quartz tube, showing the relative positions of the substrate, sample holder and furnace thermocouple.

a 45° to the mist flow by a quartz sample holder, seen in Figure 3.3.b. The deposition temperature for all depositions of SnO_2 nanotubes was 650°C, as measured by a thermocouple that was positioned behind the sample holder and controlled by an Eurotherm 3216 PID temperature controller.

Substrate preparation is an important step for the deposition of SnO_2 nanotubes, as a thin layer of Au nanoparticles is required to catalyse the nanotube growth. Si

(001) substrates were used for the deposition of non-epitaxial SnO_2 nanotubes, while vertically aligned epitaxial SnO_2 nanotubes were deposited on m-plane (10-10) sapphire. Substrates were diced to appropriate sizes, typically 10×10 mm or 10×5 mm, using a diamond scribe. This was followed with substrate cleaning, consisting of the physical cleaning of the substrate surface with a cotton bud in acetone and ultrasonic cleaning in acetone, methanol and isopropanol, before being blown dry using nitrogen gas. The substrates were then immediately mounted onto sample holders using silver tape and loaded into equipment to perform electron-beam deposition of a ~1.8 nm thick Au layer. This step is covered in more detail in Section 3.3.2. Following the Au layer deposition, the substrates were removed from the sample holders and used for Mist-CVD without further preparation.

A standard precursor solution consisting of 0.0075 M $SnCl_2$ and 0.051 M HCl dissolved in 200 mL of deionised water was used for the deposition of SnO_2 nanotubes. This solution was typically prepared 20 minutes before the solution was used for deposition. When Sb-doping of the SnO_2 nanotubes was desired, a 100 mL aqueous solution of 0.00075 M $SbCl_3$ was prepared by overnight stirring at 40 °C before an appropriate amount was added to the precursor solution to achieve the desired doping concentration.

The chemistry of formation of SnO_2 by Mist-CVD from an $SnCl_2$ solution has several proposed pathways. Importantly, a reducing component is required for the oxidation of Sn^{2+} to Sn^{4+}. Gordillo et al. proposed the following reaction for the deposition of SnO_2 from an aqueous solution of $SnCl_2$[277]:

$$\frac{1}{2}O_2(g) + SnCl_2(aq) + H_2O(l) \rightarrow SnO_2(s) + 2HCl \tag{3.1}$$

This reaction, however, requires a source of O_2 gas during deposition, which is unlikely given the use of nitrogen for both the carrier and dilution gases. Dissolved oxygen in the precursor solution is also unlikely to significantly contribute to this reaction, as the maximum dissolved molarity of oxygen from air at 25 °C is 2.56×10^{-4} M[278], significantly below the 3.75×10^{-3} M required to match the $SnCl_2$ molarity in the precursor solution. A more likely reaction pathway is a series of reactions which

involve converting $SnCl_2$ to $SnCl_4$ before deposition, with a straightforward reaction to produce SnO_2 using H_2O[29]:

$$6SnCl_2(aq) + O_2(g) + 2H_2O(l) \rightleftharpoons 4Sn(OH)Cl(s) + 2SnCl_4(aq) \tag{3.2}$$

$$SnCl_2(aq) + H_2O(l) \rightleftharpoons Sn(OH)Cl(s) + HCl(aq) \tag{3.3}$$

$$SnCl_4 + 2H_2O \rightarrow SnO_2 + 4HCl \tag{3.4}$$

Equation 3.2 shows how $SnCl_2$ can be converted to $SnCl_4$ in an aqueous solution in the presence of dissolved oxygen. Critically, this reaction occurs in solution before deposition which allows oxygen from the air to replenish dissolved oxygen used in the reaction. This reaction does, however, produce a high ratio of insoluble Sn(OH)Cl which cannot be directly converted to $SnCl_4$. In addition, $SnCl_2$ hydrolyses in water to also produce Sn(OH)Cl and HCl, seen in Equation 3.3. Fortunately, this equation shows a simple solution to promote the formation of $SnCl_4$, with the addition of HCl to the precursor solution. Following Le Chatelier's principle, the addition of HCl to the precursor solution favours the left-hand side of Equation 3.3, converting the insoluble Sn(OH)Cl back to $SnCl_2$. The decrease in concentration of Sn(OH)Cl then favours the right-hand side of Equation 3.2, leading to the production of more $SnCl_4$ and Sn(OH)Cl. With an excess concentration of HCl, this promotes the conversion of $SnCl_2$ to $SnCl_4$.

One possible way to determine if the deposition of SnO_2 follows Equation 3.4 is to follow the methods used by Uno et al., who deposited ZnO and Ga_2O_3 films by Mist-CVD with the addition of H_2O^{18} in the precursor solution[270, 279]. Secondary ion mass spectroscopy showed that the ratio of O^{16}/O^{18} in the deposited films was the same as the ratio of H_2O^{16}/H_2O^{18} in the precursor solution, indicating that the entirety of the oxygen in the films was sourced from the H_2O solvent. This approach could also be used to determine the growth mode of SnO_2 by Mist-CVD. An O^{16}/O^{18} ratio equivalent to that of H_2O^{16}/H_2O^{18} would support Equation 3.4 as the growth

mechanism, while an O^{16}/O^{18} ratio closer to half that of H_2O^{16}/H_2O^{18} would support Equation 3.1.

3.2.4 Growth Procedure of Square SnO_2 Nanotubes

This section contains the growth procedure utilised to grow square nanotubes of SnO_2 by Mist-CVD. A table showing the key parameters used for depositions of the SnO_2 nanotubes can be seen in Table 3.1. Substrates were typically prepared within two weeks before depositions, while precursor solutions were prepared 20 minutes before being introduced to the solution chamber. Before each growth, the solution chamber, transfer tubing and quartz deposition tube were all rinsed and scrubbed with deionised water to remove potential remnants of dust or other unwanted impurities. It should be noted that the centre of the quartz deposition tube was not scrubbed and only rinsed with deionised water, as it was observed that the first growths after scrubbing the centre of the tube resulted in poor nanotube formation. Following equipment cleaning, the Mist-CVD system was set up as shown in Figure 3.3 and the furnace temperature ramped to 650°C over a period of one hour. Before the first growth of each day, an aqueous solution of 0.01 M $SnCl_2$ and 0.051 M HCl was atomised and the mist flushed through the Mist-CVD system for 30 minutes to remove or cover any remaining unwanted impurities from the system. Furthermore, each solution used for deposition was run through the system for 10 minutes before the substrate was introduced to allow for stable mist generation.

Standard Mist-CVD Deposition Parameters	
Deposition Temperature	650°C
Deposition Time	2-4.5 Hours
Carrier and Dilution Gases	N_2
Carrier Gas Flow Rate	1.6 $Lmin^{-1}$
Dilution Gas Flow Rate	4.5 $Lmin^{-1}$
$SnCl_2$ Concentration	0.0075 M
HCl Concentration	0.051 M
Substrate	(001) Si or m-plane (10-10) Sapphire
Catalyst	Au, ~1.8 nm

Table 3.1: The standardised Mist-CVD deposition parameters used for the growth of square SnO_2 nanotubes.

Before the substrate was inserted into the furnace tube for deposition, the ultrasonic transducers and carrier gas were turned off and the remaining mist was allowed to run through the system. The thermocouple and exhaust were then removed from the back end of the quartz deposition tube, allowing the heated sample holder to be removed from the centre of the tube. The substrate was then placed on the sample holder and reinserted into the furnace, along with the thermocouple and exhaust. The substrate was then left in the furnace tube for 10 minutes without mist flow to allow for the furnace to return to 650 °C and for the Au nanoparticles to dewet to act as the catalyst for the growth of SnO_2 nanotubes. Following this, the ultrasonic transducers were restarted and the carrier gas valve reopened, allowing precursor solution mist to flow into the furnace tube and begin the SnO_2 nanotube deposition. Growths were typically performed for a minimum of 2 hours to achieve nanotube lengths of ~4 µm, although longer growth times up to 4.5 hours were performed to achieve longer nanotubes for the fabrication of SnO_2 nanotube devices. Once the desired deposition time had elapsed, the flow of mist was stopped by similarly disabling the ultrasonic transducers and carrier gas. The sample holder was then removed from the furnace without sample cooling, at the furnace operating temperature of 650 °C. The system could then be used for another growth or shut down, allowing the connecting tubing and solution chamber to be cleaned and stored for subsequent growths while the deposition tube was allowed to cool to room temperature.

3.3 Device Fabrication Methodologies

3.3.1 UV Photolithography

The use of light to to transfer a pattern onto another material is known as photolithography. In UV photolithography, a film of UV-light sensitive polymer known as a photoresist is applied to a substrate, and then certain areas are selectively exposed to UV light through a photolithographic mask. Depending on the type of photoresist used, UV exposure can increase or decrease the solubility of the photoresist to a specific developer. This allows the pattern present on the mask to be transferred to the substrate in the form of the remaining undeveloped photoresist. Photoresists

that increase in developer solubility after exposure are known as positive photoresists and those that decrease in solubility are known as negative photoresists. Examples of positive photoresists include AZ 1518 and polymethylmethacrylate (PMMA), while negative photoresists include AZ nLOF 2020 and SU-8. Most photoresists are typically used as masks for other processes as they possess a moderate-to-high resistvity to various material processes and can be removed through the use of specific solvents. A simplified overview of the photolithographic process for a positive photoresist can be seen in Figure 3.4.

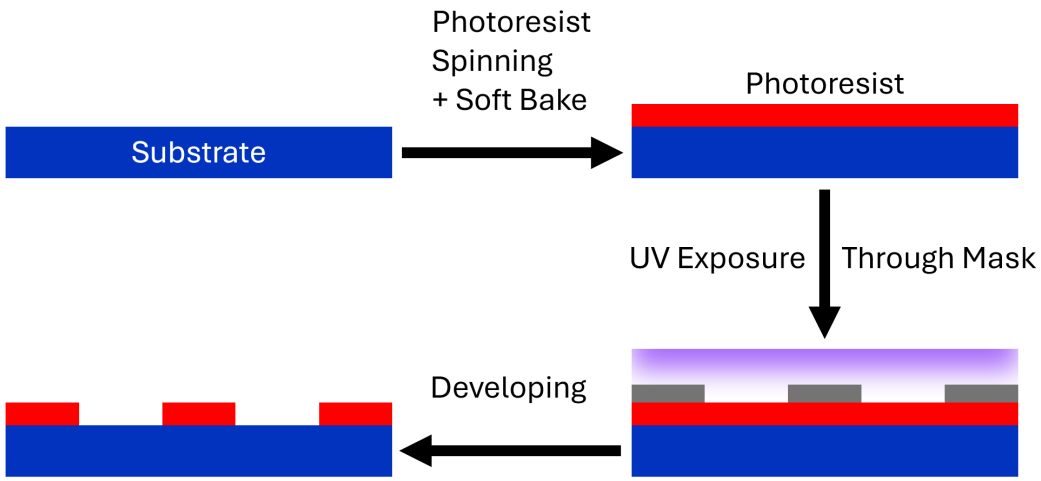


Figure 3.4: The typical photolithographic process for patterning using a simple positive photoresist.

In this work, photolithography was used in conjunction with various thin film deposition techniques to pattern metal contacts onto the square SnO_2 nanotubes. Photolithographic masks were designed using L-Edit (a hierarchical physical layout editor) and written onto Nanofilm AZ 1518 Soda Lime Low Reflective Chrome photolithography masks using a Heidelberg µPG101 laser mask writer. The laser-patterned areas were developed using AZ 326 MIF developer before the exposed Cr was removed by wet chemical etching with chromium etchant (a solution of ceric ammonium sulfate and perchloric acid). The remaining photoresist was then removed by cleaning in acetone, methanol and isopropanol, leaving a photomask which could be repeatedly used in the fabrication of metal contacts.

The standard photolithography process for contacting the SnO_2 nanotubes is as follows. Firstly, SnO_2 nanotube samples were placed inside a 10 mL vial containing

2 mL of isopropanol. The vial was sonicated in an ultrasonic waterbath for 10 s to detach some of the nanotubes from the substrate, suspending them in the isopropanol. This suspension was then dropped onto 10×10×0.4 mm SiO_2/Si substrates on a 50 °C hot plate to evaporate the isopropanol, leaving SnO_2 nanotubes scattered across the substrate. The SiO_2/Si substrate was then attached onto a larger 20×20×0.5 mm glass slide with AZ 1518 photoresist and baked in a 96 °C oven for 25 minutes. This step was performed in order to allow the entire SiO_2/Si substrate to be coated during the metal deposition process. After being removed from the oven and allowed to cool, AZ 1518 photoresist was spin-coated onto the nanotube-coated sample. A one-step spin was performed using an acceleration of 4000 RPM/s, a spin speed of 4000 RPM and a spinning time of 60 s, giving a nominal photoresist thickness of 1.8 µm. Following the photoresist spinning, an acetone-soaked cotton bud was used to remove edge beading that forms at the corners of the SiO_2/Si substrate before the sample was baked again in a 96 °C oven for 25 minutes.

UV exposure of the photoresist was performed using a Karl Suss MA6 mask aligner with the relevant designed photomask. Standard exposure parameters included a 12 s exposure to 365 nm UV light, with a hard exposure contact pressure. The exposed areas were then developed by immersing and gently agitating the sample in AZ 326 MIF developer for 24 s, before rinsing in deionised water for 30 s and drying with compressed nitrogen. The sample was then immediately mounted and loaded into the relevant thin film deposition equipment for the deposition of the metal contact layers. The relevant thin film deposition techniques are covered in more detail in subsections 3.3.2 and 3.3.3.

Following thin film deposition, a final process known as 'lift-off' was used to remove the metal covered photoresist. Samples were soaked in a glass Petri dish containing acetone for 10 minutes to dissolve the photoresist layer and to also separate the SiO_2/Si substrate from the glass slide, with the glass slide then removed from the Petri dish. After 10 minutes, the Petri dish was placed in a shallow ultrasonic waterbath and sonicated to remove the excess metal attached to the photoresist layer. The length of this sonication process varied widely with the design and thickness of the metal contacts, with some cases requiring only 30 seconds of sonication while others

needed up to 10 minutes. In each case, the metal contacts were periodically checked using an optical microscope to determine if more sonication was required. Once no excess metal remained and the metal contacts were clearly defined the lift-off process was completed. Sample cleaning was then performed by rinsing the samples in fresh acetone, methanol and isopropanol, before the samples were blown dry with nitrogen gas.

3.3.2 Electron-Beam Evaporation

Electron-beam evaporation is a physical vapour deposition (PVD) technique that is frequently used to deposit thin metal films. In this technique, a crucible of source material is heated by a high-energy electron beam, causing the material to evaporate from the crucible and condense as a film on a substrate. This high-energy electron beam is created by an electrostatic electron gun, typically a tungsten filament which generates free electrons by thermionic emission. The electron gun uses a large potential difference to accelerate the electrons while a magnetic field directs the beam into the crucible. As the electrons strike the deposition material their kinetic energy is converted into heat, which dramatically increases the temperature of the deposition material in a localised area. This process is illustrated in Figure 3.5. To facilitate the evaporation of the material, this process is performed in vacuum, typically at pressures between 1×10^{-7} to 1×10^{-5} mBar.

Electron-beam evaporation was performed for two main purposes in this work. Firstly, thin films of Au approximately ~1.8 nm thick were deposited to catalyse the growth of SnO_2 nanotubes. Two different electron-beam evaporation systems were used interchangeably for this process: an Edwards Auto 500 Sputtering/Electron-beam system utilising a 3 kW EB3 multihearth electron-beam source, and a Temescal FC-1800 evaporator. Depositions utilising the Edwards Auto 500 were performed at a potential difference of 5 kV and a emission current of 38 mA until the desired Au thickness was reached, as determined by a 6 MHz quartz crystal monitor. When depositions were performed using the Temescal FC-1800, the deposition rate was controlled instead of the emission current. These depositions were performed at a potential difference of 9 kV and a Au deposition rate of 1 Å/s, also monitored by a

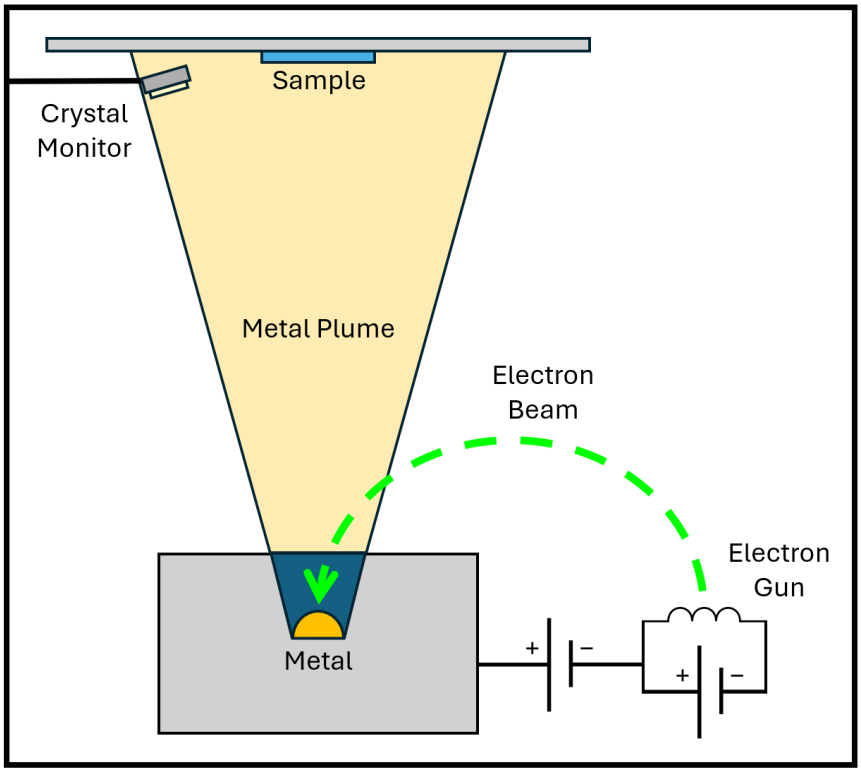


Figure 3.5: An illustrative diagram showing the electron-beam evaporation process.

6 MHz quartz crystal monitor. Figures 3.6.a and 3.6.b show scanning electron microscopy images of 1.8 nm Au films deposited by electron-beam evaporation using the two different evaporator systems, while Figures 3.6.c and 3.6.d show the Au nanoparticles following film dewetting. Both sets of films and nanoparticles show similar physical morphologies and were successfully used to catalyse the growth of square SnO_2 nanotubes, indicating a negligible difference between these two deposition systems for this process.

The second use of electron-beam evaporation was the deposition of Ti/Au layers to serve as ohmic contacts to SnO_2 nanotubes in the fabrication of electronic devices. For this process, only the Temescal FC-1800 was used. Ti depositions were similarly performed at a potential difference of 9 kV and a deposition rate of 5 Å/s. Due to the low thermal conductivity of Ti[280] the electron-beam was swept back and forth across the Ti crucible by a programmable sweep controller, achieving better melting of the Ti source material. To fully coat the SnO_2 nanotubes a Ti layer 500 nm thick was deposited. A 50 nm thick Au layer was then deposited on top of the Ti to prevent oxidation of the Ti layer. A rotating carousel allowed the changing of source

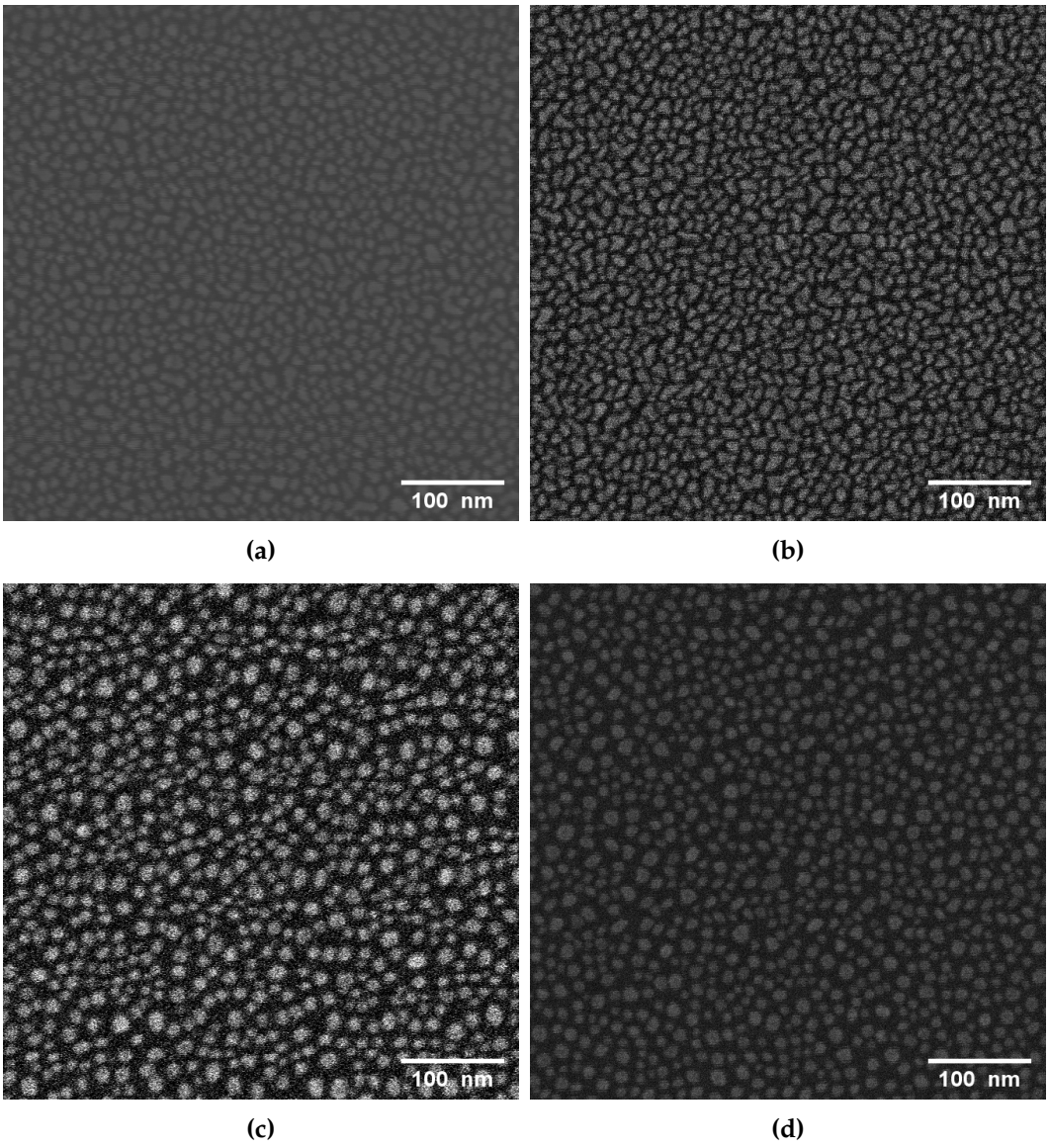


Figure 3.6: Scanning electron microscopy images of 1.8 nm thick Au films on Si substrates deposited by (a) the Edwards Auto 500 and (b) the Temescal FC-1800 . (c) and (d) show the same films after 2 minutes of dewetting in the Mist-CVD furnace at 650°C.

materials without breaking vacuum. The Au layer was deposited using a potential difference of 9 kV with a deposition rate of 2 Å/s without sweeping the electron beam.

3.3.3 Radio-Frequency Magnetron Sputtering

Another technique used to create thin films of material is known as sputtering. Sputtering is a phenomena that occurs when a material surface is bombarded with energetic ions, causing atoms of the source material to be ejected from its surface. This

process does not require the source material to be heated to the point of evaporation, and so sputtering allows a wide variety of materials to be deposited including high melting point metals and ceramics. In practice, two types of sputtering systems are commonly used: Direct current (DC) sputtering and radio frequency (RF) sputtering. Both of these two techniques have similar modes of operation. In both techniques the source material is in the form of a circular disk known as a target, which is mounted onto the cathode of a pair of planar electrodes. The substrate is then mounted above the target on the anode electrode. The system is then pumped down to vacuum, typically to pressures around 1×10^{-5} mBar. A sputtering gas, typically a noble gas such as Ar, is then fed into the deposition chamber where it is ionised into a plasma with the ions bombarding the target, causing target atoms to be ejected and deposited onto the anode substrate.

The main difference between DC and RF sputtering is the method of plasma creation. In DC sputtering a constant large negative bias voltage is applied to the cathode, causing Ar^+ ions to be constantly accelerated into the target. This produces a high sputtering rate, however, it also requires a high gas flow rate to maintain the plasma, which results in higher deposition pressures between 1×10^{-3} and 1×10^{-1} mBar. Additionally, the use of a constant applied voltage requires the target to be electrically conducting to prevent surface charging by the Ar^+ ions. In RF sputtering, a high frequency RF signal between 1-30 MHz is applied to the cathode to generate the Ar^+ plasma. This prevents surface charging of insulating targets at the cost of a lower sputtering rate.

In this work, RF magnetron sputtering was employed to deposit IrO_x/Ir layers onto the ends of SnO_2 nanotubes for the purpose of creating Schottky contacts. The Edwards Auto 500 sputtering/electron beam deposition system was used to deposit the films, where the RF plasma was generated by a Cesar Model 136 13.56 MHz power supply. A Kurt J. Lesker 3" 99.9% Ir target was used as the Ir source, with an O_2:Ar plasma as the O source. To create this O_2:Ar plasma, 99.88% O_2 and 99.999% Ar was introduced to the deposition chamber at flow rates of 6.8 and 10 sccm respectively, resulting in a processing pressure of 3.8×10^{-3} mBar. To prevent oxidative

damage to the turbomolecular high vacuum pump, N_2 gas was flowed into the output vent of the pump. The O_2:Ar plasma was initially struck at an RF power of 50 W, with 5 minutes of pre-sputtering preformed to remove any surface contaminants from the target. A moveable metal shutter plate was used to prevent any material depositing on the substrate during this pre-sputter process. Following this, the shutter was opened, allowing IrO_x to deposit on the substrate. Once 30 nm of IrO_x had been deposited, as measured by a 6 MHz quartz crystal monitor, the RF power was increased to 75 W to achieve a higher deposition rate. After the deposited material thickness exceeded 250 nm, the O_2 gas flow was shut off, resulting in the deposition of an unoxidised pure metal Ir capping layer by the 100% Ar plasma. The deposition of Ir was continued for an additional 250 nm of deposited material, yielding an IrO_x/Ir contact stack approximately 500 nm thick.

3.3.4 Surface Modification by Wet Chemical Methods

Samples of SnO_2 nanotubes underwent several different wet chemical sulfide treatments in an attempt to tune their surface band bending, building on previous work performed by Dr Liam Carroll[226]. Similar methodologies have been used in this thesis for the surface modification of the SnO_2 nanotubes and subsequent measurements using X-ray photoelectron spectroscopy.

Sulphur modification of the SnO_2 nanotubes was performed using solutions of $(NH_4)_2S$ and Na_2S. Samples modified by $(NH_4)_2S$ were immersed in 5 mL of 0.8 M $(NH_4)_2S$ in 1:1 IPA:H_2O at 40 °C for 60 minutes. The solution was agitated with a magnetic stirrer, placed away from the sample in the solution vessel in order to not physically disturb the sample. After modification, the sample was rinsed in IPA and blown dry with nitrogen gas. A similar process was adopted for the Na_2S modification, with the exception of the 0.8 M Na_2S dissolved in 1:1 EtOH:H_2O and the final rinse performed in EtOH. The modification time was also reduced to 20 minutes as the highly basic solution resulted in the delamination of the nanotubes from the substrate after 60 minutes.

3.4 Characterisation Methodologies

3.4.1 Scanning Electron Microscopy

The dimensions of the SnO_2 nanotubes cause a problem shared with other sub-micrometer scale materials. Optical imaging becomes difficult as the size of the structural features become smaller than the wavelength of light used to image them. This problem is known as the Abbe diffraction limit. The minimum resolvable feature width of an optical microscope is given by the following equation[281]:

$$d = \frac{\lambda}{2NA} \tag{3.5}$$

Where d is the minimum resolvable feature width, λ is the wavelength of the light used for imaging, and NA is the numerical aperture of the microscope. This means that if blue light with a wavelength of 400 nm is used with a microscope in air (NA = 1), the minimum resolvable feature is only 200 nm wide. To significantly decrease the minimum resolvable distance further, the wavelength used for imaging must be decreased. Electron microscopy presents a solution to the limitation of imaging by optical light. Electrons behave as waves with a wavelength that depends on their momentum. These wavelengths are much shorter than those of optical light. For example, electrons accelerated using a 10 kV source exhibit a de Broglie wavelength of 12.2 pm, allowing nanoscale features to be resolved.

Scanning electron microscopy (SEM) is a type of electron microscopy commonly used to image sub-micrometer features of electrically conducting samples. In this technique, a focused beam of electrons is rastered across a sample which causes a wide range of electrons and X-rays to be ejected from the sample surface. Inelastic scattering produces a variety of emitted electrons and photons, which includes secondary electrons and Auger electrons, bremsstrahlung X-rays, characteristic X-rays and cathodoluminescence radiation. Secondary electrons are emitted with low energies of less than 50 eV, and as such are primarily emitted from the first few nanometres of the material. This makes these secondary electrons well suited for imaging by recording the emitted intensity of secondary electrons at different points across a sample. Measuring the secondary electron flux at each beam raster point

allows a greyscale intensity to be assigned to a 2D point, creating a 2D projection of the sample surface similar to a photograph. This allows the surface of a sample to be imaged quickly in real-time, with longer scans producing images that have a resolution of a few nanometers. In practice, the minimum resolvable feature width is limited by other factors such as the beam spot size and external vibrations, rather than the electron's de Broglie wavelength.

The other emitted electrons and X-rays produce useful information when analysed. Characteristic X-rays are unique for each element, which allows the elemental composition of a sample to be determined by energy-dispersive X-ray analysis (EDX). Auger electrons are also unique for each element, although the low kinetic energy of these emitted electrons means they are only representative of the surface layers of the material. EDX can be combined with Auger electron spectroscopy to understand how the elemental composition of a material changes from the surface to the bulk. Elastic scattering of the incoming electron beam produce primarily backscattered electrons, which have a high kinetic energy and contain information that is sensitive to the atomic weight of the atoms within the sample. This produces a contrast difference in areas with different elements within them.

In this work, SEM imaging was performed using a Raith 150 electron beam lithography system, which incorporates a LEO FEG-SEM. Imaging by secondary electrons was the only imaging mode available on this machine. SEM was routinely employed to establish if SnO_2 nanotube growths were successful and to identify suitable fabricated individual SnO_2 nanotube devices. SEM images were captured using an acceleration voltage of 10 kV and a beam aperture of 30 µm. Samples were electrically grounded using metal clips or by contacting conductive layers with carbon tape.

3.4.2 Electrical Characterisation

Electrical measurements in this work were performed by an Agilent B1500A semiconductor device analyser, which uses source measurement units (SMUs) to apply variable voltages and measure the resulting currents through the tested device. Electrical contact to fabricated devices was performed by Cascade Microtech 107-158 5 µm tungsten probe tips inside a Cascade Microtech Summit 11000 manual probe

station. The probe tips were connected to the Agilent B1500A via triaxial cables, which minimise leakage currents by applying the same voltage on the central core wire and the inner shield layer, preventing leakage across the in-between dielectric. Unless specified otherwise, all measurements were performed in the dark at room temperature, with the octagonal Faraday cage of the Summit 11000 probe station blocking ambient light from reaching the sample.

The photoresponse of electrical devices fabricated from SnO_2 nanotubes was measured using a modified setup of the Summit 11000 probe station, seen in Figure 3.7. A QPhotonics UVTOP285 UV LED was used as the UV source, mounted directly above the measured sample with a vertical separation of approximately 60 mm. This LED produces UV light with a peak intensity wavelength of 285 nm and a spectral full-width half maximum of 12 nm. A hemispherical lens produces a collimated emission of photons which typically diverges from the lens at an angle of 6°. The LED was driven by a Keithley 2400 source meter at a constant forward current of 20 mA, nominally producing 600 µW of output power. Assuming an even distribution of photons across the illuminated area and a divergence of 6° from the hemispherical lens of the LED, this produced an illuminated area of 2.82 cm^2, giving an illumination power of ~200 µW cm^{-2}.

Electrical characterisation of the SnO_2 nanotube device photoresponse was performed by the Agilent B1500A in a time-measurement mode. A constant 2V bias was applied to measured devices while the current was measured over time, with a sampling interval of 75 ms. An initial measurement period of at least 100 seconds was used to determine the base current under dark conditions, known henceforth as the dark current (I_{Dark}). After this initial period of observation, the UV LED was turned on and the sample was illuminated for 100 seconds. At the end of this time, the UV LED was turned off and the persistent photoconductivity (PPC) response was observed. Re-illuminating of the sample was not performed until the current through the device returned to the dark current level.

A significant aspect of the UV photoresponse of SnO_2 nanotube devices is the photodetector recovery time, which is commonly affected by PPC. To quantify this

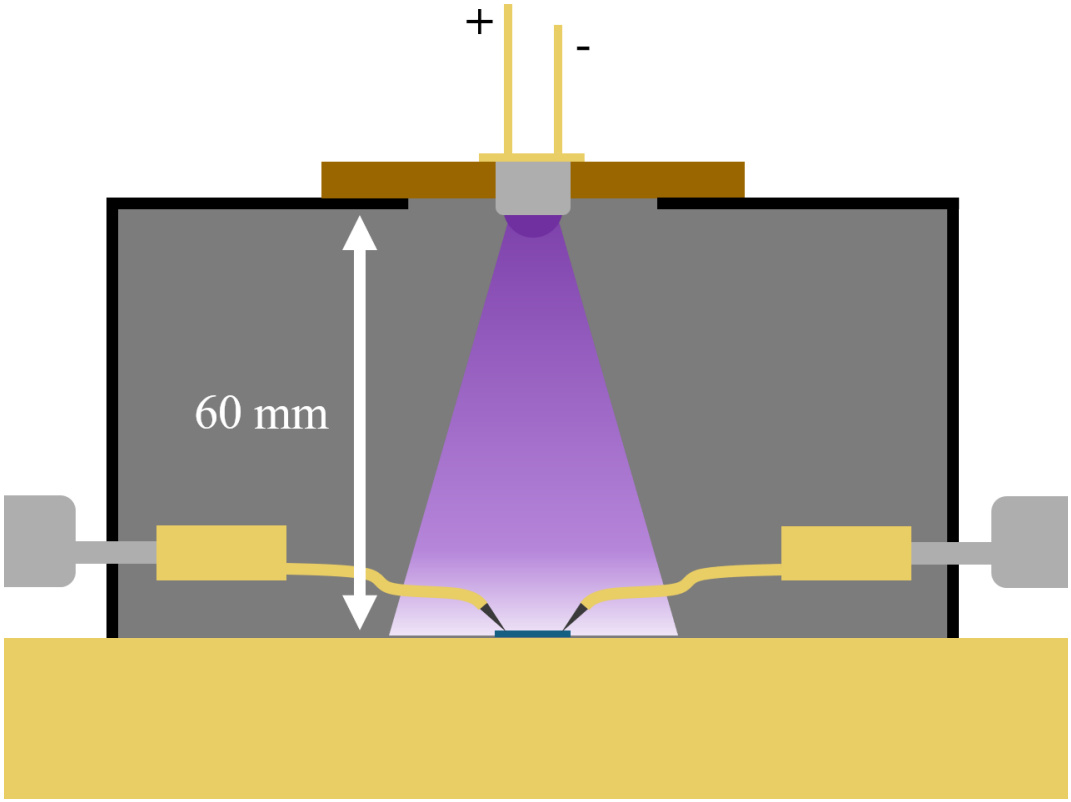


Figure 3.7: The experimental setup used to illuminate samples to 285 nm UV light and measure the resultant photoresponse.

effect, a stretched exponential function was used to model the PPC and photodetector recovery[282]:

$$I_{Ph}(t) = I(t) - I_{Dark} = I_{Ph}(0)e^{-(\frac{t}{\tau})^{\beta}} \tag{3.6}$$

Where $I_{Ph}(t)$ is the recovery current of the photodetector, I(t) is the device current after the UV exposure, τ is the decay time constant and β is the decay exponent. Often, factors that affect the PPC response of the SnO_2 nanotubes also affect the maximum photocurrent under UV illumination. To observe the current decay independently from the maximum photocurrent, the normalised current response is used:

$$I_{Norm}(t) = \frac{I_{Ph}(t)}{I_{Ph}(0)} = e^{-(\frac{t}{\tau})^{\beta}} \tag{3.7}$$

The fitting of this equation to experimental data was performed by transforming Equation 3.7 to Equation 3.8, which allows a simple linear fit to the transformed data to be used to extract values for τ and β.

$$ln(ln(\frac{1}{I_{Norm}(t)})) = \beta[ln(t) - ln(\tau)] \tag{3.8}$$

Different factors such as the photodetector operating temperature and gas ambient were varied to study their effect on the PPC of SnO_2 nanotubes. Measurements under increased operating temperatures were performed by heating the metal chuck within the Cascade Microtech Summit 11000 probe station on which the sample was placed using a Temptronic TP03000 heating system. Different gas ambients were introduced to the Faraday cage enclosure of the Cascade Microtech 11000 probe station with a constant positive pressure that displaced most of the ambient air. Measurements were performed in four other gas environments aside from air: dry N_2, dry O_2, humid N_2 and humid O_2. N_2 gas was sourced from the in-house nitrogen gas supply, while 99.9% O_2 gas (Southern Gas Co. Ltd.) was used as the O_2 source. The dry gases produced environments with a relative humidity of < 5%, while humid gas environments with relative humidities > 95% were produced by bubbling the dry gases through a water bubbler heated to 60 °C. The measurement performed in ambient air had a relative humidity of ~30%.

In the case of fabricated SnO_2 nanotube Schottky diodes, the sample was back-illuminated through the m-plane sapphire substrate to allow UV light to access the Schottky contact. In this arrangement, seen in Figure 3.8, the sample was placed on a sample holder with a hole that allowed UV light to pass through the back of the sample. A QPhotonics UVSMD275 was positioned directly below the substrate, with a separation of 2.25 mm between the substrate top surface and the top of the LED. This LED produces UV light with a peak intensity at a wavelength of 275 nm and a spectral full-width half maximum of 12 nm. A flat window lens produces a photon emission that diverges from the LED at a maximum angle of 45°. The LED was similarly driven by a Keithley 2400 source meter at a constant forward current of 20 mA, nominally producing 400 µW of output power. Assuming an even distribution of photons across the illuminated area and a divergence of 45° from the flat window lens of the LED, this produced an illumination power of ~360 µW cm^{-2}.

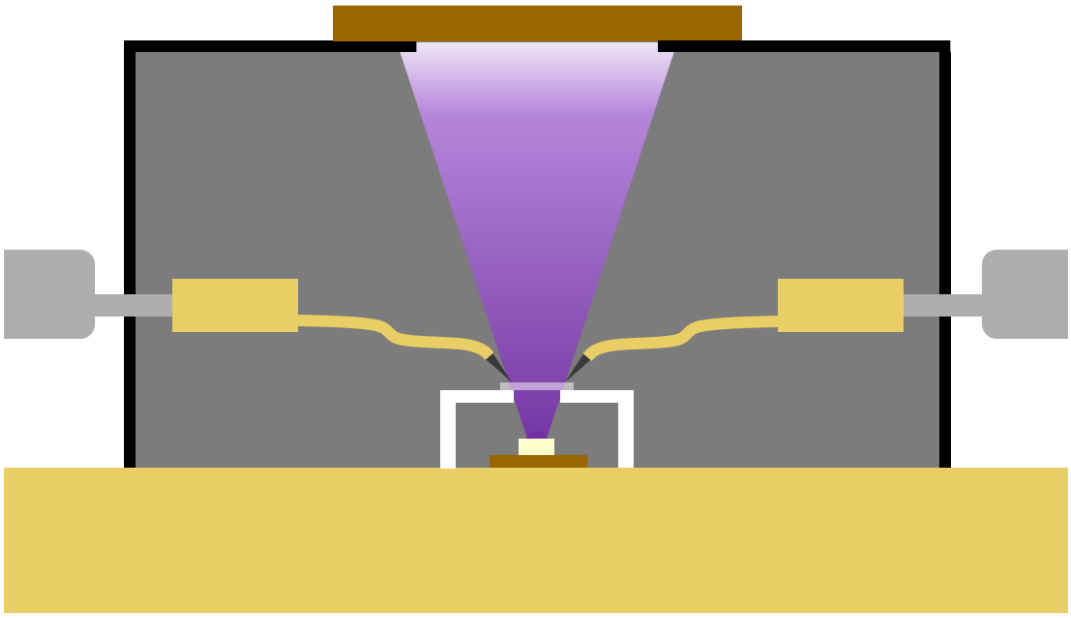


Figure 3.8: The experimental setup used to back-illuminate samples to 275 nm UV light and measure the resultant photoresponse.

3.4.3 X-Ray Photoelectron Spectroscopy

X-ray Photoelectron Spectroscopy (XPS) is a non-destructive method performed in vacuum which is used to characterise the chemical and electronic properties of material surfaces. When a material is illuminated with soft X-rays, core-level electrons are excited above the atom's vacuum level and are emitted from the sample as photoelectrons. These photoelectrons are released from the material with a certain kinetic energy, which depends on the incident X-ray energy and the energy required to remove the electron from its core level to the vacuum level, known as the electron binding energy. By using monochromated X-rays with a known energy and measuring the kinetic energies of the emitted photoelectrons, the binding energies of the electrons within a material can be found, as seen in the following equation:

$$BE = hv - KE - \phi_s \tag{3.9}$$

Where BE is the binding energy of the electrons in the material, hv is the energy of the incident photons, KE is the kinetic energy of the emitted photoelectrons and ϕ_s is the spectrometer work function[283].

The binding energies of the core-level electrons reveal a large amount of information about the material being studied. Each element has a unique set of binding energies which allows the major components of the material's elemental composition to be found. Elemental atomic sensitivity factors are also known for commonly used X-ray energies, such as the Al Kα (1486.6 eV) and Mg Kα (1253.6 eV)[283], which allow the material's stoichiometry to be determined. Furthermore, chemical effects such as bonding with other elements and oxidation state changes can result in characteristic shifts of the core electron binding energy, which allow these chemical effects to be identified within a sample, as seen in Figure 3.9.a. XPS also allows key features of the valence band density of states to be identified, and also the position of the valence band maximum with respect to the Fermi level of the material to be measured, which can be used to determine the surface band bending. An example of a typical valence band spectrum for the SnO_2 nanotubes is shown in Figure 3.9.b.

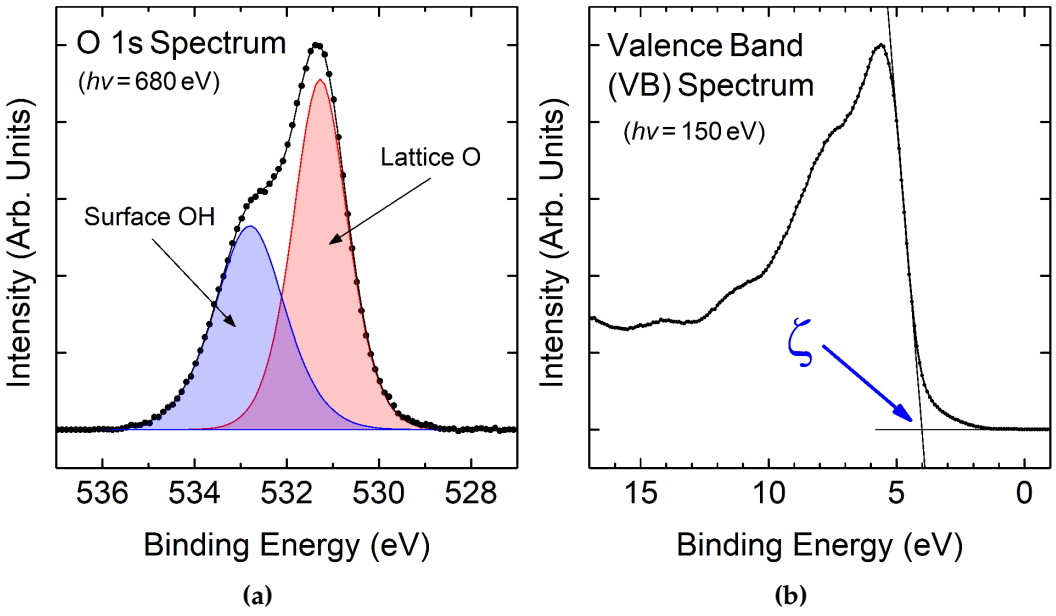


Figure 3.9: (a) O 1s XPS spectrum of undoped SnO_2 nanotubes with fitted peaks attributed to surface hydroxyl groups and lattice oxygen. (b) Valence band XPS spectrum of undoped SnO_2 nanotubes, with the position of the valence band maximum with respect to the Fermi level (ζ) extracted using the method of Chambers et al.[284].

In this thesis, all of the XPS measurements were made using synchrotron XPS. Synchrotron light sources operate by accelerating a beam of electrons close to the

speed of light through the use of a linear accelerator and booster ring, before injecting the electrons into a storage ring for use by different beamlines. For the beamline that performs XPS, a device known as an undulator oscillates the electrons in the beam, causing them to release photons with a range of energies in the X-ray spectrum. A crystal monochromator is used to select a single wavelength component, which is then passed through vertical and horizontal slits allowing a small area of monochromated X-rays to illuminate a sample within an analysis chamber. The emitted photoelectrons are then collected by a hemispherical analyser which measures the photoelectron flux at different kinetic energies.

Synchrotron XPS exhibits several advantages and disadvantages over lab-based XPS systems. Firstly, the X-ray flux is significantly higher than lab-based systems, which leads to shorter acquisition times and better signal-to-noise ratios. This higher X-ray flux, however, can lead to beam damage on the sample surface. Secondly, the angle of the crystal monochromator plates can be adjusted to change the incident X-ray energy. In lab-based XPS systems, the fixed X-ray energy allows photoelectrons with high kinetic energies to escape from deeper within the sample than photoelectrons with low kinetic energies. This is due to electrons with a higher kinetic energy having a larger inelastic mean free path (IMFP), which is the average distance an electron can travel before experiencing an inelastic collision. As the X-ray energy can be changed in synchrotron XPS, this allows each elemental core level to be probed at the same depth within the sample by adjusting the X-ray energy to a fixed amount above the binding energy of the core level being measured. This is very important for studying and understanding phenomena that occur at material surfaces such as surface band bending. The main disadvantage of this adjustment of the X-ray energy is that the atomic sensitivity factors for each different X-ray energy are often unknown, preventing the atomic stoichiometry of the sample being easily determined.

Synchrotron XPS measurements in this thesis were performed at the Australian Synchrotron in Melbourne, Australia on the soft X-ray (SXR) beamline. An elliptically polarised undulator and silicon crystal monochromator allow adjustment of

the X-ray energy between 81 - 2500 eV, with photon fluxes between $5{\times}10^{11}$ - $3{\times}10^{12}$ photons/s/200mA in the optimal energy range. All samples measured were electrically conducting and grounded to the sample holder using carbon tape. Individual core-level measurements were performed at X-ray energies of 150 eV above the core-level being measured, equivalent to an IMPF of $\lambda \approx 6.2$ Å[285]. The energy of the X-rays was calibrated using a Au reference foil using the Au $4f_{7/2}$ core level at 84 eV as the main binding energy scale reference. Core level spectra were fitted with a pseudo-Voigt Gaussian-Lorentzian function with a Gaussian:Lorentzian weighting of 30:70, with linear or Shirley backgrounds as appropriate. The energetic position of the valence band maxima with respect to the Fermi level (ζ) was extracted by linearly extrapolating the edge of the lowest binding energy emission within the valence band to the instrument background, in accordance with the method of Chambers et al. [284]. An example of this extraction can be seen in Figure 3.9.b.

3.4.4 Near-Edge X-Ray Absorption Fine Structure

Near-edge X-ray absorption fine structure (NEXAFS) is another experimental technique that is commonly performed at synchrotron light sources to investigate the chemical and structural composition of materials. NEXAFS is a form of X-ray absorption spectroscopy (XAS), and is also known as X-ray absorption near-edge structure (XANES)[286, 287]. Fundamentally, NEXAFS uses the same principle as XPS, in that X-rays are able to excite core level electrons to higher energy levels. The key difference between these two techniques is that NEXAFS measures the absorption of the X-rays as a function of the X-ray photon energy. For a photon energy smaller than the binding energy of a certain core level electron, there is relatively little absorption of the incident X-rays. If the photon energy is increased such that it is larger than the binding energy of the core level electron, the X-ray absorption sharply increases. If a continuous spectrum is obtained across this range of photon energies, then this can be used to reveal the absorption edge of the core level being investigated. Key factors such as the position of the absorption edge and its shape then provide insight into the oxidation states and coordination environments of the investigated element. Typically, most measurements are performed on the element's

K-edge (1s core level), although other transitions such as the L- (2s and 2p) or M-edges (3s, 3p and 3d) can also be studied. An example of a NEXAFS spectrum can be seen in Figure 3.10, for the case of the S K-edge in a ZnS single crystal. As this technique requires an adjustable, monochromated photon energy, the majority of XAS is performed at synchrotron light sources[286].

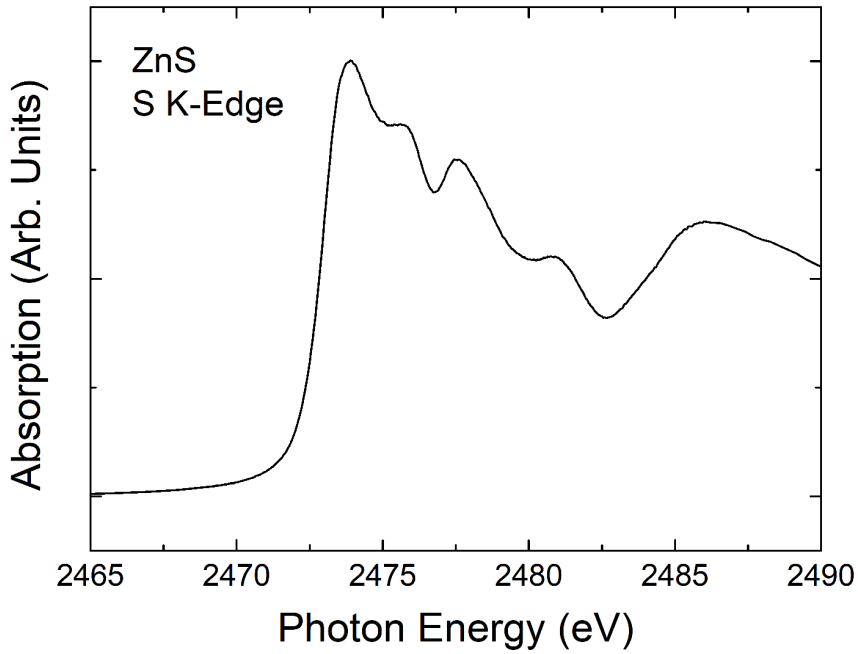


Figure 3.10: A drain current NEXAFS spectrum for the S K-edge of a ZnS single crystal, demonstrating a sharp increase in X-ray absorption when the incident photon energy is increased past the S 1s core level binding energy.

Several methods can be used to determine the X-ray absorption during XAS. The most direct method measures the transmission of the X-rays before and after the sample, allowing the absorption of the X-rays to be calculated. This method is commonly used for elements with a higher atomic number (such as transition metals), as their 1s core levels have large binding energies that require the use of hard X-rays which are very penetrating[286]. This technique is less ideal for elements with a lower atomic number, as the lower energy X-rays used are less penetrating which requires much thinner samples to be used for transmission measurements. The fluorescent yield of the sample is another method used to quantify the absorption of the incoming X-rays, and is more suitable for thicker samples and lower atomic number elements. When core level electrons are excited by the incoming X-rays, electrons in higher energy orbitals lose energy when they replace the excited electron, commonly

by emitting characteristic X-rays. The total yield of these characteristic X-rays is representative of the absorption of the incoming X-rays. A final method to determine the X-ray absorption is by observing the drain current of the sample. As the excited core level electrons are mainly emitted from the sample as photoelectrons, an incoming flow of electrons from the grounded sample holder occurs to replace the emitted photoelectrons to prevent the sample from charging. This flow of electrons into the sample is known as the drain current, and is also representative of the X-ray absorption. As the emission of photoelectrons only occurs near the surface of the sample, the absorption calculated by the drain current is typically more surface sensitive than the absorption found by transmission or fluorescence measurements[287].

NEXAFS investigations were performed in this thesis to identify different sulfur species present at the surfaces of SnO_2 nanotubes and ZnO single crystals that had undergone sulfur treatments. These studies were performed at the Australian Synchrotron on the medium energy X-ray absorption spectroscopy beamline (MEX-2). S K-edge NEXAFS spectra were captured using simultaneous fluorescence and drain current measurement modes for photon energies between at least 2465-2490 eV. All samples were electrically conducting and grounded to metal sample holder rulers with carbon tape.

3.4.5 Scanning Tunnelling Microscopy

Scanning tunnelling microscopy (STM) is a form of scanning probe microscopy that is used to image material surfaces. In vacuum, a fine tip is moved close to the surface of an electrically conducting material and a voltage bias is applied between the material and the tip, which results in a tunnelling current across the vacuum between the tip and the material surface. Given a constant voltage bias, the tunnelling current is exponentially dependent on the distance between the tip and the material surface. By setting a fixed target tunnelling current, the distance between the tip and the sample surface can be held constant to a very high accuracy. As the tip is scanned across the surface at this constant target tunnelling current, a 3-dimensional image of the surface can be mapped out.

STM was used to image the surface morphology of the side walls of SnO_2 nanotubes. A Scienta Omicron VT STM was used with a commercial tungsten STM tip to capture STM images at room temperature. A constant bias of 3.5 V was applied to the tip with a target tunnelling current of 20 pA. The sample was prepared by dispersing 0.15% Sb-doped SnO_2 nanotubes across a highly conducting n-type Si (111) substrate. The sample was then annealed at 400 °C for 10 minutes to remove any remaining organic solvents from the sample surface. After the sample was inserted into the STM vacuum, the sample was further annealed at 200 °C for one hour to remove excess water vapour adsorbed on the sample surface.

Chapter 4

Electrical Properties and Applications of SnO_2 Nanotube Resistors

4.1 Introduction

Previous work by Dr. Jonty Scott attempted to construct electrical devices that incorporated the SnO_2 nanotubes[29], which would allow the performance of the nanotubes in different applications to be assessed. These simple devices consisted of two Ti/Au ohmic contacts on each end of an individual SnO_2 nanotube, creating a nanotube-based resistor and allowing current-voltage (IV) measurements to be performed. Despite the simplicity of the devices, problems during device fabrication resulted in the detachment of the Ti/Au contact from the top of the nanotubes. This provided a poor electrical contact causing potentially unreliable IV characteristics. Furthermore, IV measurements were only performed on a small number of Sb-doped nanotubes, with no successful measurements performed on undoped nanotubes. More work was needed to optimise the fabrication process for high-quality and reliable devices to be made. This would allow the study of the effect of Sb-doping on the electrical properties of the SnO_2 nanotubes and allow the investigation of SnO_2 nanotubes for different applications.

In this chapter, individual undoped and Sb-doped SnO_2 nanotube resistors are

successfully fabricated, and their electrical properties are characterised. These devices are then investigated for their UV sensing ability, with a focus on the persistent photoconductivity that commonly occurs with SnO_2-based photodetectors. The UV response of both undoped and Sb-doped resistors is characterised, and measurements performed at different temperatures and in different gas environments provide a greater understanding of the main contributing factors of persistent photoconductivity in the devices.

4.2 Resistors of Individual SnO_2 Nanotubes

4.2.1 Fabrication and Imaging of Individual SnO_2 Nanotube Resistors

Resistors based on individual SnO_2 nanotubes were constructed using a technique similar to that of Scott[29]. Undoped or 0.15% Sb-doped SnO_2 nanotubes were grown for a period of 3 hours or more to achieve lengths of at least 6 μm long. These nanotubes were dispersed across a 10×10 mm oxidised (100) Si substrate with a ~130 nm thick oxide layer, grown by 30 minutes of wet oxidation at 950 °C. The oxidised Si substrate provides electrical isolation of the contact pads yet prevents substrate charging under electron microscopy, allowing individual nanotubes to be clearly imaged. Photolithography was then used to pattern areas for ohmic contacts onto the nanotube-coated substrate. Electron-beam deposition was used to deposit Ti/Au layers with 500 nm and 50 nm of Ti and Au respectively, for a total contact thickness of 550 nm. Importantly, this Ti/Au layer was 50 nm thicker than the 500 nm thick layer deposited by Scott, which created a stronger top contact due to a thicker connection between the material on top of the nanotube and the material attached to the substrate. This increased thickness, however, did result in a more difficult lift-off step. This lift-off step was performed in acetone with ultrasonic agitation to remove the excess metal from between the contact pads. A solvent clean in acetone, methanol and isopropanol was performed, before the samples were annealed at 400 °C for 10 minutes in nitrogen. More details of these fabrication procedures can be found in Section 3.3.

The deposited ohmic contacts were patterned in a regular grid formation, seen in

Figure 4.1.a, consisting of arrays of 50×50 µm squares separated by a ~4 µm gap. As the nanotubes were dispersed across the substrate, contacting a nanotube required a nanotube to be bridging two contacts by chance. This required the use of a SEM to identify the locations of individually contacted nanotubes and to assess the quality of the metal contacts. Given the ~22,500 contacts across the 10×10 mm substrate and large numbers of dispersed nanotubes, bridged nanotubes occurred frequently, allowing multiple bridging undoped and Sb-doped nanotubes to be identified. The additional thickness of the metal contact layers and care taken during metal lift-off also resulted in the formation of high-quality top contacts, with the top layer remaining attached to contacted nanotubes as seen in Figure 4.1.b.

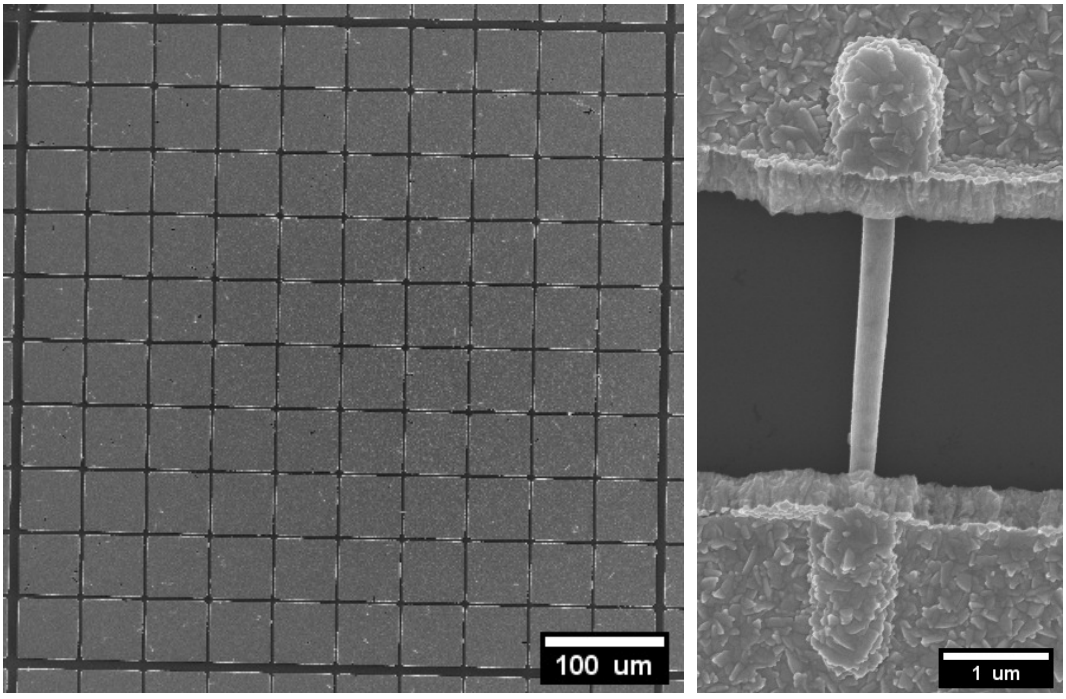


Figure 4.1: SEM images of (a) a grid of Ti/Au ohmic contacts patterned using optical lithography and deposited by electron-beam evaporation. (b) An individual, well contacted SnO_2 nanotube.

4.2.2 Electrical Measurements of Individual SnO_2 Nanotube Resistors

The successful contacting of both undoped and 0.15% Sb-doped nanotubes allowed electrical measurements to be performed on the devices. IV measurements of contacted undoped and 0.15% Sb-doped SnO_2 nanotubes can be seen in Figure 4.2. These measurements consisted of forward and reverse sweeps between -3 V and

3 V. Zero current above the probe station electrical noise floor was observed between unbridged contacts, indicating that all measured current flowed through the bridging nanotube. The undoped nanotubes show forward currents up to ~10^{-6} A at 3 V, equivalent to a total resistance around 3×10^{6} Ω. Some unideal IV characteristics are also present, with all undoped nanotubes displaying current hysteresis and noise during the forward and reverse sweeps. The maximum currents for each device at ±3 V are also generally asymmetric, indicating non-ideal ohmic contacts with a partial Schottky contact nature. Conversely, the Sb-doped nanotubes exhibit good electrical properties, with very little hysteresis, low noise and symmetrical IV curves under both positive and negative voltage bias. This indicates that the Sb-doping promotes the formation of high-quality ohmic contacts to the doped nanotubes. This is due to the thinning of the depletion region at any Schottky contact barriers with increasing carrier concentration[154], which allows electrons to tunnel across the barrier promoting ohmic conduction in the device. The increase in carrier concentration by Sb-doping has also decreased the material resistivity, resulting in an approximate two orders of magnitude increase in the maximum current to 10^{-4} A in the best Sb-doped device.

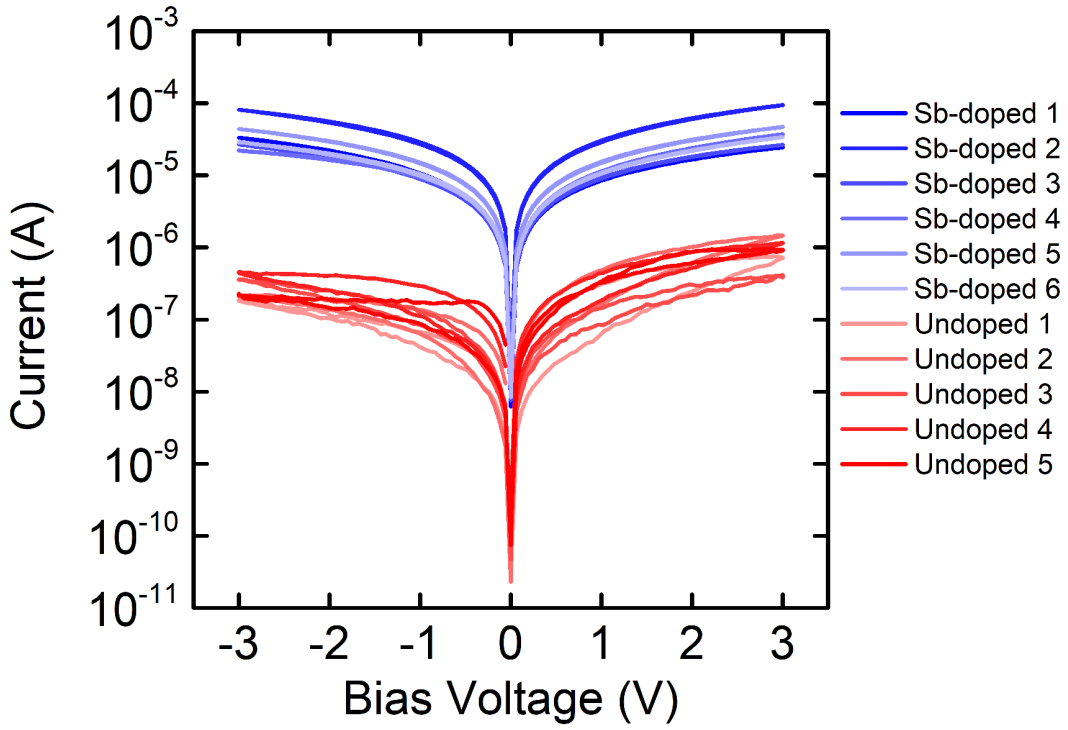


Figure 4.2: IV measurements of grid-contacted undoped (Red) and 0.15% Sb-doped (Blue) SnO_2 Nanotubes.

The significant hysteresis observed in the undoped nanotubes warranted further characterisation. Subsequent IV scans were performed on an individual undoped nanotube under an increasing maximum bias voltage, seen in Figure 4.3. The voltage step size and subsequent sweep rate were kept constant for each set of voltage sweeps. It can be seen that an increasing maximum voltage bias during the sweep results in larger current hysteresis. Increasing maximum voltages give smaller currents under stages 1 and 3 of the sweeps and the larger currents during stages 2 and 4. These changes are dependent on the increasing maximum voltages as multiple ±3 V sweeps did not change the observed hysteresis.

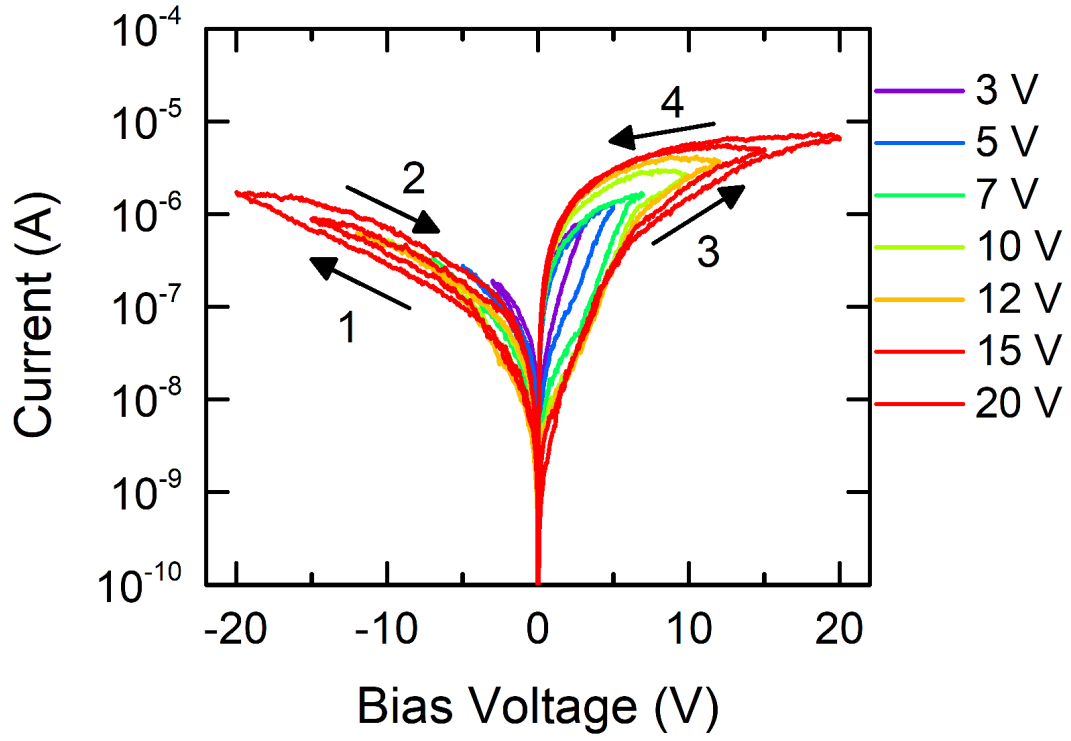


Figure 4.3: IV measurements of a single undoped SnO_2 nanotube under an increasing range of bias voltage, demonstrating significant hysteresis effects. Labelled arrows show the sweep order and direction.

These hysteresis effects have been observed before in the literature, all of which contain non-ohmic contacts to a SnO_2 channel forming back-to-back Schottky barriers. Nieh et al. produced devices constructed from Au contacts to SnO_2 nanowires for resistive memory applications[195]. The same mode of hysteresis was observed as in this work, and was explained as the migration of oxygen vacancies to the cathode electrode under voltage bias causing changes to the Schottky barrier there. Zhou

et al. used Sm-doped SnO_2 nanowires with Ag contacts to produce memristive devices[196]. They attributed their observed hysteresis to electron trapping at surface defects near the cathode under high voltages, leading to a maintained low resistance state in the device. Chen et al. characterised the resistive switching properties of Au-contacted SnO_2 nanowires, and believed their hysteresis was due to the movement of adsorbed oxygen ions caused by Joule heating under voltage bias[197]. Moisés and Chiquito found that water vapour significantly affected the memristive properties of SnO_2 nanowire networks with Ag electrodes, and proposed the migration of water vapour filling oxygen vacancies for this observation[198].

To identify if a similar conduction mechanism to that found in the literature is responsible for the hysteresis in the undoped SnO_2 nanotube resistors, the forward voltage sweep (step 3 in Figure 4.3) was transformed to plot $\ln(I/V^2)$ against $1/V$. A graph showing this transformation for each maximum bias voltage can be seen in Figure 4.4. There is a clear linear relationship between $\ln(I/V^2)$ and $1/V$ at values of $1/V$ below 0.4 V^{-1}, which is consistent with a transition in the main conduction mechanism to Fowler-Nordheim tunnelling[288, 289]. This is the same mechanism that has been shown in other memristive SnO_2 nanowire-based devices[196, 197].

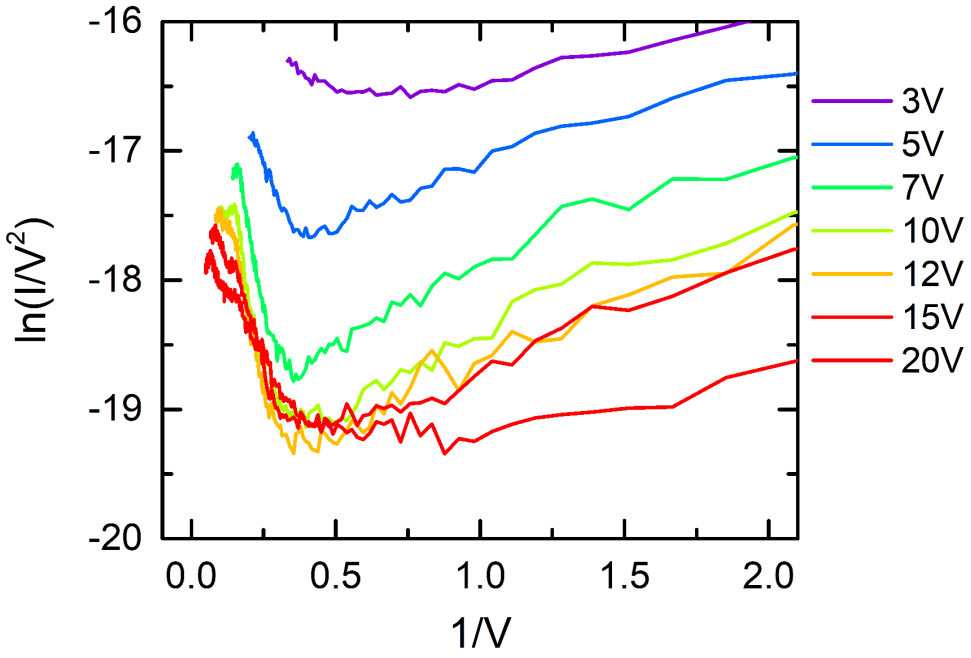


Figure 4.4: The forward voltage sweeps transformed to plot $\ln(I/V^2)$ against $1/V$, indicating a Fowler-Nordheim tunnelling mechanism dominating at small $1/V$.

The range of different proposed mechanisms for this behaviour and lack of agreement within the literature means we are unable to identify the exact mechanism that causes the behaviour in the devices fabricated here. However, we can conclude that this behaviour is due to the formation of Schottky barriers at the Ti/Au contact with the SnO_2 nanotubes, with a hysteresis behaviour that may also depend on atmospheric oxygen and water vapour. The explanation of Schottky barriers at the contacting material interface also explains why the hysteresis effect is only present in devices that use undoped nanotubes, as the Sb-doped nanotubes exhibit better ohmic contact behaviour due to their much higher carrier density and as such have little current hysteresis.

4.3 UV Photoresponse of Individual SnO_2 Nanotube Resistors

The successful fabrication of individual SnO_2 nanotube resistors allowed for potential applications to be investigated. The combination of material properties such as a wide bandgap and good electrical conductivity means that the SnO_2 nanotube resistors may be well suited for UVB photodetection. In this section, individual SnO_2 nanotube resistors are illuminated with above-bandgap UVB light to observe the UV photoresponse and persistent photoconductivity (PPC) that commonly affect SnO_2-based UV photodetectors.

The standard UV exposure procedures were as follows. Firstly, in a dark environment a 2 V bias was applied to the measured device, while the current through the device was monitored over time (known as the dark current). This was maintained until no changes in current were observed over a period of at least 100 s, removing any self-heating/hysteresis effects from the UV photoresponse. A 285 nm QPhotonics UV LED was used as the UV source, which produced an on-device illumination intensity of ~200 μW cm^{-2}. The UV LED was turned on for a period of 100 s, with current measurements occurring for an extended period after this to observe the PPC response of the device. Further details of the characterisation methodology can be seen in Section 3.

4.3.1 UV Photoresponse of Undoped SnO_2 Nanotubes

Firstly, an undoped individual SnO_2 nanotube resistor was illuminated multiple times in air to determine the effects of repeated UV exposures on the photoresponse and PPC recovery. The observed photoresponse can be seen in Figure 4.5. The first exposure to UV light produced a sharp increase in current through the device from 0.96 µA to 20.1 µA, equivalent to a sensor response of 20.9. After the UV LED is turned off, the induced photocurrent remains high for a long period, falling much more slowly than the induced photocurrent rise. This is indicative of a strong PPC effect in these devices. Subsequent UV exposures show a slight decrease in the maximum photocurrent after 100 s of UV illumination, with the fourth exposure having a maximum photocurrent of 18.2 µA and a sensor response of 7.7. The dark current and PPC were also affected with subsequent UV exposures, as the dark current failed to return to the same level as before the first exposure and the PPC appeared to be longer for the later exposures.

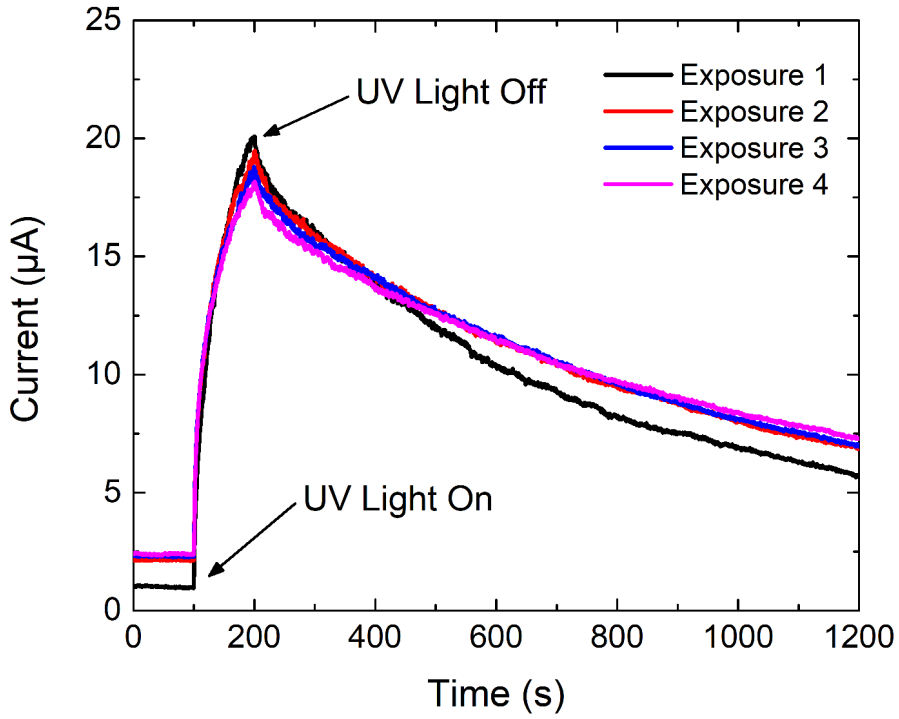


Figure 4.5: The UV photoresponse of an undoped SnO_2 nanotube resistor to repeated 100 s exposures to 285 nm UV light.

To observe further differences between the shapes of the induced photocurrent rise and PPC recovery independently from the dark current and maximum photocurrent, the photocurrent rise and PPC recovery were normalised with respect to these parameters for each UV exposure. On the Y-axis in each normalised figure, 0 represents the level of the dark current while 1 represents the photocurrent after 100 seconds of UV exposure. The normalised photocurrent rise and PPC recoveries for the repeated UV exposures to the undoped SnO_2 nanotube resistor can be seen in Figures 4.6 and 4.7 respectively. There are only minor differences between each UV exposure for the normalised current rise, with the first UV exposure rising at a slightly slower normalised rate in the first 20 seconds compared to the other exposures. There is a more pronounced effect on the normalised PPC recovery, where subsequent exposures result in an increase to the PPC recovery time.

To quantify the recovery time of the PPC recovery, a Kohlrausch stretched exponential function was fitted to the normalised PPC recovery[282]:

$$e^{-(\frac{t}{\tau})^{\beta}} \tag{4.1}$$

Where τ is the decay time constant for the PPC recovery and β is a decay exponent used to stretch the fitted exponential, as time-dependant processes often deviate from an ideal exponential decay. The values of τ extracted from the fits to the four normalised PPC recovery curves can be seen in Figure 4.8. The four repeated UV exposures result in an increase of τ from 542 s to 831 s from the first to fourth exposure, illustrating how repeated UV exposures cause an increase in the PPC recovery time.

These results indicate how the SnO_2 nanotube resistors are susceptible to unwanted PPC effects that commonly affect other SnO_2-based photodetectors. In the literature, these effects are associated with the desorption of acceptor-like surface adsorbates, which are commonly attributed to molecular oxygen ionosorbed as superoxide ions (O_2^-)[184, 185, 290]. The commonly accepted process for PPC is as follows. The natural hydroxyl groups present on SnO_2 surfaces exposed to atmosphere result in strong surface electron accumulation, downward surface band bending and emanating electric fields from the material surface[30]. Electrophilic molecules such

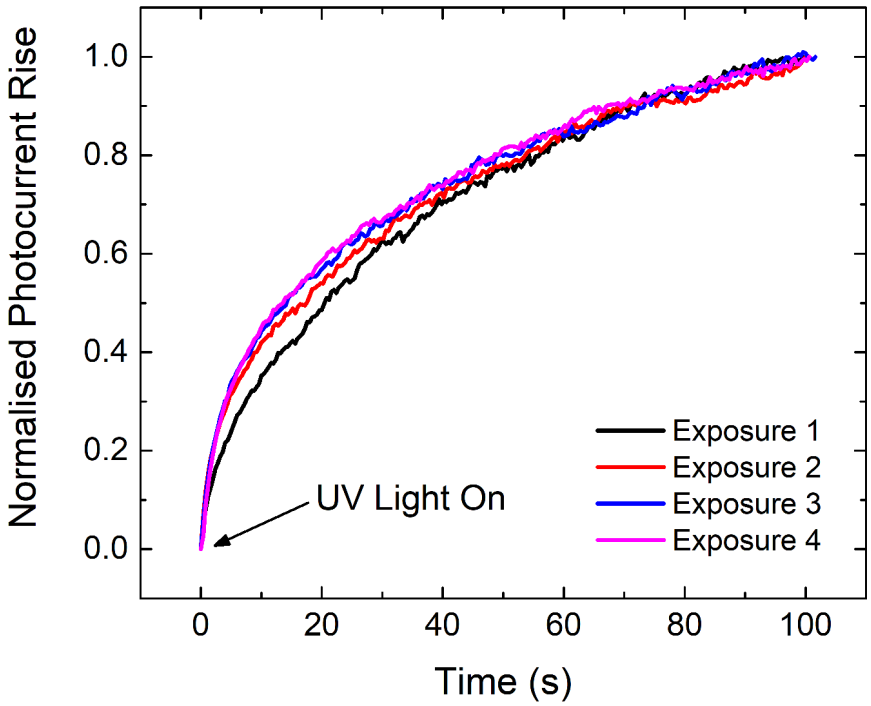


Figure 4.6: The normalised photocurrent rise for repeated UV exposures to an undoped SnO_2 nanotube resistor.

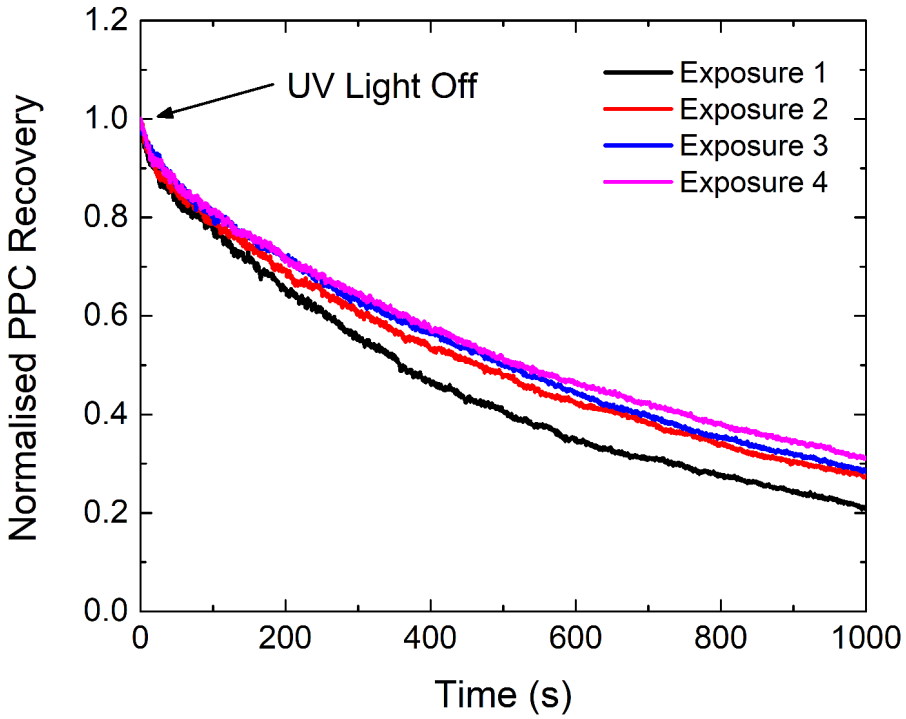


Figure 4.7: The normalised PPC recovery for repeated UV exposures to an undoped SnO_2 nanotube resistor.

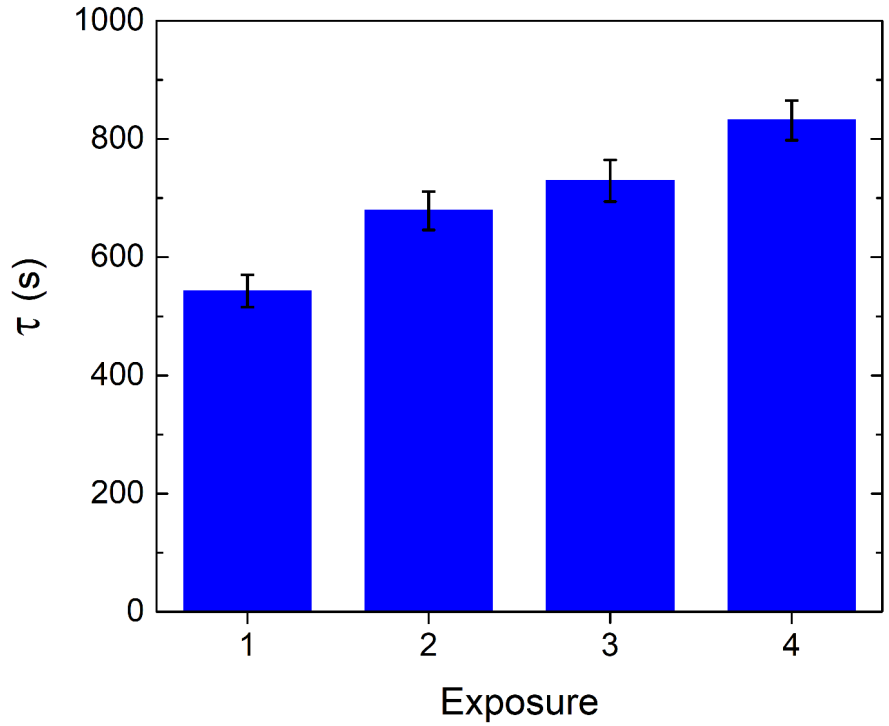


Figure 4.8: Fitted values of τ for repeated UV exposures to an undoped SnO_2 nanotube resistor.

as oxygen are attracted to the SnO_2 surface and chemisorb, capturing a surface electron[41, 170]:

$$O_2(g) + e^- \rightarrow O_2^-(ads) \tag{4.2}$$

The continued adsorption of oxygen on the SnO_2 surface then results in an eventual electron depletion and an inversion of the surface electronic bands from downward to upward band bending[291, 292]. This is the steady-state that a device in atmosphere tends to operate in. When above-bandgap UV light is illuminated on a SnO_2 photodetector, electron-hole pairs are created as electrons are promoted from the valence band to the conduction band by incident photons. Due to the built-in electric field of the upward band bending present at the SnO_2 surface, the electron-hole pairs are separated, with the photogenerated hole migrating towards the surface. The hole then recombines with an electron trapped by an adsorbed oxygen molecule which then desorbs from the surface[184, 185, 290]. The photogenerated electron, however, remains in the bulk of the material and this leads to an overall increase in conductivity. When the UV illumination is stopped, oxygen molecules begin to readsorb

onto the SnO_2 surface, recapturing conduction band electrons and re-establishing the surface depletion region. This process, however, operates on a much longer time scale than the UV-induced desorption, resulting in a long-lasting PPC effect.

4.3.2 UV Photoresponse of 0.15% Sb-doped SnO_2 Nanotubes

A similar procedure was adopted to determine the effect of Sb-doping on the UV response of the SnO_2 nanotube resistors. A 0.15% Sb-doped nanotube resistor was repeatedly exposed to 100 s of UV light in air, seen in Figure 4.9. The first UV exposure produced an increase in device current from 26.4 µA to 85.4 µA, equivalent to a sensor response of 3.2. This reduction in sensor response is largely due to the Sb-doping, as this increases the material carrier concentration and thus reduces the width of the depletion region caused by ionosorbed oxygen at the nanotube surface. As this thinner depletion region is removed by the UV-induced desorption of oxygen, there is a smaller relative increase in the conducting nanotube cross-section, resulting in a smaller sensor response. Repeated exposures to UV light in the Sb-doped device cause a much larger reduction in the maximum photocurrent compared to the undoped device, with the sixth UV exposure producing a maximum photocurrent of 64.2 µA. This is equivalent to a ~36% drop in the relative photocurrent rise between the first and sixth UV exposure. This unusual and non-ideal behaviour may be caused by the stabilisation of surface adsorbates as they readsorb after each UV exposure, in which less adsorbates are released from the surface decreasing the magnitude of the photocurrent rise. Unlike on the undoped nanotube, the post-UV dark current was able to return close to the initial dark current after all six UV exposures.

Figures 4.10 and 4.11 show the normalised photocurrent rise and PPC recovery for the 0.15% Sb-doped nanotube resistor. Compared to the undoped nanotube resistor, the photocurrent rises and saturates at its maximum much faster on the Sb-doped device, achieving an ~85% photocurrent rise after 20 seconds compared to a ~55% photocurrent rise on the undoped device. Repeated UV exposures causes only minor slowing of the normalised photocurrent rise, despite the significant drop in the

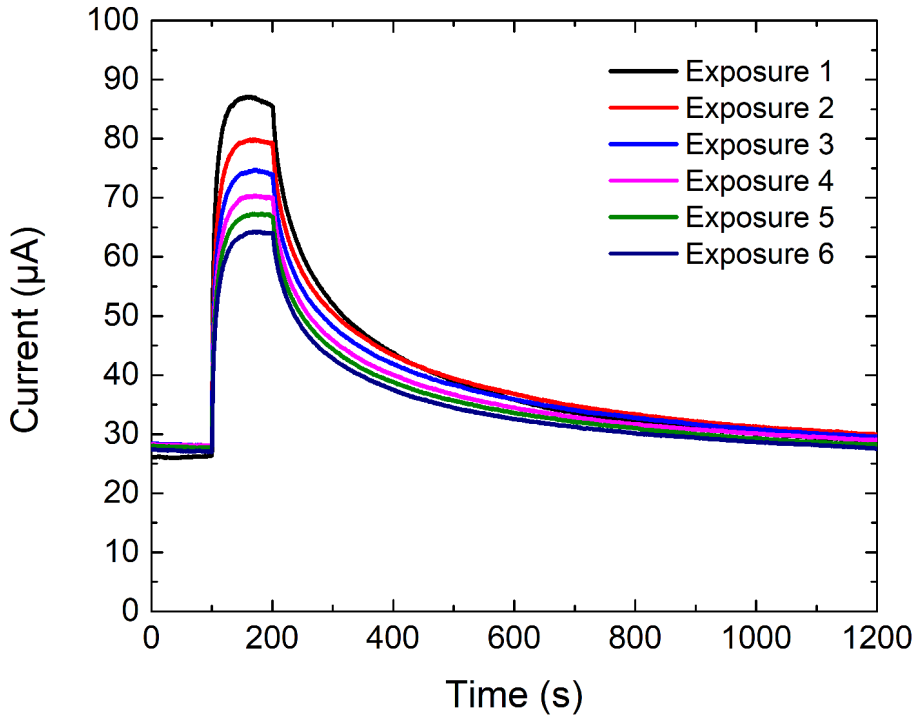


Figure 4.9: The UV photoresponse of a 0.15% Sb-doped SnO_2 nanotube resistor to repeated 100 s exposures of UV light.

maximum induced photocurrent. Similarly, the normalised PPC recovery is unaffected by multiple UV exposures, exhibiting a near-identical normalised current recovery for each exposure. The fitted values of τ, seen in Figure 4.12, span between 136 s and 148 s. These values of τ confirm the faster recovery of the Sb-doped device compared to the undoped device, which had a smallest τ of 542 s.

The faster PPC recovery of the Sb-doped device compared to the undoped device can be explained by the higher carrier concentration caused by the Sb doping. The re-adsorption of acceptor-like adsorbates when the UV light is turned off begins to cause a degree of upward band bending that depletes the SnO_2 surface of electrons, creating a depletion region at the material surface. This depletion region and its built-in electric field opposes the movement of free electrons from the bulk of the material to the material surface where they can be captured by more surface adsorbates. In a material with higher carrier concentration however, this depletion region is thinner, which allows electrons to more easily reach the SnO_2 surface and be captured by surface adsorbates, resulting in a faster PPC recovery.

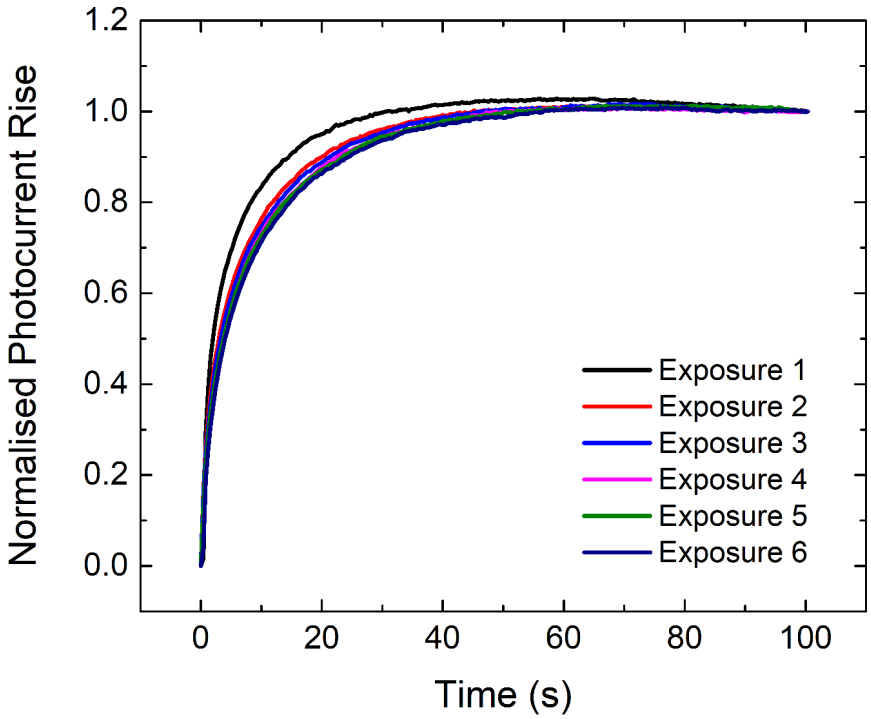


Figure 4.10: The normalised photocurrent rise for repeated UV exposures to a 0.15% Sb-doped SnO_2 nanotube resistor.

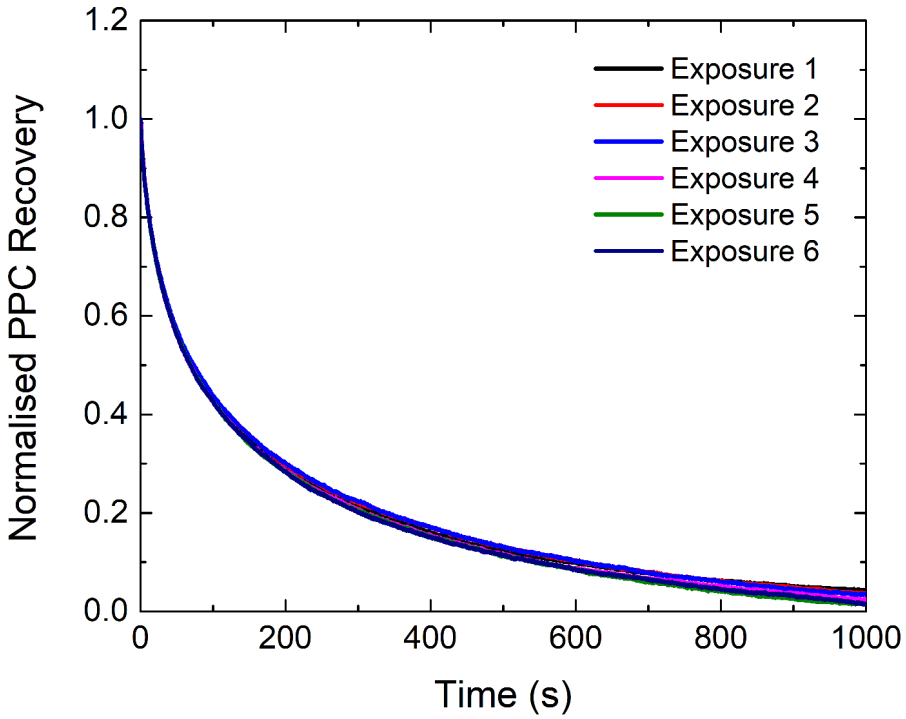


Figure 4.11: The normalised PPC recovery for repeated UV exposures to a 0.15% Sb-doped SnO_2 nanotube resistor.

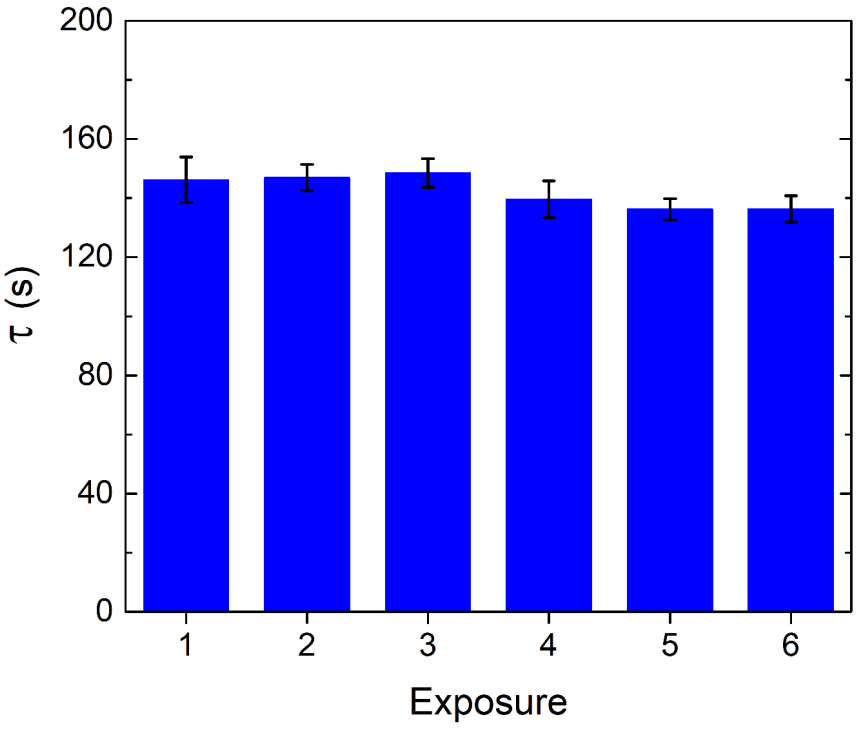


Figure 4.12: Fitted values of τ for repeated UV exposures to a 0.15% Sb-doped SnO_2 nanotube resistor.

4.3.3 Temperature Dependence of UV Photoresponse in Undoped SnO_2 Nanotubes

The temperature dependence of the photoresponse for the SnO_2 nanotube resistors was investigated by performing UV exposures at increasing substrate temperatures. UV measurements took place in air at room temperature (~23 °C), 50 °C, 100 °C, and 150 °C. The resulting photoresponses can be seen in Figure 4.13. Firstly, it can be clearly seen that the increasing temperatures result in a significant increase to the device dark current, which increases from 1.6 µA at room temperature to 10.1 µA and 15.4 µA at 100 °C and 150 °C respectively. While the conductivity of all semiconducting materials increases with increasing temperature due to the thermal ionisation of more charge carriers, an 860% increase in current over a ~125 °C temperature range is far too large to be solely attributed to this process. Instead, much of this current increase likely comes from the thermal desorption of acceptor-like surface adsorbates, primarily molecular and ionosorbed oxygen. Temperature desorption spectroscopy (TDS) studies have shown that oxygen species desorb from the SnO_2 surface in a

broad temperature range that has peaks at 80 °C and 150 °C, which have been attributed to the desorption of O_2 and O_2^- respectively[70, 293]. Water vapour also desorbs from the SnO_2 surface around 100 °C, which can affect the surface conductivity[70]. Despite the rise in dark current with temperature, the magnitude of all photocurrent rises remain relatively constant, with all UV exposures producing a 20-25 µA photocurrent rise.

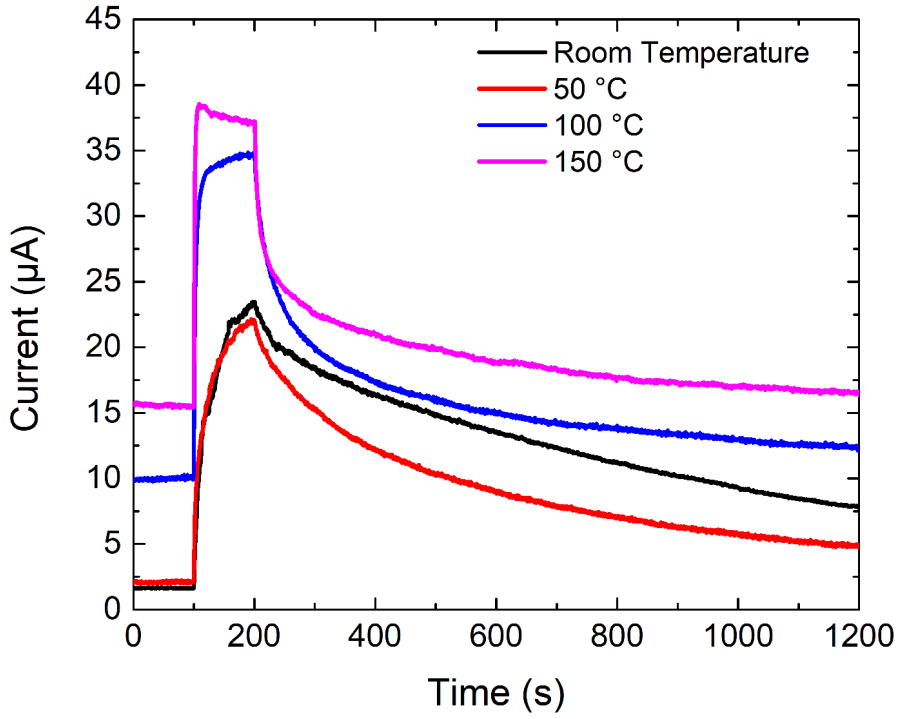


Figure 4.13: The UV photoresponse of an undoped SnO_2 nanotube resistor under increasing temperatures.

The normalised photoresponse rise and PPC recovery for the temperature series on the undoped nanotube resistor can be seen in Figures 4.14 and 4.15. It can be observed that the normalised photocurrent rise increases quickly at higher temperatures, with the exposures performed at 100 °C and 150 °C exhibiting sharp electronic rises during the first few seconds of the UV exposure. This indicates that an increasing operating temperature allows a fast electronic response to occur under UV illumination and minimises the effect of oxygen desorption, as much of the ionosorbed oxygen has already been thermally desorbed. A similar case can be seen in the normalised PPC recovery, where increasing temperatures result in a faster relative decrease in photocurrent. The fitted value of τ at each temperature can be seen

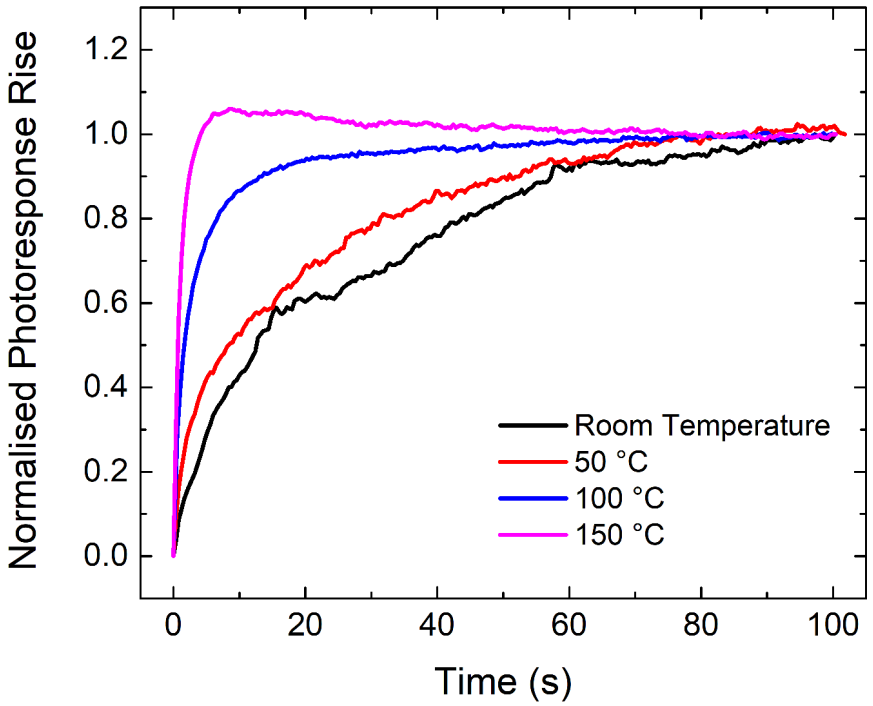


Figure 4.14: The normalised photoresponse rise of an undoped SnO_2 nanotube resistor under increasing temperatures.

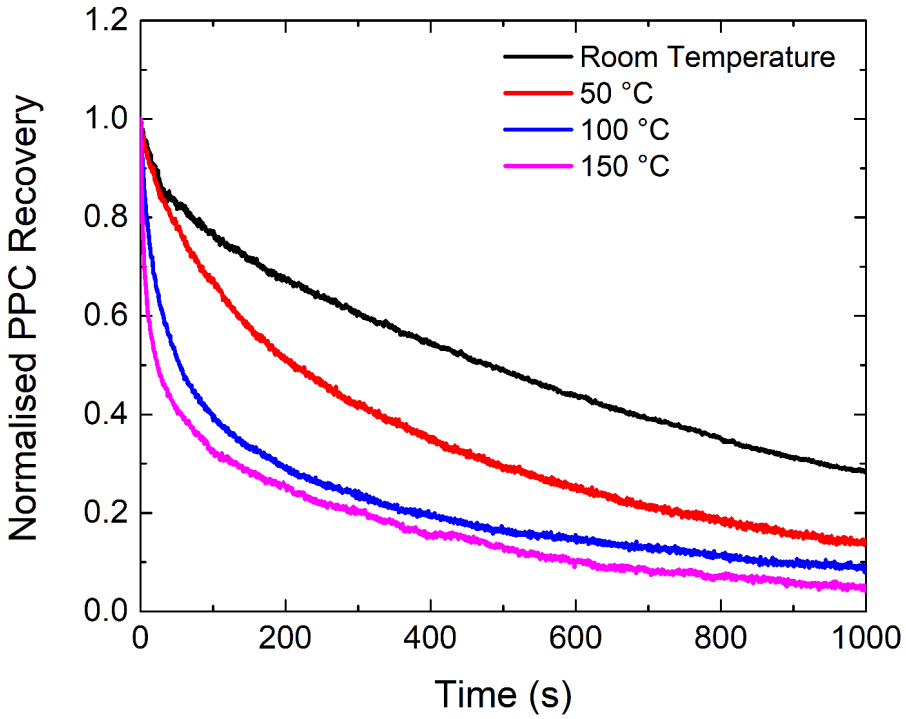


Figure 4.15: The normalised PPC recovery of an undoped SnO_2 nanotube resistor under increasing temperatures.

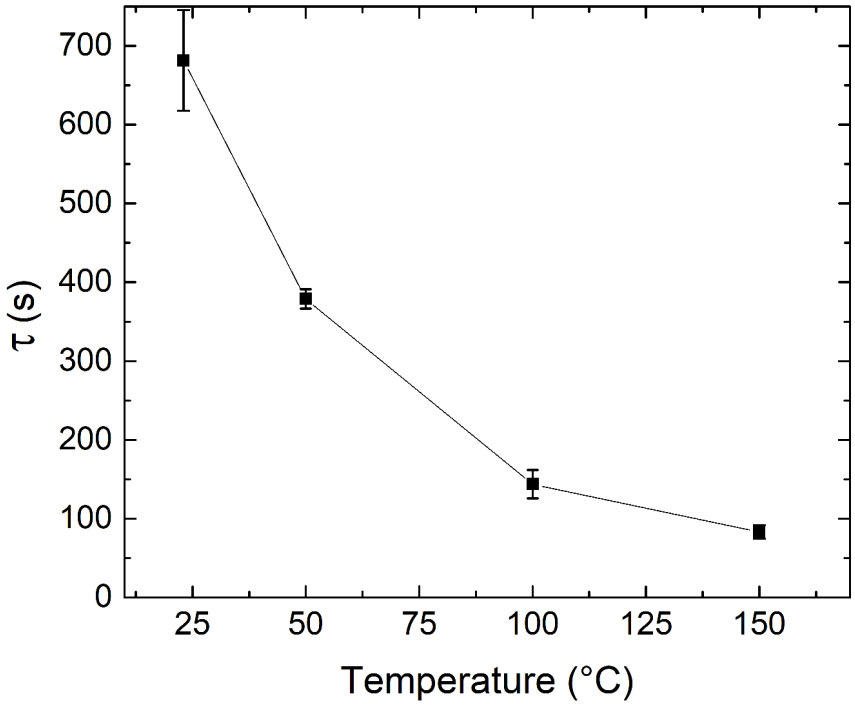


Figure 4.16: Fitted values of τ for an undoped SnO_2 nanotube resistor under increasing temperatures.

in Figure 4.16, where τ decreases from 682 s at room temperature to 83 s at 150 °C.

As the PPC recovery is observed to be a temperature dependant process, the Arrhenius equation can be used to model the temperature dependence of τ[294, 295]:

$$\tau = \tau_0 e^{\frac{E_a}{k_b T}} \tag{4.3}$$

Where τ_0 is a pre-exponential factor, k_b is the Boltzmann constant and E_a is the reaction activation energy. Figure 4.17 shows the Arrhenius plot of the log of τ against 1/T, where a clear linear relationship can be observed. Fitting the slope of this linear relationship gives an activation energy of 180 meV.

Similar values to this activation energy for the temperature dependence of PPC recoveries have been previously reported in the literature. dos Santos et al.[296] synthesised SnO_2 thin films through a sol-gel process, and found an activation energy of 190 meV for their temperature dependent PPC recoveries. Similarly, Liu et al.[297] found an activation energy of 160 meV for the temperature dependent PPC in SnO_2 nanowires. A thermal activation energy of 230 meV was observed in SnO_2

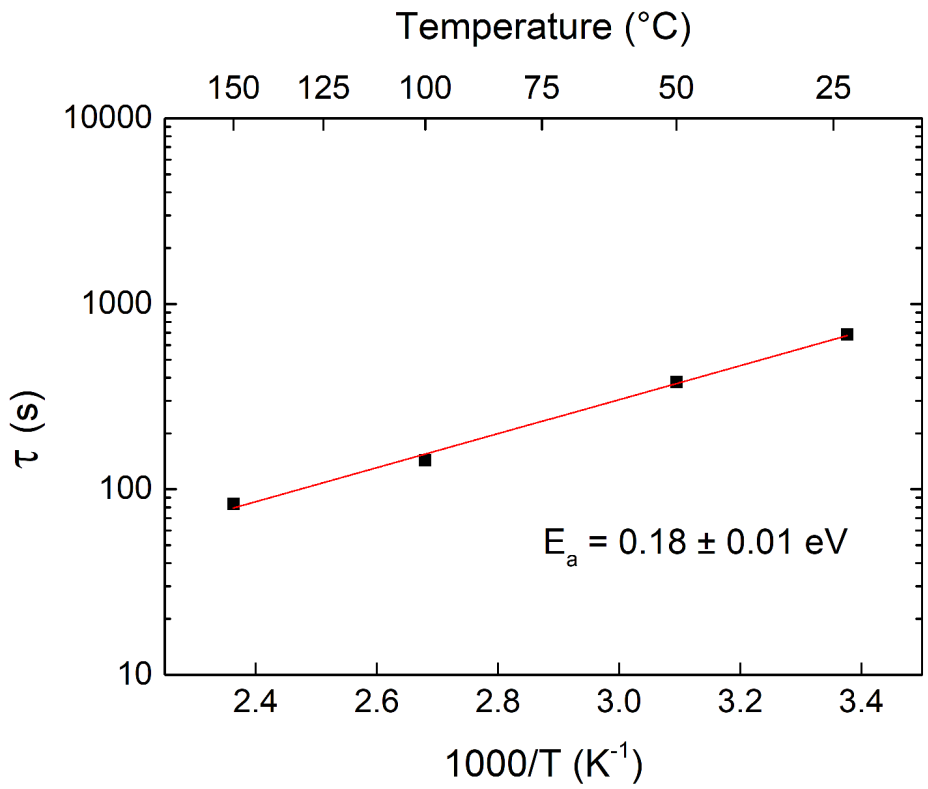


Figure 4.17: Arrhenius plot of the log of τ against 1/T on the undoped device, where a clear linear relationship allows the extraction of an activation energy for the temperature dependence.

nanobelts by Viana et al.[184]. While their study correlated the activation energy to mid-bandgap defects, the activation energy associated with the SnO_2 nanotube resistors is more likely connected to atmospheric adsorbates and a temperature dependent readsorption process, particularly given the strong ambient dependence observed in the upcoming Section 4.3.5 and the high single crystal material quality of the square SnO_2 nanotubes.

4.3.4 Temperature Dependence of UV Photoresponse in Sb-doped SnO_2 Nanotubes

The temperature dependence of the UV photoresponse was similarly investigated on the 0.15% Sb-doped nanotube resistors to determine if a similar process was occurring compared to the undoped device. Figure 4.18 shows the UV photoresponse of a 0.15% Sb-doped nanotube resistor under an increasing operating temperature, up to 150 °C. Similar to the undoped device, the dark current of the Sb-doped device increases dramatically with an increasing operating temperature, from 12.2 µA

at room temperature to 36.5 µA at 150 °C. This is again likely due to the thermal desorption of oxygen from the nanotube surface, with a smaller relative increase in dark current compared to the undoped device due to the thinner depletion region on the Sb-doped nanotubes. The amplitude of the photocurrent increase under UV illumination was observed to decrease with increasing temperature, with the absolute photocurrent increase compared to the dark current being 38.8 µA at room temperature, 26.8 µA at 50 °C, 21.0 µA at 100 °C and 14.9 µA at 150 °C. As the four exposures were performed in the order of increasing temperature, there may also be a significant contribution from the decreasing maximum photocurrent effect observed during multiple UV exposures to Sb-doped devices as described in Section 4.3.2. Consequently, we will focus on the temperature dependence of the normalised photoresponse rise and PPC recovery which did not significantly change with multiple UV exposures.

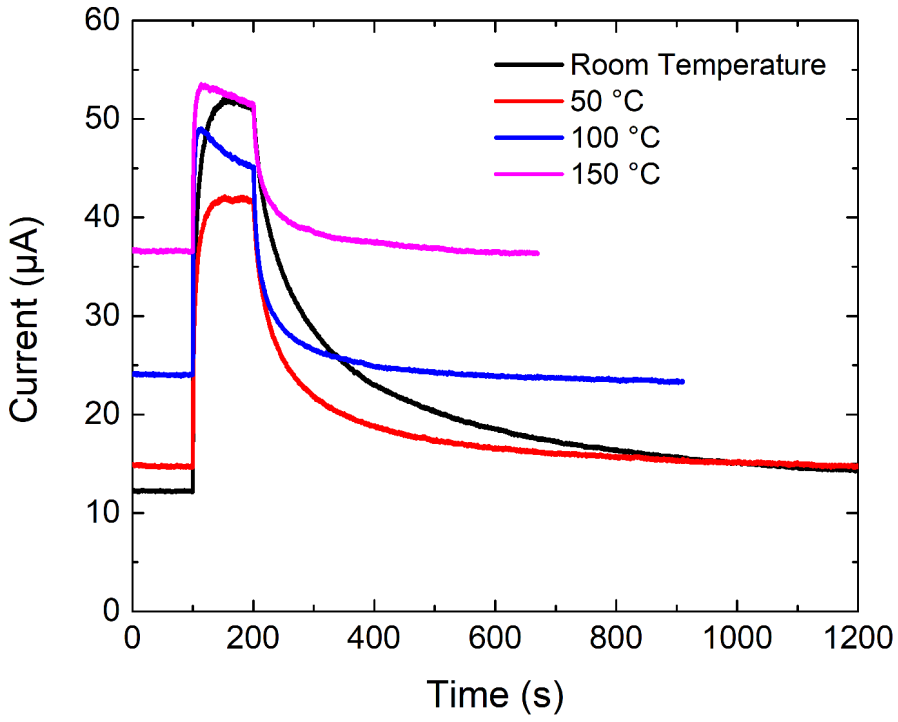


Figure 4.18: The UV photoresponse of an Sb-doped SnO_2 nanotube resistor under increasing temperatures.

The normalised photocurrent rise and PPC recovery of the Sb-doped nanotube resistor under increasing temperatures can be seen in Figures 4.19 and 4.20 respectively. The effect of the thermal desorption of surface adsorbates above 100 °C is clearly observed in the normalised rise current, where a slower photocurrent rise is present at room temperature and 50 °C while a fast electronic rise is present at 100 °C and 150 °C. Interestingly, the fast electronic response is slightly counter balanced by a slower decay in photocurrent intensity after 10 s of exposure, suggesting either the desorption of a donor species or the dynamic readsorption of an acceptor species. The normalised PPC recovery shows a similar temperature dependence to the undoped nanotubes, with an increasingly faster recovery under increasing temperatures. However, the PPC recoveries at 100 °C and 150 °C show only slightly different recovery rates. This can be seen in the plot of τ against temperature in Figure 4.21, where there is little difference between the fitted values of τ at 100 °C and 150 °C.

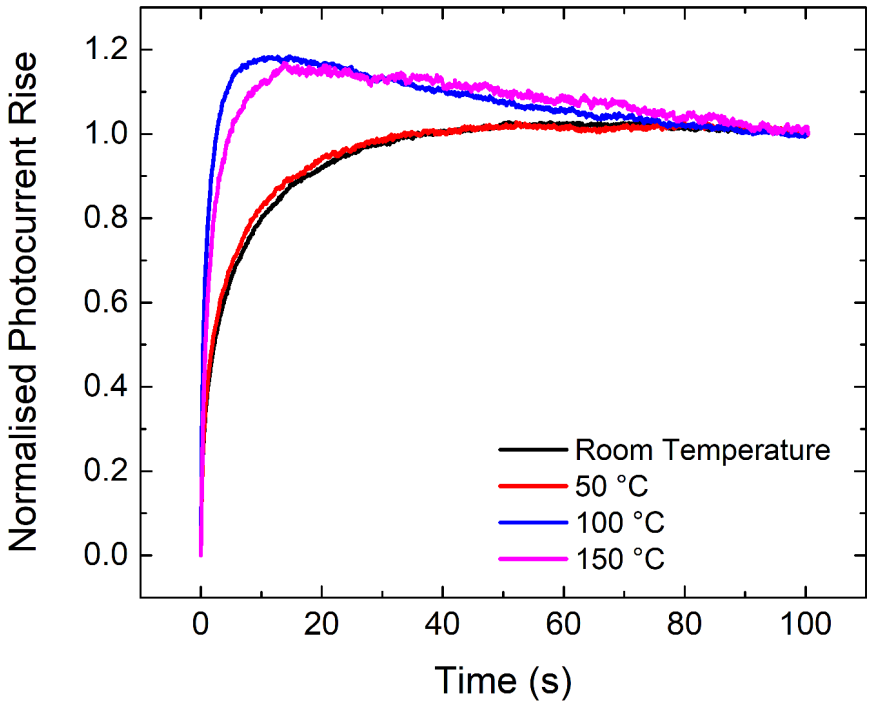


Figure 4.19: The normalised photoresponse rise of an Sb-doped SnO_2 nanotube resistor under increasing temperatures.

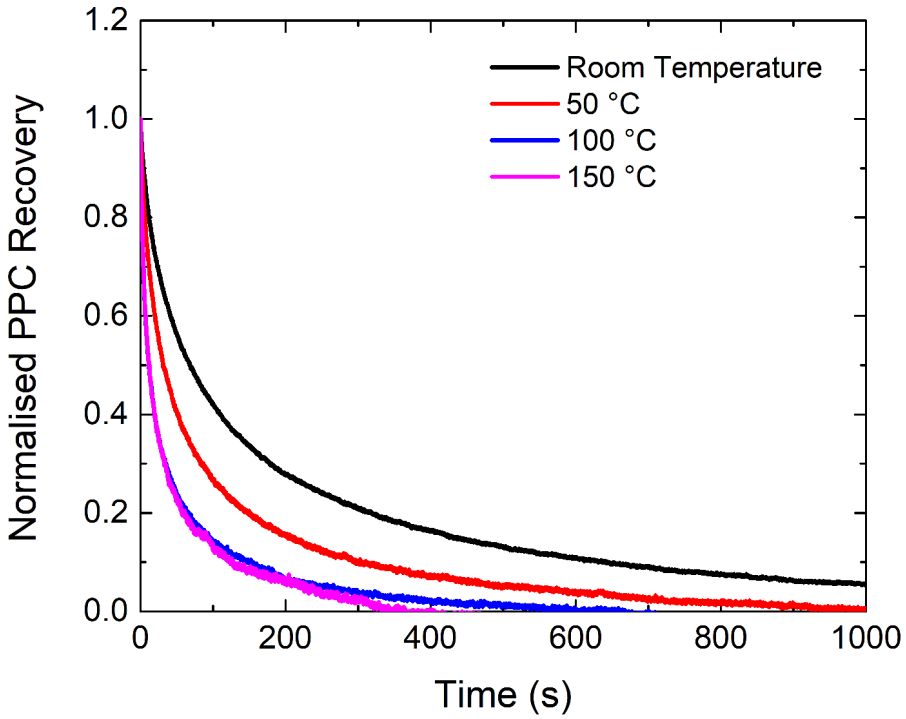


Figure 4.20: The normalised PPC recovery of an Sb-doped SnO_2 nanotube resistor under increasing temperatures.

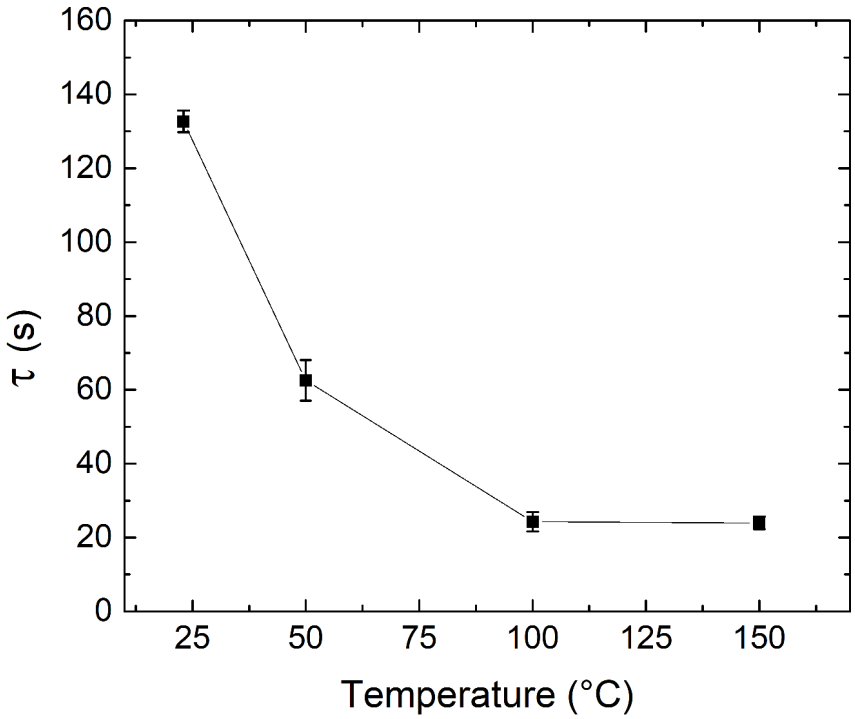


Figure 4.21: Fitted values of τ for an Sb-doped SnO_2 nanotube resistor under increasing temperatures.

The minor differences in τ between measurements performed at 100 °C and 150 °C complicate the Arrhenius plot of τ against temperature, seen in Figure 4.22. The intrinsically faster recovery time of the Sb-doped nanotube device combined with the thermal desorption of surface adsorbates likely means that electronic processes such as electron-hole recombination dominate much of the PPC recovery instead of an O_2/H_2O molecular readsorption process at higher temperatures. These electronic processes are likely less temperature dependent than a molecular readsorption process, leading to a significantly decreased overall temperature dependence when these processes dominate. Despite this transition from a readsorption-dominated process to an electronic process at higher temperatures, a potential linear trend for the three lowest temperatures within the Arrhenius plot can still be observed. A fit to these three data points yields an activation energy of 210 meV, which is close to the value of 180 meV seen on the undoped device. This suggests that a similar activation energy is present for the readsorption processes on both the undoped and Sb-doped SnO_2 nanotubes, indicating that the same adsorption mechanisms are present irrespective of the doping and material carrier concentration.

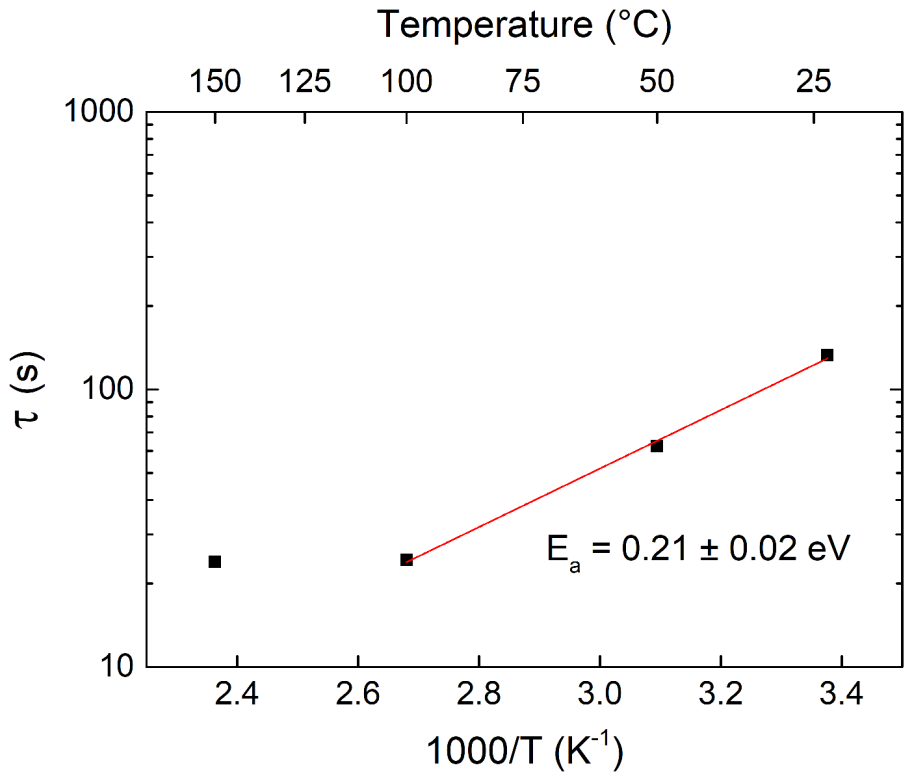


Figure 4.22: Arrhenius plot of the log of τ against 1/T on the Sb-doped device.

4.3.5 Ambient Dependence of UV Photoresponse in Undoped SnO_2 Nanotubes

The commonly accepted mechanism behind PPC effects in SnO_2 in the literature is the desorption of ionosorbed oxygen by photogenerated holes. However, much of the existing literature only performs PPC measurements in air, and compares these with measurements performed under vacuum or in inert gases such as He[184, 186, 297]. Oxygen is not the only surface adsorbate present in the atmosphere, with water vapour presenting itself as a common SnO_2 adsorbate in relatively high concentrations. The interaction of water vapour on the SnO_2 surface has been widely documented, with water vapour acting as an electron donor[70, 123, 298, 299]. This electron-donating effect has led SnO_2 nanomaterials to be used in humidity sensing applications[123, 299, 300]. Despite the clearly documented adsorbate interactions between water vapour and the SnO_2 surface and the importance of surface adsorbates in the existing theories behind the PPC mechanisms, very little research exists on the effect of water vapour on the PPC recovery of metal oxide UV photodetectors. Key results from Martinez-Gazoni et al. demonstrated that a humid N_2 environment resulted in a faster PPC recovery than in air or even a pure O_2 environment, although this study investigated the (101) SnO_2 surface with below-bandgap illumination[187]. This gives a clear gap in the existing body of knowledge related to the combined roles of oxygen and water vapour on the above-bandgap PPC in SnO_2-based photodetectors. For this reason, the UV photoresponse of the undoped SnO_2 nanotube resistors was investigated under a range of different ambient gas environments.

To produce a specific gas environment around the nanotube-based resistors, the Faraday enclosure of the Cascade Microtech 11000 probe station was filled with a positive pressure of the desired gas. While some molecules from the ambient air may remain in the probe station chamber, the positive pressure of gas resulted in an environment that was dominated by the desired gas. Measurements were performed in five different gas environments: Air, dry N_2, dry O_2, humid N_2 and humid O_2. The dry gases produced environments with a relative humidity of $< 5\%$, while the humid gas environments with relative humidities $> 95\%$ were produced by bubbling

the dry gases through a water bubbler heated to 60 °C. The measurement performed in ambient air had a relative humidity of ~30%.

Figure 4.23 shows the UV photoresponse of an undoped SnO_2 nanotube resistor under the different ambient conditions, while Figures 4.24 and 4.25 show the normalised photocurrent rise and PPC recovery respectively. The fitted values for τ for each ambient can be seen in Figure 4.26. Immediately, there are several dramatic differences in the UV photoresponse and photocurrent decay in the different ambients.

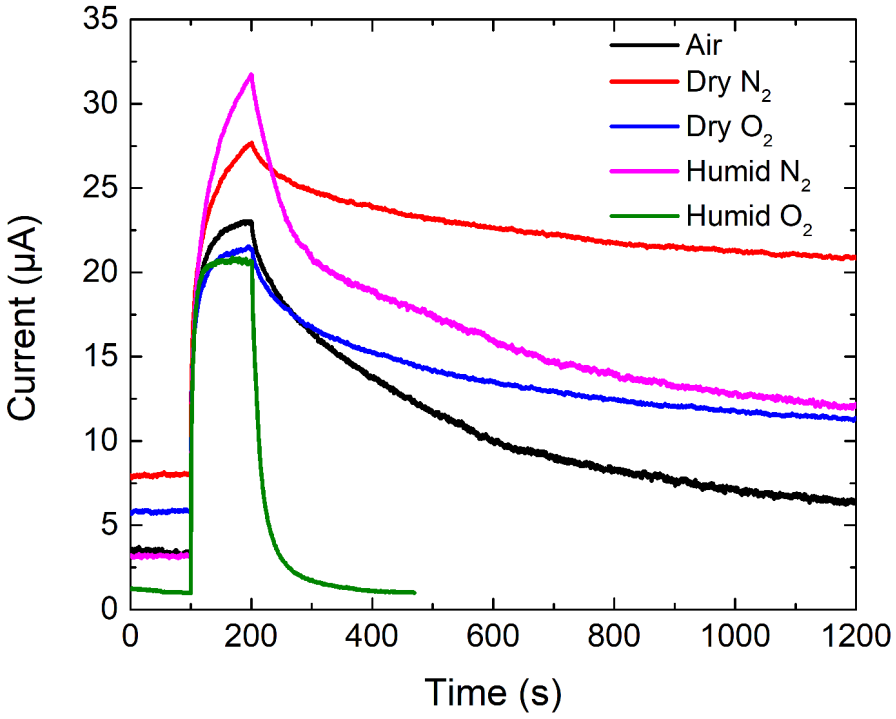


Figure 4.23: The UV photoresponse of an undoped SnO_2 nanotube resistor in different gas ambients.

Firstly, in a dry N_2 environment devoid of both water vapour and oxygen, a larger UV photoresponse than in air alongside a slower normalised photoresponse rise and significantly longer PPC recovery was observed. These observations fit well within the existing theory of the desorption of surface adsorbates. As the UV light is turned on, photogenerated holes remove adsorbed oxygen from the surface which is then displaced by the N_2 ambient. There are few oxygen molecules in the gas phase near the surface, so there is little readsorption of oxygen both when the UV light is on and after the UV light is turned off. This process gives the slower normalised photocurrent rise and significantly longer PPC recovery, with a fitted τ value of 5000 s

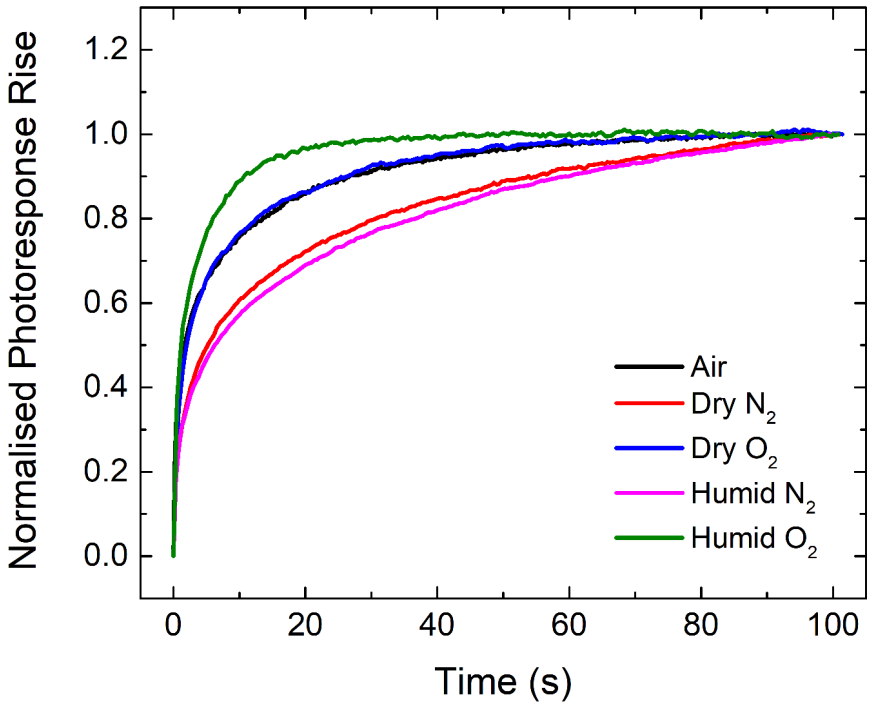


Figure 4.24: The normalised photoresponse rise of an undoped SnO_2 nanotube resistor in different gas ambients.

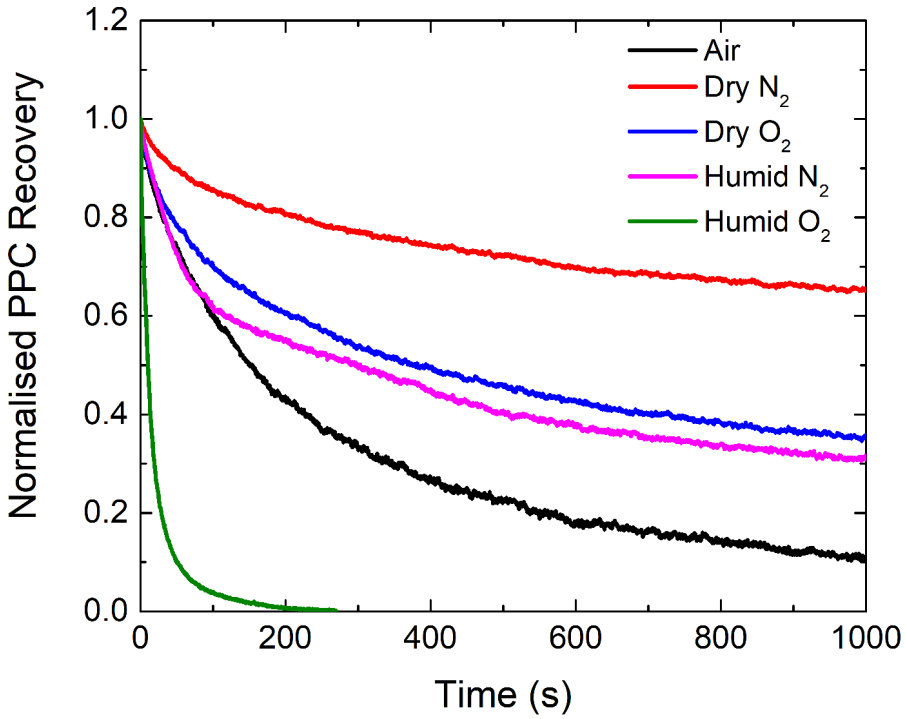


Figure 4.25: The normalised PPC recovery of an undoped SnO_2 nanotube resistor in different gas ambients.

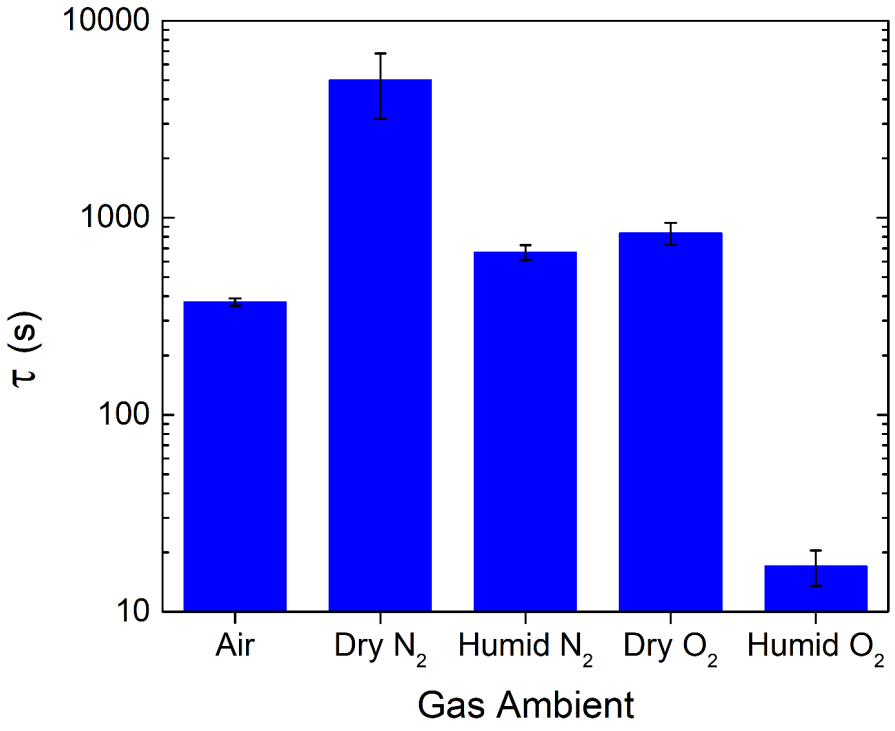


Figure 4.26: Fitted values of τ for an undoped SnO_2 nanotube resistor in different gas ambients.

compared to 374 s in air.

In a dry O_2 environment a smaller absolute UV photoresponse is observed compared to air. This is due to the dynamic process of competitive oxygen desorption with some readsorption while the UV light is on. A steady-state of adsorbed oxygen coverage will be reached when the rate of oxygen desorption matches the rate of oxygen readsorption. Under a higher partial pressure of oxygen, this results in a larger steady-state adsorbed oxygen coverage that gives a smaller maximum photocurrent compared to the UV exposure performed in air. A significant finding is observed in the normalised PPC recovery, where the PPC recovers at a slower rate in a 100% dry O_2 compared to in air. The fitted value of τ for the PPC recovery in dry O_2 is 835 s compared to 374 s in air. This finding suggests that, despite acting as a surface acceptor, additional processes are occurring that further determine the rate of the PPC recovery.

The humid N_2 environment produced the largest absolute photoresponse out of

all the gas ambients investigated here. This is explained by the absence of the acceptor oxygen species and the presence of the electron-donating water vapour, producing a donor-rich surface under UV illumination that creates a large UV photocurrent. The lack of oxygen is also present in the normalised photocurrent rise, where the current rises at a similar rate to the dry N_2 environment. Unlike the results of Martinez-Gazoni et al.[187], the presence of water vapour in an N_2 environment does not produce a photoresponse that recovers faster than in air. Instead, the PPC recovery occurs on a similar time scale to dry O_2, with τ displaying a fitted value of 669 s. Clearly, water vapour does play a role in the PPC recovery of the SnO_2 nanotube resistors as the response is still significantly faster than in dry N_2. Although, neither oxygen or water vapour is solely responsible for the faster PPC recovery in air.

The UV exposure performed in humid O_2 reveals that the combined effect of both water vapour and oxygen significantly impacts the UV photoresponse and PPC recovery of the undoped SnO_2 nanotube resistors. The normalised photocurrent rise shows the fastest rise to steady-state out of all the different gas ambients measured, indicating that the rate of UV induced desorption and readsorption equalise at a faster rate in humid O_2. The normalised PPC recovery occurs significantly faster in humid O_2, with a fitted value of τ of only 17 s, which is over 20 times faster than the recovery in air. This remarkable result clearly illustrates the co-operative effect of water and oxygen on the minimisation of the PPC in the SnO_2 nanotube resistors.

To further explore the effect of humid O_2 on the UV photoresponse of the nanotube resistors, multiple UV exposures were performed in humid O_2 followed by multiple exposures in a humid N_2 environment. The measured UV photoresponse can be seen in Figure 4.27, while the fitted values of τ for each exposure can be seen in Figure 4.28. The fast PPC recovery time is maintained throughout each UV exposure in humid O_2, with τ decreasing slightly from 26 s to 16 s. Switching the gas ambient from humid O_2 to humid N_2 still results in a relatively fast PPC recovery following the first UV exposure, however, τ increases with each subsequent exposure, increasing from 34 s for the first exposure to 103 s for the fourth. Furthermore, all the exposures performed in humid N_2 following the humid O_2 gas environment

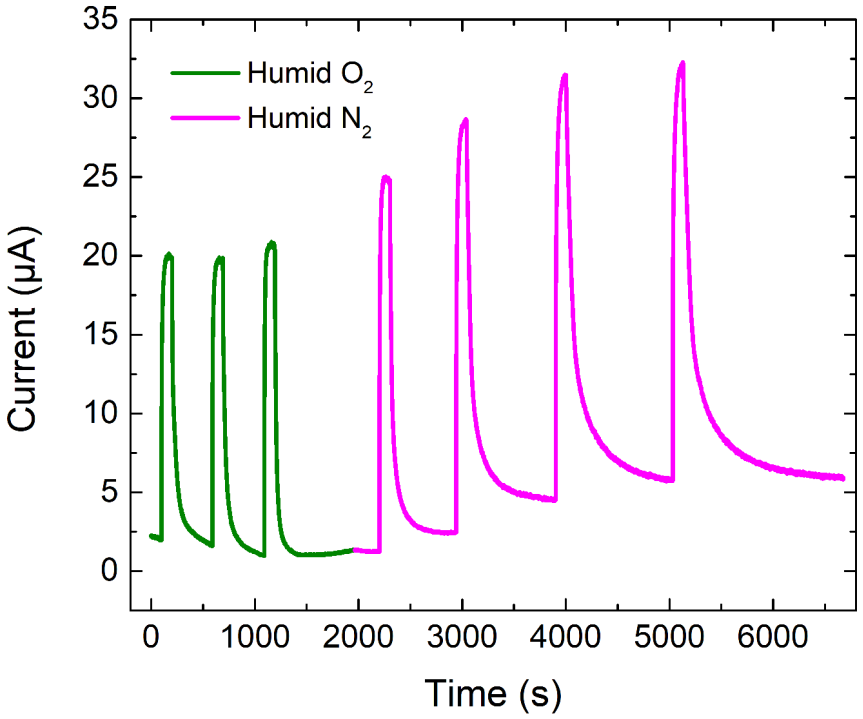


Figure 4.27: The UV photoresponse of an undoped SnO_2 nanotube resistor to multiple exposures in a humid O_2 and then humid N_2 environment.

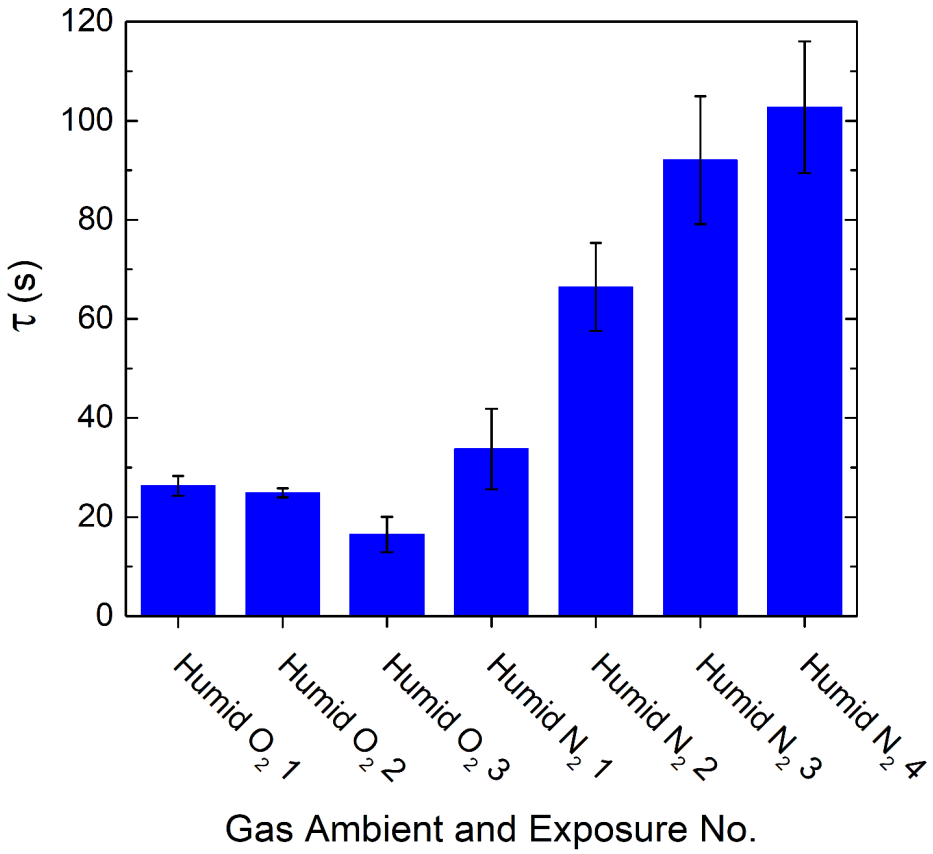


Figure 4.28: The fitted values of τ for each repeated UV exposure in the humid O_2 and humid N_2 environment.

were faster than the original humid N_2 measurement, where the humid N_2 ambient was introduced when the sample was originally resting in an air ambient rather than humid O_2. This difference in the UV photoresponse between these two initial conditions suggests that water vapour helps to retain the PPC recovery properties from the original gas environment before the water vapour is introduced.

4.3.6 Model for the Co-Operative Role of Oxygen and Water Vapour on the PPC Recovery

A model will now be proposed that explains the co-operative role of oxygen and water vapour on the UV-induced PPC of the SnO_2 nanotube resistors. At the nanotube surface, a large coverage of terminating hydroxyl (OH) groups results in downward band bending of the electronic bands at the surface resulting in strong electron accumulation. These terminating OH groups arise from the chemisorption of water vapour onto an exposed in-plane Sn atom, which dissociates into two OH groups by donating a H atom to a nearby bridging oxygen site[95, 96]. This produces a terminal OH on the in-plane Sn and a bridging OH on the bridging oxygen. The bridging OH group likely becomes ionised and donates an electron to the surface conduction band[95, 301, 302]:

$$H_2O + Sn_{5c} + O_{br} \rightarrow (Sn_{5c}^{\delta+} - OH^{\delta-}) + O_{br}H^+ + e^- \tag{4.4}$$

These surface OH groups are very thermally robust, only being removed by heating at temperatures above 300-400 °C[30, 70] and are unlikely to be affected by UV illumination.

In air, this electron rich surface and the polar OH surface terminating groups attract additional layers of water vapour and oxygen, which physisorb onto the surface. In particular, the adsorbed O_2 molecules capture the free surface electrons forming superoxide ions (O_2^-)[41, 170], depleting the surface and reversing the downward band bending to upward band bending. This creates the surface depletion region widely reported on SnO_2 in atmospheric conditions[184, 186, 291, 292]. Under high coverages of both water vapour and charged O_2^-, a strong hydrogen-bonded network is formed which inhibits the complete desorption of O_2 and H_2O

by UV light. This effect is illustrated in Figure 4.29, where oxygen molecules are kept near the SnO_2 surface by the hydrogen bonded network. As there is a minimised effect of adsorbate desorption and readsorption, faster electronic-type processes such as electron-hole recombination dominate the PPC recovery process which gives rise to the faster PPC recovery times observed in humid O_2 ambients.

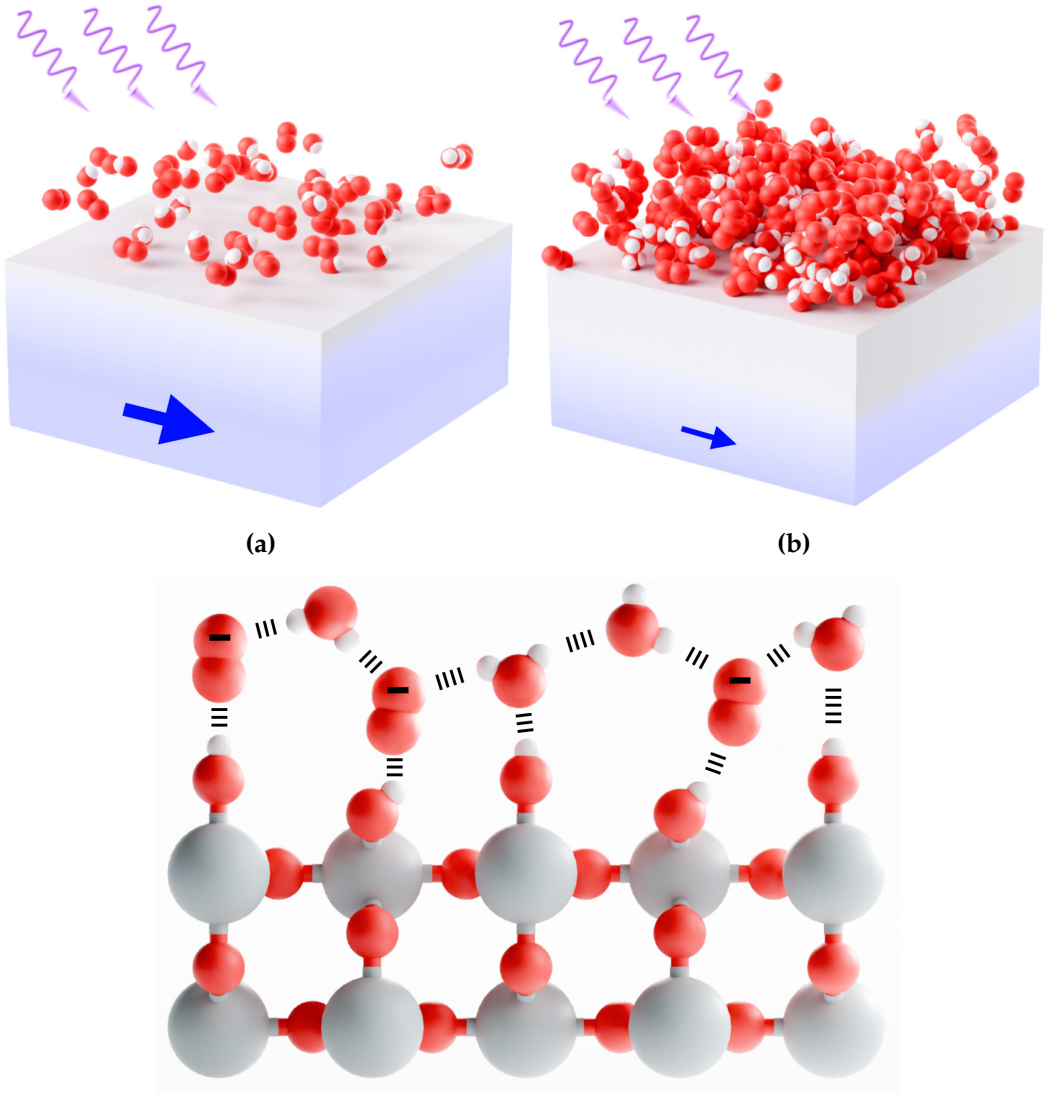


Figure 4.29: The PPC model under UV illumination for (a) partial O_2^-/H_2O coverage (such as in air) and (b) high O_2^-/H_2O coverage (such as in humid O_2). (c) Illustration of the hydrogen-bonded network at high coverages on the hydroxylated SnO_2 (110) surface. All images used with permission, originally created by Dr. Maxime Le Ster.

Hydrogen-bonded networks of physisorbed water have previously been shown to exist on the SnO_2 surface. Kittaka et al. showed that hydrogen bonds between surface hydroxyl groups and physisorbed water stabilise a hydrogen-bonded network

at high water coverages[303]. Similarly, Morishige et al. have reported the formation of 2-D monolayers of hydrogen-bonded physisorbed water on hydroxylated SnO_2 surfaces[304]. Computational results by Bandura et al. found that multilayers of water are stabilised on the SnO_2 (110) surface by hydrogen bonding between each adsorbed layer[305].

In a similar fashion, the interactions of H_2O with O_2^- have been studied both theoretically and experimentally. Computational studies have shown that there is a strong electrostatic interaction between H_2O and O_2^- due to the large difference in electronegativity between the hydrogen and negatively charged oxygen ions[306–310]. In solution, O_2^- can form up to four direct hydrogen bonds with water, observed experimentally as superoxide tetrahydrate by cluster ion spectroscopy[311, 312].

These well documented interactions between the SnO_2 surface, H_2O and O_2^- means it is likely that O_2^- ions adsorbed on the SnO_2 surface are able to be included in the hydrogen-bonded network of physisorbed water molecules on the hydroxylated SnO_2 surface. This would increase the binding energy of the adsorbed O_2^- and reduce the removal of O_2 away from the SnO_2 surface during UV illumination. The increased availability of O_2 and O_2^- near the SnO_2 surface leads to the faster photocurrent rise and PPC recovery times. This process is further supported by the repeated exposures in humid O_2 and humid N_2 ambients. As the humid O_2 environment is replaced by humid N_2, the ionosorbed O_2^- remains in the hydrogen-bonded network at the nanotubes surface. This leads to the fast initial PPC recoveries in the humid N_2 ambient. Over time and with repeated UV exposures, the ionosorbed O_2^- is removed from the hydrogen-bonded network, leading to the increasing PPC recovery time in the humid N_2 ambient.

4.3.7 Ambient Dependence of UV Photoresponse in Sb-doped SnO_2 Nanotubes

To confirm the same co-operative effects of oxygen and water vapour occurs in the PPC of Sb-doped nanotubes, the UV photoresponse of the 0.15% Sb-doped SnO_2 nanotube resistors was measured in the same five gas environments as the undoped device. Figure 4.30 shows the UV photoresponse in different gas ambients for the Sb-doped device, while Figures 4.31, 4.32 and 4.33 show the normalised photocurrent rise, PPC recovery and fitted values of τ, respectively.

Similar to the previous results on the Sb-doped device, it is more difficult to interpret trends from the absolute photocurrent increase due to the decreasing maximum photocurrent effect with repeated UV exposures. In this case, the UV exposures were performed in different ambients in the following order: air, dry N_2, humid N_2, dry O_2, and finally humid O_2. In between each gas ambient measurement, the chamber was opened to air and the induced photocurrent was allowed to recover to the dark current level. This order of UV exposures may explain some findings, such as how the humid N_2 and humid O_2 exposures had smaller absolute photocurrent rises than their dry counterparts, which is different to the case on the undoped device.

Despite the differences in the magnitude of the photocurrent rise, similar trends to those seen in the undoped device are observed in the normalised photocurrent rise. Dry and humid N_2 show similar slower photocurrent rises, while there is little difference between the measurements performed in dry O_2 and air. The measurement performed in humid O_2 initially has a fast photocurrent rise to ~75% of the maximum photocurrent followed by a slower, nearly linear rise throughout the remaining UV exposure. This may be indicative of a fast initial electronic-type response due to the stabilisation of the hydrogen-bonded network followed by a smaller and slower desorption-type response.

The normalised PPC recoveries also show similar ambient dependence compared to the undoped device, albeit with a faster PPC recovery for each environment following the trend of faster PPC recoveries on the 0.15% Sb-doped device. The longest

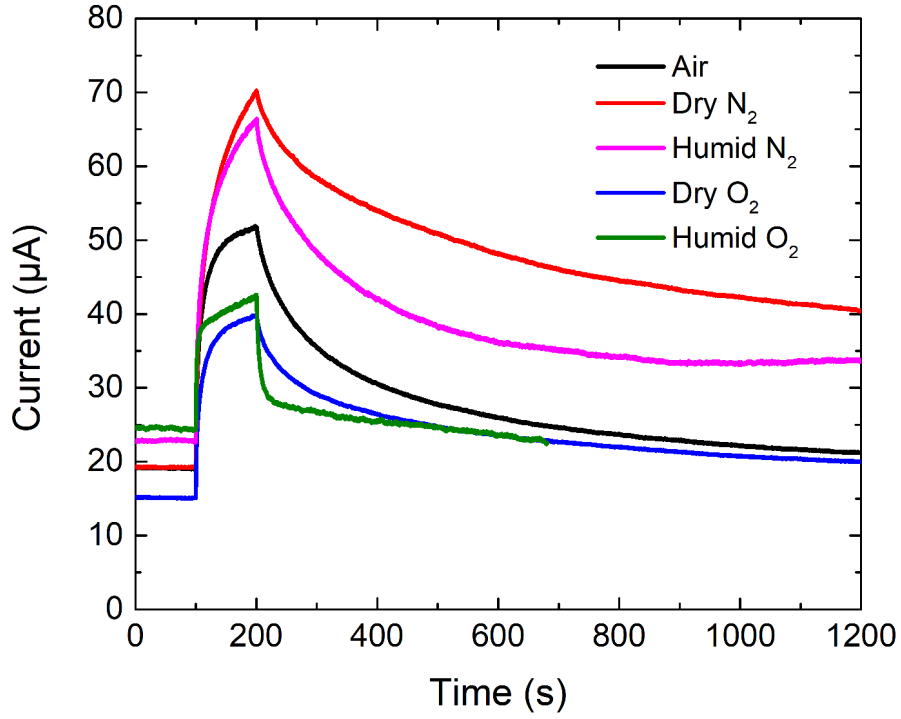


Figure 4.30: The UV photoresponse of an Sb-doped SnO_2 nanotube resistor in different gas ambients.

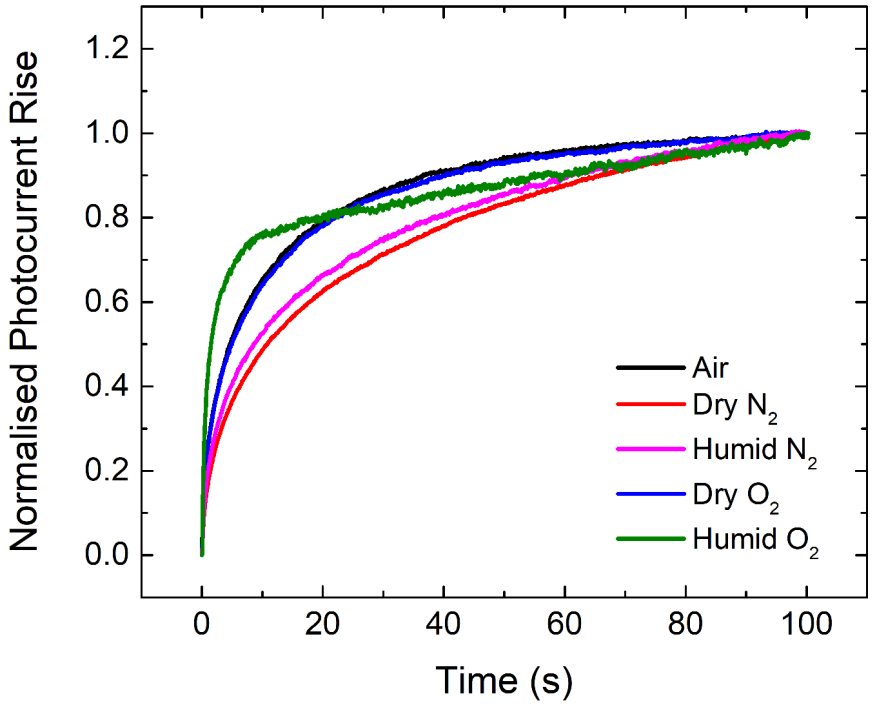


Figure 4.31: The normalised photoresponse rise of an Sb-doped SnO_2 nanotube resistor in different gas ambients.

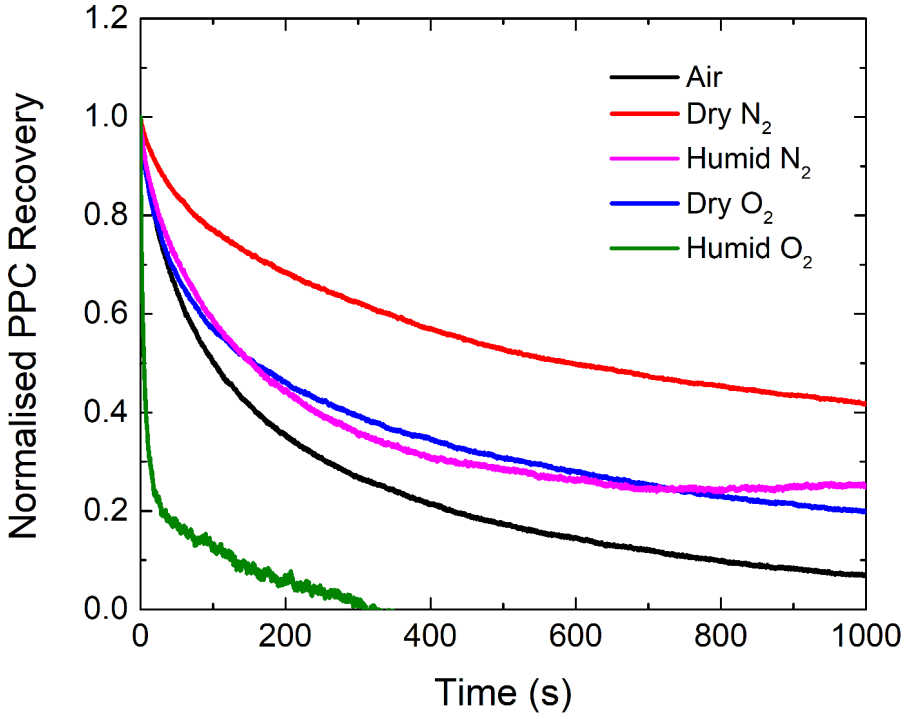


Figure 4.32: The normalised PPC recovery of an Sb-doped SnO_2 nanotube resistor in different gas ambients.

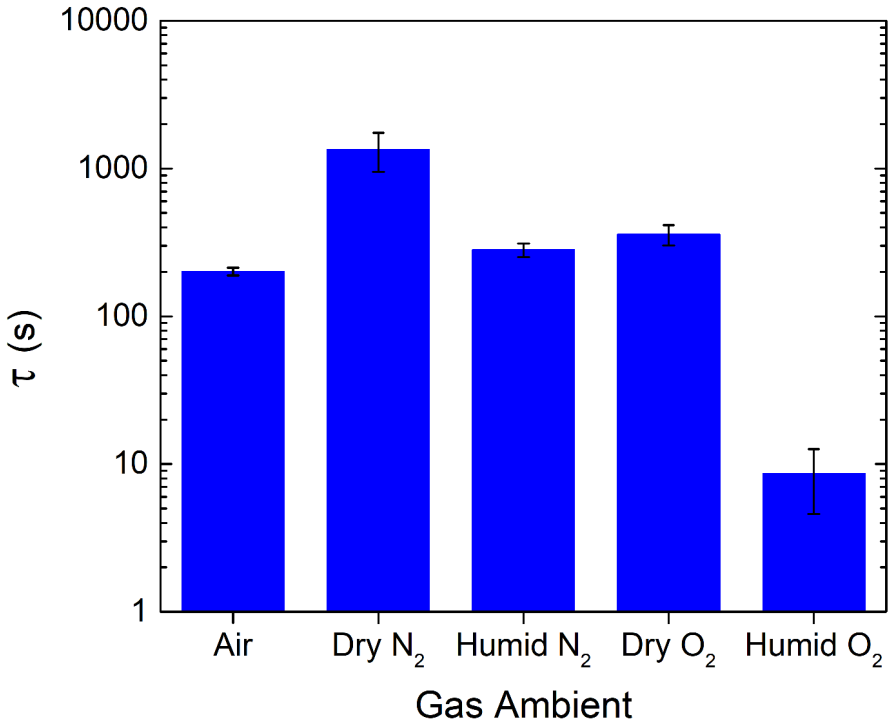


Figure 4.33: Fitted values of τ for an Sb-doped SnO_2 nanotube resistor in different gas ambients.

recovery occurs in dry N_2, with a τ value of ~1300 s. The dry O_2 and humid N_2 ambients again show similar recovery periods, both of which are slower than exposures performed in air. The fastest PPC recovery occurs in the humid O_2 environment, where a response similar to that of the photocurrent rise is observed. There is an initial fast electronic-type drop to ~20% of the PPC recovery which is then followed by a slower gas-readsorption like decay. This electronic-type fall exhibits a τ value of 8.6 s, the fastest decay observed under all measured conditions. These results indicate that the co-operative effect of H_2O and O_2 is repeated on the Sb-doped SnO_2 nanotubes.

4.4 Summary and Conclusions

Individual undoped and 0.15% Sb-doped SnO_2 nanotubes have been contacted with robust Ti/Au contacts to create individual SnO_2 nanotube resistors. This successful contact fabrication allowed for the electrical characterisation of these devices to be performed. The effect of Sb-doping on the SnO_2 nanotubes were quantified by IV measurements, which produced a 100-fold increase in the measured current compared to undoped devices. High quality and almost ideal ohmic contacts were formed on the Sb-doped nanotubes with very low hysteresis. Undoped SnO_2 nanotubes, however, did not form such high quality ohmic contacts and demonstrated asymmetrical IV curves caused by Schottky barriers at each contact. These non-ideal contacts also caused significant current hysteresis in the undoped devices, which increased in magnitude under an increasing bias voltage. Analysis of the hysteresis from the IV measurements found that it likely arises from Fowler-Nordheim tunnelling across the Schottky barriers, although a lack of consistent mechanisms present in the literature prevented the exact cause of this behaviour to be identified.

The UV photodetection properties of the fabricated undoped and Sb-doped SnO_2 nanotube resistors was investigated as a potential application of these devices and as a means of investigating the surface chemistry of the nanotubes. Both undoped and Sb-doped devices demonstrated significant PPC effects following illumination with 285 nm UV light, with the Sb-doped device exhibiting a significantly faster PPC recovery compared to an undoped device. Repeated UV exposures on an undoped

device showed good repeatability in the maximum observed photocurrent, although the PPC recovery time increased with each UV exposure. The Sb-doped device, on the other hand, demonstrated a poor repeatability in the maximum photocurrent, which decreased with each subsequent UV exposure. The Sb-doped device did however demonstrate strong repeatability with respect to its normalised PPC recovery.

To further investigate the roles of atmospheric adsorbates which are commonly attributed to cause the PPC in SnO_2-based UV photodetectors, UV measurements were performed at different temperatures and in different ambients for both the undoped and Sb-doped devices. Increasing operating temperatures up to 150 °C affected both the undoped and Sb-doped devices similarly, with a sharper photoresponse rise and faster PPC recovery observed. These faster response times were attributed to the thermal desorption of surface adsorbate species, minimising the UV-induced O_2/H_2O desorption and readsorption process. Fitting an Arrhenius equation to the observed temperature dependence provided activation energies of 180 mev and 210 meV for the undoped and Sb-doped devices respectively, which agrees well with existing values in literature.

UV measurements performed in different ambient gas environments revealed a range of unique findings not present in the literature. The slowest PPC recoveries and largest photocurrents were observed in a dry N_2 environment devoid of both H_2O and O_2 due to the lack of readsorbing surface adsorbates. Interestingly, despite O_2 desorption and readsorption being solely attributed to causing PPC in SnO_2-based photodetectors, UV measurements performed in dry O_2 environments showed a slower PPC recovery than in air. Measurements in a humid N_2 environment also demonstrated a PPC recovery slower than in air. A humid O_2 gas environment illustrated the co-operative role of both H_2O and O_2 on the minimisation of the PPC recovery time, with dramatically faster PPC recovery times for both the undoped and Sb-doped devices observed in this ambient. A potential mechanism was proposed which involved the formation of a strongly hydrogen-bonded network between adsorbed H_2O and O_2^- ions which inhibits the desorption of O_2^- from the SnO_2 surface under UV illumination. This mechanism allows faster electronic-type responses to dominate the UV photoresponse and PPC recovery which gives rise to

the dramatically faster recovery times.

Chapter 5

Advanced Devices of SnO_2 Nanotubes and their Applications

5.1 Introduction

Resistor-based devices are the simplest devices that can be created using the square SnO_2 nanotubes. While these devices have been shown to perform reasonably well in UV sensing applications, more advanced devices such as field-effect transistors (FETs) and Schottky diodes offer numerous advantages in different applications. In particular, the large range of electrical currents these devices produce under certain operational conditions can be used to provide increased sensitivity in UV and gas sensing. These different device arrangements also control the device current through means other than the SnO_2 nanotube resistivity, which means advanced devices fabricated from the SnO_2 nanotubes may be less environmentally sensitive than the resistor based devices. Finally, the IV characteristics of FETs allow the determination of certain fundamental electrical characteristics of the channel material, such as an estimated carrier concentration and field-effect electron mobility. This may allow further electrical characterisation of the SnO_2 nanotubes.

An additional problem that has presented itself throughout the fabrication of individual nanotube devices is the low device yield due to the large numbers of contact pads and dispersed nanotubes. This results in a labour-intensive process that required SEM imaging to identify potentially functional devices for electrical testing.

As this process is not scalable nor appropriate for the production of commercial devices, alternative device structures are needed to produce reliable, high-yield devices that maintain the nanoscale benefits of the SnO_2 nanotubes.

In this chapter, field-effect transistors and Schottky diodes are constructed using individual SnO_2 nanotubes and are electrically characterised. The SnO_2 nanotube FETs allow for the approximation of several important material characteristics, while the SnO_2 nanotube Schottky diodes are tested for their improved UV sensing properties. In addition to these individual-nanotube devices, resistive devices that contain a large number of bridging SnO_2 nanotubes are fabricated, and their electrical properties are characterised. These multi-nanotube devices are then incorporated into ionic-gated FETs which are investigated for neuromorphic applications.

5.2 Field-Effect Transistors of Individual SnO_2 Nanotubes

The field-effect transistor structure is an expansion of the previously fabricated resistor devices. In addition to the two contacts that allow current to flow through an individual SnO_2 nanotube, a third isolated contact is required to produce an electric field across the nanotube that modulates the device current. This requirement can be easily achieved by using the p-Si/SiO_2 substrate that provides an insulating base for nanotube contacting. By applying a voltage bias to the underlying p-Si bulk, the required electric field is produced across the SiO_2 dielectric and SnO_2 nanotube. This means that, with the selective removal of specific areas of the SiO_2 layer, contacts to dispersed SnO_2 nanotubes and the underlying p-Si bulk can be made in one contact deposition step. An illustrative diagram of the cross-section of the device structure can be seen in Figure 5.1. This device structure is known as a back-gated FET and is commonly used to construct nanowire-based FETs in the literature[140, 143, 144].

5.2.1 Fabrication of Individual SnO_2 Nanotube FETs

A similar methodology to that used in Section 4.2.1 was used to produce the individual SnO_2 nanotube FETs. A 10×10 mm p-type (100) Si substrate with a resistivity of 1-10 Ωcm was thermally oxidised at 950 °C in a humid O_2 environment for 15 minutes.

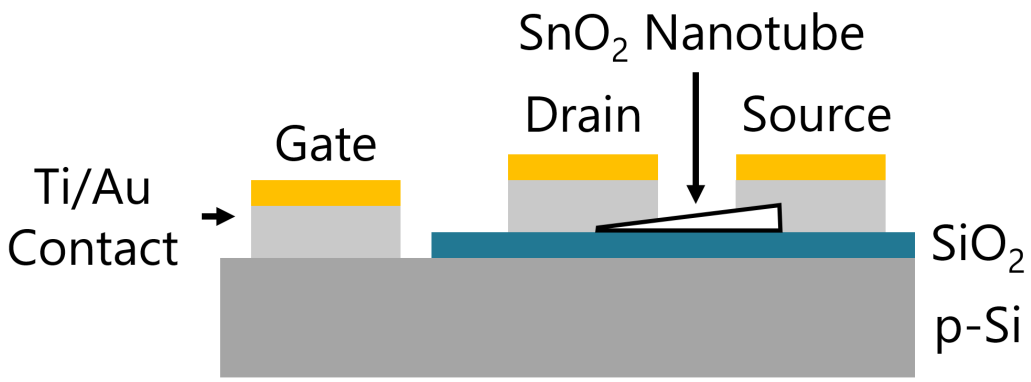


Figure 5.1: Device cross-section of the fabricated SnO_2 nanotube FETs, formed by back-gating contacted SnO_2 nanotubes through the conducting Si substrate. All contacts are Ti/Au.

This produced a surface SiO_2 layer ~77 nm thick. To access the underlying Si substrate, photolithography was used to expose 1.5×1.5 mm squares in each corner of the substrate with the remaining surface covered in AZ1518 photoresist. Wet chemical etching of the exposed SiO_2 areas was then performed in an NH_4F:HF buffer solution for 60 s, after which the sample was rinsed thoroughly in deionised water. The remaining photoresist was then removed by cleaning in acetone, methanol and isopropanol. The remaining steps used to fabricate contacts to the SnO_2 nanotubes from Section 4.2.1 were repeated. This involved the key steps of nanotube dispersal, photolithographic contact patterning, Ti/Au metal deposition and excess metal liftoff. This produced devices which contained electrical contacts to isolated SnO_2 nanotubes as well as separate contacts to the underlying conducting Si substrate. A final 400 °C anneal in a nitrogen ambient for 10 minutes was then performed to improve the quality of the ohmic contacts to the SnO_2 nanotubes. Both undoped and 0.15% Sb-doped SnO_2 nanotube FETs were fabricated using this method.

5.2.2 Electrical Characteristics of Undoped Individual SnO_2 Nanotube FETs

Fabricated FETs were characterised by IV measurements of their output and transfer characteristics. Output characteristics were measured by sweeping V_{DS} from 0 to 3 V under a series of constant V_{GS}, while transfer characteristics were obtained by sweeping V_{GS} from -10 to 10 V under a series of constant V_{DS}. All measurements

were performed in the dark in air, and zero current above the probe station electrical noise floor was recorded across unbridged drain-source contacts or across the gate.

Output and transfer characteristics for a typical undoped individual SnO_2 nanotube FET can be seen in Figures 5.2 and 5.3 respectively. This device demonstrates the expected behaviour for a n-type FET, with current enhancement under positive gate voltages and current depletion with negative gate voltages. The output characteristics show the two distinct regions of on-current for a FET, with a linear region at small V_{DS} followed by a saturation region at higher V_{DS}.

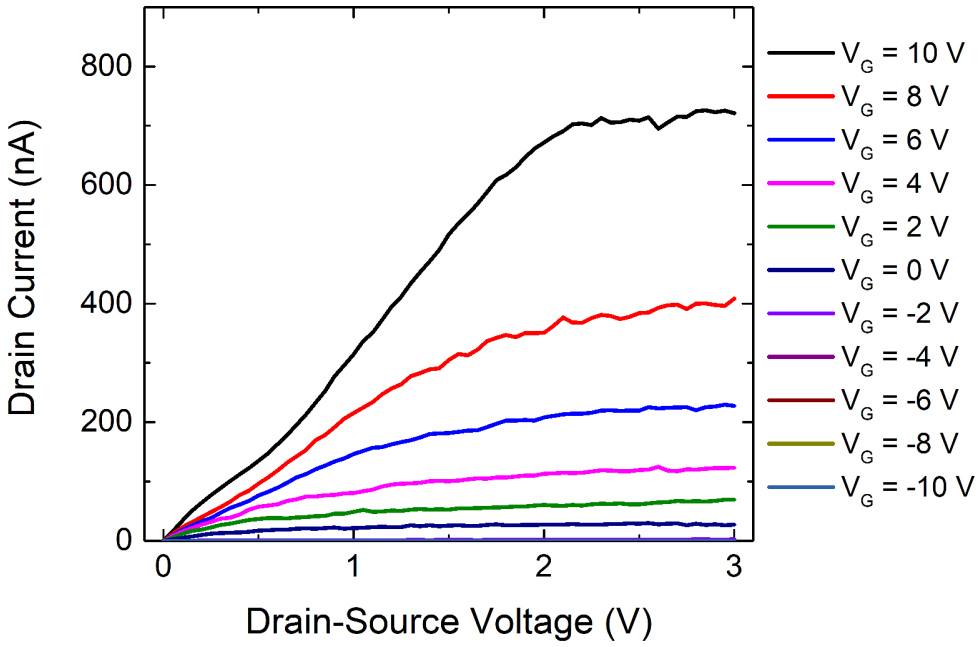


Figure 5.2: The output characteristics of a back-gated individual undoped SnO_2 nanotube FET.

Several FET characteristics are able to be extracted from the transfer characteristics in Figure 5.3, and are presented in Table 5.1. Large on/off current ratios are observed, with values greater than 10^6 observed at $V_G = \pm 10$ V. The device threshold voltage (V_{Th}) was determined from the x-intercept of $(I_{DS})^{1/2}$-V_G, and displayed a small average value of -1.96 V across the different V_{DS} biases. The threshold voltage was observed to shift throughout the transfer characteristic sweeps, and this phenomenon will be further discussed in Section 5.2.4. The subthreshold slope was extracted from $dV_G/d[\log(I_{DS})]$, with an average value of 187 mV/decade across the

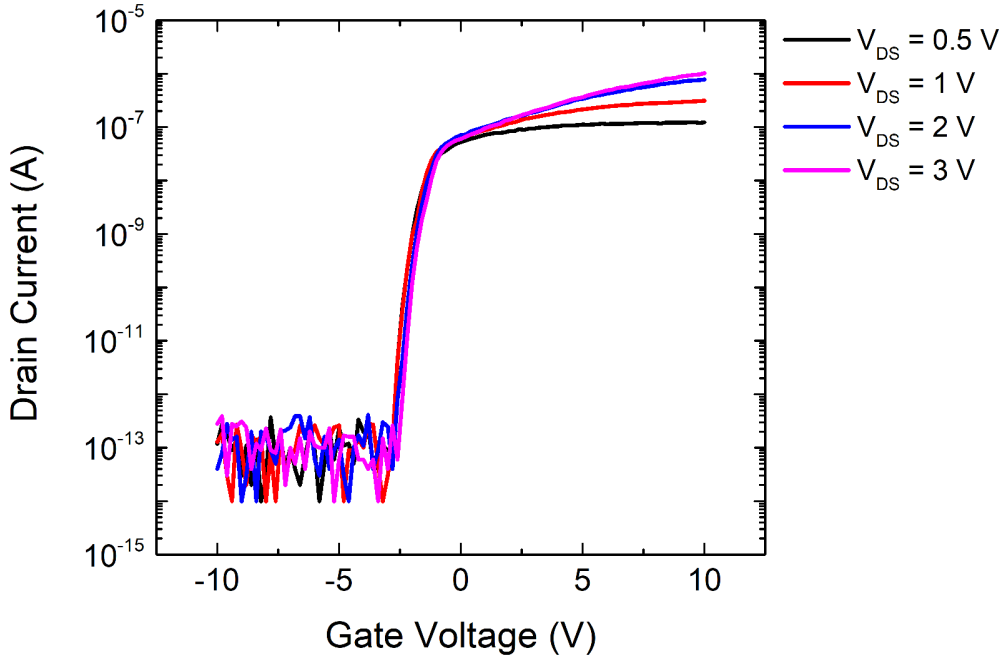


Figure 5.3: The transfer characteristics of a back-gated individual undoped SnO_2 nanotube FET.

Table 5.1: Transistor characteristics extracted from each of the transfer sweeps at different V_{DS}.

V_{DS} (V)	On/Off Ratio (V_G = ±10 V)	Threshold Voltage, V_{Th} (V)	Subthreshold Slope (mV/decade)
0.5	4.69×10^5	-2.16	176
1	1.20×10^6	-2.02	187
2	2.99×10^6	-1.91	204
3	3.92×10^6	-1.73	181

different V_{DS} biases. This subthreshold slope is larger than the theoretical minimum value of ~60 mV/decade[140, 154], although it is comparable to other reported values for SnO_2 FETs in the literature[143, 147]. The subthreshold slope and threshold voltage could be further improved by a different gate structure, such as a conformal top-gate or gate-all-around design, as the top of the nanotube is significantly further away from the gate compared to the bottom of the nanotube.

5.2.3 FET Gate Capacitance Simulations

An important parameter of the SnO_2 nanotube FETs that can be determined from the device transfer characteristics is the field-effect mobility, μ_{FE}. This value can be determined from the device transconductance (g_m) by the following equations[144]:

$$g_m = \frac{\partial I_{DS}}{\partial V_G}|_{V_{DS}=Const} \tag{5.1}$$

$$\mu_{FE} = \frac{g_m L^2}{C_g V_{DS}} \tag{5.2}$$

Where L is the length of the nanotube channel and C_g is the gate capacitance to the nanotube. The length of the nanotube channel can be determined using SEM, however, the gate capacitance cannot be easily measured due to the small nanotube size and relatively large areas of the 50×50 µm square contacts. Groups that have previously reported SnO_2 nanowire transistors typically calculate the gate capacitance using a simple cylinder-on-plane model[140, 143, 147], however the unique square geometry of the SnO_2 nanotubes prevents this approach from being used here.

To estimate the gate capacitance of the measured SnO_2 nanotube FET, the gate capacitance per length was simulated using a 2D electrostatic model in Ansys Maxwell. The width of the nanotube halfway across the channel was ~186 nm, which was used as the nanotube width within the simulation. The gate oxide thickness was matched to the 77 nm measured SiO_2 layer in the device, and a -5 V gate bias relative to the nanotube was used for the electrostatic simulation.

Figure 5.4 shows the 2D electrostatic model used in the simulation alongside the simulated electric field strengths within the surrounding air and SiO_2 dielectrics. The bottom of the nanotube can be seen to experience larger electric field strengths, due to its close proximity to the conducting substrate and the SiO_2 layer with its higher relative permittivity. The lower relative permittivity of the air and longer distance from the substrate means the sides and top of the nanotube experience lower electric field strengths, which decrease in intensity moving from the sides to the top of the nanotube. Areas of increased electric field strength are also observed near the sharp corners of the nanotube. Solving for the capacitance per unit length yielded

a value of 146 aF μm^{-1}, which agrees well with the ~100aF μm^{-1} computed capacitances in similar nanowire-based devices[147]. Given the length of the exposed nanotube was determined to be 2.14 µm by SEM, this gave a total gate capacitance of ~312 aF for the measured device.

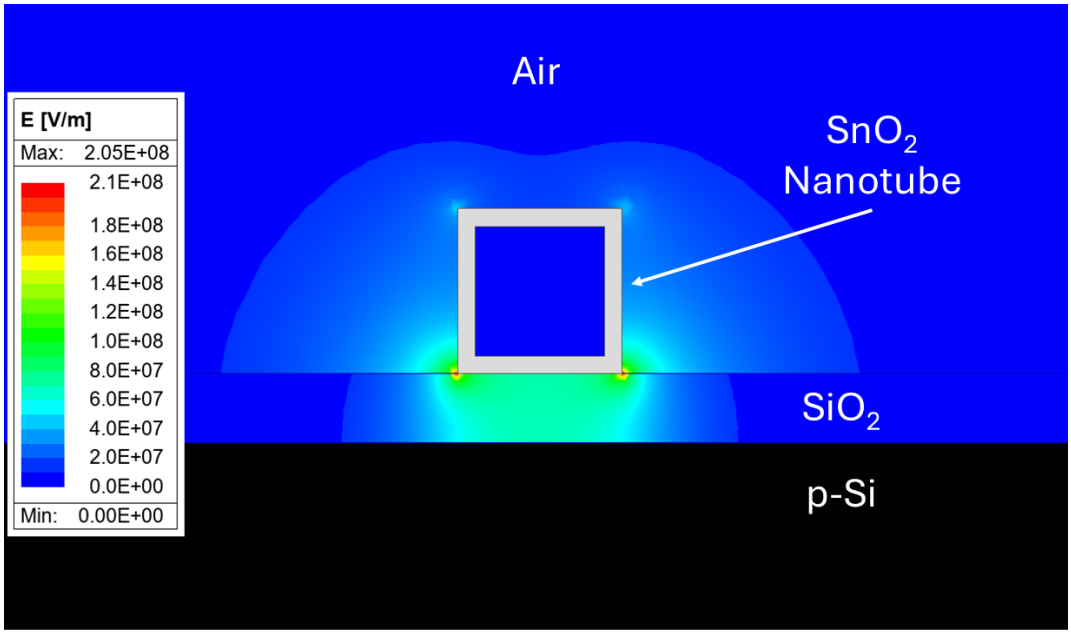


Figure 5.4: Simulated electric field strength for a 186 nm diameter SnO_2 nanotube transistor on a 77 nm thick SiO_2 layer with a -5 V gate bias.

Table 5.2 shows the calculated transconductance and field-effect mobility for the measured undoped SnO_2 nanotube FET. The average field effect mobility across the four measurements was 6.13 $cm^2V^{-1}s^{-1}$, which is similar to mobility values reported in other undoped SnO_2 nanowire FETs[143, 313], and higher than values reported for thin-film transistors of sputtered SnO_2[314]. Despite the similar field-effect mobilities to other undoped devices, these values are still significantly smaller than other field-effect mobilities reported for lightly-doped SnO_2 nanowire FETs, which typically range between 100-200 $cm^2V^{-1}s^{-1}$[140, 147, 315]. In addition, the calculated field-effect mobility is significantly lower than the ~30 $cm^2V^{-1}s^{-1}$ measured by Hall-effect for undoped SnO_2 thin films grown in the same Mist-CVD setup by Scott[29].

There are two factors that are likely to contribute to errors within the calculation of the field-effect mobility: contact resistance at the drain-source contacts, and interface states between the gate and the channel. Both of these factors are not taken into account in Equation 5.2, and neglecting these factors if they are present leads to

Table 5.2: Calculated values of the undoped SnO_2 nanotube FET transconductance, field-effect mobility and modulated carrier concentration at V_G = 0 V.

V_{DS} (V)	Transconductance, g_m (nS)	Field-Effect Mobility, μ_{FE} ($cm^2V^{-1}s^{-1}$)	Modulated Carrier Concentration, Δn (cm^{-3})
0.5	23.5	6.90	5.10×10^{16}
1	31.9	4.68	4.54×10^{16}
2	90.6	6.64	2.04×10^{16}
3	129	6.31	1.44×10^{16}

the field-effect mobility being underestimated[144]. It was demonstrated in Chapter 4.2.2 that Ti/Au contacts to undoped SnO_2 nanotubes forms unideal Schottky contacts that affects the electrical measurements of the devices. As a result, the undoped SnO_2 nanotube transistors likely have significant contact resistance that contributes to an underestimation of the field-effect mobility. In addition, the upcoming Section 5.2.4 demonstrates that there exists significant gate interface states and charge trapping that occurs within the SnO_2 nanotube transistors, Which likely also contributes to an underestimation of the field-effect mobility. Whilst the actual field-effect mobility within the SnO_2 nanotubes is likely larger than that estimated here, the calculated value can still be considered a lower bound in these devices.

Despite the likely underestimation of the actual field-effect mobility, the calculation of the modulated carrier concentration in the channel by the gate can still be performed. This is given by the following equation[144]:

$$\Delta n = \frac{I_{DS}L}{q\mu_{FE}V_{DS}A} \tag{5.3}$$

Where q is the elementary charge and A is the cross-sectional area of the nanotube. When calculated at small V_{DS} and with V_G = 0 V, Δn approximates the carrier concentration of the intrinsic channel assuming $I_{DS,on} >> I_{DS,off}$. With increasing V_{DS}, however, channel length modulation causes increasing current saturation that decreases the calculated value of Δn. Calculated values of Δn for each transfer characteristic sweep voltage can be seen in Table 5.2. Importantly, since μ_{FE} scales with I_{DS} due to the dependence on the device transconductance, Δn at small V_{DS} is likely close to the true carrier concentration of the SnO_2 nanotubes.

5.2.4 Charge Trapping at the Gate of the SnO_2 Nanotube FETs

Throughout the measurement of the transfer characteristics of the undoped SnO_2 back-gated FETs, it was observed that the position of the threshold voltage and sub-threshold slope would change across multiple sweeps. This suggested that charge trapping was occurring at the gate, affecting the transfer characteristics of the measured devices. To further characterise the degree of charge trapping, gate transfer sweeps were performed between V_G = ±20 V with different sweep directions. Additionally, voltage sweeps were preformed after biasing the gate for 2 minutes at ±20 V to observe the range of threshold voltage movement under the charge trapping. All measurement were performed in the dark and in air, with V_{DS} = 2 V.

Figure 5.5 shows the transfer characteristics of the back-gated SnO_2 nanotube FET under the different gate voltage sweeps. The threshold voltage from each measurement has been extracted and can be seen in Table 5.3. It is clear that significant charge trapping is occurring at the gate of the nanotube FET, as the threshold voltage shifts significantly towards the bias applied at the beginning of the sweep. This charge trapping also appears to be a dynamic process, as the threshold voltage shifts by an increasing amount with a longer time exposed to the initial bias. The on-current is also affected by the charge trapping, with a lower current when there is more charge trapping due to the 2 minute gate voltage biases. Clearly, this charge trapping will affect values used in the calculation of the field-effect mobility and modulated carrier concentration, such as the maximum value of g_m and I_{DS} at V_G = 0 V.

Charge trapping and gate hysteresis has been reported previously for transistors that utilise a thermally grown SiO_2 layer as the gate dielectric to a non-epitaxial channel, such as in other nanowire-based transistors and organic thin-film transistors[313, 316–321]. In these studies, the observed hysteresis is generally attributed to water vapour acting as a charge trap between the gate and the channel[313, 317–319]. Kim et al.[319] proposed that two types of water vapour act as charge traps in carbon nanotube FETs. The first type involves the adsorption of water vapour onto the nanotube surface, while the second type involves the adsorption of water vapour onto the hydroxylated SiO_2 surface near the nanotube. The same group has

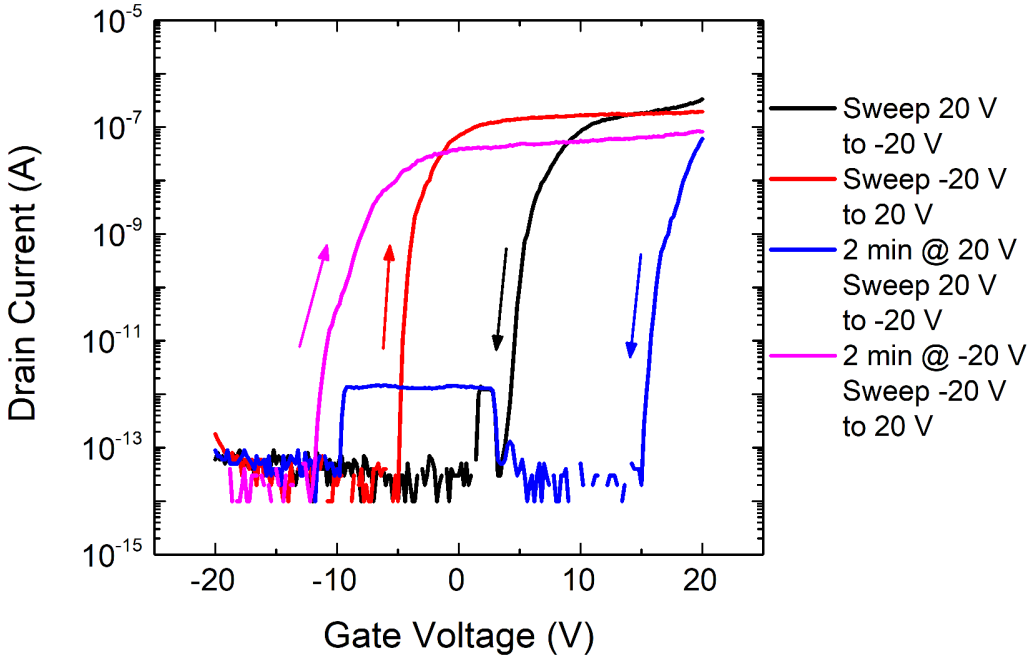


Figure 5.5: Observed shifting in the threshold voltage and subthreshold slope with changes in the sweep direction and 2 minute voltage holds at V_G = ±20 V due to gate charge trapping. V_{DS} = 2 V for all sweeps.

Table 5.3: The position of the measured device threshold voltage from each gate voltage sweep.

Gate Voltage Sweep	Threshold Voltage, V_{Th} (V)
Sweep 20 V to -20 V	5.20
Sweep -20 V to 20 V	-4.24
2 min @ 20 V, Sweep 20 V to -20 V	15.8
2 min @ -20 V, Sweep -20 V to 20 V	-9.32

also performed similar analyses to Ge nanowire FETs[317], finding a similar dependence on water vapour for the observed hysteresis, although they reported a larger dependence on water vapour adsorbed to the nanowire surface compared to the carbon nanotube. Similar dependencies of water adsorption onto material surfaces in FETs have been reported for In_2O_3[318] and ZnO[322]. Given the strong documented interactions of water vapour on SnO_2 nanowire FETs[123, 140] and the significant interaction of water vapour and oxygen observed earlier in this thesis, there is strong evidence that water vapour is the likely cause of the significant charge trapping observed in the fabricated SnO_2 nanotube FETs.

5.2.5 Electrical Characteristics of Sb-doped Individual SnO_2 Nanotube FETs

In addition to the back-gated FETs that utilised individual undoped SnO_2 nanotubes, FETs were also fabricated using 0.15% Sb-doped SnO_2 nanotubes. Figures 5.6 and 5.7 show the output and transfer characteristics for a typical Sb-doped device, respectively. All fabricated devices were unable to be pinched-off due to the increased carrier concentration, even at large negative gate voltages up to -50 V. Gate voltages larger in magnitude than 50 V resulted in the dielectric breakdown of the insulating SiO_2 layer, which follows given the breakdown field strength of SiO_2 is ~8.8 MV cm^{-1}[323], equivalent to a breakdown voltage of 67.8 V across the 77 nm thick SiO_2 layer. Despite being unable to be pinched off, carrier enhancement and depletion still occurred, with a maximum enhancement/depletion current ratio of 1.77 at $V_G = \pm 50$ V.

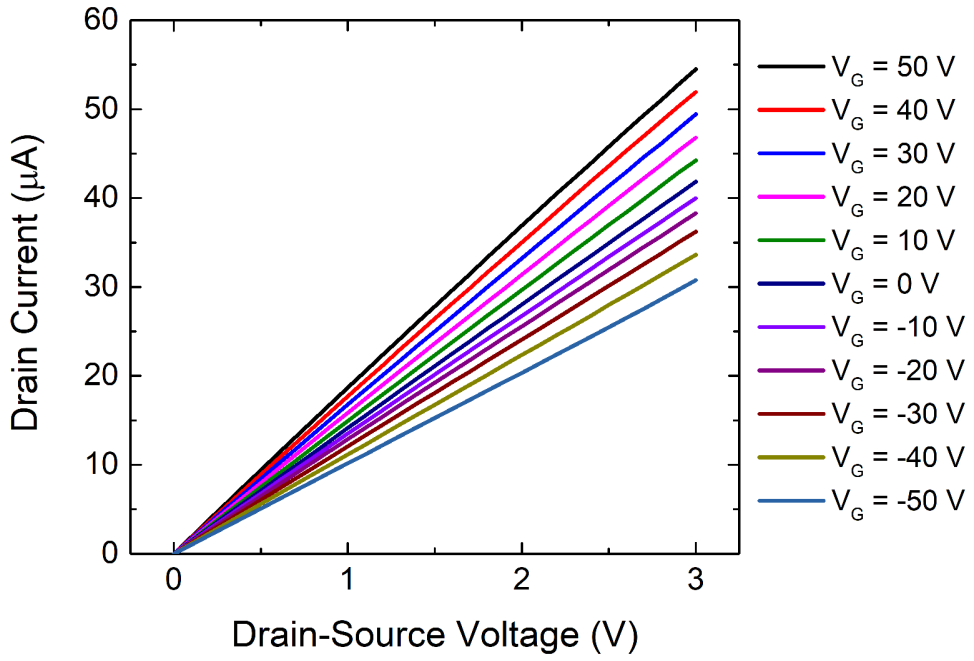


Figure 5.6: The output characteristics of a back-gated individual 0.15% Sb-doped SnO_2 nanotube FET.

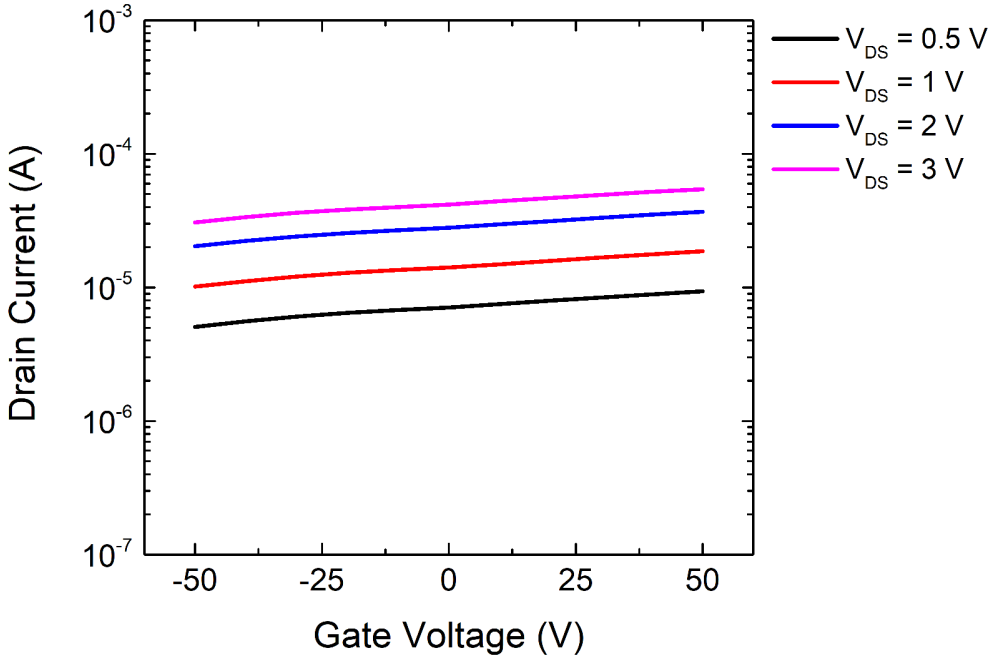


Figure 5.7: The transfer characteristics of a back-gated individual 0.15% Sb-doped SnO_2 nanotube FET.

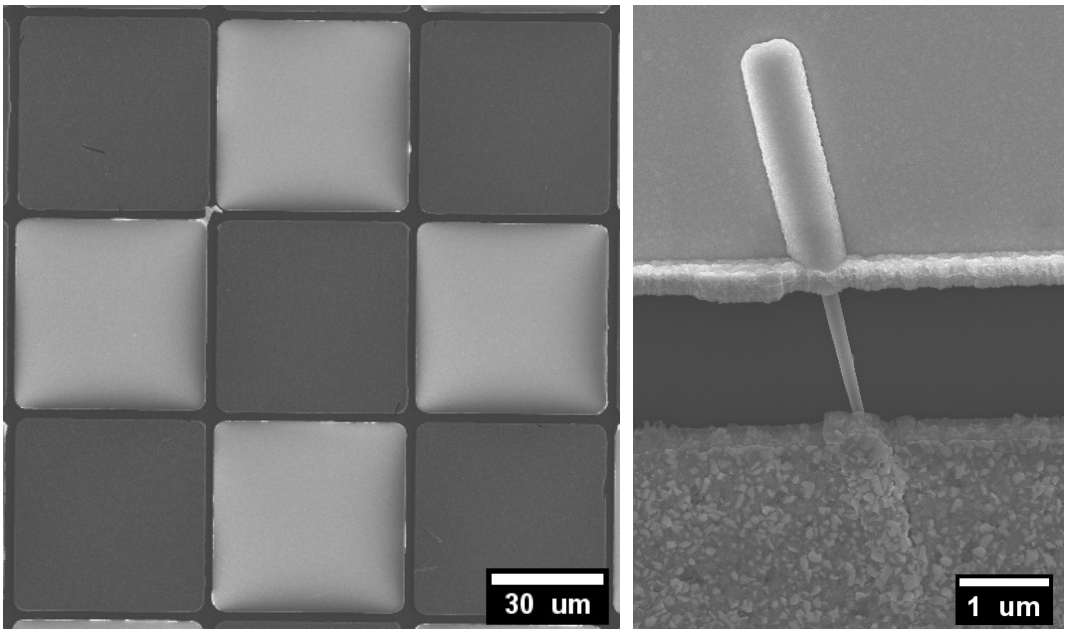
5.3 Schottky Diodes of Individual SnO_2 Nanotubes

The primary difference between resistor-based devices and Schottky diodes is the replacement of one of the Ti/Au ohmic contacts with a metal Schottky contact that intentionally produces a large Schottky barrier at the metal-semiconductor junction. This large barrier is particularly useful in applications such as UV sensing, where the built-in potential of the barrier efficiently separates photogenerated electron-hole pairs. This results in a large sensor response under UV illumination. High-quality Schottky contacts have historically been difficult to form to SnO_2 due to the strong downward band bending that occurs at the material surface[30, 94, 163], and as a result Schottky diodes in literature tend to show poor diode characteristics[158, 159, 161, 162, 324–327]. However, oxidised noble metal Schottky contacts have been previously demonstrated to produce large Schottky barrier heights to metal oxides such as SnO_2[163], β-Ga_2O_3[164, 165] and ZnO[166, 167]. Among these reported Schottky contact materials, IrO_x has demonstrated a significant increase in Schottky barrier height compared to a plain Ir contact[165, 166] combined with a high thermal stability[328]. For these reasons, Schottky diodes that utilise the SnO_2 nanotubes and an

IrO_x Schottky contact were fabricated to improve the UV sensing performace of the SnO_2 nanotubes.

5.3.1 Fabrication of Individual SnO_2 Nanotube Schottky Diodes

A two-stage photolithography and metal deposition process was used to create ohmic and Schottky contacts to individual SnO_2 nanotubes to create Schottky diodes. Similar to previous methodologies, undoped or 0.15% Sb-doped SnO_2 nanotubes were dispersed across an insulating SiO_2/Si substrate with a ~130 nm thick SiO_2 layer. Ohmic contacts to one end of the scattered nanotubes were then deposited using photolithographic patterning of a checkerboard-type pattern and the electron-beam evaporation of 500 nm and 50 nm of Ti and Au respectively. Following a standard lift-off and solvent cleaning procedure, areas for the Schottky contacts were then opened using photolithography by rotating the checkerboard mask by 90° and aligning to pre-patterned alignment marks. Schottky contacts to the exposed end of the SnO_2 nanotubes were then deposited by RF magnetron sputtering of a 250 nm thick IrO_x layer followed by a 250 nm thick Ir capping layer. Further details of this sputtering process can be seen in Section 3.3.3 Following a final series of lift-off and



(a) **(b)**

Figure 5.8: SEM images of (a) a grid of checkerboard-patterned Ti/Au ohmic contacts (dark grey) and IrO_x/Ir Schottky contacts (light grey). (b) An individual, well contacted SnO_2 nanotube Schottky diode.

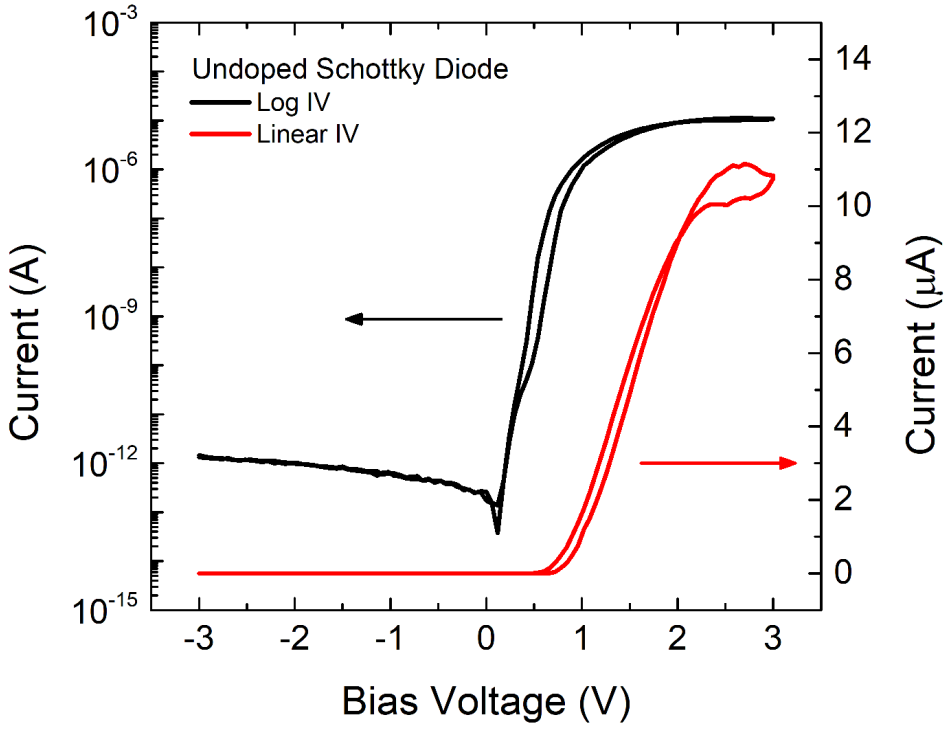
solvent cleaning steps, the fabricated devices were annealed at 400 °C in a nitrogen environment for 10 minutes. Similar to previous individual nanotube devices, SEM was used to identify well-contacted devices for measurement. SEM images showing the checkerboard-style patterning and an example of an individual SnO_2 nanotube Schottky diode can be seen in Figure 5.8.

5.3.2 Electrical Characteristics of Individual SnO_2 Nanotube Schottky Diodes

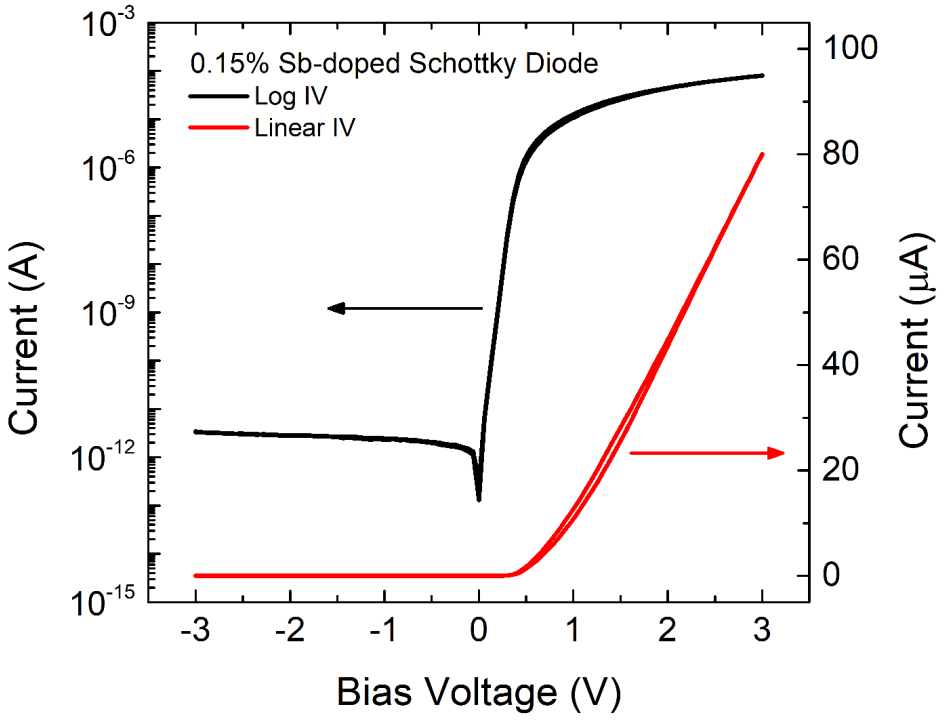
Electrical measurement were performed on the SnO_2 nanotube Schottky diodes immediately following the final annealing step. The IV measurements for a typical undoped and 0.15% Sb-doped SnO_2 Schottky diode can be seen in Figure 5.9. Both diodes exhibit very good electrical properties, with low reverse bias leakage currents around 10^{-12} A and relatively high forward currents above 10^{-5} A at ±3 V. This provides a current rectification ratio above 10^7 for each device. These properties demonstrate that the IrO_x layer has created a large Schottky barrier at the SnO_2 surface with relatively few defects.

An interesting observation is seen in the IV measurements of the Schottky diode constructed from an undoped SnO_2 nanotube. The forward current at 3 V is ~11 μA, which is larger than the highest forward currents observed with the undoped SnO_2 nanotube-based resistors. This suggests that the IrO_x/Ir contact has a much lower contact resistance compared to the usual Ti/Au ohmic contact. Figure 5.10 shows a comparison between a sputtered IrO_x/Ir contact and an electron-beam deposited Ti/Au contact where both top capping layers have been removed from the top of the nanotube by ultrasonic agitation. The IrO_x/Ir contact has many grains that appear strongly attached to the sides of the SnO_2 nanotube, while the Ti/Au contact has virtually no side-contact with the SnO_2 nanotube. Even when the top capping metal layer is not removed, the IrO_x/Ir contact has a much larger effective contact area compared to an equivalent Ti/Au contact, which results in a lower overall contact resistance.

There are several minor differences between the IV measurements of the undoped and 0.15% Sb-doped SnO_2 nanotube Schottky diodes. Firstly, the current begins to saturate at voltages above 2 V in the undoped device while the current continues to



(a)



(b)

Figure 5.9: IV measurements of typical (a) undoped and (b) 0.15% Sb-doped SnO_2 nanotube Schottky diodes on log (black) and linear (red) scales.

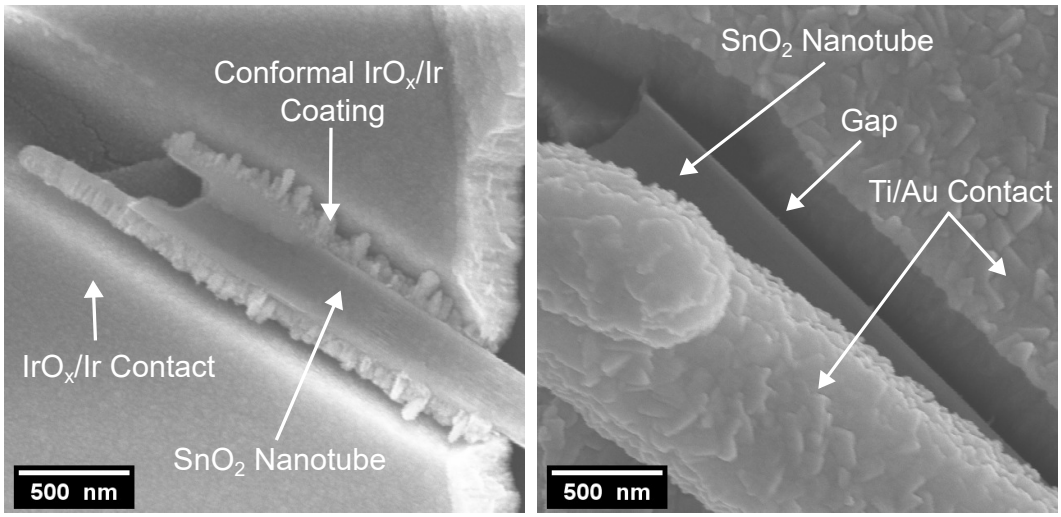


Figure 5.10: SEM images of (a) an IrO_x/Ir Schottky contact and (b) a Ti/Au ohmic contact to SnO_2 nanotubes. The top of both contacts have been detached by ultrasonic agitation, showing the difference in conformal coatings on the sides of the nanotubes.

increase linearly in the Sb-doped device. The forward current in the undoped device also exhibits more hysteresis compared to the Sb-doped device. These differences are likely due to the Schottky barriers present in the Ti/Au contacts to the undoped SnO_2 nanotube, which are not present in the Sb-doped devices.

Several key figures of merit for Schottky diodes are able to be extracted from the recorded IV measurements. In Schottky diodes, the primary means of current transport is by the thermionic emission of electrons over the Schottky barrier[154]. The equation of this thermionic emission is as follows:

$$I = AA^*T^2e^{(-\frac{q\Phi_B}{kT})}\{e^{[\frac{q(V-IR_S)}{\eta kT}]} - 1\} \tag{5.4}$$

Where A is the area of the Schottky contact, A* is the effective Richardson constant and R_S is the series resistance of the Schottky diode. The important figures of merit that can be extracted from this equation are the Schottky barrier height, Φ_B, and the ideality factor, η. The Schottky barrier height influences parameters such as the forward voltage drop of the diode and the reverse bias leakage current, while the ideality factor is influenced by the density of defects and variations/inhomogeneities within the Schottky contact. An ideal diode has an ideality factor close to 1 and a large Schottky barrier height.

The values of Φ_B and η for a range of undoped and 0.15% Sb-doped SnO_2 nanotube Schottky diodes have been extracted from the measured IV characteristics using Equation 5.4, and can be seen in Figure 5.11. Clearly, the undoped Schottky diodes have more inhomogeneous Schottky contacts compared to the Sb-doped devices, as shown by the wide span of ideality factors across multiple devices. The Sb-doped devices have a much more consistent and high quality Schottky contact, with only a narrow span of ideality factors from 1.04 to 1.13, compared with 1.04 to 1.53 for the undoped devices. It can also be seen that the undoped Schottky diodes have a higher average barrier height than the Sb-doped devices, with average barrier heights of 0.82 eV for undoped devices compared to 0.67 eV for doped devices. This effect is known as image force lowering and is due to the creation of image charges in the IrO_x layer that attracts electrons close to the metal-semiconductor interface in the SnO_2 nanotube. These image charges reduce the effective height of the Schottky barrier. As the Sb-doped devices have a higher carrier concentration than the undoped devices, more image charges are formed and Sb-doped devices experience a larger reduction in barrier height.

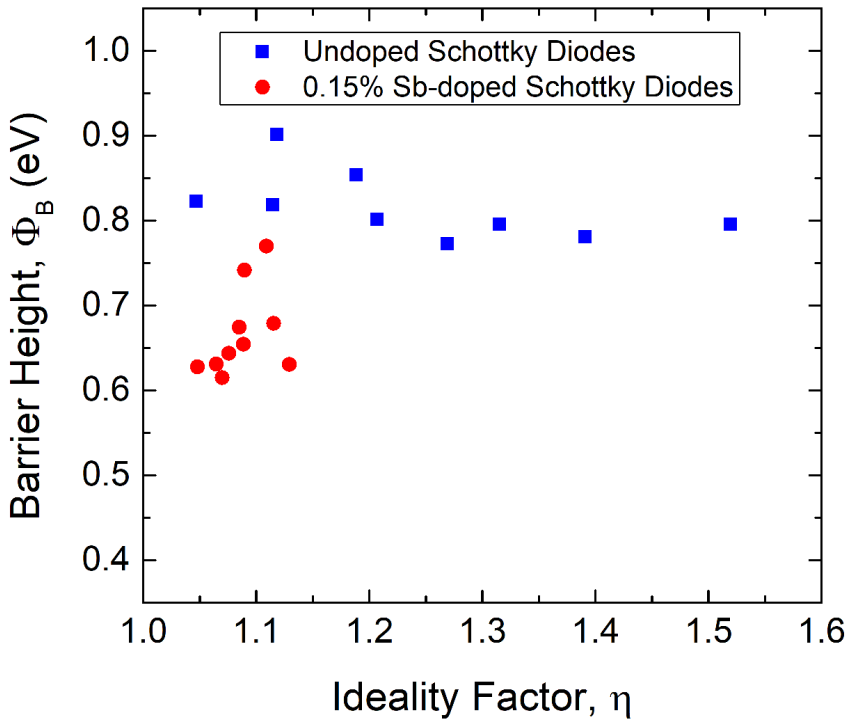


Figure 5.11: Calculated barrier heights and ideality factors for the undoped and 0.15% Sb-doped individual SnO_2 nanotube Schottky diodes.

Table 5.4: Diode properties from SnO_2-based thin-film (TF) and nanowire (NW) devices in literature compared to the diodes fabricated in this work from square SnO_2 nanotubes (NT).

Structure	Barrier Height Φ_B	Ideality Factor η	Rectification Ratio	Refs.
SnO_2-TF/Ni	0.64 eV	4.11	~1000	[162]
SnO_2-TF/p-Si	0.73-0.84 eV	3.21-4.85	~10	[324]
SnO_2-TF/p-Si	0.87 eV	2.1	~500	[325]
Sb:SnO_2-TF/Pt	-	-	~1000	[161]
SnO_2-TF/Au	0.60 eV	1.78	~1000	[159]
SnO_2-NW/Pt	-	-	~1000	[158]
SnO_2-NW/p-Si	-	20.8	~10^4	[326]
SnO_2-NW/p-CNT	-	13	~100	[327]
SnO_2-TF/AgO_x	0.89 eV	1.1	~10^7	[163]
SnO_2-NT/IrO_x	0.82 eV (Av.)	1.24 (Av.)	~10^7	This work
Sb:SnO_2-NT/IrO_x	0.67 eV (Av.)	1.09 (Av.)	>10^7	This work

The Schottky diodes fabricated from individual SnO_2 nanotubes and an IrO_x Schottky contact demonstrate significantly better diode properties than those reported for other SnO_2-based heterojunction and Schottky diodes[158, 159, 161–163, 324–327]. Table 5.4 shows a comparison of the SnO_2 nanotube Schottky diodes fabricated in this work with other SnO_2 diodes in the literature. The diodes reported in the literature typically suffer from large ideality factors and small rectification ratios despite similarly reported barrier heights. Importantly, the devices fabricated in this work offer improved characteristics compared to the best thin-film SnO_2 diodes in the literature, and are significantly better than the reported nanostructured SnO_2 devices. The only device with comparable characteristics, fabricated by Dang et al., also utilises a RF-sputtered oxidised noble metal Schottky contact to a Mist-CVD grown thin film of SnO_2[163]. This shows that the device structure and deposition methodologies used in these devices produces the most ideal diode characteristics.

5.3.3 Effects of Air Exposure Post-Annealing on the SnO_2 Nanotube Schottky Diodes

Despite the successful fabrication of undoped and Sb-doped SnO_2 nanotube Schottky diodes, a severe ageing effect was observed that negatively influenced the electrical properties of the devices. When left in air for several days after the final annealing step, a large decrease in the forward current and barrier height was observed, along with an increase in the reverse bias current. When the devices were reannealed at 400 °C for 10 minutes in a nitrogen environment and measured immediately afterwards, the ageing effect were reversed and the as-fabricated electrical properties were restored, with a decrease in the reverse bias current and an increase in forward bias current. The IV measurements from four undoped SnO_2 nanotube Schottky diodes that demonstrate these effects can be seen in Figure 5.12. Furthermore, when comparing the IV measurements after each annealing step, it can be seen the second annealing step has resulted in a slightly higher barrier in these devices. This increase in barrier height is represented by the decrease in reverse bias current and rightward-shift of the logarithmic increase in current for small forward bias voltages.

The significant differences between the measurements performed immediately following annealing and after six days in air suggests that atmospheric adsorbates such as water vapour or oxygen may be responsible for the ageing effect. Firstly, it is unlikely that the effect is related to the IrO_x/Ir Schottky contact material as previous reports have demonstrated stable Schottky contacts to Ga_2O_3[164] and ZnO[166, 167]. As a result, it is likely that an interaction with the SnO_2 surface is responsible for this effect. It is known to be difficult to produce high-quality Schottky contacts to SnO_2 due to the downward band bending caused by electron-donating surface hydroxyl groups[30, 94]. Scott et al. demonstrated that annealing the SnO_2 nanotubes at 400 °C removed surface oxygen species such as hydroxyl groups and in-plane oxygen atoms[30]. The annealing process may remove the electron-donating groups near or within the Schottky contact at the SnO_2 surface, resulting in the excellent IV characteristics measured immediately following annealing. As the device is left in air, these electron-donating groups readsorb at the SnO_2 surface of the Schottky

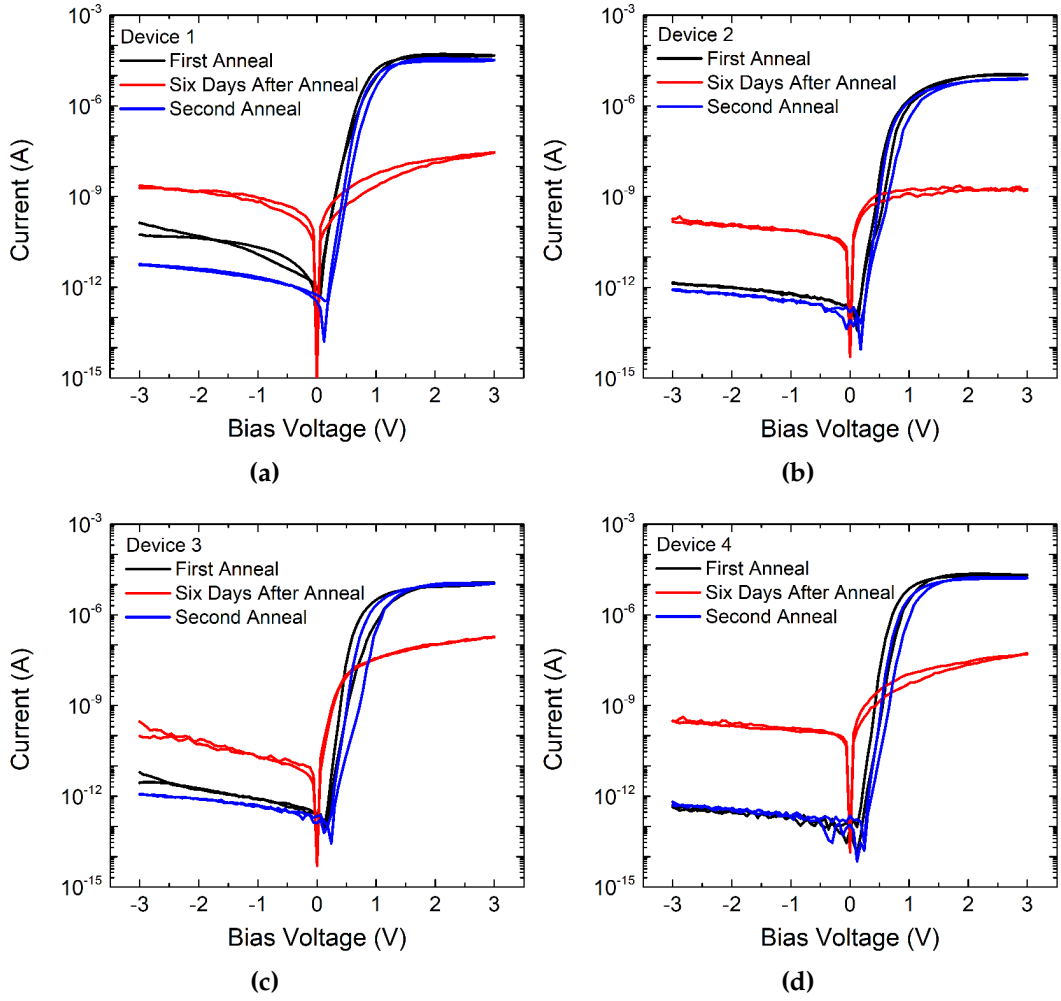


Figure 5.12: IV measurements of multiple undoped SnO_2 nanotube Schottky diodes following an initial annealing (Black), after six days in air (Red) and after a second reannealing (Blue).

contact, lowering the Schottky barrier height and resulting in the less ideal Schottky contact. The effect of air and water/oxygen exposures on the annealed SnO_2 nanotube surfaces is studied further in Chapter 6.2.

In an attempt to mitigate or remove the ageing effect that affects the SnO_2 nanotube Schottky diodes, the sample of undoped devices was encapsulated with a layer of high molecular weight polymethyl methacrylate (PMMA) immediately following reannealing. Devices were then remeasured while encapsulated after six days. The measured IV characteristics for the same four undoped devices can be seen in Figure 5.13. After six days in air while encapsulated the reverse bias current remained low, and in some cases decreased to the nose level of the probe station. The forward bias current, however, was severely affected by the PMMA encapsulation,

with large current hysteresis and a significant decrease in the maximum forward current. Following these measurements, the PMMA layer was removed with a solvent clean in acetone, methanol and isopropanol to observe if the Schottky diodes would return to the annealed state. The measured IV characteristics show the forward currents increase near to the currents observed following annealing. However, this was accompanied by a significant increase in the reverse bias leakage current and a decrease in the Schottky barrier height. These results indicate that PMMA encapsulation is not an effective method to prevent the negative ageing effect of the

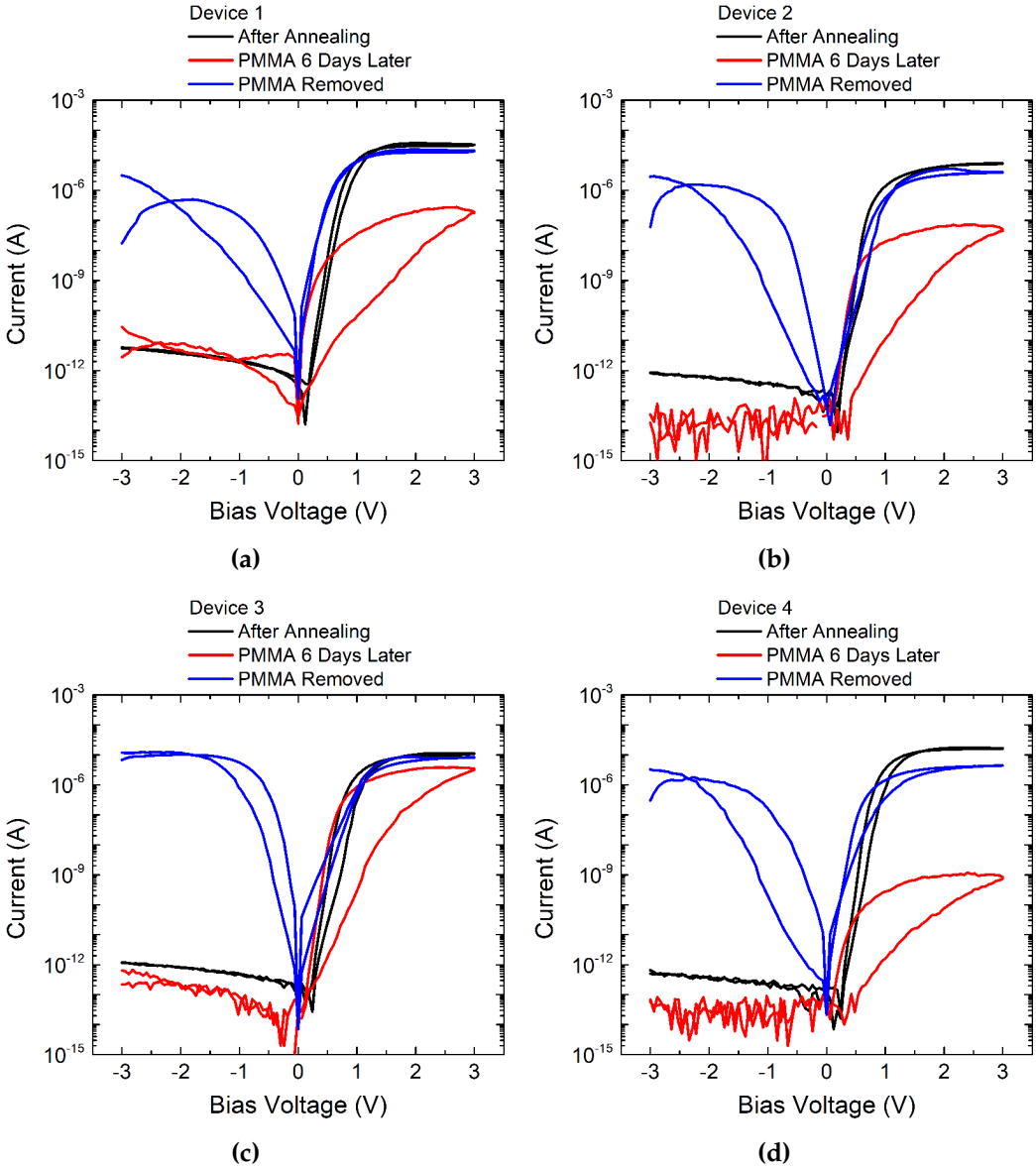


Figure 5.13: IV measurements of multiple undoped SnO_2 nanotube Schottky diodes following annealing (Black), after six days encapsulated in PMMA (Red) and after the removal of the PMMA layer (Blue).

SnO_2 nanotube Schottky diodes.

5.3.4 UV Sensing of Individual SnO_2 Nanotube Schottky Diodes

The high quality Schottky diodes produced immediately following 400 °C annealing are ideal candidates for the effective sensing of UV light. The reverse bias current through the device is limited by the Schottky contact, and electron-hole pairs that are created near this junction are readily separated by the built-in potential. This means that the SnO_2 nanotube Schottky diodes may be affected less by the PPC observed earlier for the SnO_2 nanotube resistors.

As the UV light is required to enter the interface between the Schottky contact and SnO_2 nanotube, the sample is required to be back illuminated due to the thickness of the IrO_x/Ir contact. 0.15% Sb-doped SnO_2 nanotube Schottky diodes were fabricated on insulating m-plane sapphire substrates that have low light absorption at 275 nm, allowing for the Schottky diodes to be illuminated through the substrate. A QPhotonics 275 nm surface-mount LED was positioned 2.25 mm below the substrate surface and produced an optical output power of ~360 μW cm^{-2}. The tested

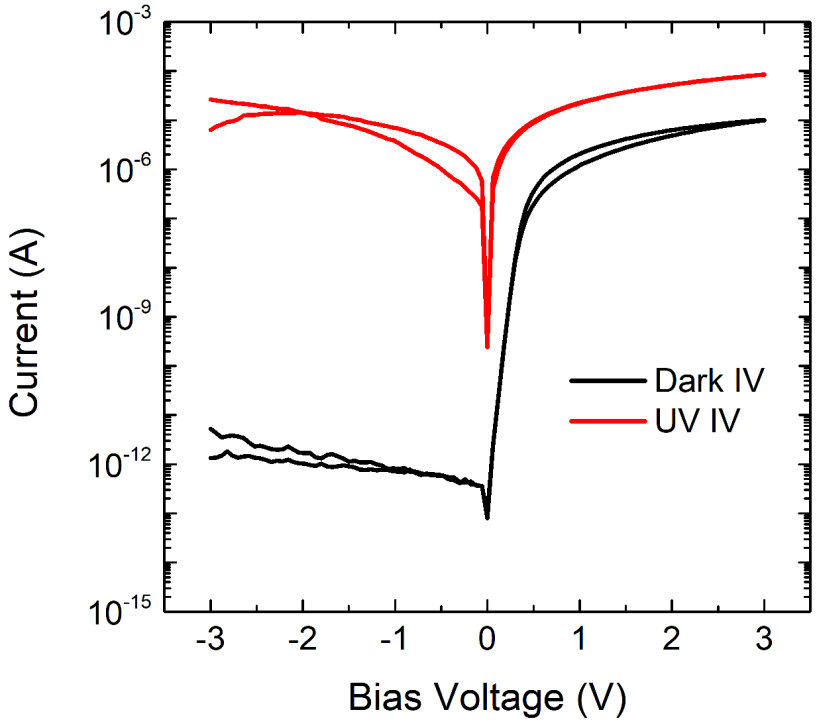
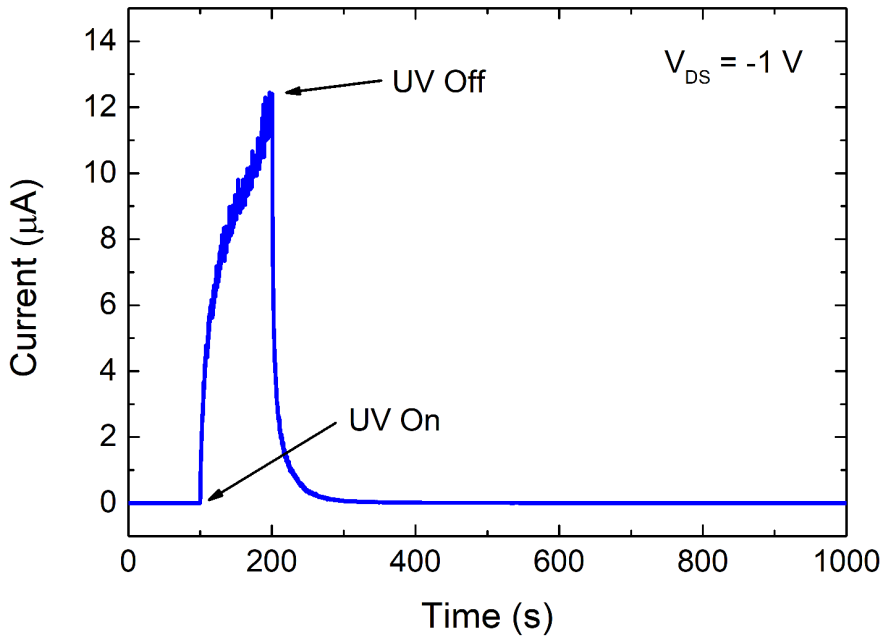
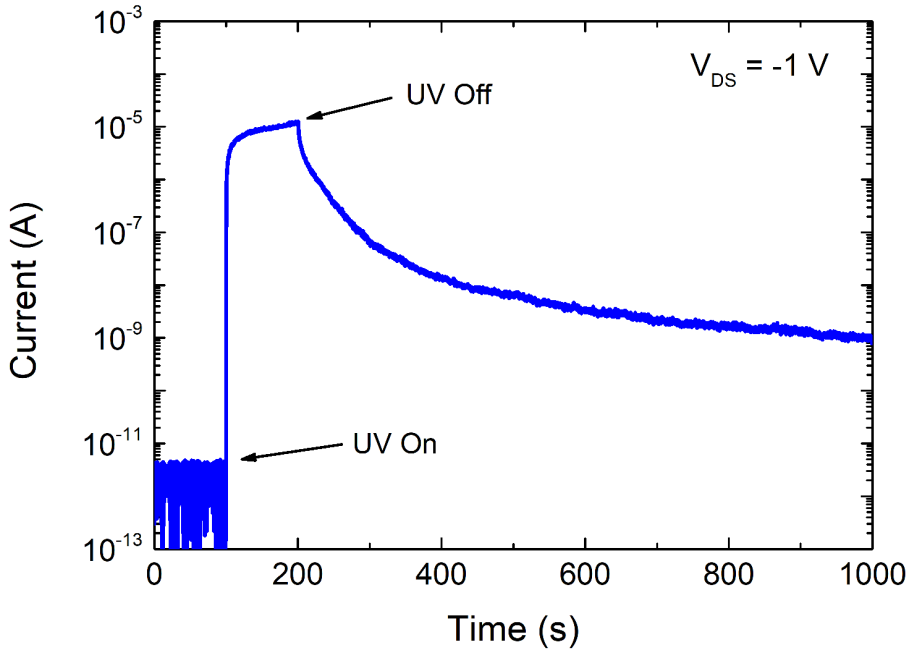


Figure 5.14: IV measurements of a 0.15% Sb-doped SnO_2 nanotube Schottky diode in the dark (Black) and during 275 nm UV back-illumination (Red).



(a)



(b)

Figure 5.15: UV photoresponse measurements of a 100 s back-illumination of a 0.15% Sb-doped SnO_2 nanotube Schottky diode on (a) linear and (b) log scales.

Schottky diode was annealed immediately prior to the measured IV characteristics and UV exposures.

Figure 5.14 shows the IV measurements of an Sb-doped SnO_2 nanotube Schottky diode before and during UV back-illumination. There is a significant increase in the reverse bias current under UV illumination, with a sensor response of ~10^7 under negative bias voltages. The forward currents additionally increase by an order of magnitude under UV illumination.

Figure 5.15 shows the transient UV photoresponse of the same Sb-doped SnO_2 nanotube Schottky diode when exposed to the 275 nm UV light for 100 s under a negative bias voltage. In the linear-scale view of the UV photoresponse (Figure 5.15.a) the current increases sharply when the UV light is turned on, although the photocurrent is not observed to saturate after 100 s of UV exposure. Very little PPC is observed in the SnO_2 nanotube Schottky diode, with a fast recovery to near 0 A when the UV light is turned off. Fitting a stretched-exponential function to this UV recovery provides a τ value of 3.4 s, which is significantly smaller than the 136 s observed for the Sb-doped SnO_2 nanotube resistor tested in similar atmospheric conditions. Clearly, the Schottky contact is allowing for the efficient separation of electron-hole pairs and is likely the primary limiting factor in the device reverse bias current. The log-scale view of the UV photoresponse (Figure 5.15.b) does illustrate how some PPC effects still remain within the SnO_2 nanotube Schottky diode, as while the current has decreased by 4 orders of magnitude after 800 s it is still 2.5 orders of magnitude larger than the current before UV illumination.

5.4 Multi-Nanotube Interdigitated Finger Resistors

The previous electronic devices fabricated from individual SnO_2 nanotubes have demonstrated good electrical characteristics across multiple device configurations. However, a common problem exists with the fabrication techniques used to create these devices. All these devices are fabricated by dispersing a large number of SnO_2 nanotubes onto an insulating substrate, followed by the deposition of thousands of metal contacts. While this strategy does create good electronic devices, the overall device yields are very low, with many metal contacts remaining isolated with no

bridging SnO_2 nanotubes. Furthermore, the use of SEM to identify and record the locations of well-contacted SnO_2 nanotubes is very labour intensive and not scalable for commercial devices. This is a common theme with nano-scaled electronic devices, where individual devices demonstrate strong performances within different applications yet the fabrication methodologies are difficult to up-scale for commercial production[329]. As a result, a device configuration with high yields that still maintains the nano-scale benefits of the SnO_2 nanotubes is desirable for device applications.

5.4.1 Fabrication of Multi-Nanotube Interdigitated Finger Resistors

The approach used to increase the yield of SnO_2 nanotube-based devices involved increasing the effective area that SnO_2 nanotubes could bridge across two metal contacts. This was achieved through the design of a set of interdigital electrodes (Figure 5.16.a). In this design, a total of 10 interdigital fingers with a width of 39 µm and overlapping length of 368 µm are separated by a 5 µm gap to allow SnO_2 nanotubes to bridge the two electrodes. For the previous individual SnO_2 nanotube devices, individual 50×50 µm square pads had overlapping sides totalling 200 µm. For these interdigital finger devices, the total length of the interdigital area is 3312 µm, over 1500% larger than the individual square contact pads.

A modified fabrication procedure was used to produce interdigitated finger resistors that incorporated the SnO_2 nanotubes. When the standard fabrication procedure of nanotube dispersal, photolithographic patterning, contact deposition and lift-off was followed, the large interdigital area made the complete lift-off of all excess metal difficult. As any excess metal within the interdigital area resulted in an electrical short between the two electrodes, the fabrication procedure needed to be modified to optimise an efficient lift-off process. A significant factor that affects the lift-off process is the thickness of the deposited metal layers, with thicker layers being more difficult to completely lift-off. In previous devices, effectively contacting the SnO_2 nanotubes required Ti and Au layers 500 nm and 50 nm thick respectively. This total thickness of material was deemed too thick to achieve a consistent lift-off with the interdigital contact, and so the total thickness of deposited material was

reduced to 150 nm. This change, however, resulted in the lift-off of all the dispersed SnO_2 nanotubes during the ultrasonic agitation in the lift-off process. The process order of nanotube dispersal and metal contact formation were subsequently swapped in the fabrication methodology, with the metal contacts deposited first, followed by the dispersal of the SnO_2 nanotubes. While this bottom-contact of the SnO_2 nanotubes does likely produce lower quality contacts to individual nanotubes compared to the top-contacting method used previously, the large number of bridging SnO_2 nanotubes compensates this problem.

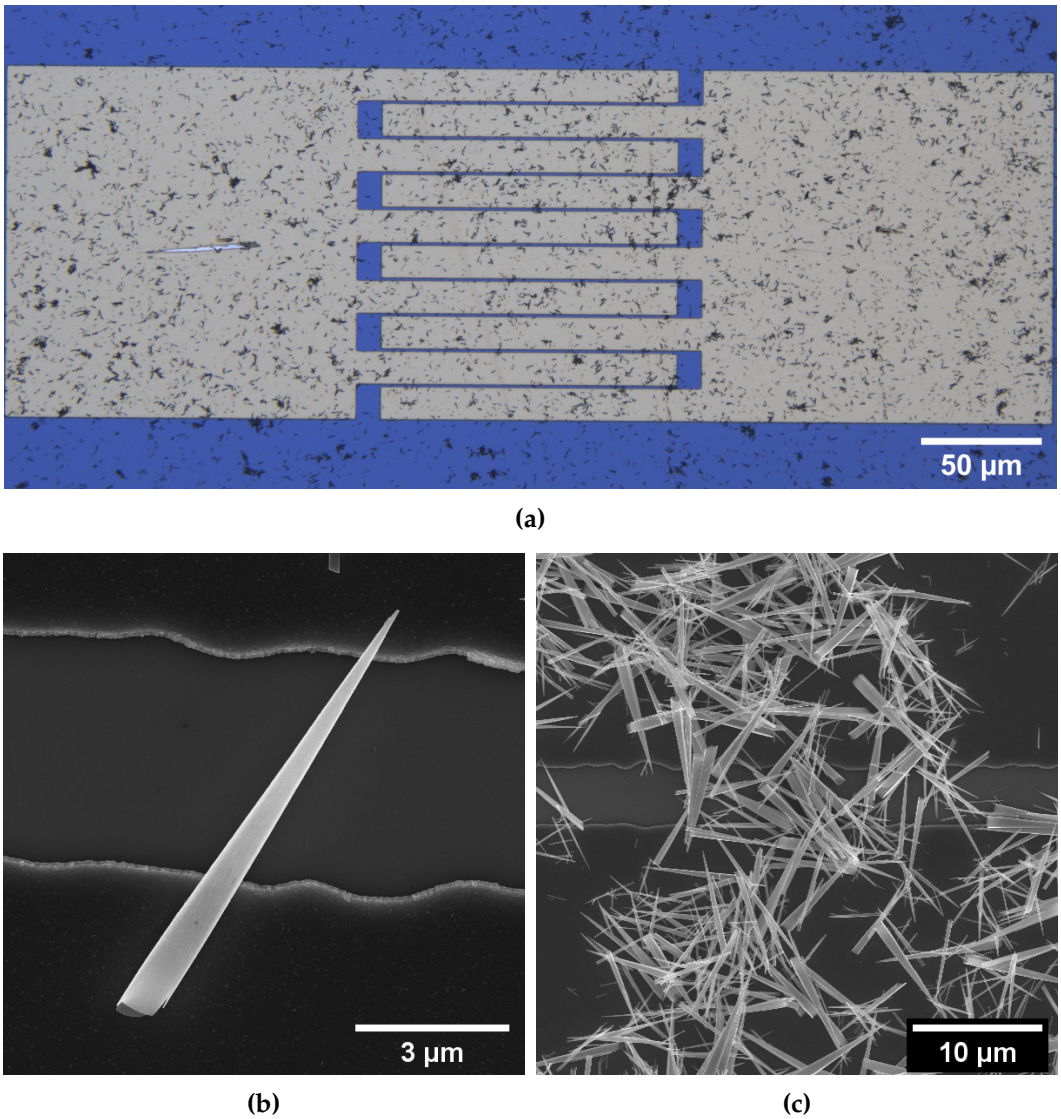


Figure 5.16: (a) Optical micrograph of a fabricated SnO_2 multi-nanotube interdigitated finger resistor with Ti contacts. A range of different SnO_2 nanotube bridging arrangements were observed under SEM, including (b) individual nanotubes and (c) large clusters of multiple nanotubes.

Several different samples of multi-nanotube interdigitated finger resistors were produced using undoped and 0.15% Sb-doped SnO_2 nanotubes alongside two different metal contacts. As the SnO_2 nanotubes contact the top of the metal electrodes in this device, the top metal layer forms the metal-semiconductor junction. This means the typical Ti/Au contact now produces an Au-SnO_2 junction rather than a Ti-SnO_2 junction. Au has been shown to produce Schottky barriers in contacts to SnO_2[159, 160, 330], although other studies report low contact resistances using Au contacts[331, 332]. Ti is known to form ohmic contacts with SnO_2[158, 160], although the exposed surface oxidises in air to form a surface layer of TiO_2[333]. To account for these potential problems with the contact metal, multiple samples of multi-nanotube interdigitated finger resistors were produced with contacts of Ti and Ti/Au. Samples fabricated with Ti contacts were deposited with 150 nm of Ti while samples with Ti/Au contacts were produced with 100 nm of Ti and 50 nm of Au. All samples were annealed for 10 minutes at 400 °C in nitrogen following the final SnO_2 nanotube dispersal.

Fabricated multi-nanotube interdigitated finger resistors were characterised by SEM to observe the variety of nanotube bridges across the metal contacts. Different nanotube configurations were observed bridging the contacts, ranging from individual nanotubes (as in Figure 5.16.b) to large, complex clusters of SnO_2 nanotubes (Figure 5.16.c).

5.4.2 Electrical Characteristics of Multi-Nanotube Interdigitated Finger Resistors

Figure 5.17 shows the IV characteristics of a selection of multi-nanotube interdigitated finger resistors for each doping and contact type. Importantly, the yield of devices was significantly improved, with the majority of fabricated devices exhibiting relatively strong electrical conductivity. Currents of 10^{-11} A were recorded across unbridged contacts, indicating that all current recorded through measured devices flowed through bridging SnO_2 nanotubes. Each type of fabricated resistor shows a range of maximum currents due to the unique numbers and configurations of bridging SnO_2 nanotubes across the metal contacts. The undoped multi-nanotube devices

show a significant increase in the maximum current compared to the individual-nanotube resistors. The best undoped individual-nanotube resistor demonstrated a maximum current around 1 µA at 3 V while the best device shown here has a maximum current above 10 µA at 3 V. Interestingly, the 0.15% Sb-doped multi-nanotube interdigitated finger resistors do not show a similar increase in current despite the increased carrier concentration compared to the undoped nanotubes. This suggests that the current through these devices is being limited more by the contact resistance compared to the resistance of the nanotubes themselves. This is further supported by the fact that all measured devices show some form of current hysteresis in their

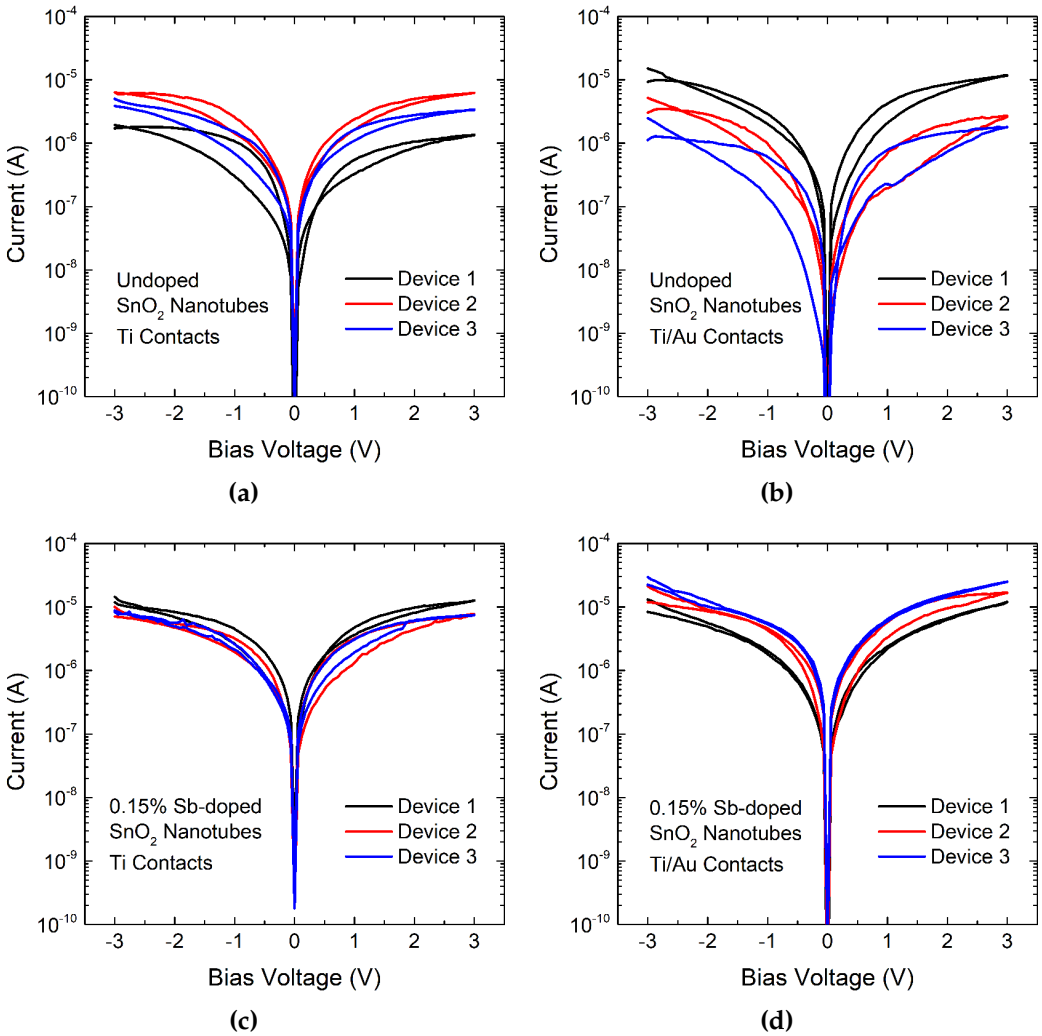


Figure 5.17: IV measurements of a range of multi-nanotube interdigitated finger resistors with different nanotube doping and contact materials. (a) Undoped nanotubes with Ti contacts, (b) undoped nanotubes with Ti/Au contacts, (c) Sb-doped nanotubes with Ti contacts, and (a) Sb-doped nanotubes with Ti/Au contacts.

IV characteristics, suggesting that Schottky barriers are present at all the contacts in these devices. Interestingly, both contact metals produce similar currents despite the reported differences in ohmic contact formation to SnO_2.

To characterise the IV properties of the multi-nanotube interdigitated finger resistors independently of each device's bridging nanotube configuration, a statistical analysis was performed on the maximum current at 3 V in a large number of devices. A box plot showing the maximum current at ±3 V for all measured devices across the four samples can be seen in Figure 5.18. All measurements on each sample span multiple orders of magnitude, illustrating how the device current is still highly dependant on the nanotube bridging configuration rather than the doping or contact metal. These different materials do still affect the average electrical properties however, as the undoped SnO_2 nanotube devices show lower mean and median maximum currents compared to the Sb-doped devices. The Sb-doped devices also show a smaller difference between the upper and lower quartiles in the data, meaning the Sb-doped materials tend to form more consistent contacts on average compared to the undoped devices. A consistent difference can also be seen between devices with Ti and Ti/Au contacts, where devices with Ti/Au contacts show higher mean and

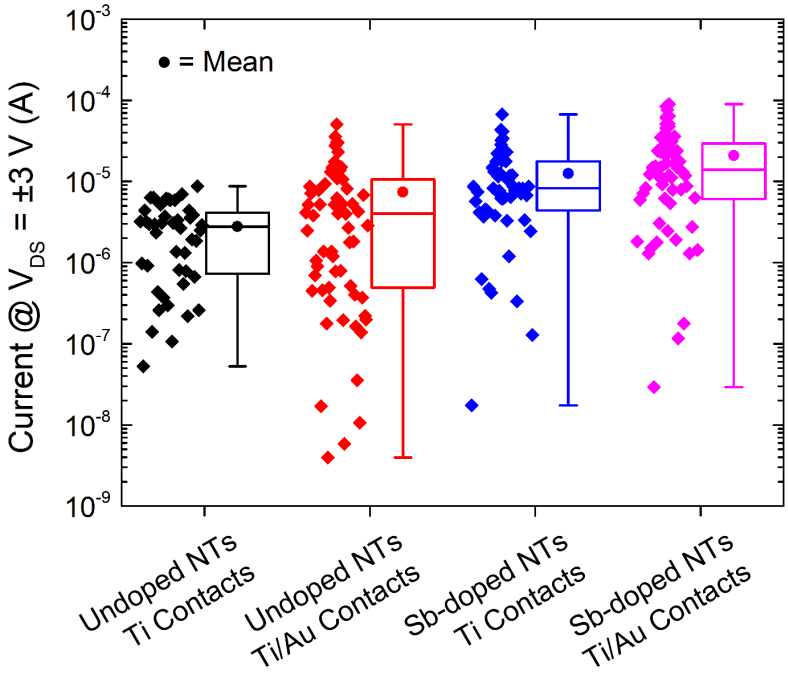


Figure 5.18: Box plot of the maximum current at ±3 V for all measured multi-nanotube interdigitated finger resistors.

median maximum currents compared to devices with Ti contacts. Future devices that utilise the multi-nanotube interdigitated finger arrangement should use Ti/Au contacts as the contact material to maximise the average current through fabricated devices.

5.5 Ionic-Gated Interdigitated Finger Transistors

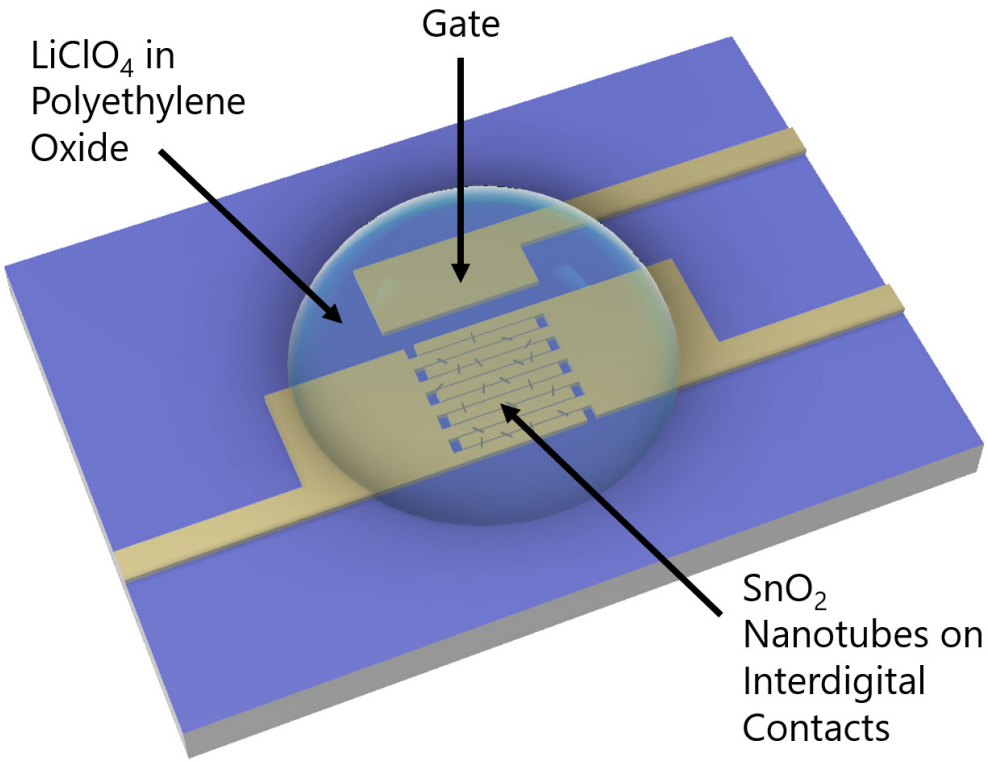
Section 5.4 demonstrated that multi- SnO_2 nanotube devices could be fabricated by the bottom-contacting of SnO_2 nanotubes with Ti/Au interdigitated fingers. These produced devices with a high device yield and good electrical characteristics that did not require the use of a SEM to identify functional devices. This interdigitated finger contact arrangement is more suitable for the larger-scale production of electronic devices, although suitable applications need to be identified to produce relevant devices.

In the developing technologies of neuromorphic computing and artificial synapses, ionic-gated transistors are emerging as low-power devices that can replicate several synaptic behaviours such as short- and long-term plasticity[201, 334, 335]. These transistors operate without a conventional dielectric separating the gate and conducting channel. Instead, an ion-conducting medium is used between the gate and channel, using materials such as polyethylene oxide combined with $LiClO_4$. When a charge is applied to the gate, ions of the opposite charge migrate towards the gate while ions of the same charge migrate at the channel. This results in the accumulation of ions at the gate and at the channel, forming an electric double layer (EDL). As there is no dielectric to isolate the ions from the channel surface, very large electric fields are produced by this EDL, even when the initial ion accumulation is produced by small gate voltages. This produces devices which can operate at small voltages and with a low power consumption. Furthermore, in biological synapses much of the synaptic behaviour is theorised to be caused by the accumulation of Ca^{2+} ions in nerve terminals after a presynaptic stimulus. Ionic-gated transistors operate with an analogous ion accumulation, and have been previously demonstrated to exhibit synaptic behaviours[201, 334, 335]. As a result, combining SnO_2 nanotubes

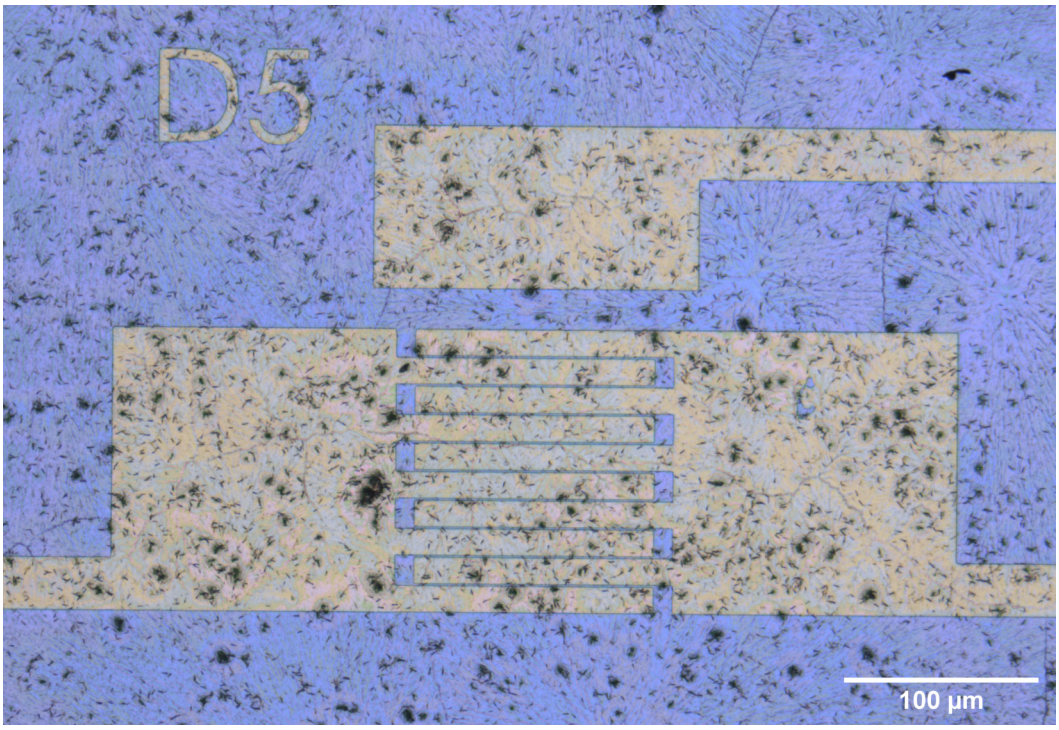
contacted by interdigitated fingers with an ionic-gated device structure could create scalable devices with potential applications in neuromorphic systems.

5.5.1 Fabrication of Ionic-gated Interdigitated Finger Transistors

Ionic-gated transistors that incorporate the SnO_2 nanotubes on an interdigitated finger arrangement were fabricated to create artificial synapses for neuromorphic applications. An illustrative diagram of the fabricated transistors can be seen in Figure 5.19.a while an optical micrograph of a fabricated device is shown in Figure 5.19.b. These devices were fabricated by the following method. Firstly, standard photolithographic patterning, metal deposition and lift-off were performed on an insulating substrate to deposit Ti/Au contacts with Ti and Au thicknesses of 100 nm and 50 nm respectively. Three contacts were patterned for each device: two interdigitated finger drain-source contacts and a third contact that acts as the transistor gate. Undoped SnO_2 nanotubes were then dispersed across the drain-source contacts to act as the transistor channel. The samples were then annealed for 10 minutes at 400 °C in nitrogen. The precursor solution of the polymer electrolyte was prepared by dissolving polyethylene oxide (PEO) and $LiClO_4$ in acetonitrile at ratios of 3% weight and 0.0268 M respectively. This gave a PEO ether oxygen : $LiClO_4$ ratio of 20:1. 10 µL of the precursor polymer electrolyte solution was pipetted on top of fabricated devices using a micropipette, before the sample was heated to 95 °C on a hot plate to remove excess acetonitrile and dissolved water. The sample was slowly cooled back to room temperature at a rate of 2 °C/min to allow the PEO to crystallise without the formation of significant grain boundaries.



(a)



(b)

Figure 5.19: (a) An illustrative diagram of the ionic-gated interdigitated finger transistors. (b) Optical micrograph of a fabricated ionic-gated SnO_2 nanotube interdigitated finger transistor.

5.5.2 Electrical Characteristics of Ionic-gated Interdigitated Finger Transistors

The electrical measurements of the ionic-gated interdigitated finger transistors required an adapted measurement methodology to obtain good IV characteristics. As the gating effect in these devices is caused by the diffusion of ions within the polymer medium and the formation of an EDL, a slow sweep rate is required to allow the ions to diffuse and accumulate at the nanotube surfaces. This slower sweep rate was implemented with a 5 s hold of the starting voltages before the beginning of the IV measurement and a 1 s delay between the application of each measurement voltage and the current measurement. In this case, the 60 mV step size between each voltage measurement gave an effective sweep rate of 60 mV s^{-1}. This sweep rate was maintained for both the output and transfer characteristics measurements. The current through a fabricated device without the dispersed SnO_2 nanotubes was measured to observe the leakage current through the polymer electrolyte, which was found to be 3×10^{-9} A at 2 V across the drain-source contacts.

Figures 5.20 and 5.21 show the output and transfer characteristics respectively of a typical ionic-gated interdigitated finger transistor. The expected behaviour of an n-type transistor is observed, with the output characteristics showing strong current enhancement and depletion with small changes in gate voltage. Potential current saturation is also beginning to occur at higher gate voltages. The transfer characteristics show a steep subthreshold swing at low drain-source voltages and large on/off ratios above 10^4. These extracted values at each drain-source voltage can be seen in Table 5.5. At a higher drain-source voltage of 3 V the leakage current across the drain-source contact increases rapidly, which leads to a smaller on/off ratio and shallower subthreshold swing.

As the diffusion and accumulation of the Li^+ and ClO_4^- ions to form the EDL is time-dependant, double sweeps of the transfer characteristics show large amounts of gate hysteresis that is sweep rate dependent. Figure 5.22 shows double sweeps of the transfer characteristics of the ionic-gated interdigitated finger transistor at 60 mV s^{-1} and 6 mV s^{-1} voltage sweep rates. Significantly more hysteresis is observed for the faster sweep rate, illustrating the time-dependant nature of the diffusion processes.

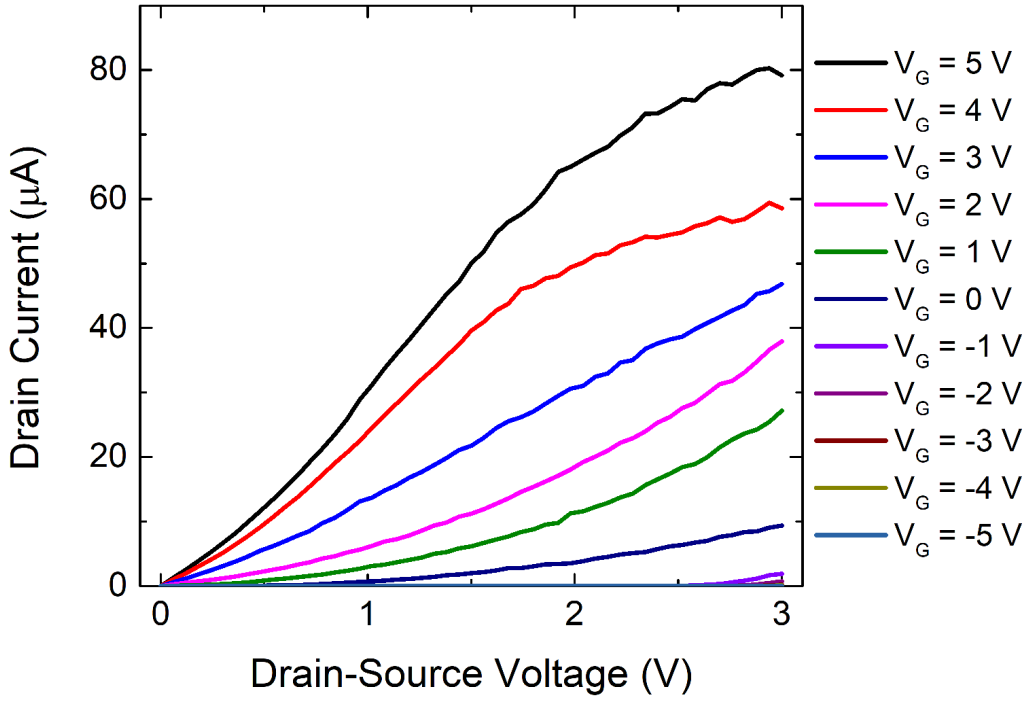


Figure 5.20: The output characteristics of an ionic-gated interdigitated finger transistor that incorporates undoped SnO_2 nanotubes.

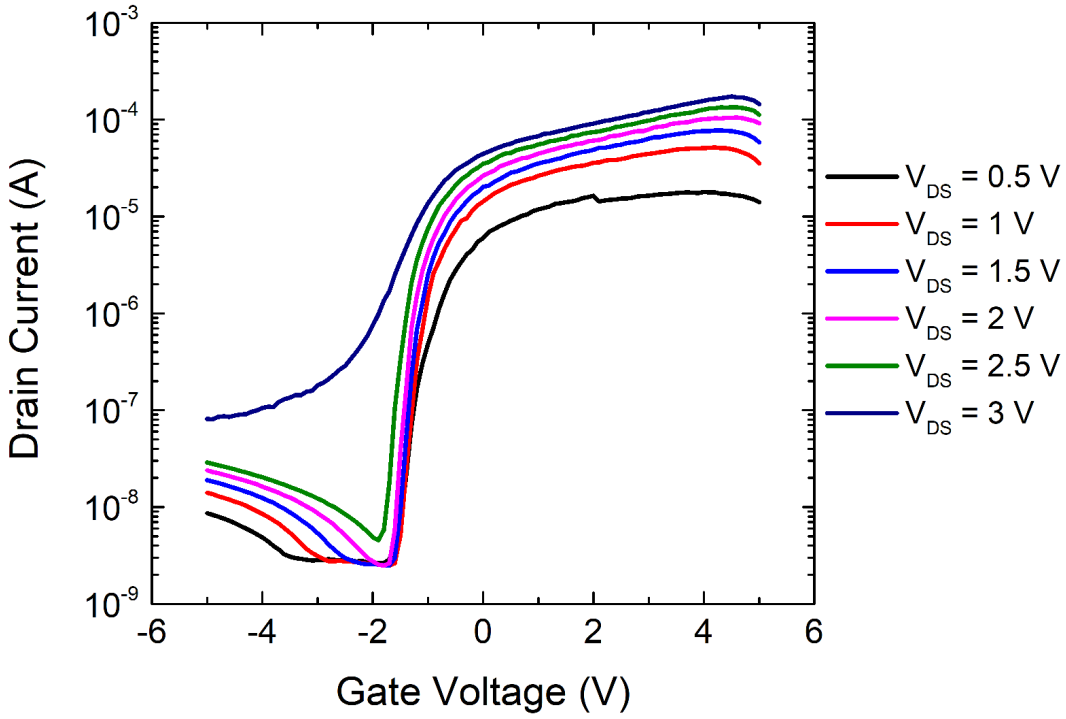


Figure 5.21: The transfer characteristics of an ionic-gated interdigitated finger transistor that incorporates undoped SnO_2 nanotubes. The gate voltage is swept from +5 V to -5 V.

Table 5.5: Transistor characteristics extracted from each of the transfer sweeps at different V_{DS}.

V_{DS} (V)	On/Off Ratio (V_G = ±2 V)	Subthreshold Slope (mV decade^{-1})
0.5	6.22×10^3	215
1	1.38×10^4	168
1.5	1.88×10^4	161
2	2.19×10^4	146
2.5	1.53×10^4	177
3	1.22×10^2	692

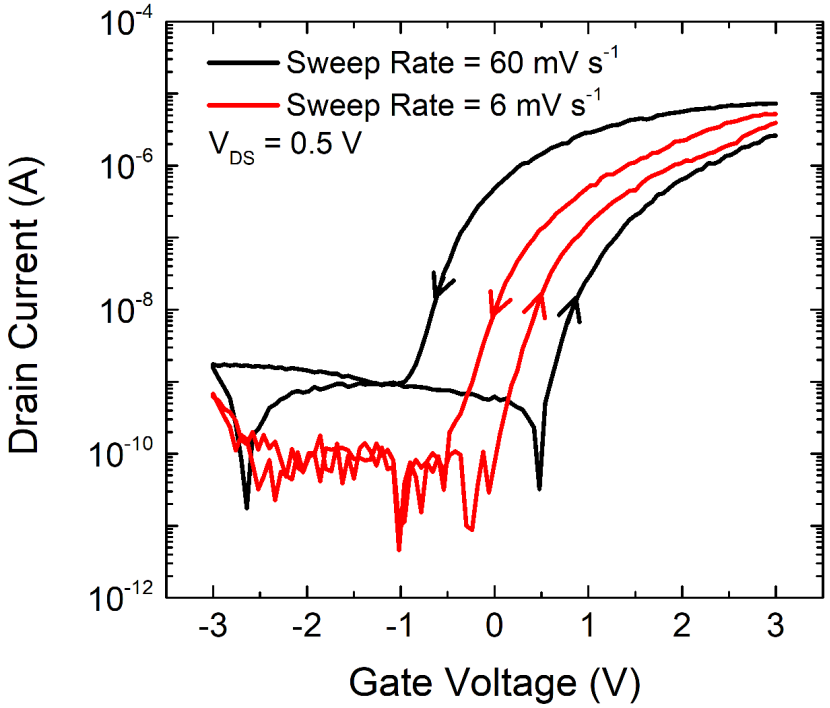


Figure 5.22: Transfer characteristics of an ionic-gated interdigitated finger transistor that demonstrates gate hysteresis that depends on the voltage sweep rate and direction.

5.5.3 Neuromorphic Measurements of Ionic-gated Interdigitated Finger Transistors

The processing of temporary information in neuromorphic systems relies on a process known as short-term plasticity (STP), where characteristics of previous signals through artificial synapses are remembered for a short time[336]. The simplest version of STP is known as excitatory postsynaptic current (EPSC), where a short presynaptic voltage pulse results in a persistent postsynaptic current that decays over

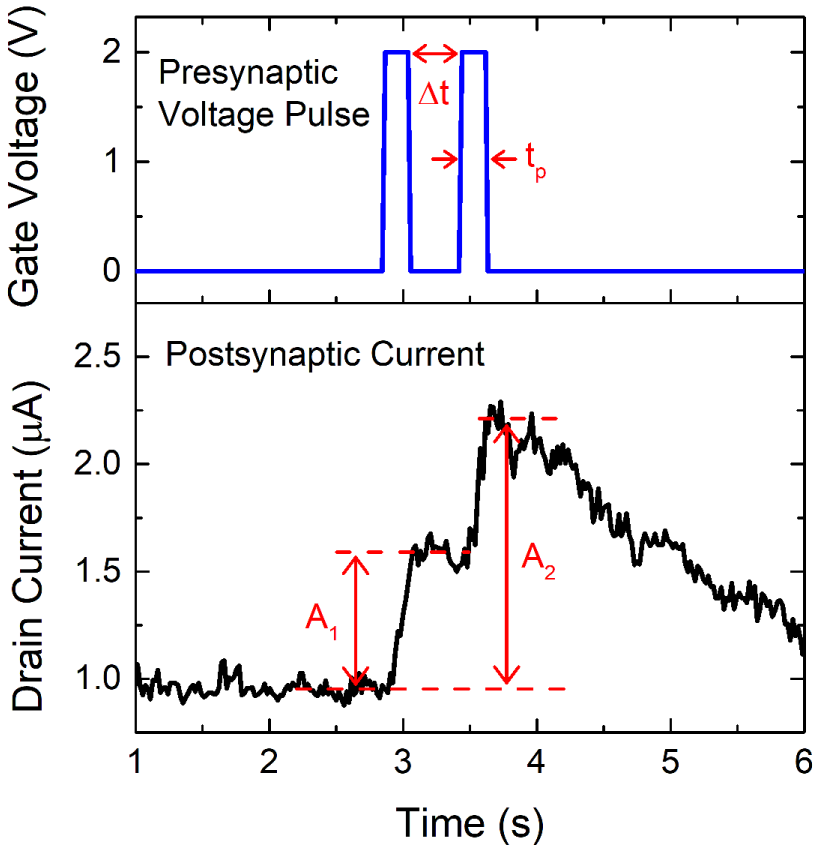
time. This phenomena can then lead to a range of other synaptic behaviours such as paired-pulse facilitation (PPF) and spike-rate-dependent-plasticity (SRDP)[336]. The time-dependent ion diffusion produced by applied gate voltages to the ionic-gated interdigitated finger transistors means these devices can emulate these synaptic behaviours, which leads to potential uses in neuromorphic applications.

Figure 5.23.a shows a pair of 2 V, ~200 ms presynaptic voltage pulses to the gate of an ionic-gated interdigitated finger transistor and the resultant postsynaptic current response in the drain current with an applied drain-source voltage of 2 V. Clear EPSC and PPF is observed, as seen by the significant persistent postsynaptic current and the large difference in amplitude between peaks A_1 and A_2. During the positive gate voltage pulse, Li^+ ions are strongly driven to the nanotube surfaces by the applied electric field, accumulating in an EDL and quickly increasing the current through the nanotubes by field-enhancement. When the gate voltage returns to zero, no external electric fields are present, and so the accumulated Li^+ ions are slowly repelled from the nanotube surface through Coulombic repulsion with other ions. This slow diffusion of ions away from the nanotube surfaces leads to the observed EPSC and PPF. A problem that presents itself throughout the neuromorphic testing is the relatively large amount of noise that occurs in the device current. This noise arises from the poor contacts to the SnO_2 nanotubes, and can lead to increased margins of error in the PPF ratio A_2/A_1.

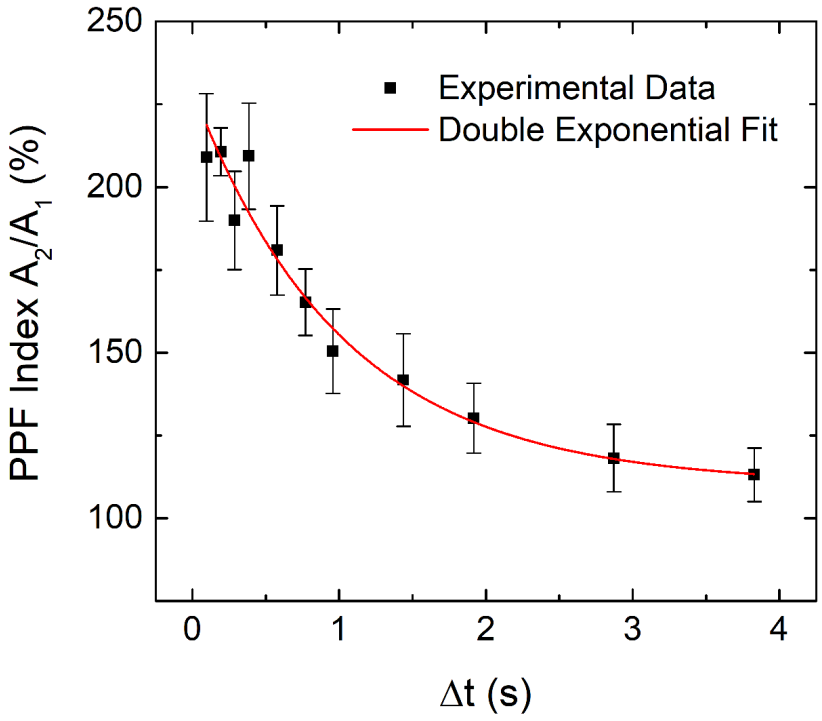
The PPF index of the ionic-gated interdigitated finger transistor with different inter-pulse spacing (Δt) can be seen in Figure 5.23.b. The PPF index is observed to decrease with an increasing Δt, which is consistent with the behaviour of artificial synapses. A maximum PPF index of 212% is obtained at small Δt, which is similar to that obtained for ionic-gated SnO_2 nanofibers[201]. PPF indices are commonly fit using a double exponential decay of the following form[201, 334, 337]:

$$PPFIndex = 100\% + C_1 e^{-\frac{t}{\tau_1}} + C_2 e^{-\frac{t}{\tau_2}} \tag{5.5}$$

Where C_1 and C_2 are the initial facilitation amplitudes for each exponential phase and τ_1 and τ_2 are the characteristic decay times for each exponential phase. Fitting Equation 5.5 to the experimental PPF data provides C_1, τ_1, C_2 and τ_2 values of 117%,



(a)



(b)

Figure 5.23: (a) A pair of presynaptic voltage pulses and the resulting postsynaptic current of an ionic-gated interdigitated finger transistor. (b) The PPF index of an ionic-gated interdigitated finger transistor with varying inter-pulse spacing. Error bars represent one standard deviation across 10 measurements.

1.01 s, 12% and 31.5 s respectively. These initial facilitation amplitudes and characteristic decay times indicate that most of the PPF can be well fitted by a single exponential decay curve with a short decay time, demonstrating a majority of STP. However, the smaller amplitude yet longer decay time component suggests a degree of longer-term plasticity is present within the device characteristics.

STP is also able to be modulated using trains of presynaptic voltage pulses of different inter-pulse spacings. When the presynaptic inter-pulse spacing is significantly shorter than the PPF characteristic decay time, multiple sequential pulses result in an accumulation of the EPSC. Figure 5.24 shows three different presynaptic pulse trains with inter-pulse spacings that are shorter, approximately the same as, and longer than the 1.01 s characteristic decay time of the PPF index, alongside the postsynaptic current response. The EPSC amplitude following the first voltage pulse is designated A_1 while the EPSC amplitude following the last voltage pulse is designated A_{Sat}. Significant accumulation of the EPSC is observed for the pulse train with a short inter-pulse spacing, with A_{Sat}/A_1 = 1490%. Less accumulation is observed for larger inter-pulse spacings, as Li^+ ions have more time to diffuse away from the nanotube surfaces. This behaviour is similar to that seen in biological synapses[338], and effectively emulates spike-rate-dependent-plasticity[334, 339]. Furthermore, all three measurements show current saturation after multiple presynaptic voltage pulses, which is analogous to desensitization observed in biological synapses where repeated stimuli results in a reduced sensitivity of the synaptic response[338, 340].

Further evidence of spike-rate-dependant-plasticity is seen in Figure 5.25, which shows the postsynaptic current response to nine presynaptic voltage pulses in which the inter-pulse spacing decreases with each pulse from 5.7 s to 100 ms. Similar PPF is observed as in Figure 5.24, however there is no saturation of the postsynaptic current due to the increasing frequency of the presynaptic voltage pulses. This indicates that the postsynaptic output remains sensitive to changes in presynaptic pulse frequency even while having been previously facilitated. When combined with the results shown in Figures 5.23 and 5.24, these findings illustrate the potential to produce a range of synaptic-like behaviours using the ionic-gated interdigitated finger

transistors. The spiking outputs of these devices are of a suitable form for hardware-implemented spiking neural networks[341].

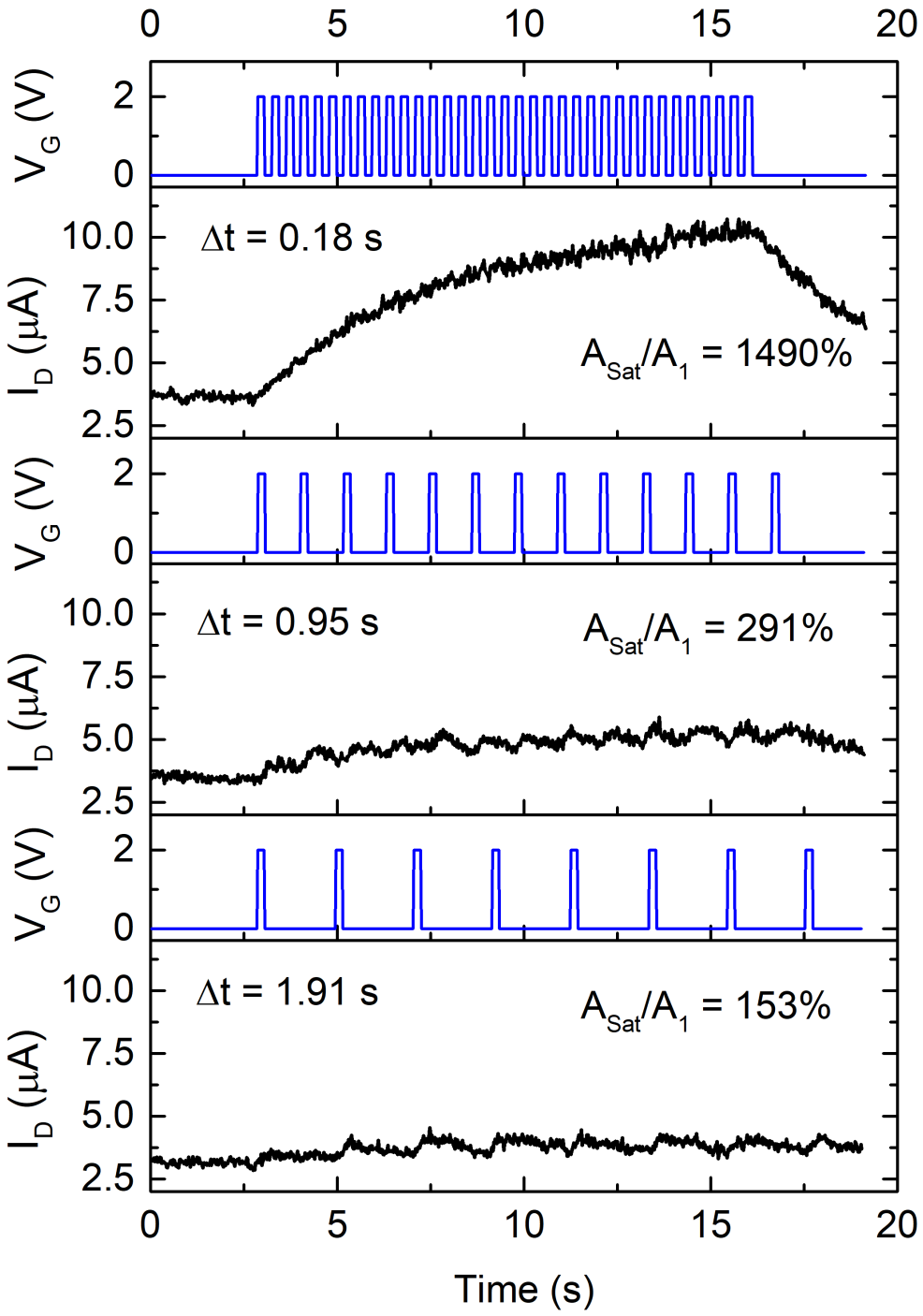


Figure 5.24: Trains of presynaptic voltage pulses with different inter-pulse spacings alongside the postsynaptic current response, demonstrating PPF and adaptation.

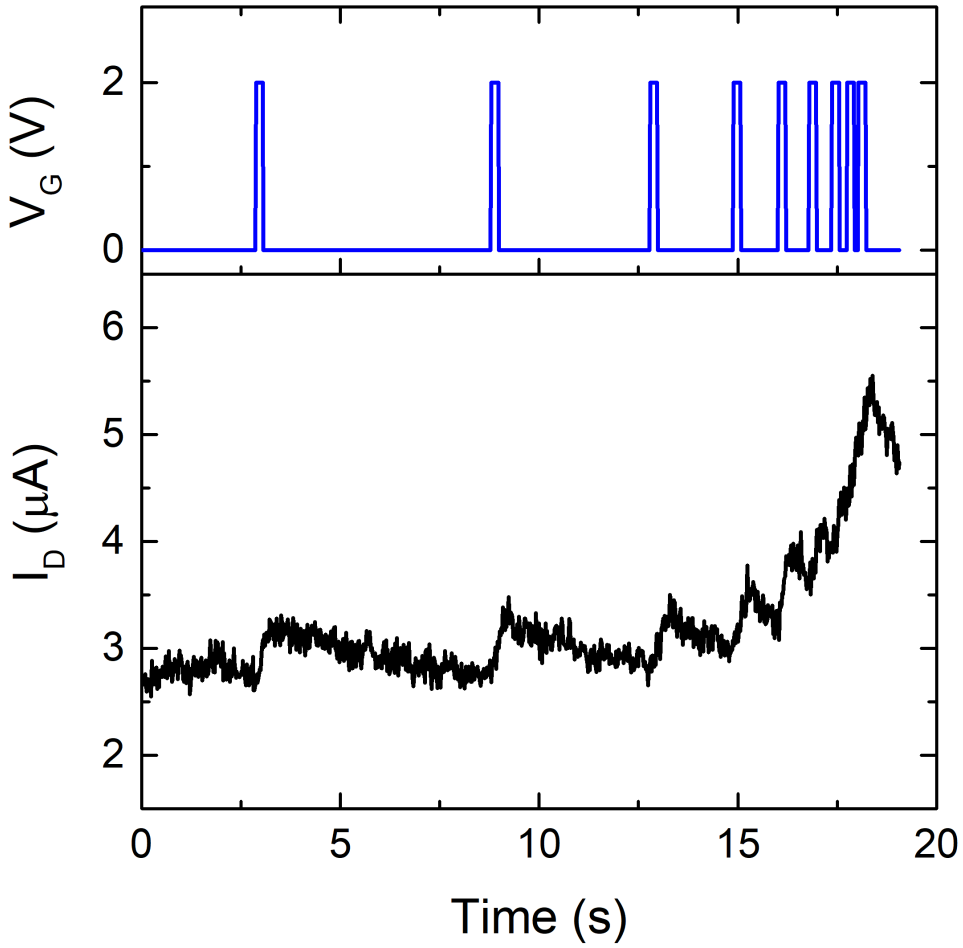


Figure 5.25: A series of presynaptic voltage pulses with a shrinking interpulse spacing alongside the postsynaptic current response.

5.6 Summary and Conclusions

Undoped SnO_2 nanotube field-effect transistors have been successfully fabricated by back-gating individual nanotubes through the conducting p-Si/SiO_2 substrate, with the SiO_2 layer acting as the gate dielectric. These devices demonstrated good n-type FET characteristics with a maximum on/off ratio of ~4×10^6, average subthreshold slope of 187 mV/decade and a small average threshold voltage of -1.96 V, comparable to other results seen in literature. A 2D electrostatic simulation found an individual SnO_2 nanotube FET exhibited a gate capacitance of 146 aF μm^{-1}, which allowed a field-effect mobility to be calculated from the FET transfer characteristics. The calculated average field-effect mobility of 6.13 $cm^2V^{-1}s^{-1}$ is similar to that found in other

undoped SnO_2 FETs, but is smaller than the 30 $cm^2V^{-1}s^{-1}$ found by Hall-effect measurements for Mist-CVD deposited undoped SnO_2 thin films. This difference was explained by the poor ohmic contact created by the Ti/Au contacts to the undoped nanotubes and by interface states at the device gate. The carrier concentration of the undoped SnO_2 nanotube was estimated to be 5×10^{16} cm^{-3}. Large amounts of gate hysteresis was observed under an extended gate voltage bias, which was explained by charge trapping most likely caused by water vapour between the gate dielectric and the nanotube channel. Finally, 0.15% Sb-doped SnO_2 nanotube FETs demonstrated current enhancement and depletion, however the carrier concentration was too high to pinch off these devices, even under large gate bias voltages.

Individual undoped and 0.15% Sb-doped SnO_2 nanotube Schottky diodes were fabricated using an IrO_x/Ir Schottky contact. Devices measured immediately following 400 °C annealing demonstrated excellent electrical characteristics that outperformed all existing nanostructured devices in the literature, with properties that were similar to the best thin-film Schottky diodes. Large average barrier heights of 0.82 eV and 0.67 eV were found for the undoped and Sb-doped devices respectively, with the difference in barrier height explained by the image-force lowering effect which increases with carrier concentration. Large current rectification ratios greater than 10^7 and small ideality factors of 1.24 and 1.09 were also observed for the undoped and Sb-doped devices respectively. Despite the excellent initial electrical properties, a severe ageing effect was observed after leaving the fabricated devices in air for six days, which resulted in a lowering of the Schottky barrier and a drop in the forward current. Reannealing the devices restored the excellent electrical properties, which suggested that atmospheric adsorbates play an important role in this effect. PMMA encapsulation was not effective at preserving the devices' electrical properties. The UV sensing performance of a recently-annealed Sb-doped SnO_2 nanotube Schottky diode was tested by the back-illumination of a fabricated device with 275 nm UV light. A large sensor response of ~10^7 was observed under negative bias voltages, and was accompanied by the shortest observed PPC recovery with a fitted τ value of 3.4 s. PPC effects were still observed to some degree, with a persistent reverse bias current that remained orders of magnitude larger than the initial

dark current after 800 s.

To address the low yield and labour-intensive process required to create individual SnO_2 nanotube devices, multi-SnO_2 nanotube resistors were fabricated by bottom-contacting multiple nanotubes using a large interdigital area. Due to the bottom-contact nature of these devices, two contact materials were tested: Ti only and Ti/Au. The yield of conducting devices was significantly higher compared to the individually contacted SnO_2 nanotube resistors, with the majority of the fabricated devices exhibiting conductivity through bridging SnO_2 nanotubes. These devices demonstrated IV characteristics that were generally symmetrical, however some current hysteresis was present in all devices and was more pronounced in undoped devices. Both sets of contact materials likely created poor contacts to the SnO_2 nanotubes due to the non-conformal bottom contact, as indicated by the small average difference in current between undoped and Sb-doped devices. Instead, much of the current variation was more likely associated with different SnO_2 nanotube bridging arrangements. Despite Au reportedly forming Schottky contacts to SnO_2, Ti/Au contacts produced devices with more current at 3 V in both undoped and Sb-doped devices compared to resistors that utilised Ti contacts.

To demonstrate further applications of the interdigitated finger devices, $LiClO_4$/PEO ionic gated SnO_2 nanotube transistors were fabricated for use in potential neuromorphic applications. Fabricated devices exhibited good n-type transistor characteristics with on/off ratios up to ~2.2×10^4 and an average subthreshold slope of 173 mV/decade. The transfer characteristics were shown to be highly dependent on the sweep rate and sweep direction, due to the time-dependent nature of the ion diffusion through the polymer medium. Neuromorphic testing of the ionic-gated transistors demonstrated several synaptic behaviours such as excitatory postsynaptic current, paired-pulse facilitation, spike-rate-dependent-plasticity and adaptation. These findings demonstrate the potential use of the ionic-gated SnO_2 nanotube transistors as artificial synapses in spiking neuromorphic systems, although more work is required to reduce the noise within these devices during operation.

Chapter 6

Band Bending Studies of the Square SnO_2 Nanotubes

6.1 Introduction

Earlier work performed by Scott et al. established that the SnO_2 nanotubes exhibit large downward band bending that results in a strong electron accumulation layer at the nanotube surfaces[29, 30]. Unlike in other metal oxides, this downward band bending was maintained upon heating to higher temperatures due to the formation of electron-donating in-plane oxygen vacancies[71]. This was despite the thermal removal of surface hydroxyls which were initially responsible for the strong downward band bending.

Work presented earlier in this thesis demonstrated that the thermal annealing of SnO_2 nanotube-based Schottky diodes significantly improved the diode characteristics, however, these improved characteristics regressed when the devices were left exposed to the atmosphere for an extended period. This presented a gap in knowledge of what occurs when an annealed SnO_2 surface is exposed to water vapour and oxygen present in the atmosphere. Furthermore, both oxygen and water vapour were shown to be highly important in the minimisation of persistent photoconductivity in the sensing of UV light by contacted SnO_2 nanotubes. Large amounts of hysteresis were also observed in SnO_2 nanotube based devices such as resistors and FETs that were exposed to the atmosphere. With these observations, it was clear that more research was required concerning the interaction of water vapour and

oxygen on the SnO_2 nanotube surfaces, particularly following important annealing processes.

In addition to the heating studies performed by Scott, work by McNeill and Carroll found that the surface band bending of metal oxides could be controlled by a variety of surface chemical treatments[226, 233]. In particular, sulfur treatments produced large valence band shifts on ZnO and, in certain conditions, significantly reduced the downward band bending of the SnO_2 nanotubes. The responsible mechanisms for these band bending shifts were still poorly understood, and warranted further investigation. Additionally, sulfur passivation of other semiconductors such as InAs and GaAs has been shown to inhibit surface oxidation and the formation of terminating hydroxyl groups[342, 343], suggesting that sulfur treatments may be more atmospherically stable than annealing treatments.

Two different investigations were undertaken to understand the interaction of the SnO_2 nanotubes with water vapour and oxygen, and to further investigate the mechanism and effectiveness of sulfur treatments on the band bending at the nanotube surfaces. Firstly, synchrotron XPS was used to investigate the recovery of surface hydroxyls and oxygen vacancies following high temperature heating, and the effect this has on the surface band bending. This involved performing a series of heating and H_2O/O_2 dosing experiments to observe the changes in the O 1s and valence band XPS spectra. Secondly, the effect of sulfur treatments on the SnO_2 nanotubes was investigated further, utilising SEM, synchrotron XPS and NEXAFS to investigate the sulfur species present on the nanotube surfaces and their effect on the surface band bending. In addition, sulfur-treated ZnO was investigated with NEXAFS to understand the different effects of the sulfur treatment on ZnO and SnO_2.

6.2 Recovery of Surface Hydroxyls and Oxygen Vacancies Following High Temperature Heating

To further understand the role of water and oxygen on the recovery of surface hydroxyls and oxygen vacancies, investigations were undertaken using synchrotron XPS on the soft X-ray beamline at the Australian Synchrotron. The vacuum system

of the soft X-ray beamline consists of an XPS analysis chamber which is connected to a series of other chambers, including a preparation chamber. This preparation chamber allows the heating and gas dosing of samples without removing the samples from the vacuum system and exposing them to atmosphere. Utilising this system, undoped and 0.15% Sb-doped SnO_2 nanotubes were treated by a range of heating and gas dosing experiments, seen in the steps below. O 1s and valence band spectra were captured after each step to characterise the state of the surface hydroxyl coverage and surface band bending.

1. XPS spectra were captured on the as-grown samples to serve as a reference for the following treatments.
2. Samples were heated in vacuum at a temperature of 600 °C for 30 minutes to remove the surface hydroxyls and induce the formation of oxygen vacancies.
3. Samples were dosed with dry O_2 gas at a pressure of 1×10^{-5} mBar for 30 minutes, for a total O_2 dose of 13,500 Langmuir.
4. Samples were dosed with de-oxygenated water vapour at a pressure of 1×10^{-5} mBar for 30 minutes, for a total H_2O dose of 13,500 Langmuir.
5. The samples were vented to atmosphere and removed from the vacuum equipment for a period of 10 hours, allowing the surface hydroxyls to recover before being reintroduced to vacuum.
6. Samples were heated again in vacuum at a temperature of 600 °C for 30 minutes.
7. Samples were dosed simultaneously with dry O_2 gas and de-oxygenated water vapour at a total pressure of 2×10^{-5} mBar for 30 minutes, for a total O_2/H_2O dose of 27,000 Langmuir.
8. Samples were exposed to a high pressure of deoxygenated water vapour within the vacuum system load-lock, by venting the load-lock using the deoxygenated water source. A maximum gauge pressure of 0.93 mBar was observed and the venting was performed for a total of 20 minutes. This gave an H_2O dosing on the order of 8×10^{8} Langmuir.

9. Samples were heated in vacuum at a temperature of 120 °C for 30 minutes to observe if residual H_2O was removed from the samples.

10. The samples were vented to atmosphere within the load-lock for a period of 20 minutes before being reintroduced to vacuum, to observe the degree of re-hydroxylation that occurs in a shorter venting period.

11. The samples were vented to atmosphere and removed from the vacuum equipment for a period of 10 hours, before being reintroduced to vacuum.

6.2.1 O 1s and Valence Band Spectra of Undoped SnO_2 Nanotubes Throughout the Heating/Dosing Series

O 1s Core-level Spectra

The O 1s spectra of the undoped SnO_2 nanotubes throughout the heating/dosing series can be seen in Figure 6.1. Similar to previous heating series performed on the SnO_2 nanotubes, three peaks were used to fit the O 1s spectrum[29, 30]. These three peaks were attributed to lattice oxygen (~531.1 eV, red), in-plane oxygen vacancies (~531.9 eV, grey) and surface hydroxyl groups (~532.6 eV, blue) in accordance with previous literature[30, 344–346]. The ratios of the hydroxyl and oxygen vacancy peaks compared to the lattice oxygen peak throughout the heating/dosing series are shown in Figure 6.2.

On the as-grown undoped sample, only the lattice O and hydroxyl peaks are required to achieve a good fit to the observed spectrum, with no substantial contribution from oxygen vacancies. This indicates a largely vacancy-defect free surface with partial surface hydroxylation, as expected for atmosphere-exposed SnO_2. Upon heating to 600 °C for 30 minutes, the hydroxyl/lattice oxygen ratio decreases by >50% and the contribution from oxygen vacancies increases, due to the thermal desorption of surface hydroxyls and in-plane lattice oxygen. When the undoped SnO_2 nanotubes are dosed with 13,500 Langmuir of dry O_2 gas, there is no significant change in the ratio of lattice oxygen to the surface hydroxyls. However, the contribution from oxygen vacancies decreases by ~30%, suggesting the self-repair of some of the oxygen vacancies under an O_2 environment. Unlike the O_2 dosing, H_2O dosing

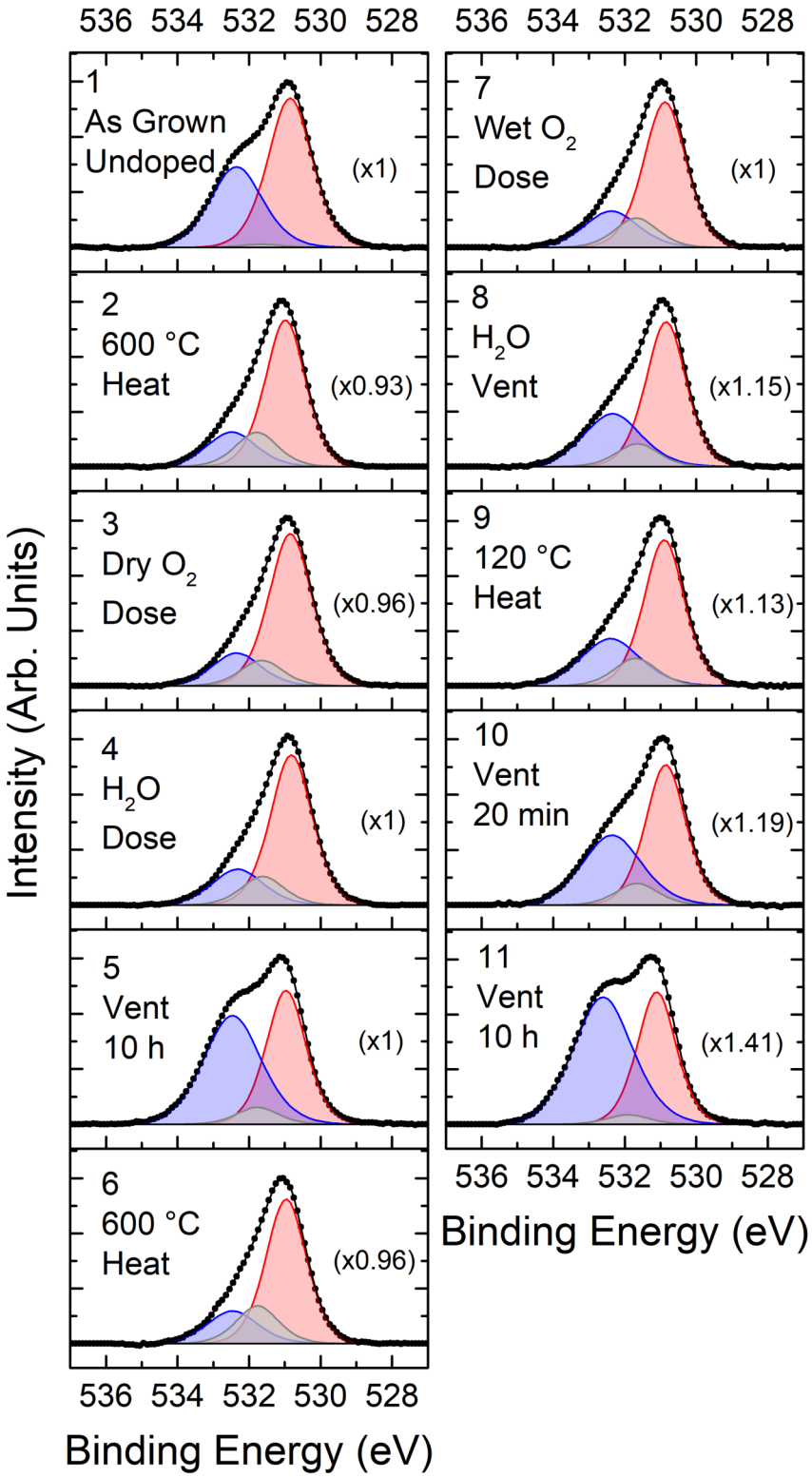


Figure 6.1: O 1s core level XPS spectra for the series of heating and O_2/H_2O dosing of undoped SnO_2 nanotubes. Three fitted components are attributed to lattice oxygen (red), surface hydroxyl groups (blue) and oxygen vacancies (grey). All spectra were captured with a photon energy of $h\nu$ = 680 eV.

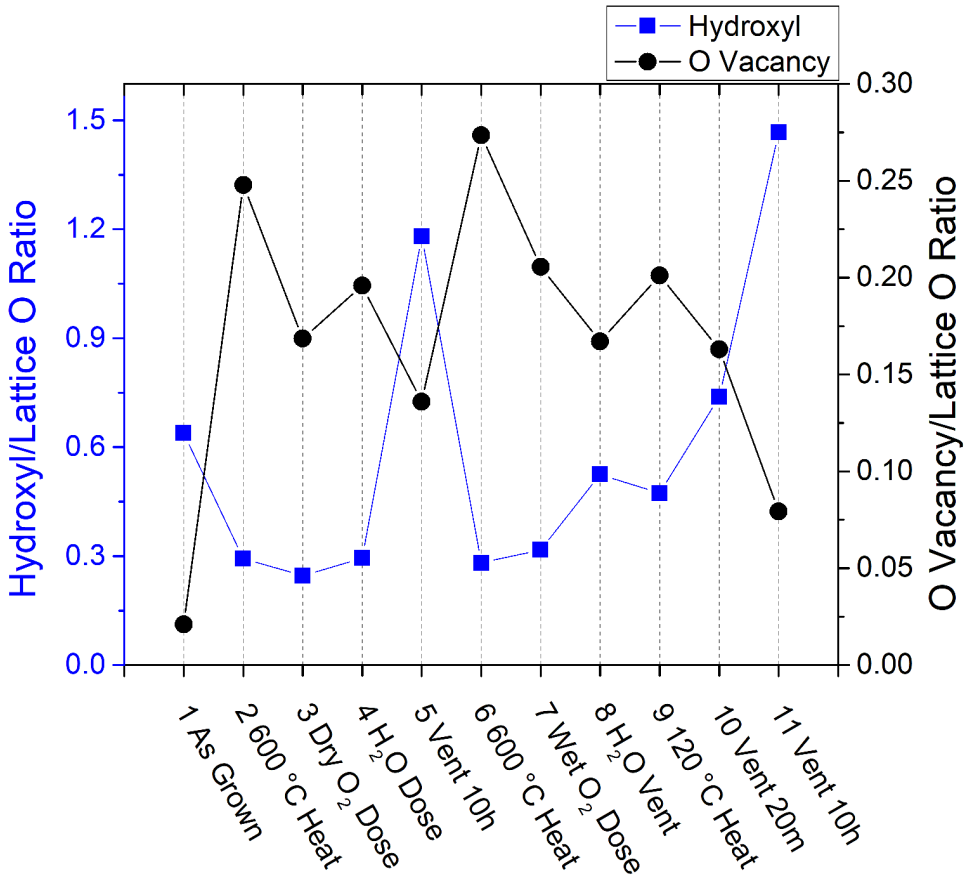


Figure 6.2: Area ratios of the surface hydroxyl and oxygen vacancy peaks compared to the lattice oxygen peak from the O 1s spectra of undoped SnO_2 nanotubes for each heating/dosing treatment.

does not increase the hydroxyl/lattice oxygen ratio, suggesting that water vapour at low pressures is insufficient to recover the surface hydroxyl groups. The difference between these two groups may be due to the differences in the desorption temperatures of molecular O_2 and H_2O from the SnO_2 surface in vacuum. Gercher and Cox showed that the defective SnO_2 surface with in-plane and bridging oxygen vacancies desorbs molecular H_2O at -73 °C in vacuum[347]. On the other hand, chemisorbed O_2^- desorbs from the SnO_2 surface around 150 °C[293]. Computational studies of the formation of surface hydroxyls on (110) oriented SnO_2 show that hydroxyl formation occurs first by the adsorption of molecular H_2O onto an exposed Sn_{5c} site, before the molecule dissociates into two surface hydroxyls using a nearby bridging oxygen[95, 96]. If the molecular water is unable to first adsorb onto the SnO_2 surface then it is unable to dissociate, limiting the formation of surface hydroxyls. Similarly, if the SnO_2 surface is fully reduced with few bridging oxygens, then the dissociation

of water to hydroxyl groups would also be unfavourable. These factors likely contribute to the H_2O dosing producing no increase of the surface hydroxyl coverage.

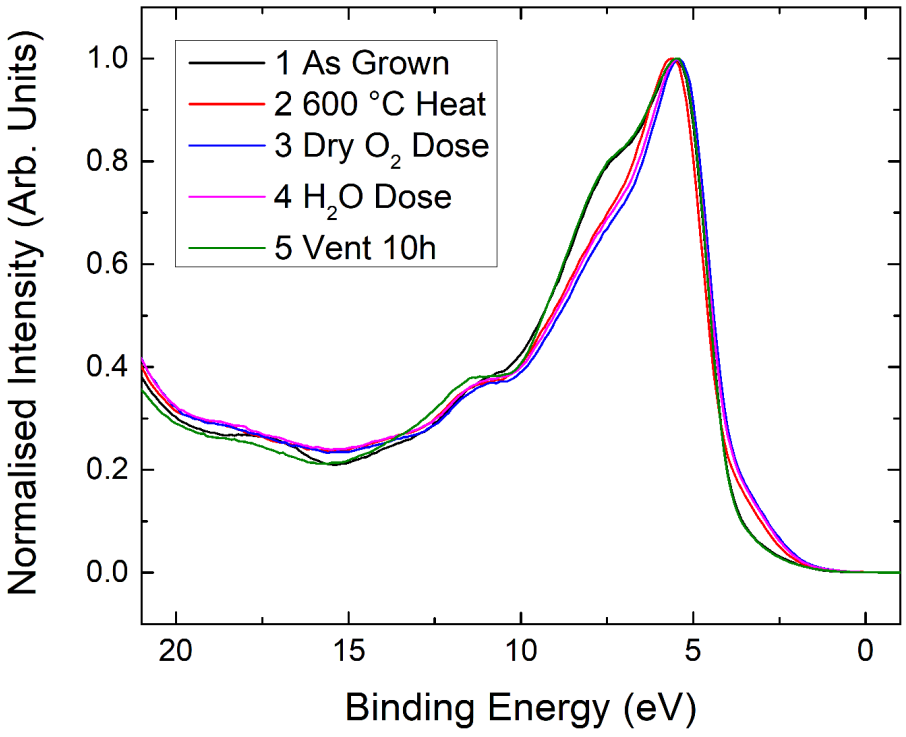
Following the O_2 and H_2O dosing, the sample was removed from the vacuum system and exposed to atmosphere for 10 hours to allow the surface hydroxyls and oxygen vacancies to recover. This resulted in a significant increase in the hydroxyl/lattice oxygen ratio which was larger than on the as-grown sample. The relative contribution from oxygen vacancies significantly decreased, however some contribution still remains when compared to the as-grown sample. Reheating the sample to 600 °C again causes a significant decrease in the hydroxyl contribution and an increase of oxygen vacancies. The wet O_2 dose produces an O 1s spectrum and component fits that are almost identical to that following the first H_2O dose, indicating that there is no difference between the separate and combined dosing of O_2 and H_2O on the undoped SnO_2 nanotubes. Exposing the undoped sample to a high pressure of deoxygenated water through the H_2O vent did produce an increase in the surface hydroxyl coverage alongside a slight decrease in the number of oxygen vacancies. This shows that H_2O is capable of increasing the concentration of surface hydroxyls on the SnO_2 surface, although a higher H_2O pressure is required for this to occur. Heating to 120 °C only has a minor effect on the surface hydroxyls and oxygen vacancies, with small decreases and increases, respectfully. The shortened vent to atmosphere for 20 minutes produces an increase in the surface hydroxyl concentration, with a minor decrease in oxygen vacancies. A larger relative increase in surface hydroxyls is observed when compared to the H_2O vent, although this can be explained as 20 °C air with a relative humidity of 20% has a vapour pressure of ~4.7 mBar, five times that of the maximum pressure observed during the H_2O vent. This higher water vapour pressure likely contributes to the recovery of more hydroxyls from the 20 minute vent to atmosphere compared to the 20 minute H_2O vent. Finally, the last 10 hour vent produced a significant increase of the surface hydroxyl groups, with a larger hydroxyl/lattice oxygen ratio than the as-grown sample and first 10 hour vent step. The proportion of oxygen vacancies also decreases, indicating the oxidation of the reduced and defected surface caused by the 600 °C heating. These

last two venting steps show that the processes of hydroxyl and oxygen vacancy recovery takes several hours to fully occur, with a 20 minute venting step insufficient to return the surface to a steady-state condition. Even after the 10 hour vent, the undoped SnO_2 surface is different when compared to the as-grown sample, with higher proportions of surface hydroxyls and oxygen vacancies. This suggests that more recovery processes could occur with longer atmospheric exposure, leading to a complete recovery time longer than 10 hours.

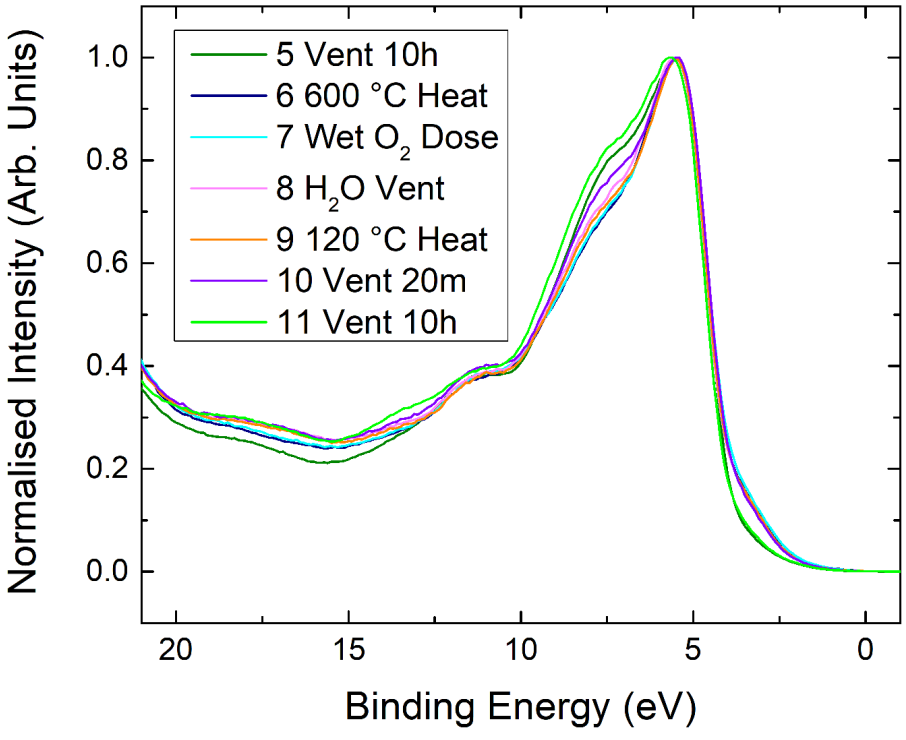
Valence Band Spectra

The valence band spectra of the undoped SnO_2 nanotubes throughout the heating/dosing series can be seen in Figure 6.3, with steps 1-5 and 5-11 shown separately for clarity in Figures 6.3.a and 6.3.b respectively. All valence band spectra have been normalised by the largest peak at ~6 eV, which is attributed to O 2p bonding within the SnO_2 lattice[348]. Throughout the series of heating and dosing only two significant changes in the valence band are observed. The first is the contribution at ~7.5 eV, which represents $pp\,\sigma$ hybridisations between Sn 5p and O 2p orbitals[348] and is affected by the removal of surface oxygen species such as bridging oxygen and surface hydroxyls. Firstly, the contribution at ~7.5 eV changes from the as-grown sample with a decrease in intensity following 600 °C heating. Neither O_2 or H_2O dosing produce significant changes in the contribution intensity, although venting to atmosphere for 10 hours causes the contribution to return to the same position as on the as-grown sample. A similar trend is observed for the second round of heating and dosing, which sees the contribution intensity drop following the second 600 °C heating and no change following the combined O_2 and H_2O dosing. The H_2O vent only produces a minor increase in the contribution intensity, with the 20 minute vent to atmosphere causing a larger recovery of the surface oxygen species. The longer 10 hour vent to atmosphere causes an increase of the contribution that is higher than the as-grown sample, likely due to the increased number of surface hydroxyls observed in the O 1s spectrum.

The second main change in the valence band structure is observed below the valence band edge at ~3 eV. This contribution arises from Sn 5s - O 2p bonding from Sn^{2+}, indicating the reduction of the SnO_2 surface to SnO[348, 349]. This feature



(a) VB, $h\nu$ = 150 eV



(b) VB, $h\nu$ = 150 eV

Figure 6.3: Valence band XPS spectra for the series of heating and O_2/H_2O dosing of undoped SnO_2 nanotubes. The series has been split across two subfigures for clarity, with (a) showing the first five spectra until the first 10 hour vent of the samples, and (b) showing the remaining spectra from the first 10 hour vent to the second 10 hour vent.

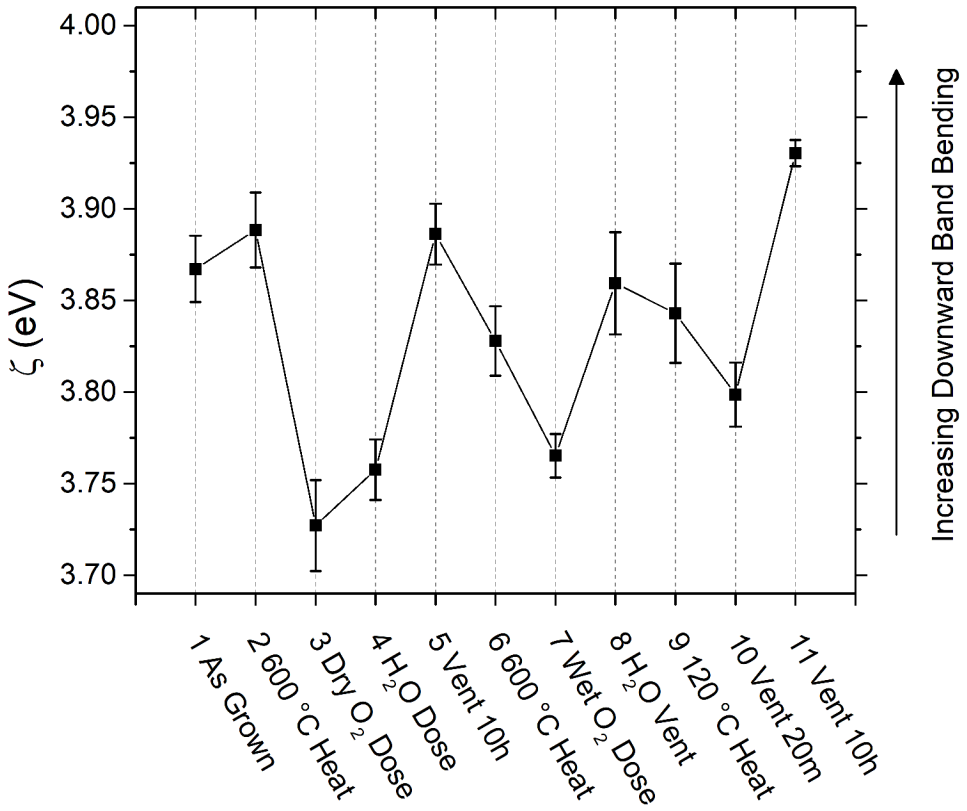


Figure 6.4: The value of ζ throughout the heating/dosing series of the undoped SnO_2 nanotubes, extracted from the valence band spectra using the method by Chambers et al.[284]

arises following both 600 °C heatings and remains present throughout the multiple rounds of O_2/H_2O dosing, only returning to the as-grown position following each 10 hour vent. This shows the number of Sn atoms that are converted from the (+4) to (+2) oxidation states by the 600 °C heating do not change throughout the dosing experiments, including the H_2O vent where the concentration of surface hydroxyls increase. Interestingly, even the 20 minute vent to atmosphere does not produce a change in this feature, showing a long exposure to atmospheric oxygen over the period of several hours is required to oxidise the SnO_2 surface.

To investigate the changes in surface band bending produced by the change in surface oxygen species, the energetic distance between the valence band maximum and Fermi level (ζ) was extracted from the valence band spectra using the method by Chambers et al.[284], which involves linearly extrapolating the edge of the lowest binding energy emission within the valence band to the instrument background. The values of ζ for each step in the heating/dosing series on the undoped SnO_2

nanotubes can be seen in Figure 6.4. The as-grown undoped sample has an initial ζ value of 3.87 eV, which is indicative of downward band bending and electron accumulation as this value is larger than the ~3.6 eV bandgap of SnO_2. Heating to 600 °C does not reduce the downward band bending, with the removal of surface hydroxyls and the production of in-plane oxygen vacancies resulting in a slight increase of ζ to 3.89 eV. The dry O_2 dosing of the SnO_2 nanotubes does reduce the downward band bending by 0.16 eV as ζ is reduced to 3.73 eV. Molecular O_2 is a known acceptor to SnO_2, and O_2 adsorption onto the nanotube surfaces over Sn_{5c} sites will deplete some of the surface charge. Furthermore, any repair of the oxygen vacancy defects will also remove charge carriers from the surface. This charge depletion produces the reduction of downward band bending and the shift of the valence band edge. H_2O dosing produces an only minor increase on the position of the valence band edge, with a slight increase possibly due to the adsorption of molecular H_2O or the displacement of adsorbed O_2. Venting to atmosphere allows the recovery of surface hydroxyls, and results in ζ increasing from 3.76 to 3.89 eV.

The 600 °C reheating of the undoped sample produces a reduction of the downward band bending by 0.06 eV, indicating that in this case the removal of surface hydroxyls has resulted in a net loss of surface charge despite the counteracting formation of oxygen vacancies. Wet O_2 dosing shifts the valence band to a near identical position to that following the first H_2O dose, reinforcing how there is little difference between separated and combined O_2/H_2O dosing. The H_2O vent results in an increase of ζ from 3.77 to 3.86 eV, likely by the increased formation of surface hydroxyl groups by the higher H_2O pressure. This is reinforced by the 120 °C heating, which would desorb molecularly adsorbed H_2O but not remove surface hydroxyls. Only a small decrease to ζ is observed, aligning with the O 1s spectra which shows only a small decrease in the hydroxyl contribution. Despite the relative increase in surface hydroxyl coverage, the 20 minute vent to atmosphere causes a slight reduction in the downward band bending, potentially due to oxygen chemisorption or the repair of in-plane and bridging oxygen vacancies which must occur before the surface can be fully hydroxylated. Finally, the last 10 hour vent to atmosphere produces strong

downward band bending, consistent with the repair of surface oxygen vacancies allowing a highly hydroxylated surface to form. The downward band bending in this last step is stronger than on the as-grown sample, which is consistent with the higher proportion of hydroxyls as seen in the O 1s spectra.

Despite the observed changes of the O 1s and valence band spectra throughout the heating/dosing series, it should be noted that the shifts in the position of the valence band edge are small compared to those observed in other metal oxides. ζ spans from 3.73 to 3.93 eV throughout this series, a range of only 0.2 eV. O-polar ZnO, on the other hand, has been shown to produce a significant ~0.7 eV shift from downward band bending to upwards band bending upon heating to 650 °C[350]. This change was strongly correlated to the removal of surface hydroxyl groups with no observed increase to oxygen vacancies. Similarly, $(\bar{2}01)$ β-Ga_2O_3 experiences an upward band bending shift of 0.7 eV after 600 °C heating[351]. Comparisons with these oxides illustrate the resilience of the downward band bending on the SnO_2 nanotubes throughout a range of heating and dosing treatments. That is not to say, however, that small reductions of surface band bending do not significantly affect the surface charge density. Earlier studies on the SnO_2 nanotubes showed that a strengthening of the downward band bending by 0.14 eV increased the surface charge density by 60%[30]. Reductions of the downward band bending on InN surfaces by 0.15 eV equated to a reduction of the surface charge density by 30%[231]. These results show that the small changes observed in the surface band bending on the undoped SnO_2 nanotubes can still markedly affect the surface charge density, affecting the surface conductivity and environmental sensitivity of the nanotubes.

6.2.2 O 1s and Valence Band Spectra of 0.15% Sb-doped SnO_2 Nanotubes Throughout the Heating/Dosing Series

O 1s Core-level Spectra

The O 1s spectra of the 0.15% Sb-doped SnO_2 nanotubes throughout the heating/dosing series can be seen in Figure 6.5, with the relative ratios of surface hydroxyls and oxygen vacancies compared to lattice oxygen seen in Figure 6.6.

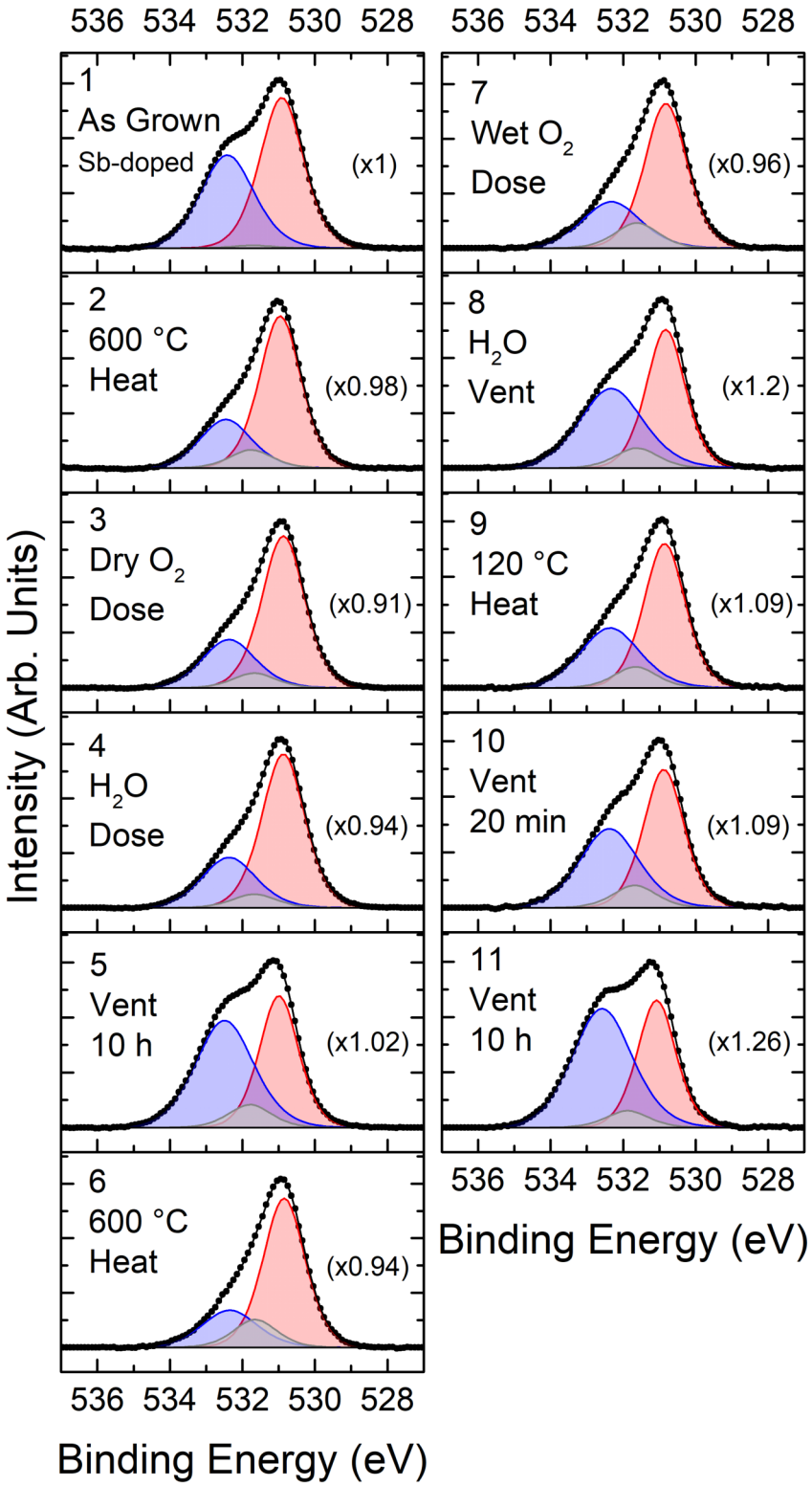


Figure 6.5: O 1s core level XPS spectra for the series of heating and O_2/water dosing of 0.15% Sb-doped SnO_2 nanotubes. Three fitted components are attributed to lattice O (red), surface hydroxyl groups (blue) and O vacancies (grey). All spectra were captured with a photon energy of $h\nu$ = 680 eV.

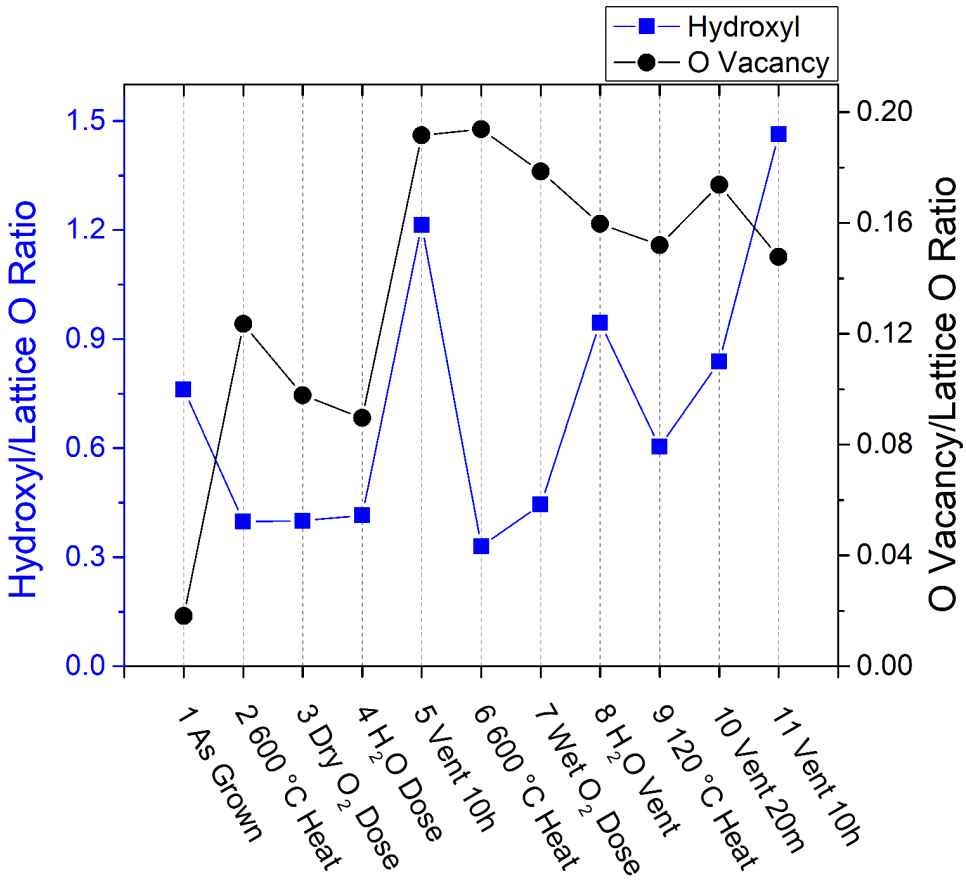


Figure 6.6: Area ratios of the surface hydroxyl and O vacancy peaks compared to the lattice O peak from the O 1s spectra of 0.15% Sb-doped SnO_2 nanotubes for each heating/dosing treatment.

In general, similar trends to the undoped nanotubes are observed in the surface hydroxyl coverage, with a decrease in the proportion of surface hydroxyls following 600 °C heating. This hydroxyl/lattice oxygen ratio of ~0.4 is slightly higher than on the undoped sample which has a ratio of ~0.3. Like on the undoped sample, hydroxyl coverages do not change following separated or combined O_2/H_2O dosing. The H_2O vent does, however, produce a significant increase in the surface hydroxylation which reduces following 120 °C heating. All the steps involving venting to atmosphere produce increasing hydroxyl coverages, with the final 10 hour vent resulting in a ratio of ~1.5, similar to the undoped sample.

The contribution of oxygen vacancies in the Sb-doped sample appears to be less significant than on the undoped sample. Following the first 600 °C heating the oxygen vacancy/lattice oxygen ratio increases only to 0.12, about half that of the 0.25 observed on the undoped sample. This value slightly decreases following the O_2

and H_2O dosing. Unexpectedly, the fitting to the first 10 hour vent indicates an increased contribution from oxygen vacancies. This opposes what occurs in the undoped sample and the conventional theory, where it is expected that the oxygen vacancy contribution would decrease due to the repair of vacancies in an environment with high oxygen pressure. The remaining heating, dosing and venting steps do not significantly change the oxygen vacancy/lattice oxygen ratio, with this value decreasing to only ~0.15 following the final 10 hour vent.

There are several limitations of the fitting and analysis techniques that may contribute to the unexpected behaviour of the oxygen vacancies throughout the heating/dosing series on the Sb-doped nanotubes. Firstly, it should be noted that the relative concentration of oxygen vacancies on the Sb-doped sample is much lower than on the undoped sample, increasing the relative potential error within the XPS fitting. Secondly, the peak attributed for oxygen vacancies sits in the centre of a generally broad spectra, which raises the potential for error within the XPS fitting. Thirdly, this analysis compares the peak ratio of oxygen vacancies to lattice oxygen as the lattice oxygen peak remains largely unchanged throughout this series. However, the lower binding energy tail of the hydroxyl peak overlaps with the lattice oxygen peak, and results in a reduced size of the lattice oxygen peak in samples with a high hydroxyl contribution. This affects the oxygen vacancy/lattice oxygen ratio, increasing the relative contributions from oxygen vacancies. These different sources all may contribute to the unexpected behaviour of the relative oxygen vacancy contribution through the heating/dosing series.

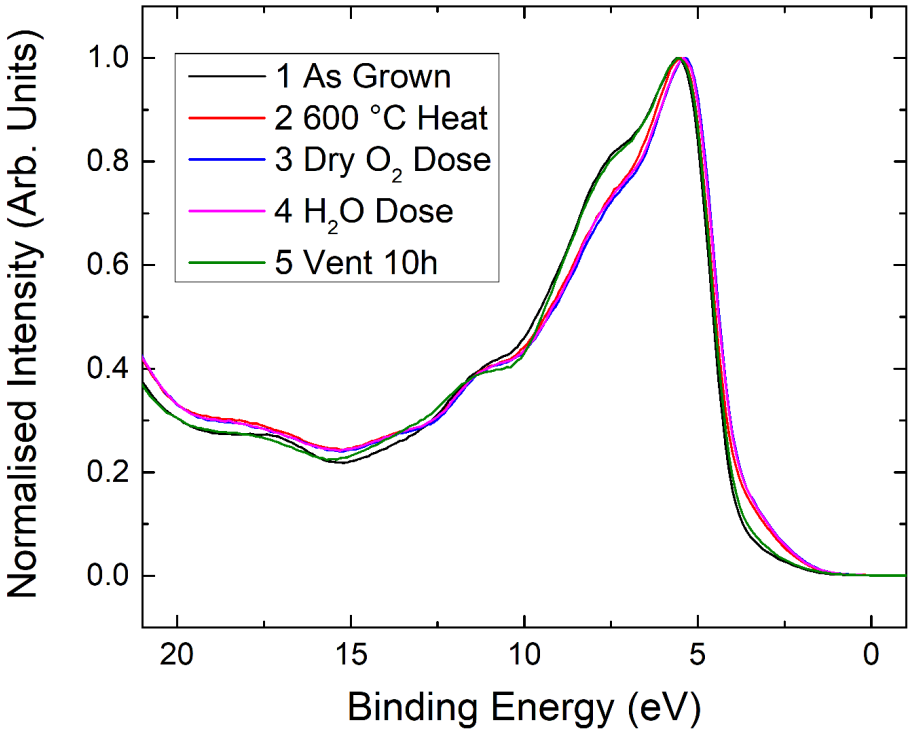
Despite the limitations of the oxygen vacancy peak fitting in the O 1s spectra of the Sb-doped nanotubes, several inferences can still be drawn from the data. Firstly, the lower proportion of oxygen vacancies in the Sb-doped nanotubes compared to the undoped nanotubes suggests that oxygen vacancies exhibit a higher formation energy in Sb-doped SnO_2. This is supported by Kilic and Zunger, who showed that the defect formation energy of oxygen vacancies increases as the Fermi level approaches the conduction band minimum[59]. The Sb-doping of the SnO_2 nanotubes produces a Fermi level that is closer to the conduction band minimum, therefore causing an increase in the oxygen vacancy defect formation energy. The formation of fewer

oxygen vacancies is also supported by the relative changes in the hydroxyl contribution. The hydroxyl/lattice oxygen ratio following the H_2O vent on the undoped nanotubes increased from ~0.2 to ~0.5, while the Sb-doped nanotubes exhibited a larger increase from ~0.4 to ~0.9. Several computational studies have shown that the formation of surface hydroxyls requires surface bridging oxygen to support water dissociation[95, 96]. As the Sb-doped sample is able to form more surface hydroxyls following the high pressure water vent, this suggests the surface is less reduced with fewer oxygen vacancies allowing more water to dissociate into surface hydroxyls.

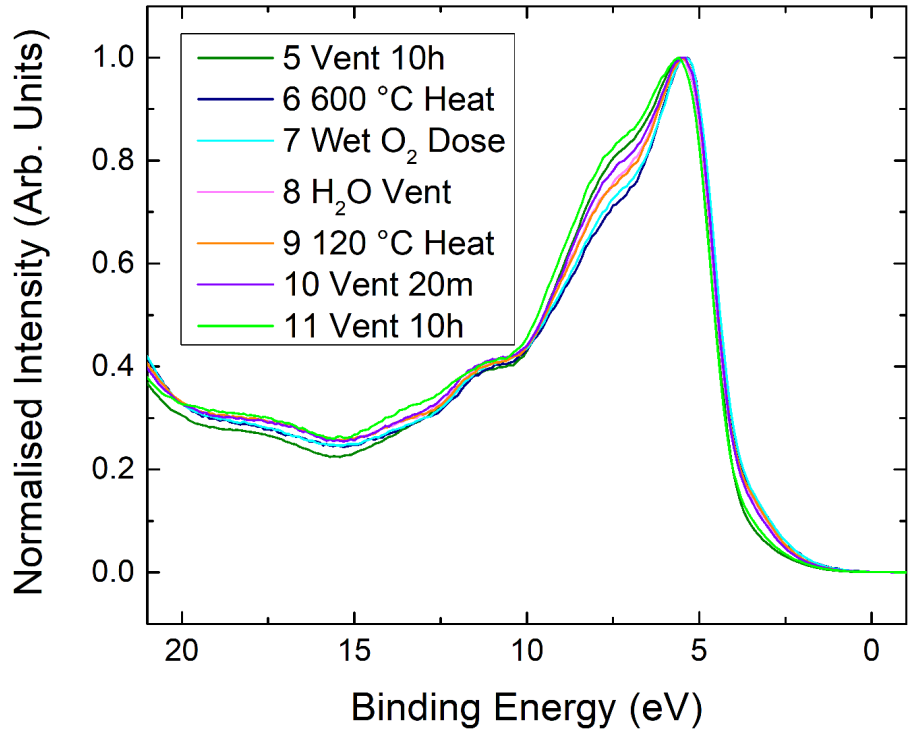
Valence Band Spectra

The valence band spectra of the 0.15% Sb-doped SnO_2 nanotubes throughout the heating/dosing series can be seen in Figure 6.7, with steps 1-5 and 5-11 shown separately for clarity in Figures 6.7.a and 6.7.b respectively. Similar to the undoped nanotubes, only two significant features change throughout the series: The Sn 5p - O 2p related bonding feature at ~7.5 eV which includes surface hydroxyl contributions, and the Sn 5s - O 2p bonding feature at ~3 eV that represents a reduction of the SnO_2 surface. As on the undoped nanotubes, heating causes a reduction of the feature intensity at ~7.5 eV as surface hydroxyls are removed, while the ~3 eV feature increases with the surface reduction. The separated and combined O_2/H_2O dosing has little effect on either of these two features, with the features only returning to the as-grown position following a 10 hour vent to atmosphere. The H_2O vent, however, does produce an increase of the feature at ~7.5 eV which is consistent with the larger observed increase of surface hydroxyls as seen in the O 1s spectra.

The values of ζ were also extracted from the valence band edge for each step in the heating/dosing series and can be seen in Figure 6.8. Following the first 600 °C heating the value of ζ decreases by 0.12 eV. This is different to what occurs on the undoped nanotubes in which there is no large change in ζ following the first heating step. This is due to the lower number of oxygen vacancies which are formed on the Sb-doped nanotubes, which provide less charge compensation when the electron donating surface hydroxyls are removed. This results in a lower surface charge density and weakened downward band bending. As on the undoped nanotubes, O_2 dosing results in a lower ζ due to oxygen adsorption and the potential repair of some



(a) VB, $h\nu$ = 150 eV



(b) VB, $h\nu$ = 150 eV

Figure 6.7: Valence band XPS spectra for the series of heating and O_2/water dosing of 0.15% Sb-doped SnO_2 nanotubes. The series has been split across two subfigures for clarity, with (a) showing the first five spectra until the first 10 hour vent of the samples, and (b) showing the remaining spectra from the first 10 hour vent to the second 10 hour vent.

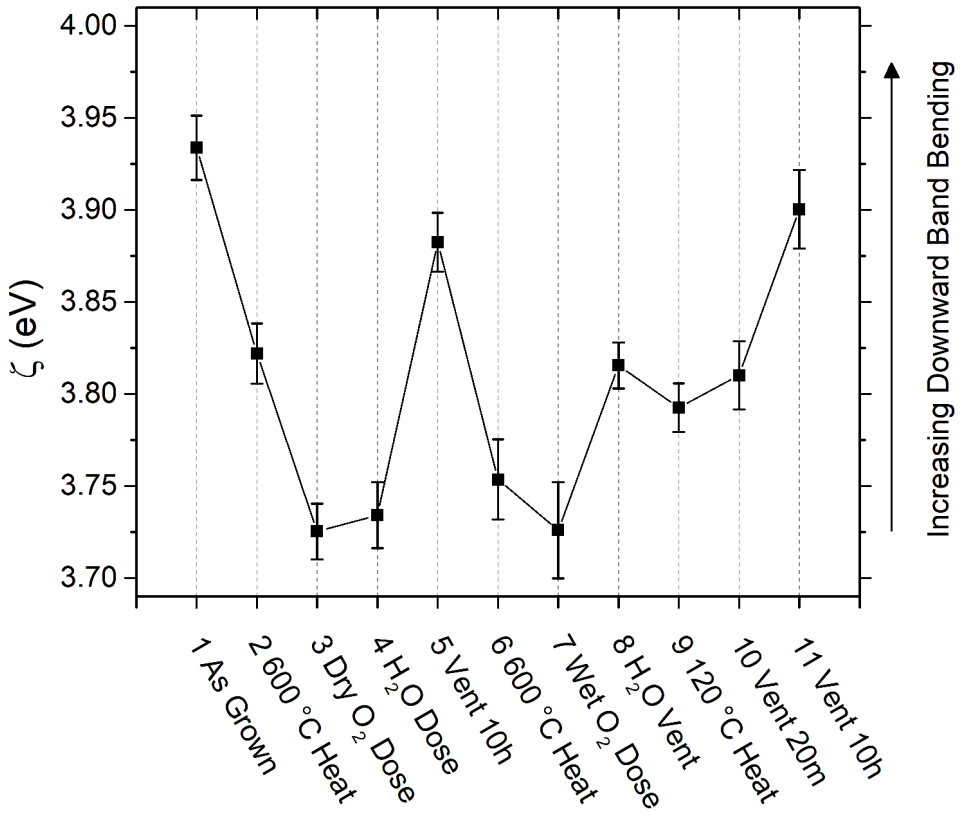


Figure 6.8: The value of ζ throughout the heating/dosing series of the 0.15% Sb-doped SnO_2 nanotubes, extracted from the valence band spectra using the method by Chambers et al.[284]

oxygen vacancies. The small H_2O dose has little effect on the surface band bending, however, the H_2O vent produces a significant increase of ζ due to the large recovery of surface hydroxyls. Similarly, both 10 hour vents to atmosphere cause a strengthening of the downward band bending due to the recovery of surface hydroxyls.

6.3 Sulfur Treatment of the SnO_2 Nanotubes

The previous sulfur treatments of various metal oxides produced promising results of significant surface band bending shifts, particularly in the case of ZnO. To further understand the effect of sulfur treatments on the SnO_2 nanotubes, two different wet chemical sulfur treatments were performed on both undoped and 0.15% Sb-doped nanotube samples. The first treatment immersed the samples in a 0.8 M Na_2S solution with a 1:1 EtOH:H_2O solvent at 40°C for 20 minutes. The second treatment involved the immersion of the samples in a solution of 0.8 M $(NH_4)_2S$ in 1:1 IPA:H_2O at 40°C for 60 minutes. Despite the identical S^{2-} ion concentration between the two solutions, each solution has a different pH. The Na_2S solution is highly alkaline, as the free S^{2-} ion strips hydrogen off water producing HS^- and OH^-. A similar process occurs within the $(NH_4)_2S$ solution, except the NH_4^+ ion acts as a pH buffer and results in a solution that is less alkaline. To investigate the effect of each solution on the chemical and electronic properties of the SnO_2 nanotubes, the treated samples were analysed by SEM, XPS and NEXAFS.

6.3.1 Scanning Electron Microscopy of the Sulfur Treated Nanotubes

The sulfur-treated SnO_2 nanotubes were imaged by SEM to determine the degree of visible sulfur coverage across the nanotubes. Figure 6.9 shows undoped and 0.15% Sb-doped SnO_2 nanotubes that have undergone sulfur treatments compared to untreated references.

$(NH_4)_2S$ treated nanotubes (Figures 6.9.b and 6.9.e) show minimal differences compared to the reference samples, aside from small particles occasionaly observed on the nanotube surface. The Na_2S treated nanotubes (Figures 6.9.c and 6.9.f) show a larger amount of sulfur particles on the nanotube surfaces, although coverage remains relatively poor, with sparsely spaced patches of sulfur compounds on both the undoped and 0.15% Sb-doped nanotubes. Large areas of both $(NH_4)_2S$ and Na_2S treated nanotubes appear to have no obvious modification, although it is difficult to determine if there is a monolayer coverage of sulfur species by SEM.

During SEM imaging of the Na_2S treated nanotubes, some large particles of sulfur

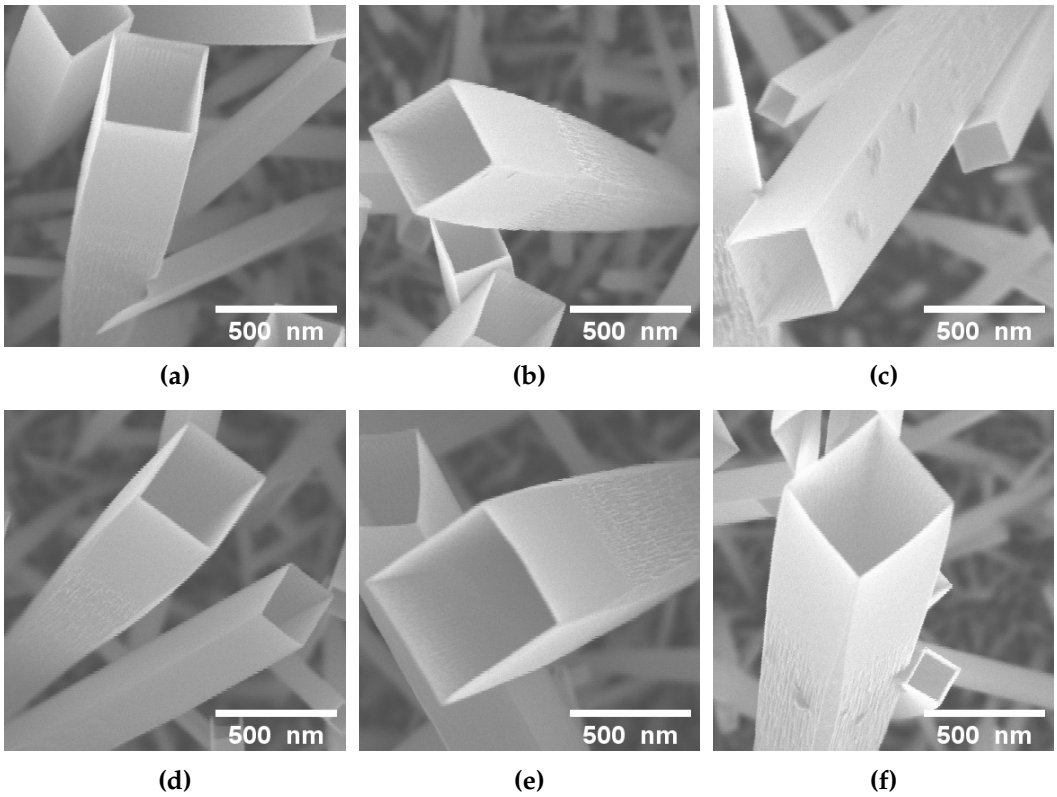


Figure 6.9: Scanning electron microscopy images of (a-c) undoped and (d-f) 0.15% Sb-doped SnO_2 nanotubes that have undergone sulfur treatments. (a,d) Untreated reference, (b,e) 1 hour $(NH_4)_2S$ treated and (c,f) 20 minute Na_2S treated.

species up to ~500 nm in length were observed. Furthermore, these large particles appeared to be sensitive to the electron beam, with decomposition occurring under extended electron beam exposure. Figure 6.10 shows a sulfur particle attached to an SnO_2 nanotube under increasing electron beam exposure. The size and shape of the sulfur particle changes dramatically, with the particle decreasing in size and becoming rounder as it decomposes. In addition, a region of dark contrast can be seen spreading across the nanotube surface, increasing the surface roughness. This area of contrast is most likely to be sulfur species spreading across the nanotube surface. Secondary electron emission intensity is primarily related to the atomic number of the elements present in the material, with increasing emission from higher atomic number elements. As the sulfur species migrate and cover the SnO_2 surface, the Sn atoms with atomic number 50 are masked by the surface S with atomic number 16. This prevents higher intensity secondary electron emission from the Sn atoms, resulting in a decrease in measured brightness under secondary electron imaging.

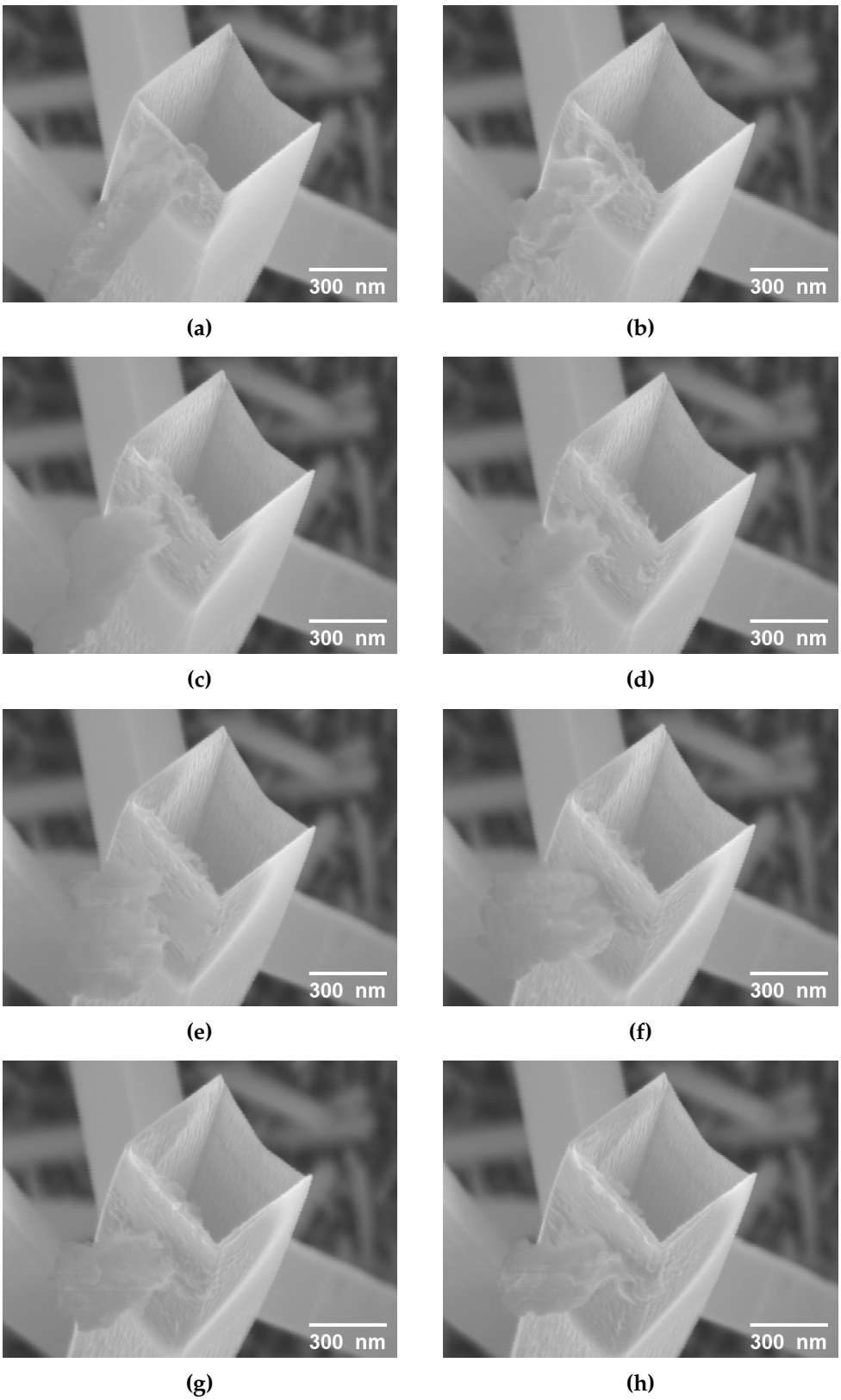


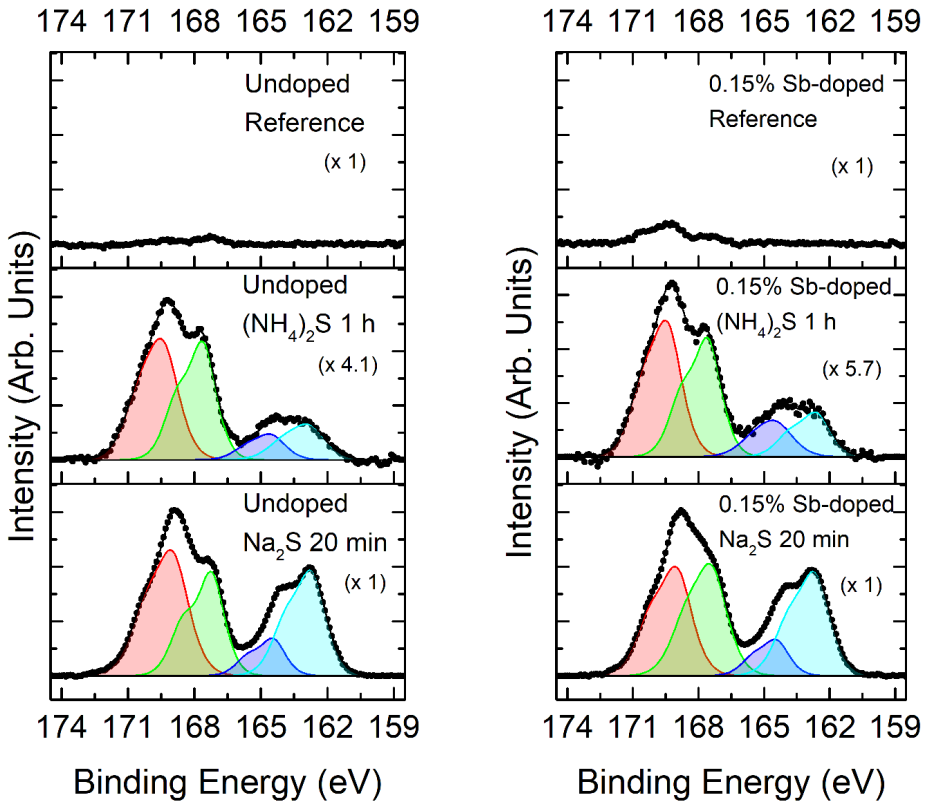
Figure 6.10: (a)-(h) Scanning electron microscopy images of an Na_2S treated undoped SnO_2 nanotube, showing the spread of sulfur species across the nanotube surface under increasing electron-beam exposure.

6.3.2 XPS of Sulfur Treated Nanotubes

The changes in surface chemistry and electronic band bending of the sulfur-treated SnO_2 nanotubes were investigated by core-level and valence band synchrotron XPS. The following sections show and analyse the S 2p, O1s, Sn3d and valence band scans of reference, 1 hour $(NH_4)_2S$ treated and 20 minute Na_2S treated undoped and 0.15% Sb-doped SnO_2 nanotubes.

S 2p

The S 2p spectra of the sulfur-treated SnO_2 nanotubes can be seen in Figure 6.11, with fitting parameters shown in Table 6.1. Four fitted doublets of S $2p_{3/2}$ and S $2p_{1/2}$ were required to achieve good fitting envelopes, attributable to four sulfur environments. Each doublet was constrained with an orbital splitting of 1.2 eV, a S $2p_{3/2}$:S $2p_{1/2}$ intensity ratio of 2:1 and a fixed FWHM between the two orbitals.



(a) S 2p, $h\upsilon$ = 320 eV **(b)** S 2p, $h\upsilon$ = 320 eV

Figure 6.11: S 2p core level XPS spectra for the reference, 1 hour $(NH_4)_2S$ treated and 20 minutes Na_2S treated (a) undoped and (b) 0.15% Sb-doped SnO_2 nanotubes. The spectra have been fit with four doublets attributed to, from left to right, surface SO_4^{2-} (red), SO_3^{2-}, bridging polysulfide (blue) and terminal polysulfide (cyan) species.

Table 6.1: S $2p_{3/2}$ XPS fitting parameters of the sulfur-treated SnO_2 nanotubes.

S $2p_{3/2}$							
		Undoped			0.15% Sb-doped		
Treatment	Component	Position (eV)	FWHM (eV)	Area (%)	Position (eV)	FWHM (eV)	Area (%)
$(NH_4)_2S$	SO_4^{2-}	169.40	1.64	41.6	169.40	1.55	40.5
	SO_3^{2-}	167.59	1.39	37.0	167.55	1.43	34.3
	Bridging S_x^{2-}	164.52	1.52	8.3	164.41	1.80	11.6
	Terminal S_x^{2-}	162.8	1.74	13.1	162.58	1.53	13.5
Na_2S	SO_4^{2-}	168.96	1.70	36.8	168.97	1.62	30.7
	SO_3^{2-}	167.22	1.37	26.2	167.38	1.67	31.5
	Bridging S_x^{2-}	164.44	1.30	8.8	164.42	1.36	8.6
	Terminal S_x^{2-}	162.68	1.51	28.2	162.63	1.61	29.2

From the XPS data in Figure 6.11 it is clear that both sulfur treatments have increased the amount of sulfur species present on the nanotube surface compared to the untreated references. However, the Na_2S treatment resulted in a peak intensity 5.7× larger than the $(NH_4)_2S$ treatment, despite the same concentration of S^{2-} in each of the precursor solutions. The positions of all four fitted doublets occur in similar locations within each spectrum (Table 6.1), indicating that each treatment has produced the same types of sulfur environments. These four environments have been attributed to three different sulfur species: Sulfates (SO_4^{2-}) at ~169 eV, sulfites (SO_3^{2-}) at ~167.4 eV and polysulfides (S_x^{2-}) at ~164.4 eV and ~162.6 eV. The two environments associated with polysulfides are due to different oxidation states of the bridging (0) and terminating (1-) sulfur atoms in polysulfide chains.

Looking at the proportion of each sulfur species in Table 6.1, it is observed that virtually all the sulfur present on the nanotube surfaces has been oxidised to some degree from the original S^{2-} present in the starting solutions. The majority of the sulfur has been oxidised to either SO_4^{2-} or SO_3^{2-}, with these contributions accounting for 62.2-78.6% of the total sulfur signal across all four treated samples. There are, however, differences in the formation of polysulfides between the $(NH_4)_2S$ and Na_2S

treatments with the Na_2S treated samples exhibiting 60% more relative polysulfide contributions than the $(NH_4)_2S$ treated samples.

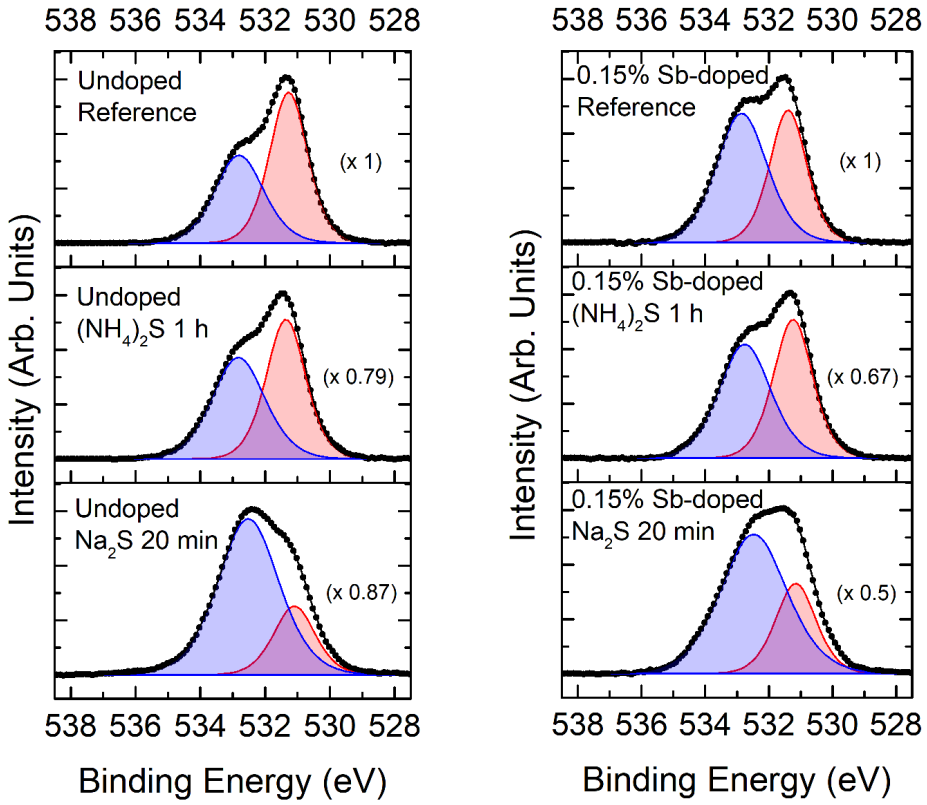
Several other sulfur species were considered when assigning sulfur species to the S 2p doublets. No formation of SnS or SnS_2 was observed in the S 2p spectra as these species occur at lower binding energies of ~161.1 eV[352] and ~161.6 eV[353], respectively. Thiosulfate ($S_2O_3^{2-}$) is a species that could be contained within the S 2p spectrum as the two oxidation states of sulfur present in the compound produce S $2p_{3/2}$ binding energies of ~168.6 and ~162.7 eV[354]. These binding energies lie within the peak areas of the sulfur treated nanotubes S 2p spectra, which could obscure their contribution. Thiosulfate was eventually discounted as a potential contributor as no thiosulfate peaks were observed during NEXAFS studies, presented in section 6.3.4. Elemental sulfur was the last considered sulfur species as this appears in the range of 163.6-164.0 eV[355], which is within the range of peaks of the lower binding energy species. However, elemental sulfur evaporates in vacuum at temperatures above 295 K[355], leading to the expectation that no elemental sulfur would be present on the nanotube surface when investigated by XPS.

O 1s

The O 1s spectra of the sulfur-treated SnO_2 nanotubes can be seen in Figure 6.12, with fitting parameters shown in Table 6.2. All spectra show broad peaks with well defined shoulders, allowing two fitted peaks to give a sufficient fit to the raw data. These two peaks were attributed to lattice O and surface O, with the surface O component containing primarily surface OH and SO_x groups.

The $(NH_4)_2S$ treated nanotubes show only minor differences when compared to the untreated references. The position and FWHM of the fitted peaks is similar across all four samples, with peak splitting of ~1.5 eV. The intensity of the O 1s spectra has increased in both the undoped and 0.15% Sb-doped samples following the $(NH_4)_2S$ treatment, however, the area proportion of the two fitted peaks remains similar. This suggests there have been only minor changes to the O1s chemical species and the increase in intensity is attributable to the removal of contaminants such as adventitious carbon during the $(NH_4)_2S$ treatment. The surface O component of these four spectra are likely dominated by surface OH groups, with minimal

contributions from SO_4^{2-} and SO_3^{2-} due to the low intensity observed in the S 2p spectra.



(a) O 1s, $h\nu$ = 680 eV **(b)** O 1s, $h\nu$ = 680 eV

Figure 6.12: O 1s core level XPS spectra for the reference, 1 hour $(NH_4)_2S$ treated and 20 minutes Na_2S treated (a) undoped and (b) 0.15% Sb-doped SnO_2 nanotubes. The red component is attributed to lattice O while the blue component is attributed to surface O species such as OH and SO_x.

The Na_2S treatment has produced a significantly different O 1s environment, with a much larger contribution from surface O species. Alongside an overall increase in signal intensity, the area contribution of surface O species has increased from 42-57% to 77-71%. This coincides with a significant increase in the surface O FWHM to 2.22-2.46 eV, a general shift to lower binding energies in the fitted peaks and a decrease in the peak splitting to 1.3-1.4 eV. These changes indicate that different surface O groups are contributing to the Na_2S treated samples O 1s spectra. These contributions likely come from SO_4^{2-} and SO_3^{2-} groups present on the nanotube surfaces. Wahlqvist and Shchukarev reported that the O emission peaks within SO_4^{2-} are found at binding energies between 531-532.5 eV, with Na_2SO_4 occurring at 531.9 eV[356]. Attempts were made to fit a third peak around this binding energy while

Table 6.2: O 1s XPS fit parameters of the sulfur-treated SnO_2 nanotubes.

O1s							
		Undoped			0.15% Sb-doped		
Treatment	Component	Position (eV)	FWHM (eV)	Area (%)	Position (eV)	FWHM (eV)	Area (%)
Reference	Lattice O	531.27	1.39	57.2	531.39	1.36	42.4
	Surface O	532.80	1.79	42.8	532.85	1.9	57.6
$(NH_4)_2S$	Lattice O	531.36	1.43	50.4	531.24	1.45	47.8
	Surface O	532.83	1.93	49.6	532.76	1.92	52.2
Na_2S	Lattice O	531.1	1.5	23.0	531.15	1.5	28.3
	Surface O	532.52	2.22	77.0	532.48	2.46	71.7

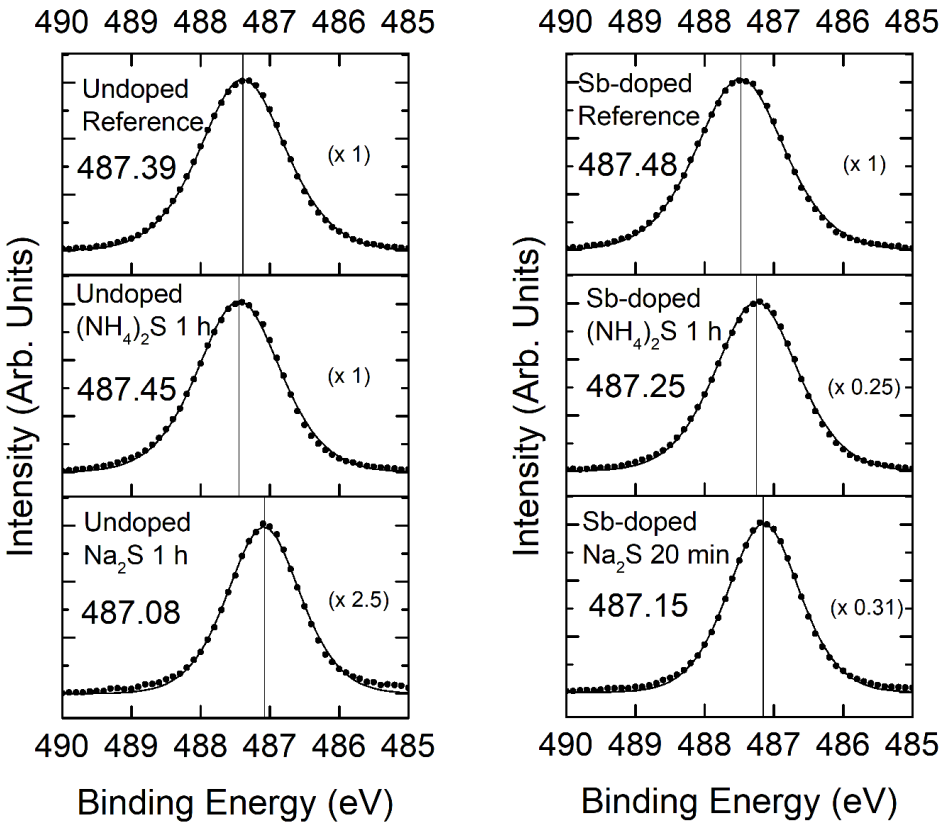
constraining the lattice O - OH splitting to 1.5 eV, however this resulted in generally poor fits or an intense minimisation of one of the three peaks, owing to the ambiguity of fitting multiple peaks under a broad spectrum. An additional complication is that the Na_2S solution is highly basic, which leads to the expectation that the amount of surface OH would increase, as observed in other metal oxides[351, 357, 358]. This means that the change in the size of the surface O peak cannot be solely attributed to SO_4^{2-} and SO_3^{2-}, and the relative increases of OH and SO_x cannot be separately determined.

Sn $3d_{5/2}$

Figure 6.13 shows the XPS data and fits to the Sn $3d_{5/2}$ peak, with Table 6.3 showing the fitting parameters and relative change in the surface band bending as determined by the shift in the Sn $3d_{5/2}$ position. Both sulfur-treated 0.15% Sb-doped samples show an increase in intensity of the Sn $3d_{5/2}$ signal, which is attributed to the removal of adventitious carbon, as similar to the O 1s spectra. The undoped $(NH_4)_2S$ treated sample shows no attenuation in the signal intensity, with little change in the peak position and FWHM when compared to the undoped reference. The 0.15% Sb-doped $(NH_4)_2S$ treated sample shows a larger shift to lower binding energy, indicating that the $(NH_4)_2S$ treatment has affected the surface Sn environment.

The Na_2S treated samples show a more convincing shift to lower binding energy, with ΔV_{bb} values of +0.31 and +0.33 eV for the undoped and 0.15% Sb-doped samples compared to their untreated counterparts. Both samples also show a reduction

in their FWHM, indicating more changes in the Sn local environment. The undoped Na_2S treated sample shows more signal attenuation compared to the untreated and $(NH_4)_2S$ treated undoped samples, which is likely due to the higher coverage of sulfur species alongside Na^+ ions present on the nanotube surfaces.



(a) Sn 3d, hv = 635 eV **(b)** Sn 3d, hv = 635 eV

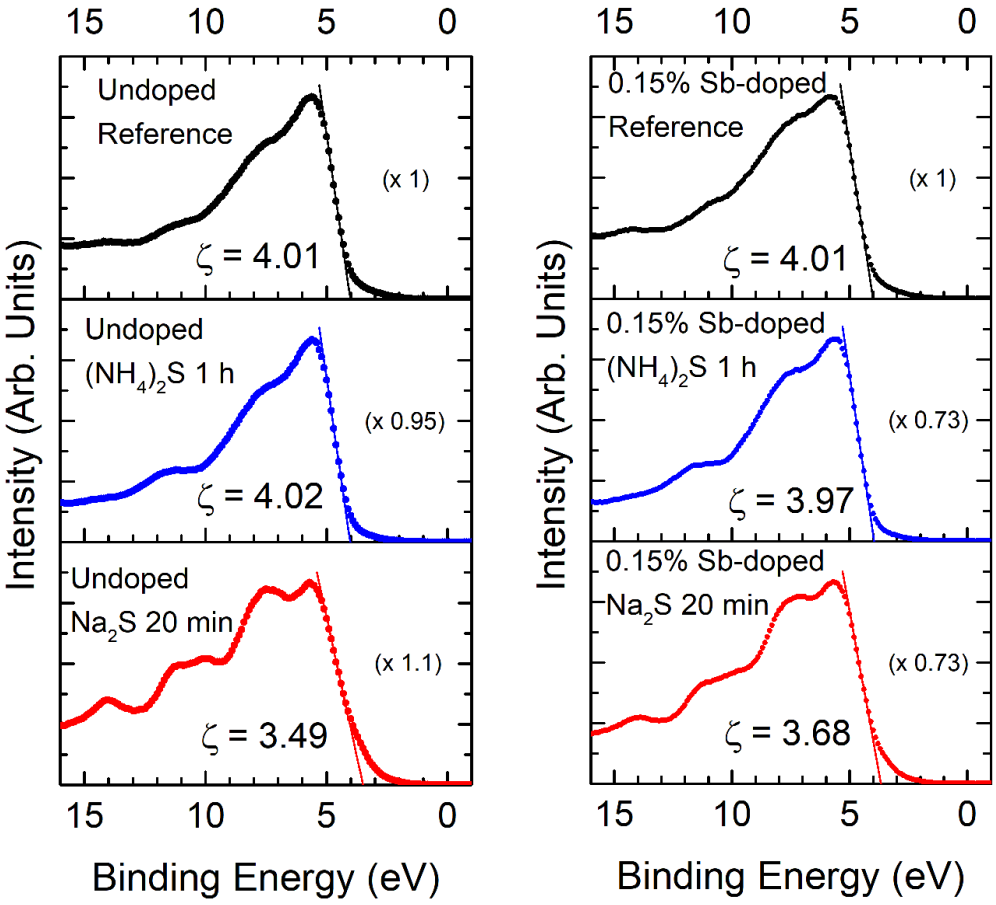
Figure 6.13: Sn $3d_{5/2}$ XPS spectra for the reference, 1 hour $(NH_4)_2S$ treated and 20 minutes Na_2S treated (a) undoped and (b) 0.15% Sb-doped SnO_2 nanotubes. The vertical lines show the centre position of the fitted spectrum in eV.

Table 6.3: Sn $3d_{5/2}$ XPS fit parameters of the sulfur-treated SnO_2 nanotubes.

		Undoped			0.15% Sb-doped		
Treatment	Component	Position (eV)	FWHM (eV)	ΔV_{bb} (eV)	Position (eV)	FWHM (eV)	ΔV_{bb} (eV)
Reference	Sn $3d_{5/2}$	487.39	1.45	-	487.48	1.48	-
$(NH_4)_2S$	Sn $3d_{5/2}$	487.45	1.40	-0.06	487.25	1.39	+0.23
Na_2S	Sn $3d_{5/2}$	487.08	1.24	+0.31	487.15	1.22	+0.33

Valence Band

Figure 6.14 shows the valence band spectra for the reference and sulfur-treated SnO_2 nanotubes. The solid, straight lines show the linear fit to the valence band edge used to extract the value of ζ. Both the undoped and 0.15% Sb-doped $(NH_4)_2S$ treated samples show few differences to the reference samples, with a slight accentuation of the peak at ~11.5 eV. There is no change in the value of ζ for these samples, indicating that the $(NH_4)_2S$ treatment has not affected the surface band bending of the nanotubes.



(a) VB, $h\nu$ = 150 eV **(b)** VB, $h\nu$ = 150 eV

Figure 6.14: Valence band XPS spectra for the reference, 1 hour $(NH_4)_2S$ treated and 20 minutes Na_2S treated (a) undoped and (b) 0.15% Sb-doped SnO_2 nanotubes. The straight, solid line shows the linear fit to the valence band edge used to extract the value of ζ

The Na_2S treated samples show multiple differences compared to the reference samples, with increases of the peaks at ~7.5 and ~11.5 eV, and new peaks appearing at ~10 and ~14 eV which are more pronounced on the undoped sample. The peak

increasing at ~7.5 eV has been shown to shown to follow the coverage of OH groups on the surface of the SnO_2 nanotubes, with decreasing intensity under heating at higher temperatures as OH groups are removed[29, 30]. This supports the theory presented with the O 1s spectra that the basic solution of Na_2S has increased the OH coverage on the nanotube surfaces. The peaks appearing at ~10 and ~14 eV are attributed to SO_4^{2-}, as these are the positions of the two largest peaks reported in valence band spectra of SO_4^{2-} compounds[359].

The values of ζ have changed considerably for the Na_2S treated samples compared to the untreated references. On the undoped Na_2S treated sample, the value of ζ has decreased from 4.01 eV to 3.49 eV, equivalent to a ΔV_{bb} of +0.52 eV. A smaller shift is observed on the 0.15% Sb-doped sample, with a shift to 3.68 eV observed, equivalent to a ΔV_{bb} of +0.33 eV. These finding show that, under standard XPS investigation conditions, the Na_2S treatment is capable of producing band bending shifts that oppose the strong downward band bending present on the untreated nanotube surfaces.

Two key issues, however, present themselves with these findings. Firstly, the position of the highest intensity peak of the valence band at 5.7 eV does not shift with the change in ζ, and instead, the change is caused by the shallowing of the valence band edge. This implies a non-homogeneous coverage of sulfur species across the nanotube surfaces, with much of the nanotube surface remaining untreated. A smaller signal contribution from treated areas would produce the shallowing of the valence band edge while the majority untreated areas would maintain the positions of the main valence band peaks. This uneven coverage was observed during SEM imaging of the treated nanotubes, presented in Section 6.3.1, with much of the nanotube surfaces appearing untreated with patchy areas of visible sulfur species.

The second issue exists with the methodology used to gather the XPS data. The standard XPS conditions involve capturing all core levels and valence band data from the same spot on the sample to ensure the same species are present within each measurement. However, Dr Liam Carroll observed that the core level and valence band measurements of sulfur-treated samples changed over extended exposures to the high-flux X-ray beam[226]. To account for changes over the time exposed to the

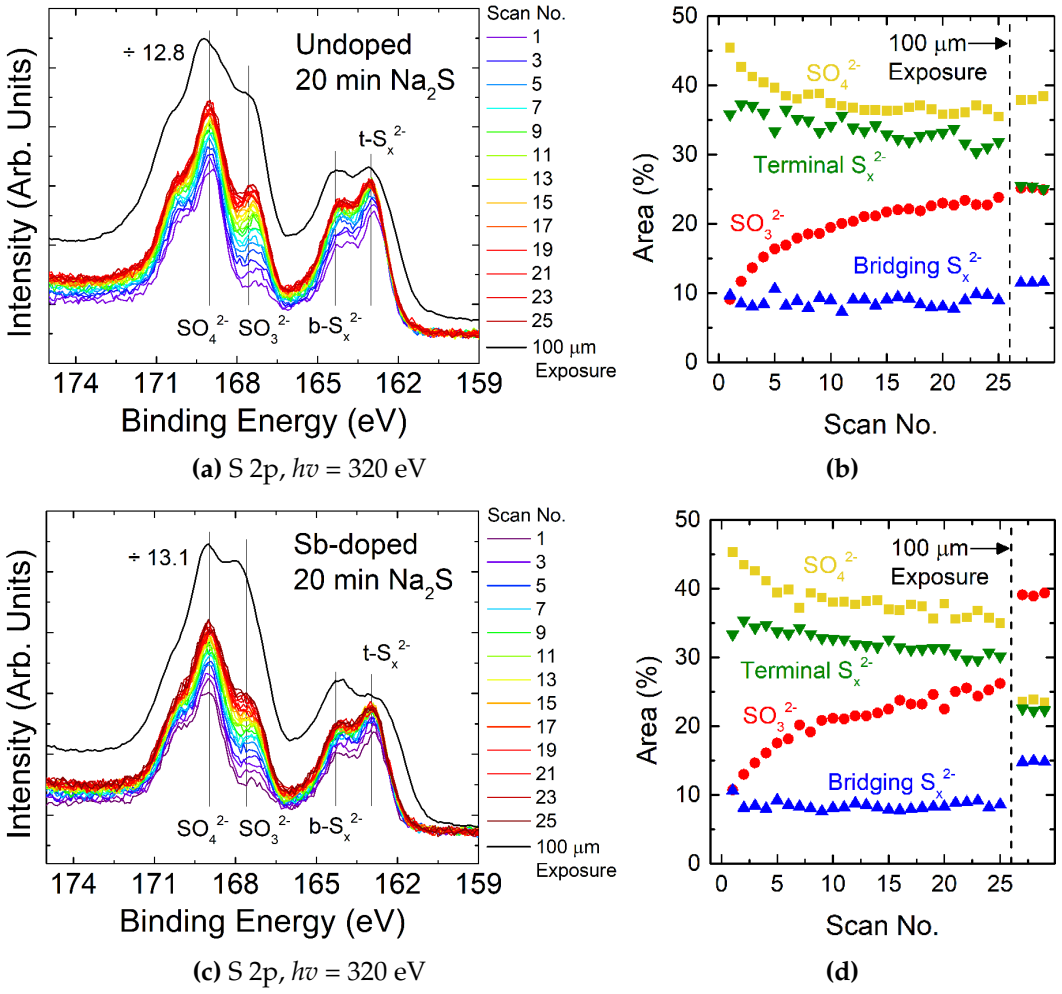
X-ray beam, additional XPS measurements were undertaken on the Na_2S treated samples, with careful considerations for previous X-ray exposures.

6.3.3 The Effect of Beam Exposure to Na_2S Treated Samples During XPS

It was established at the end of Section 6.3.2 that additional measurements were required to understand the effect of the X-ray beam on the surface sulfur species and valence band edge. S 2p and valence band XPS measurements were performed on previously unexposed areas of the Na_2S treated samples to observe changes in these spectra caused by the incident X-rays and photoelectron-rich environment. The soft X-ray beamline at the Australian Synchrotron has adjustable vertical slits which trim the beam spot size and limit the incoming X-ray flux. For standard XPS measurements, these slits are separated by a gap of 10 µm which produced an X-ray flux of 1.75×10^{11} photons/s. The gap may, however, be increased to achieve a higher incoming X-ray flux. For the following beam exposure measurements, a series of multiple scans were performed at a vertical slit spacing of 10 µm to observe how the XPS spectra changed under standard measurement conditions. When subsequent scans produced no further change in the XPS spectra, the slit spacing was increased to 100 µm, which increased the X-ray flux to 2×10^{12} photons/s. XPS measurements were recorded with a 100 µm slit spacing after 10 minutes of exposure to observe the final state of the XPS spectra.

Figures 6.15.a and 6.15.c show the evolution of the S2p spectra for the Na_2S treated undoped and 0.15% Sb-doped nanotubes under an increasing X-ray dose. The four sulfur environments from Section 6.3.2 have been fitted to each scan, and the evolution of the areas of each contribution can be seen in Figures 6.15.b and 6.15.d. There is an overall increase in the intensity of the S 2p spectrum for both the undoped and Sb-doped nanotubes, which may be attributed to the removal of surface contaminants such as adventitious carbon by the high-flux X-ray beam. Alternatively, large particles of sulfur species were observed to migrate across the nanotube surfaces under SEM imaging (Section 6.3.1). This process could also occur under the high flux of the X-ray beam, increasing the surface area of the observed sulfur species resulting in a larger S 2p signal. Both undopped and Sb-doped nanotubes show changes in the

proportion of each contributing sulfur species throughout the increasing number of scans. Initially, the majority of the contributions come from SO_4^{2-} and terminal S_x^{2-}, while the contributions from SO_3^{2-} and bridging S_x^{2-} only contribute ~10% of the total peak area each. Under the increasing X-ray dose arising from multiple scans at a 10 µm slit spacing, the relative area of SO_3^{2-} increases by ~15% while the contributions from SO_4^{2-} and terminal S_x^{2-} reduce by ~10% and ~5%, respectively. The contributions from bridging S_x^{2-} remain relatively unchanged.



(a) S 2p, $h\nu$ = 320 eV **(b)**

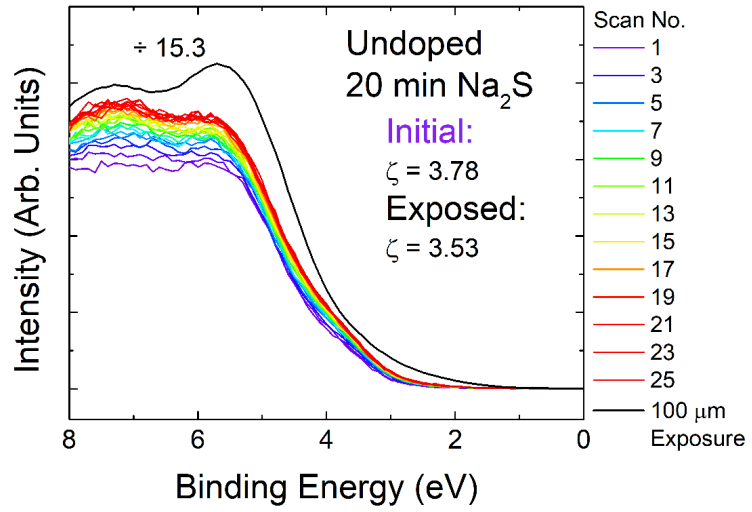
(c) S 2p, $h\nu$ = 320 eV **(d)**

Figure 6.15: S 2p XPS spectra of (a) undoped and (c) 0.15% Sb-doped 20 minute Na_2S treated nanotubes taken over 25 scans at a 10 µm slit spacing, with each scan being performed at a higher cumulative X-ray dose over the previous scan. Final scans were also performed after an extended exposure at a slit spacing of 100 µm to observe the steady-state S environment. The effective areas of each contribution from each scan for the undoped and Sb-doped samples is shown in (b) and (d) respectively.

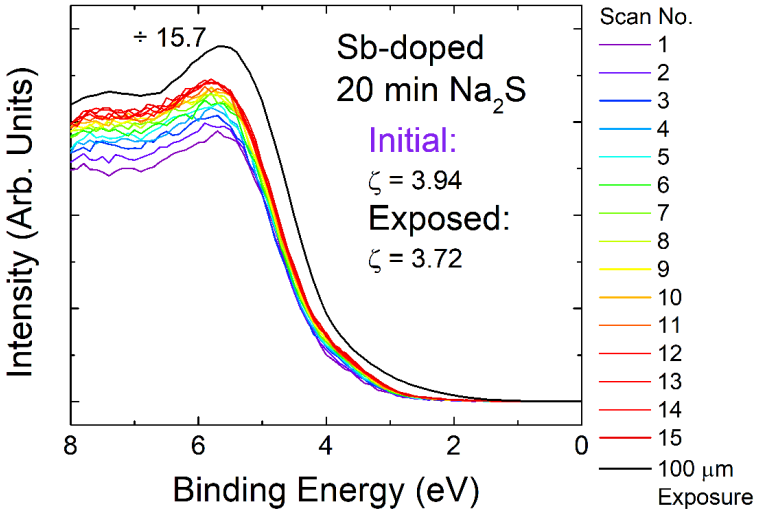
Following a long exposure with the slit spacing increased to 100 µm, some differences appear between the undoped and Sb-doped samples. The undoped sample shows only minor differences following the 100 µm exposure, with a reduction of the terminal S_x^{2-} contribution by 5% and slight increases in the contribution of SO_4^{2-}, SO_3^{2-} and bridging S_x^{2-}. The Sb-doped sample shows a more significant difference following the 100 µm exposure, with SO_3^{2-} and bridging S_x^{2-} contributions increasing while the SO_4^{2-} and terminal S_x^{2-} contributions decrease.

Figure 6.16 shows the valence band spectra of the Na_2S treated undoped and Sb-doped nanotubes under an increasing X-ray dose produced by multiple scans. Similar to the S 2p spectra, the intensities of the valence band spectra increase with an increasing X-ray dose, indicating that surface contaminants are being removed by the X-ray beam. The position of the valence band edge can also be seen to be shifting to lower binding energy with an increasing X-ray dose. The calculated values of ζ from the initial 10 µm slit spacing scan and the final scan following the 100 µm slit spacing exposure can be seen in Table 6.4. Initially, the valence band spectra of the Na_2S treated nanotubes exhibit relatively small band bending shifts compared to the reference sample, which increase following the X-ray exposures by 0.25 and 0.22 eV for the undoped and Sb-doped nanotubes, respectfully. The exposed values of ζ closely match those presented for the Na_2S treated nanotubes in Section 6.3.2.

These results show how the sulfur species present on the nanotube surfaces from the Na_2S treatment are reduced by the high flux X-ray beam, with a significant reduction of the SO_4^{2-} species to SO_3^{2-}. X-ray induced reduction has been previously reported for other sulfate species. Heske et al. reported the X-ray induced reduction of $CdSO_4$ to CdS whilst performing X-ray emission spectroscopy[360]. Weinhardt et al. found that $CaSO_4$ largely reduced to $CaSO_3$ under X-ray exposure, with only a small amount of CaS produced[361]. Although these reports confirm sulfate reduction occurs under high X-ray fluxes, neither propose a reaction pathway for the process to occur.



(a) VB, $h\nu$ = 150 eV



(b) VB, $h\nu$ = 150 eV

Figure 6.16: Valence band XPS spectra of (a) undoped and (b) 0.15% Sb-doped 20 minute Na_2S treated nanotubes taken over multiple scans at a 10 µm slit spacing, with each scan being performed at a higher cumulative X-ray dose over the previous scan. Final scans were also performed after an extended exposure at a slit spacing of 100 µm to observe the steady-state S environment.

Table 6.4: Zeta values and the relative band bending changes of the Na_2S treated SnO_2 nanotubes before and after long exposures to the X-ray beam.

	Undoped		0.15% Sb-doped	
Sample	ζ (eV)	ΔV_{bb} (eV)	ζ (eV)	ΔV_{bb} (eV)
Reference	4.01	-	4.01	-
Na_2S Initial	3.78	0.23	3.94	0.07
Na_2S Exposed	3.53	0.48	3.72	0.29

In addition to the reduction of the surface sulfur species, the same process produces a significant upwards shift in the surface band bending. This is similar to the results of Carroll[226], who showed the reduction of SO_4^{2-} on Na_2S treated SnO_2 nanotubes produced upwards band bending shifts. A key difference, however, between these results and those of Carroll concerns the relative proportion of the terminal sulfide species in polysulfides. In the beam exposure studies performed by Carroll, the upwards shift in band bending was attributed to an observed increase in sulfide species within the S 2p spectrum. In that model, surface sulfide species act as electron acceptors which deplete the surface charge of the SnO_2 nanotubes, similar to the case of Na_2S treated ZnO[233]. In the present study, the proportion of terminal sulfide is found to decrease during the X-ray reduction, yet the upwards shift in band bending still occurs. This indicates that in this case the upward shift in band bending is likely related to the X-ray induced reduction of SO_4^{2-} or a X-ray mediated interaction of SO_4^{2-} to the SnO_2 surface rather than the production of terminal sulfide.

6.3.4 NEXAFS Characterisation of Surface Sulfur Species

Analysis of the S 2p spectra of Na_2S treated nanotubes by synchrotron XPS revealed a number of different sulfur species present on the nanotube surfaces. The considerable overlap between the observed doublets and the relative difficulty in assigning exact sulfur species to each doublet required the use of additional analysis techniques. NEXAFS studies were undertaken on the MEX-2 beamline at the Australian Synchrotron to further characterise the sulfur species present on the Na_2S treated SnO_2 nanotubes. Figure 6.17 shows the S K-edge fluorescence of undoped and 0.15% Sb-doped SnO_2 nanotubes treated in an Na_2S solution for 5 minutes. A range of other sulfur-containing compounds were analysed as reference samples to compare with the edge features of the treated nanotubes. These included two single crystals of SnS grown by the Bridgeman and chemical vapor transport methods, a flux-grown SnS_2 single crystal, and powders of $ZnSO_4$ and NaS_2O_3. Reference data for two other sulfur species were retrieved with permission from the ESRF ID21 sulfur XANES database: Na_2SO_3 + Na_2SO_4[362, 363] and elemental sulfur[364]. No

beam damage or X-ray induced reduction in any of the sulfur-related signals was observed on any of the measured samples, owing to the lower X-ray flux of the MEX-2 beamline.

The S K-edge fluorescence spectra of the Na_2S treated SnO_2 nanotubes in Figure 6.17 shows three edge features present with peaks centred at 2472.4 eV, 2478.55 eV and 2482.8 eV. There are no differences in the peak positions between the undoped and 0.15% Sb-doped nanotubes, indicating that the same sulfur species are present on both sets of nanotube samples. The ratio of the three peaks does differ slightly, however, suggesting slightly different amounts of sulfur oxidation have occurred. The highest intensity peak at 2482.8 eV can be confidently attributed to SO_4^{2-}, as this value shows good agreement with the $ZnSO_4$ powder reference along with other SO_4^{2-} NEXAFS data in literature[363]. The peak with the second highest intensity at 2478.55 eV is likely an SO_3^{2-} species, as the peak position aligns reasonably well with the SO_3^{2-} reference as well as agreeing with peak positions for SO_3^{2-} reported in literature[363, 365].

The third and smallest peak in the S K-edge is observed at 2472.4 eV and appears close to the edge found on both SnS single crystal references along with the elemental sulfur reference. Figure 6.18 shows a magnified view of this peak compared to the Bridgeman grown SnS crystal and an elemental sulfur reference. Despite the similarity in the peak position between the Na_2S treated samples and the SnS crystal, the leading edge of the SnS crystal appears ~0.8 eV before the leading edge of the Na_2S treated samples. Furthermore, the SnS peak is much broader and is a different shape compared to the peaks on the Na_2S treated samples. It is unlikely that the Na_2S treatment is reducing the surface to SnS. Instead, the Na_2S treated peak shape and position matches the elemental sulfur reference much more closely, with better alignment of the leading edge. While the peak could be representative of elemental sulfur, it is more likely indicative of the bridging S atoms in longer chain polysulfides which appear at similar positions of NEXAFS spectra[366–368]. This is further supported by the XPS studies in 6.3.2.

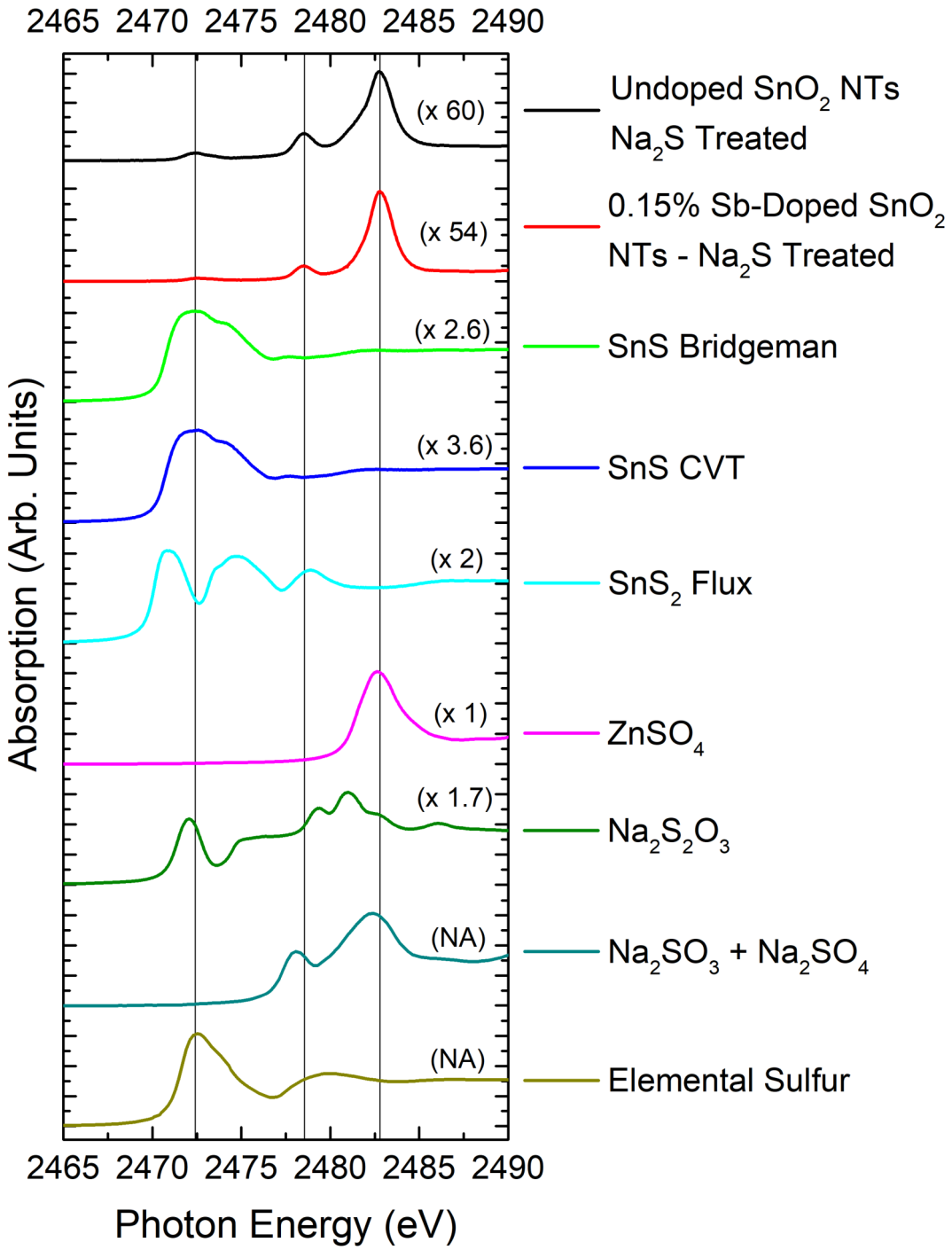


Figure 6.17: NEXAFS S K-edge fluorescence spectra of undoped and 0.15% Sb-doped SnO_2 nanotubes Na_2S treated for 5 minutes compared to other common sulfur species. The Na_2SO_3 + Na_2SO_4 and elemental sulfur references were retrieved from the ESRF ID21 sulfur XANES database[362–364]. Vertical lines show the three observed peaks from the Na_2S treated nanotubes at 2472.4 eV, 2478.55 eV and 2482.8 eV respectively.

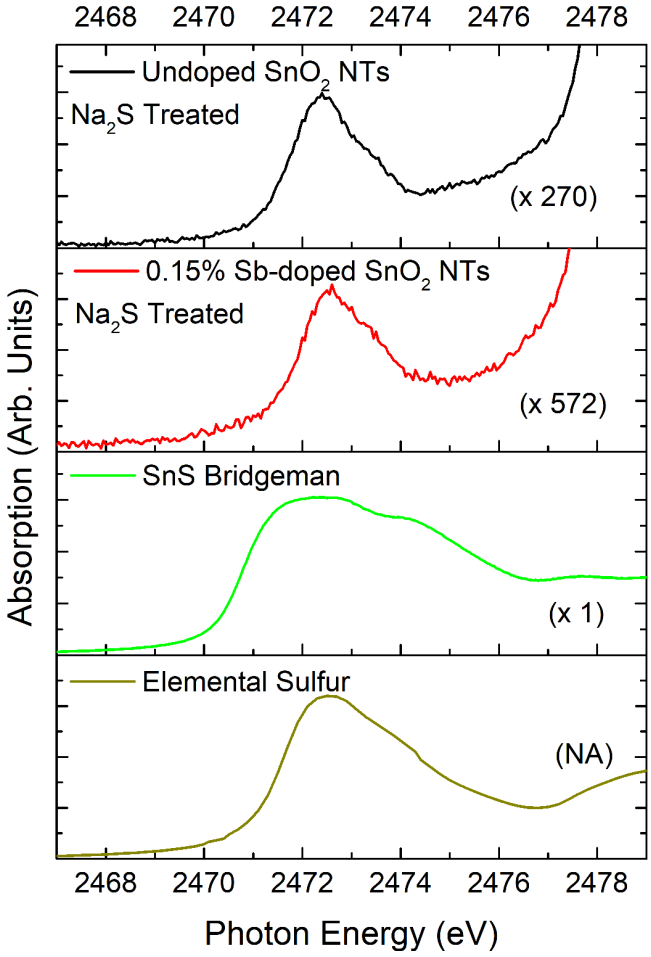


Figure 6.18: NEXAFS S K-edge fluorescence spectra of undoped and 0.15% Sb-doped SnO_2 nanotubes Na_2S treated for 5 minutes compared to the Bridgeman-grown SnS crystal and elemental sulfur, centred on the first edge present in all samples at ~2472.4 eV.

6.3.5 NEXAFS Characterisation of Na_2S Treated ZnO

Earlier XPS studies performed by Dr Alexandra McNeill and Dr Liam Carrol demonstrated that ZnO experiences a significant band bending shift when treated with an Na_2S solution[226, 233]. Unlike on SnO_2, the majority of the sulfur species on Na_2S treated ZnO analysed by XPS were in the form of sulfides. To further characterise the sulfur species present on Na_2S treated ZnO and to understand the differences in Na_2S treatment of SnO_2 and ZnO, NEXAFS studies were undertaken on Na_2S treated bulk ZnO.

Figure 6.19 shows the S K-edge of bulk a-plane and r-plane ZnO crystals that were Na_2S treated for 2 hours, compared with ZnS single crystal and powder references. All four spectra show a large main edge feature that peaks at 2473.8 eV alongside

a series of similar near-edge structures, indicating that Zn-S bonds are forming at the ZnO surface with sulfur in the 2- oxidation state. A small feature is observed at 2482.8 eV on the treated ZnO samples (indicated by *) which does not arise on the ZnS references, which is attributed to a small amount of oxidation of the sulfur to SO_4^{2-}. Despite this small amount of oxidation, it is clear from the size of the sulfide edge that the majority of the sulfur species are in the form of sulfide.

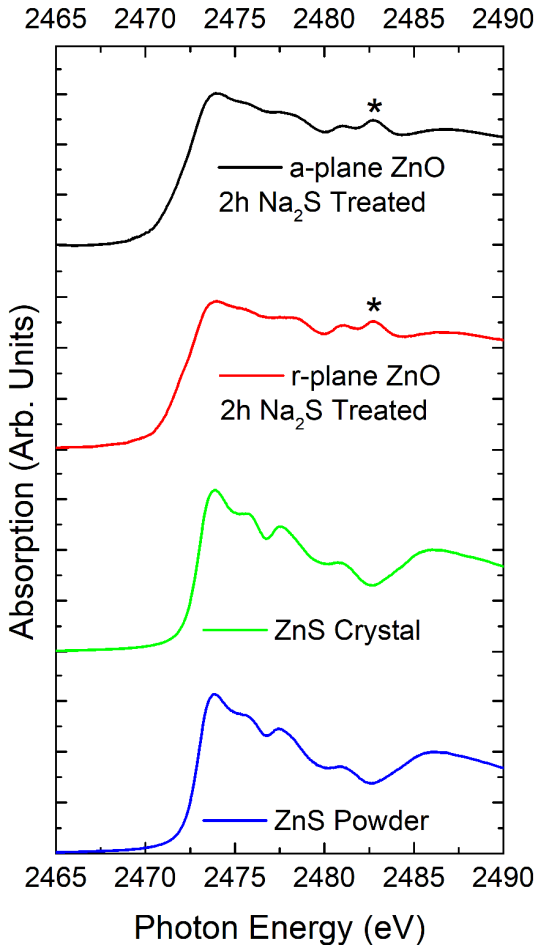


Figure 6.19: NEXAFS S K-edge drain current spectra of a-plane and r-plane bulk ZnO Na_2S treated for 2 hours compared to single crystal ZnS and ZnS powder.

The Na_2S treatment of ZnO likely forms a layer of ZnS at the surface of the bulk crystal. A common method for producing ZnO@ZnS core-shell nanoparticles involves the sulfurisation of the outer layers of ZnO nanoparticles using Na_2S solution, with this methodology being widely reported in literature[369–374]. She et al. completely converted ZnO nanoparticles to ZnS through stirring in a 2 M Na_2S aqueous solution for 2 hours at 90 °C, confirming the transformation through X-ray

diffraction and infra-red spectroscopy[375]. XPS studies performed by Guo et al. proposed that the interface between ZnO and ZnS produced a significant offset in the band alignment of the two materials[369]. A reproduction of their XPS valence band spectra and band alignment diagram can be seen in Figure 6.20. This offset in the band alignment and position of the Fermi level in the two materials produces a significantly decreased value of Zeta in ZnO with a surface layer of ZnS. These findings closely match the XPS studies performed by McNeill[233], which indicates that the observed shift in valence band position is the result of the conversion of surface ZnO to ZnS and the resultant band alignment, rather than a change in surface band bending.

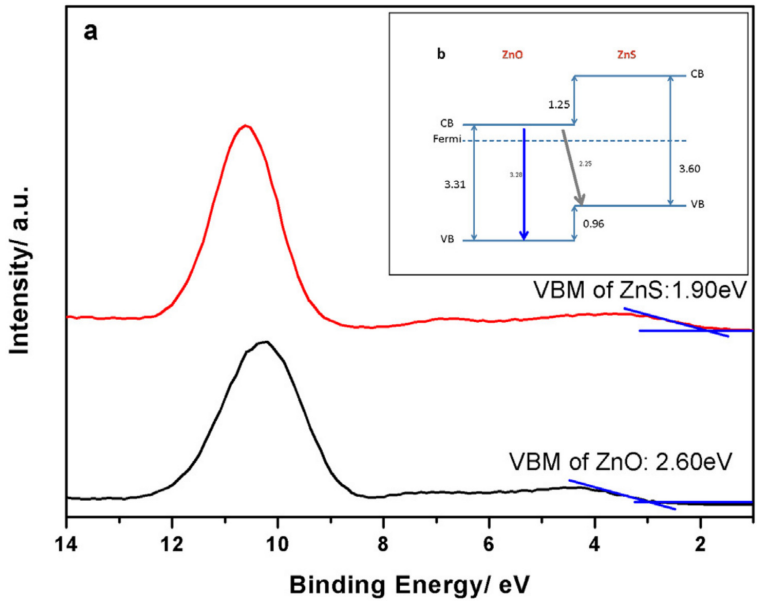


Figure 6.20: (a) The XPS valence band spectra of bulk ZnO and ZnS. (b) The proposed band offset at the ZnO/ZnS interface. Reproduced with permission from [369].

The difference in the effect of the Na_2S treatment between ZnO and SnO_2 likely arises from the difference in wet chemical resistance between the two materials. ZnO is easily etched by a wide range of acids and bases[376–379], whereas SnO_2 requires high etchant concentrations or added reducing metals in order to achieve moderate etching rates[380–382]. The higher chemical resistance of SnO_2 prevents a similar anion-exchange process that converts ZnO to ZnS from occurring. As a result, only sulfur species deposited from the Na_2S solution are present on the nanotube surfaces.

6.4 Summary and Conclusions

Synchrotron XPS has been used to study the removal and recovery of surface hydroxyls and oxygen vacancies from undoped and 0.15% Sb-doped SnO_2 nanotubes following high temperature heating and gas dosing. On undoped nanotubes, 600 °C heating removed much of the electron donating surface hydroxyl layer, although this did not result in a weakening of the downward band bending due to the formation of oxygen vacancies compensating the loss of surface hydroxyls. Combined or separated O_2/H_2O dosing produced little to no changes to the O 1s spectra, with only a slight weakening of the downward band bending due to oxygen adsorption. Venting with H_2O produced a small increase in the surface hydroxyls and downward band bending, likely limited by the removal of bridging oxygen required for water dissociation on the SnO_2 surface. Venting to atmosphere for 10 hours produced a recovery of the surface hydroxyls and the repair of oxygen vacancies, returning the downward band bending to its strong initial condition. Venting for a period of 20 minutes allowed only a partial recovery of the surface hydroxyls, demonstrating how the natural process of hydroxyl recovery occurs over the time scale of several hours.

0.15% Sb-doped nanotubes showed a similar yet slightly different trend with respect to the recovery of surface hydroxyls and oxygen vacancies. 600 °C heating removed the surface hydroxyl groups, although a higher defect formation energy likely resulted in less oxygen vacancies being formed. This provided less surface charge compensation and resulted in a net weakening of the downward band bending. Similar to the undoped nanotubes, separated and combined O_2/H_2O dosing produced no significant change in the surface hydroxyl density. The formation of fewer oxygen vacancies did allow the H_2O vent to increase the hydroxyl coverage by a larger amount compared to the undoped sample, due to more available bridging oxygen allowing for water dissociation. Similar full recoveries in the surface hydroxyl layer and downward band bending as with the undoped sample were observed for 20 minute and 10 hour vents to atmosphere.

SEM, synchrotron XPS and NEXAFS have been used to study the effect of two

sulfur treatments on the band bending of undoped and 0.15% Sb-doped SnO_2 nanotubes. A 1 hour $(NH_4)_2S$ treatment produced no changes to the surface band bending of the SnO_2 nanotubes, with no visible changes observed under SEM and only a weak sulfur environment observed using synchrotron XPS. A 20 minute Na_2S treatment under X-ray exposure produced large shifts of the valence band edge, indicating significant weakening of the downward band bending. This was observed alongside a complicated sulfur environment, largely made up of SO_4^{-2}, SO_3^{-2} and S_x^{-2} species. SEM imaging of the Na_2S treated samples showed patches of sulfur particles attached to the nanotube surfaces that spread across the nanotube surfaces under prolonged electron-beam exposure. In a similar fashion, the S 2p and valence band XPS spectra of Na_2S treated samples were observed to change with increasing X-ray irradiation, demonstrating X-ray induced reduction of SO_4^{-2} to SO_3^{-2} that resulted in a large shift of the valence band to lower binding energy, indicating a significant weakening in downward band bending.

Finally, NEXAFS was used to investigate Na_2S treated bulk ZnO to further understand the differences in band bending changes following the treatment on ZnO and SnO_2. It was found that the surface of the bulk ZnO crystals is likely converted to ZnS by Na_2S treatment, with conversions of ZnO to ZnS by Na_2S treatment also reported in the literature. This results in a band offset between the two materials that produces a valence band edge much closer to the Fermi level, which can easily be mistaken as a surface band bending shift. The same chemical treatments do not produce a similar conversion on SnO_2, likely due to the higher chemical stability of the material.

Chapter 7

Further Growth and Characterisation of the SnO_2 Nanotubes

7.1 Introduction

Through the undertaking of this body of work, several additional studies have been completed which reveal more information concerning the growth and properties of the square SnO_2 nanotubes and related nanostructures. In this chapter, the growth of the SnO_2 nanotubes is further optimised by varying two parameters not previously studied in detail (i.e, the carrier and dilution gas flow rates). In addition, a number of growths with different forms of contamination resulted in the unintentional deposition of SnO_2 nanospirals, which were also previously observed by Scott[29]. These growths give valuable additional information on the conditions that lead to the formation of these spiral structures. The surfaces of the SnO_2 nanotubes are characterised by scanning tunnelling microscopy to reveal further information about the different surface morphologies that are present. Finally, observations of large nanotube bending provides insight into the unique mechanical properties of these square nanostructures.

7.2 Nanotube Growth Optimisations and Observations

7.2.1 Gas Flow Rate Optimisation

During the initial stages of trying to reproduce the growth of the square SnO_2 nanotubes by Mist-CVD for their incorporation into electronic devices, significant difficulties were encountered in producing successful growths. When using the previous optimised growth conditions provided by Scott[29], the majority of the samples contained no square nanotubes and instead consisted of short, rod-like structures as seen in Figure 7.2.a. Growths with minor adjustments to parameters such as the deposition temperature, $SnCl_2$ concentration and Au layer thickness did not produce significant improvements in nanostructure morphology. However, in almost all of these unsuccessful growths square SnO_2 nanotubes were observed to grow in a thin strip at the bottom of the growth substrate, where the bottom lip of the sample holder extends past the height of the substrate. The bottom lip of the sample holder produces a small eddy current that significantly affects the incoming mist flow (Figure 7.1), diverting much of the mist from this area of the substrate. This effect typically results in a lower growth rate in this area. This indicated that the mist

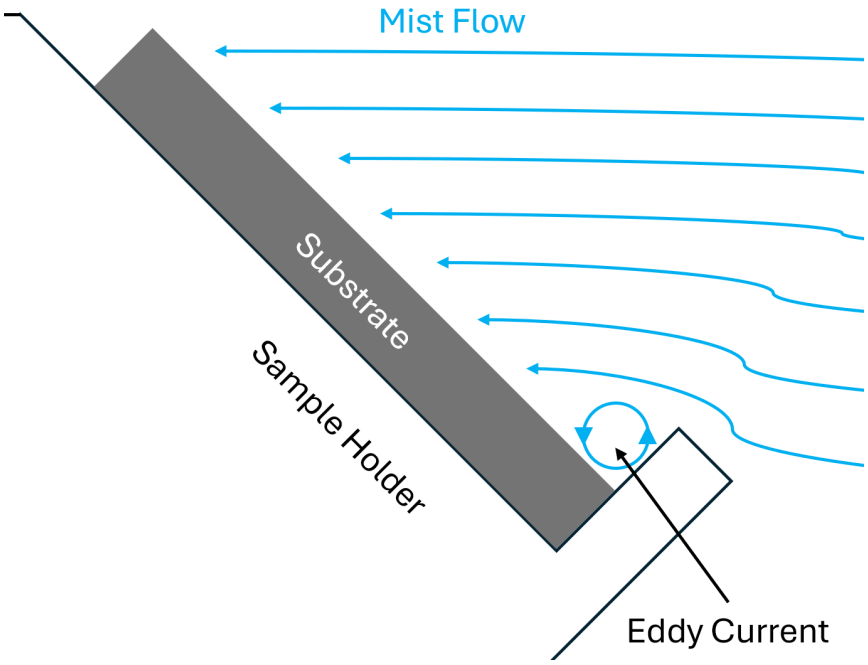
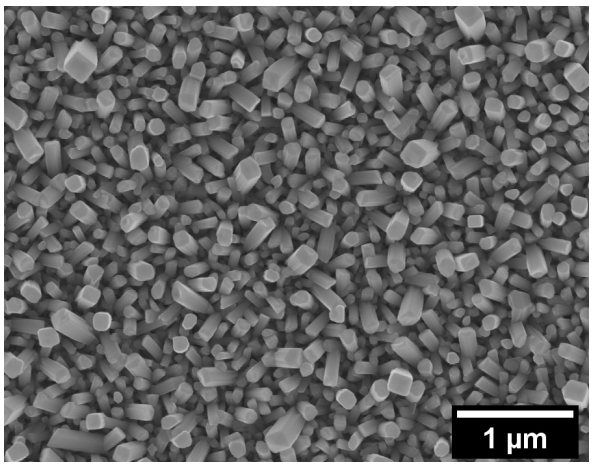


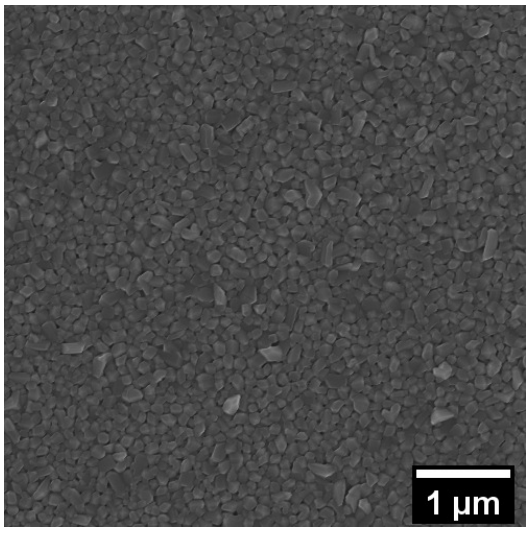
Figure 7.1: Diagram illustrating the effect of the bottom lip of the Mist-CVD sample holder on the mist flow across the growth substrate, with the produced eddy current at the bottom of the substrate.

flow (which is determined by the carrier and dilution gas flow rates) was likely to be a key factor in the successful growth of the square SnO_2 nanotubes. For this reason, a series of SnO_2 nanotube growths were performed with different carrier and dilution gas flow rates to determine the optimal rates for nanotube deposition.

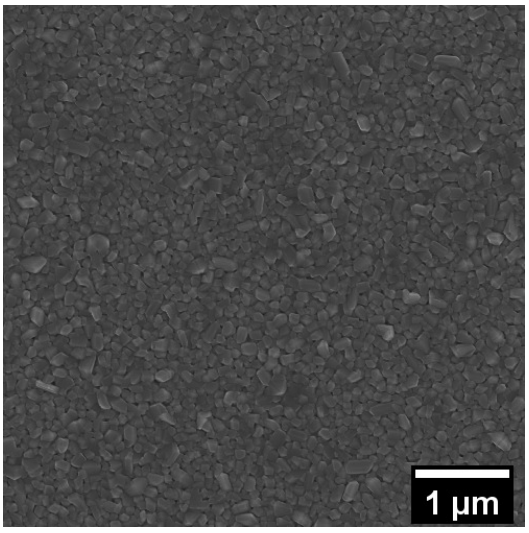
Figure 7.2 shows the surface morphologies at the centre of the growth substrate for a series of different dilution and carrier gas flow rates, with only one of the flow rates changed for each growth. All other growth parameters remained at the optimised values reported by Scott[29], and the series was performed on Au-coated Si (001) substrates for a growth time of 80 minutes per sample. As described earlier, the growth performed using the previous carrier and dilution gas flow rates of 2.5 $Lmin^{-1}$ and 4.5 $Lmin^{-1}$ produced a rough film with short, rod like structures. When either of these gas flow rate were separately increased, as in Figures 7.2.b and 7.2.c, this resulted in a rough, granular film, with no visible nanostructures across the majority of the growth substrate. This indicates that high gas flow rates are not conducive to the production of the square SnO_2 nanotubes. When the dilution gas flow rate was reduced from 4.5 $Lmin^{-1}$ to 3.0 $Lmin^{-1}$, no SnO_2 nanotubes were observed at the centre of the growth sample, with the morphology similar to that produced by the initial growth conditions. Finally, when the carrier gas flow rate was reduced to 1.6 $Lmin^{-1}$, this resulted in the successful deposition of the square SnO_2 nanotubes, indicating that a lower carrier gas flow rate is a key factor in nanotube formation. This was confirmed by repeated growths using these new flow rates that successfully produced square SnO_2 nanotubes, while a subsequent growth using the previous flow rates formed no nanotubes. This lower carrier gas flow rate, however, does result in a lower growth rate that requires increased deposition times to create longer SnO_2 nanotubes.



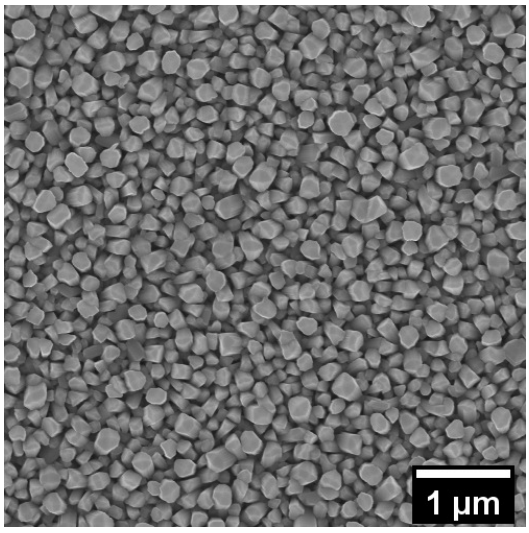
(a) Carrier: 2.5 $Lmin^{-1}$, Dilution: 4.5 $Lmin^{-1}$



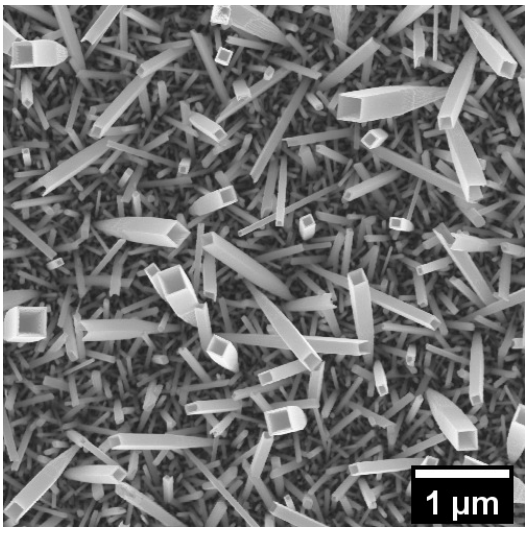
(b) Carrier: 2.5 $Lmin^{-1}$, Dilution: 7 $Lmin^{-1}$



(c) Carrier: 5 $Lmin^{-1}$, Dilution: 4.5 $Lmin^{-1}$



(d) Carrier: 2.5 $Lmin^{-1}$, Dilution: 3 $Lmin^{-1}$



(e) Carrier: 1.6 $Lmin^{-1}$, Dilution: 4.5 $Lmin^{-1}$

Figure 7.2: SEM images of the surface morphologies of SnO_2 nanotube growth attempts using different carrier and dilution gas flow rates. (a) Flow rates initially optimised by Scott[29]. (b-e) Variations applied in this thesis, including (e) the successful growth of square SnO_2 nanotubes.

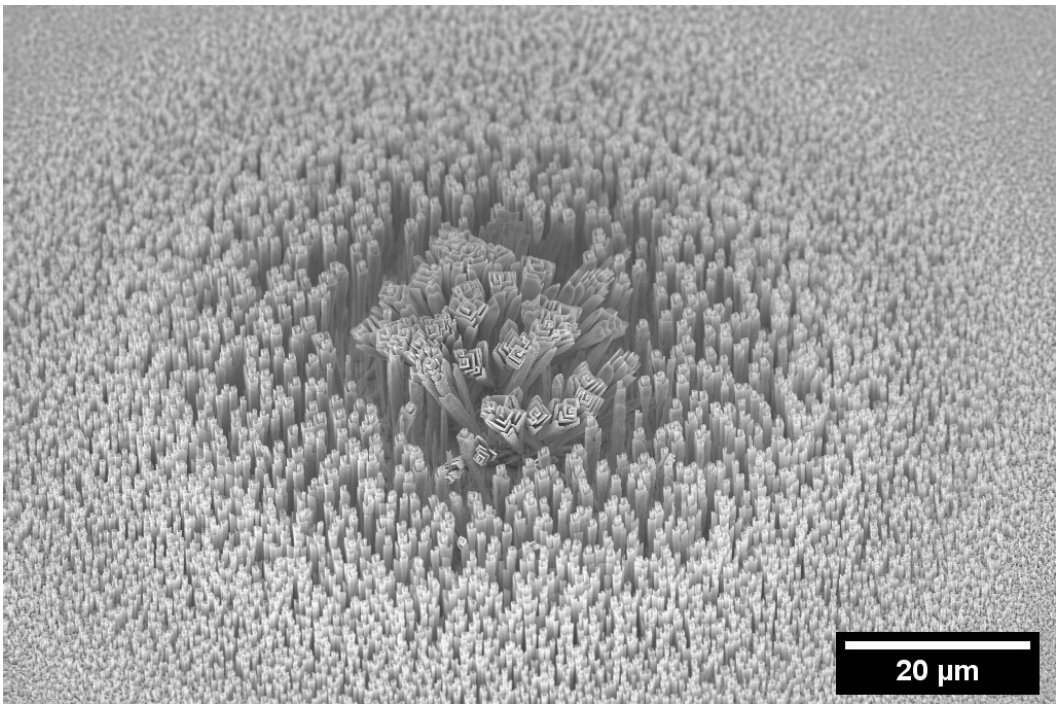
7.2.2 Contamination in Nanotube Growths

In the thesis of Scott, a second type of square nanostructure was observed following the deposition of SnO_2 by Mist-CVD[29]. This nanostructure was in the form of SnO_2 nanospirals, and was observed near contaminant particles on the growth substrate. As these contaminants were likely unintentionally introduced carbon species, it was proposed that carbon was incorporated into deposited SnO_2, producing a growth instability that created the nanospiral structures. These structures were only observed on one occasion however, and so limited information could be obtained from this singular occurrence. Throughout the numerous growths of square SnO_2 nanotubes in the present work, nanospiral-type structures were observed on several occasions and in different circumstances that provides some insight into their formation.

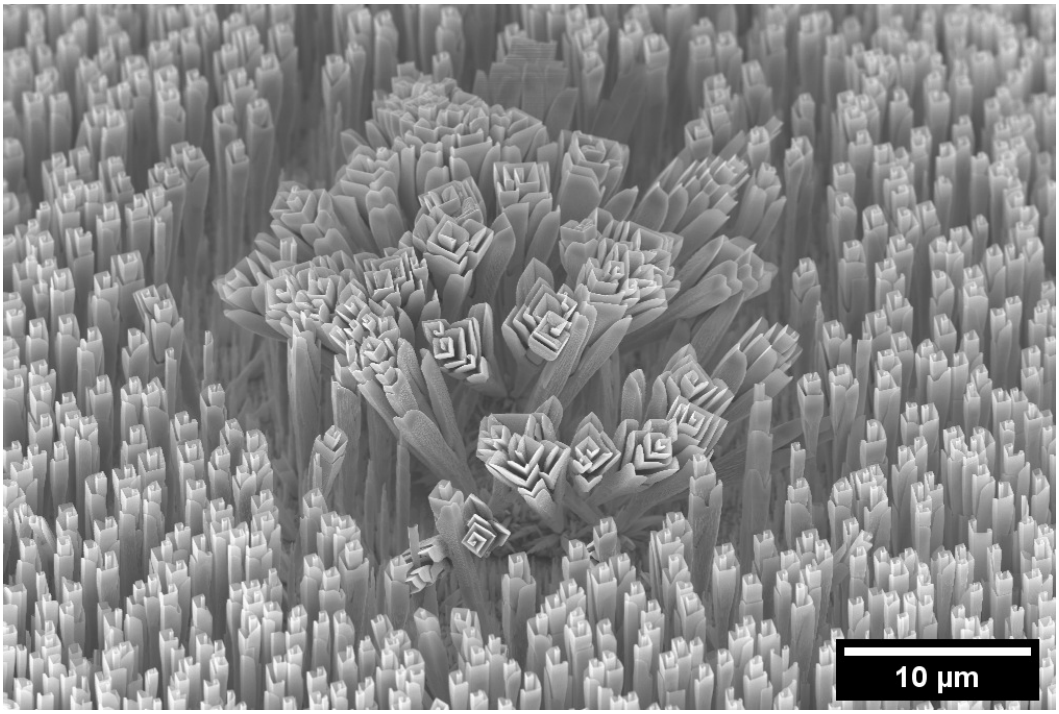
Figure 7.3 shows the formation of nanospirals on and surrounding a large contaminant on an m-plane sapphire substrate following a growth of SnO_2 nanotubes. Surprisingly, in this case the contaminant was located on an area where no Au nanoparticles were present to catalyse the nanotube growth, suggesting that the contaminant itself has acted as a catalyst that produces the growth of these nanostructures. This is further supported by the fact that the nanospirals are tallest near the contaminant and radially decrease in height away from it. In addition to the SnO_2 nanospirals, the contaminant has produced a large, unique structure that appears analogous to a bouquet of flowers.

Contaminant particles on the growth substrate surface are not the only contaminant that can produce nanospiral structures. Figure 7.4 shows a combination of SnO_2 nanotubes and nanospirals deposited on a Si substrate. In this growth, a new glass solution reservoir and silicone tubing was used to contain the precursor growth solution, which likely resulted in some degree of contamination of the growth solution. This suggests that the content of the precursor solution may also be able to induce the formation of these nanospiral structures, albeit again in an uncontrolled manner.

As suggested by Scott, carbon is the likely contaminant that affects the growth of these SnO_2 nanospirals. The majority of dust and particulate matter in indoor environments consists of either organic or elemental carbon[383], and despite cleanliness precautions contamination can still unintentionally occur. However, given



(a)



(b)

Figure 7.3: (a,b) SEM images of SnO_2 nanospirals that surround the original location of a contaminant on an m-plane sapphire growth substrate. A large flower type structure remains in the original contaminant location.

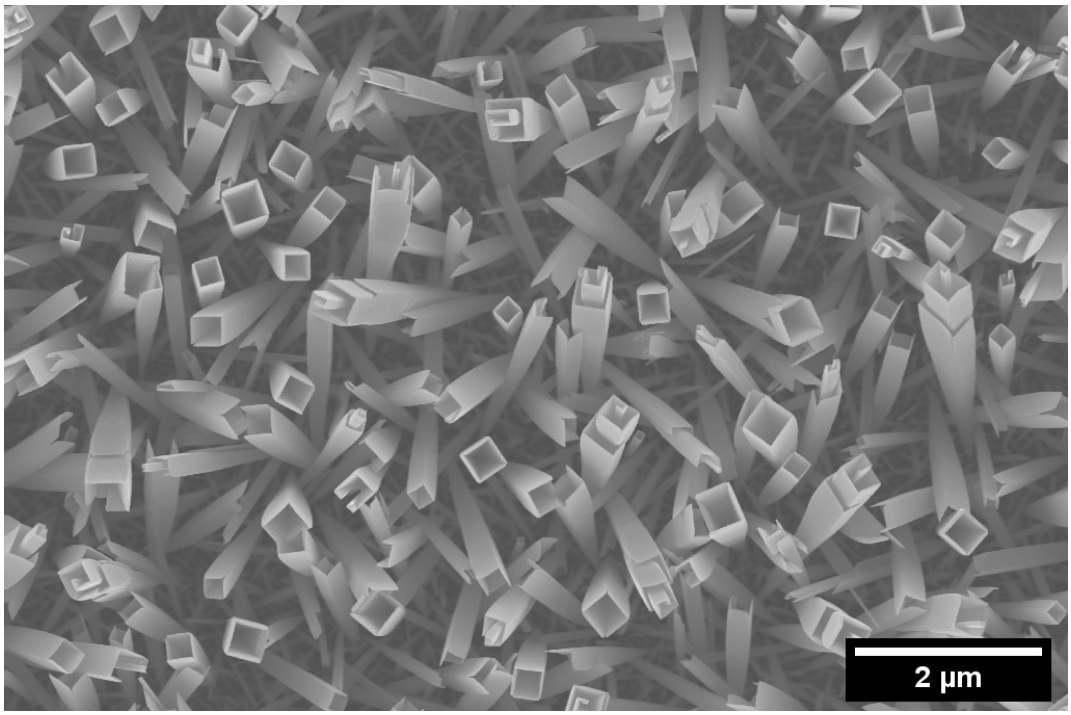


Figure 7.4: SEM image of SnO_2 nanotubes and nanospirals deposited on a Si substrate, likely due to contamination in the precursor solution from a new solution reservoir.

that the SnO_2 nanospirals in Figure 7.4 were produced by the likely contamination of the precursor solution, this may present an opportunity for future investigations into the intentional inclusion of carbon species into the precursor solution to enable the growth of SnO_2 nanospirals. There is evidence in the literature that points to carbon as a catalyst in the formation of other SnO_2 nanostructures, such as the mixing of graphite and SnO_2 powders for the deposition of SnO_2 nanowires[118, 384–386]. In addition, organic solvents are commonly used in the precursor solutions of Mist-CVD, and have been demonstrated to produce different film morphologies compared to solutions that use water as the solvent[387, 388]. A simple experiment that could be conducted in the future is the addition of a small amount of organic solvent such as methanol or isopropanol into the precursor solution, which would serve as the carbon precursor for potential nanospiral growth.

7.3 Further Characterisation of the SnO_2 Nanotubes

7.3.1 Scanning Tunnelling Microscopy of the SnO_2 Nanotubes

In collaboration with Dr Maxime Le Ster of the University of Lodz, scanning tunnelling microscopy (STM) was used to further characterise the structure of the SnO_2 nanotube surfaces. To achieve this, 0.15% Sb-doped SnO_2 nanotubes were dispersed across a highly conducting n-Si substrate which was then annealed at 400 °C to remove any organic solvents from the sample surface. Imaging of the sample surface by STM allowed the surface structure of an individual nanotube to be investigated. This nanotube had a diameter of approximately 315 nm at its top end, as measured by a line profile of the nanotube height as seen in Figure 7.5. Some artifacts can be seen near the top end of the nanotube. As the dispersed nanotubes were only held in place by van der Waal forces this is most likely due to a small shift in position due to the interaction of the STM tip with the top end of the nanotube.

When performing SEM imaging of the SnO_2 nanotubes, such as in Figure 7.6.a, two distinct surface morphologies are typically observed on the nanotube surface. Firstly, the surface of the outer walls near the top of the nanotube appear to be very smooth, with no observable steps or defects. However, further down the nanotube, the surface appears to be textured with long lamellar-like terraces running down the length of the nanotube. Similar lamellar-like terrace structures can also be seen on the interior walls of the SnO_2 nanotubes, indicating that a significant part of the surface area of the nanotubes consists of this structure.

Both the lamellar-like terraces and ultra-smooth regions were able to be resolved by STM, and a higher resolution image of these regions can be seen in Figure 7.6.b. The root-mean-square (RMS) surface roughness (ρ) of the two regions clearly illustrates the differences between these two surface morphologies. For the lamellar-like areas, $\rho \approx 1.2$ nm, while the ultra-smooth region has an atomically smooth surface with $\rho \approx 0.3$ nm. These two regions likely arise due to difference in the density of mist vapour that is able to reach different parts of the nanotube during the material deposition. Deposition during Mist-CVD growth is likely governed by the Leidenfrost effect, in which a mist particle experiences a heat-material exchange with the

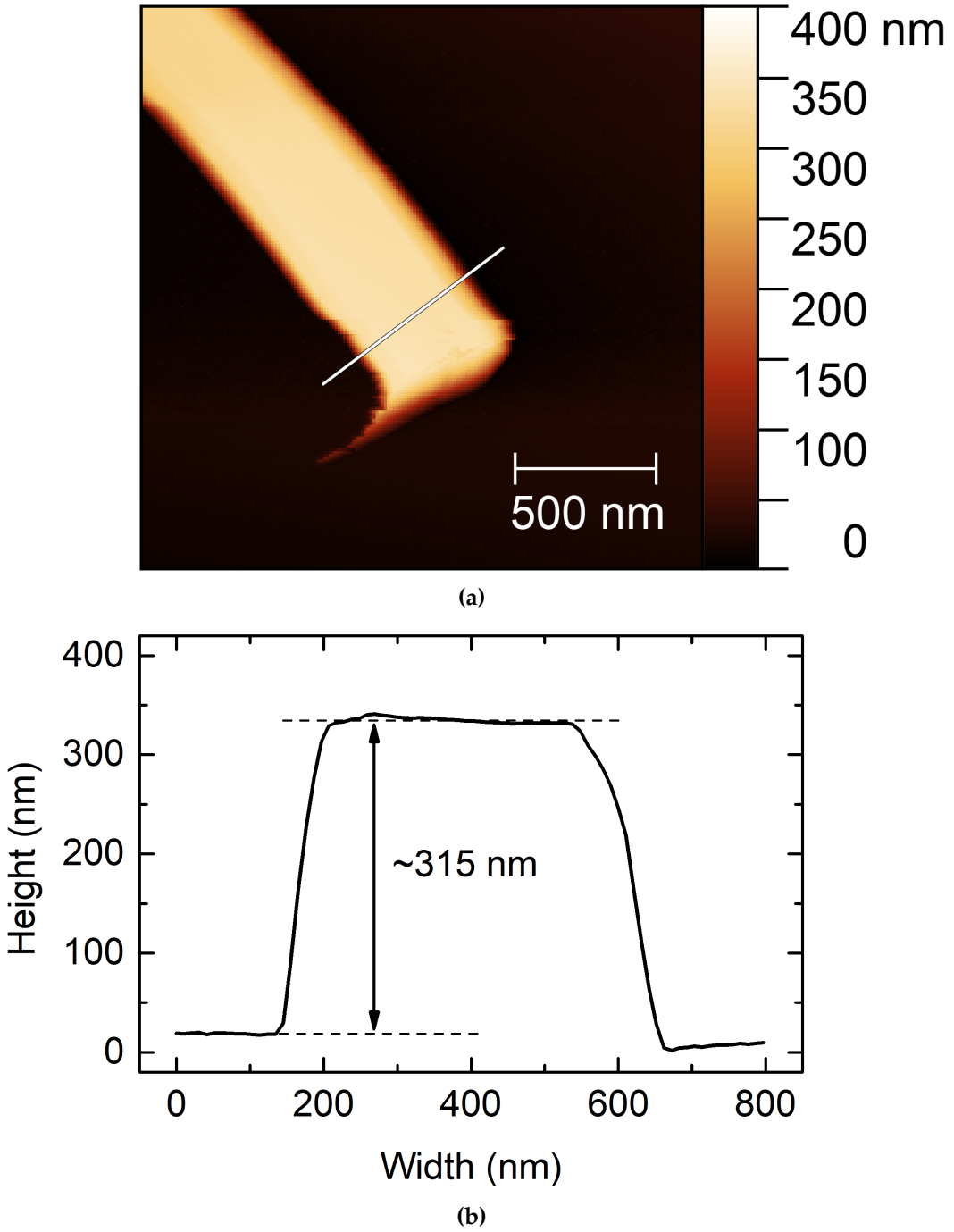
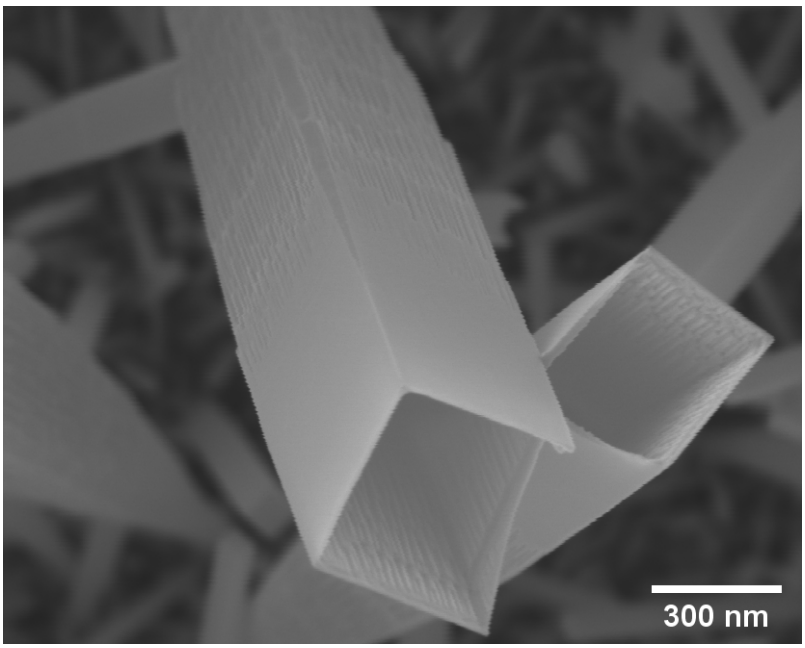
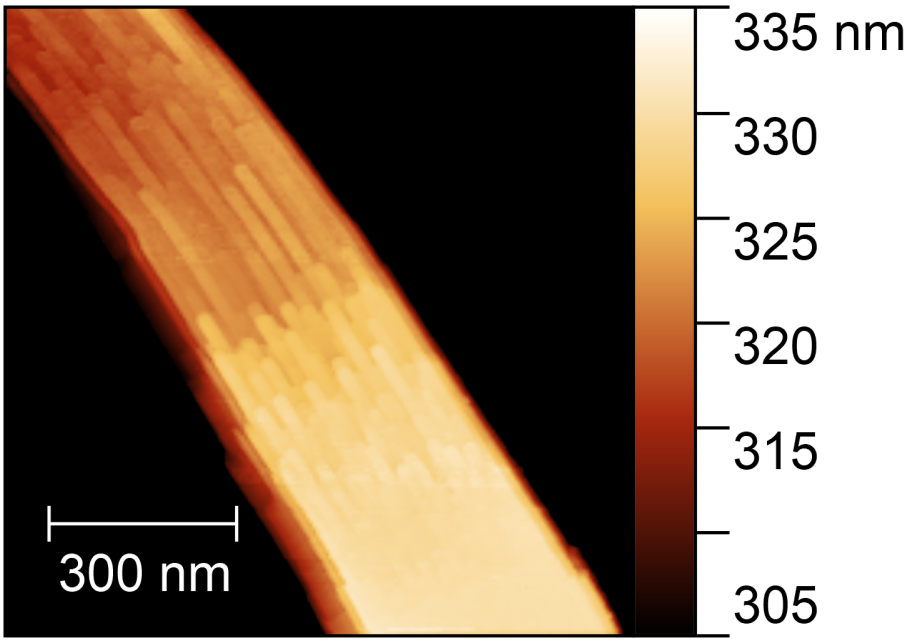


Figure 7.5: (a) STM image of an individual SnO_2 nanotube. (b) A line profile across the top end of the nanotube, showing a nanotube diameter of ~315 nm. The line profile was taken across the line shown in (a).

hot substrate during growth. As the top of the nanotubes are the closest to these mist particles, they receive a large material flux that enables the formation of large, complete (110) surfaces. Regions further down the nanotube experience a lower material flux that does not enable the complete formation of (110) planes, and so these surfaces form the lamellar-like terrace morphology. As the inside of the nanotube is



(a)



(b)

Figure 7.6: (a) SEM image of an SnO_2 nanotube with the ultra-smooth and lamellar-like terrace surface morphologies clearly visible. (b) STM image of an SnO_2 nanotube primarily showing the terrace structures down the length of the nanotube. The ultra-smooth region can also be seen at the bottom of the image.

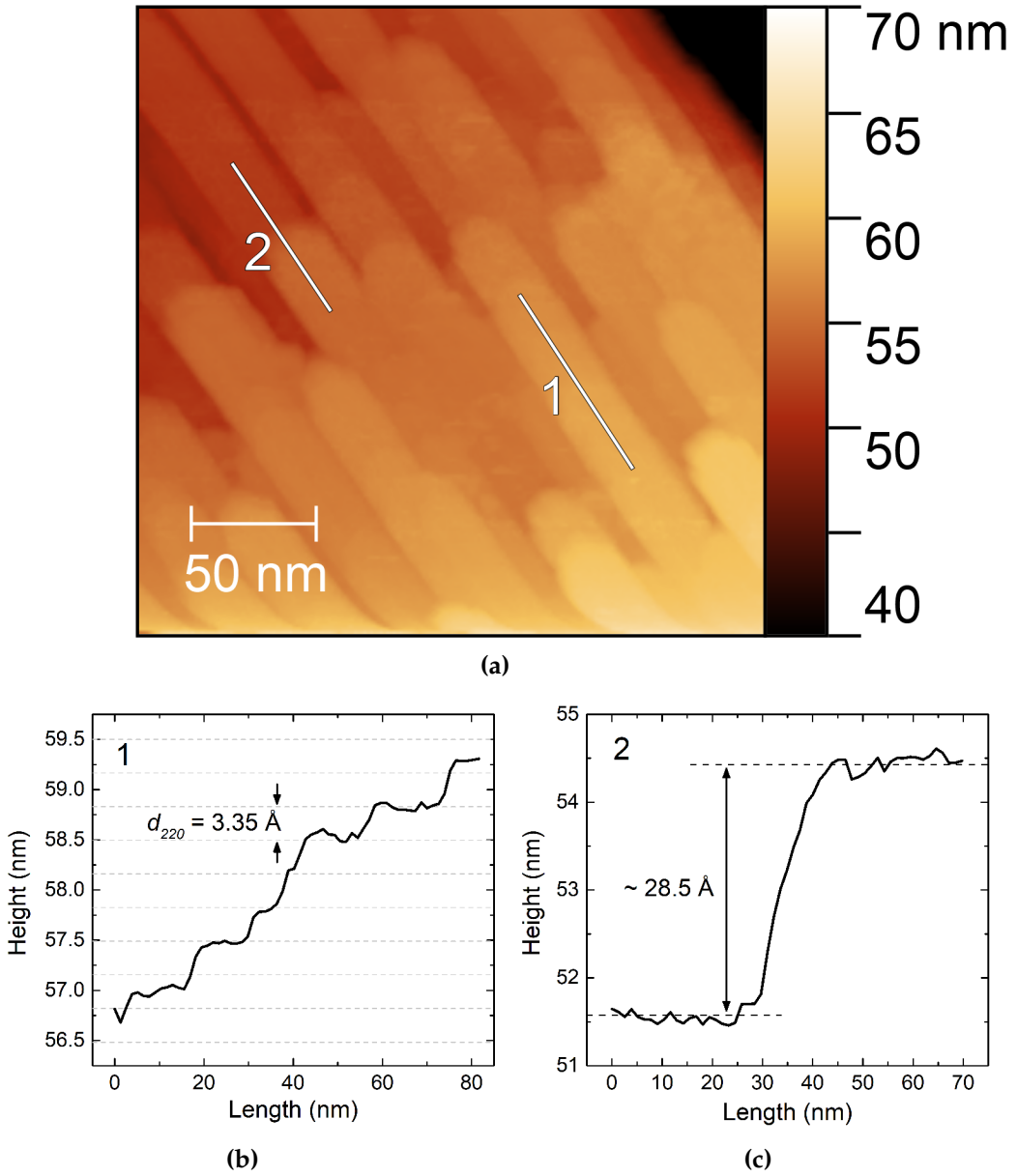


Figure 7.7: (a) STM image of the lamellar-like terrace structures of an SnO_2 nanotube. (b) A line profile along (1) showing atomic steps between successive (110) planes, with dashed lines representing the theoretical d_{220} spacing. (c) The step height between two terraces, extracted along line (2).

an enclosed space, this likely also reduces the material flux inside the nanotube.

A closer view of the lamellar-like terrace region can be seen in Figure 7.7. The lamellar-like terraces appear to have a remarkably uniform width between 25-30 nm. Extracting a line profile along the top of a single terrace (Figure 7.7.b) reveals several atomic steps between 3-4 Å in height, which is in excellent agreement with the step height between two successive (110) surface planes (d_{220} = 3.35 Å). The steps

between two terraces are significantly larger, with the measured step in Figure 7.7.c showing a step height of ~28.5 Å.

7.3.2 Mechanical Properties of the SnO_2 Nanotubes

As a hard, crystalline material, bulk SnO_2 is typically a brittle material that is not capable of experiencing large strains without breaking. However, nanostructured materials often have different mechanical properties compared to their bulk counterparts. Figure 7.8 shows an individual SnO_2 nanotube that has been coated at each end with Ti/Au contacts. However, during ultrasonic agitation in the liftoff procedure the top end of the nanotube has become detached from its contact and caught against the side of the metal layer, resulting in significant bending of the nanotube. This bending produces a significant strain within the nanotube, consisting of a compressive force on the upper wall and a tensile force on the lower wall of the nanotube. Despite the large degree of bending the nanotube appears intact with no visible cracking or fractures.

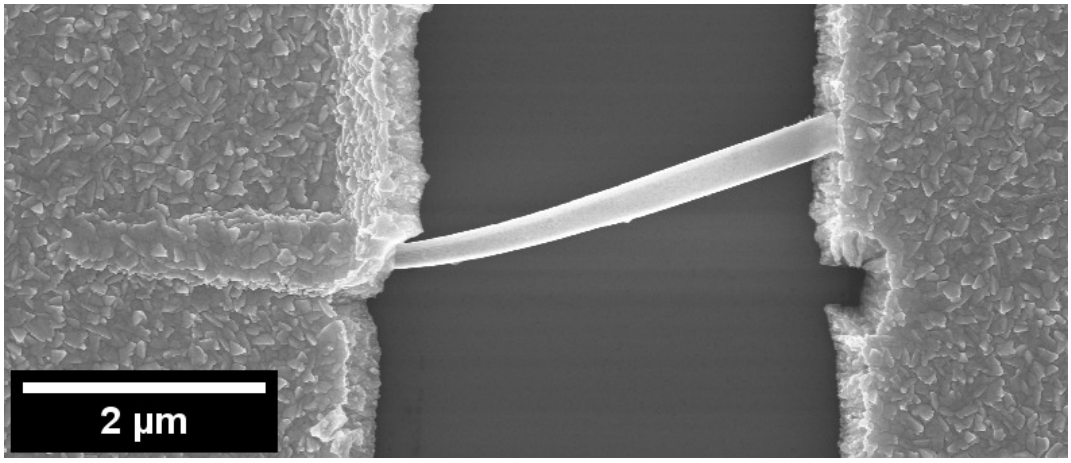


Figure 7.8: An SnO_2 nanotube caught on the side of a Ti/Au contact following ultrasonic-assisted lift-off, resulting in significant bending of the nanotube.

The bending of the nanotube in Figure 7.8 can be modelled as a simple cantilever beam with a single load at the end. This approach has been used to determine the mechanical properties of other nanostructured materials, such as ZnO nanowires[389] and SiC nanorods[390]. Using this model and assuming the nanotube is bending purely elastically, the maximum stress (σ_{max}) and strain (ϵ_{max}) within the bending

nanotube can be calculated by the following equations[389]:

$$\epsilon_{max} = \frac{3d\delta}{2l^2} \tag{7.1}$$

$$\sigma_{max} = \epsilon_{max} E = \frac{3d\delta E}{2l^2} \tag{7.2}$$

Where d is the nanotube diameter, l is the exposed length of the nanotube, δ is the deflection of the nanotube and E is the Young's modulus of SnO_2. Given the measured nanotube length of 4.33 µm, diameter of 200 nm and deflection of 1.3 µm, this gives a maximum strain of ϵ_{max} = 2.1%. Using the reported Young's modulus of ~450 GPa for the c-axis of SnO_2[391], this equates to a maximum stress of σ_{max} = 9.5 GPa in the bending nanotube.

The ability to withstand such a large strain is unusual for a ceramic material like SnO_2. The yield strains (maximum strain before fracture) of bulk glass and Al_2O_3 are reported to be in the range of 0.07-0.1%[392], which is significantly smaller than the 2% strain reported here. However, the ability to withstand large strains has been previously reported for other nanostructured materials. Chen and Zhu found yield strains of 4-7% for ZnO nanowires, which are significantly larger than the value of the bulk material[389]. Increased fracture strengths compared to the bulk have also been reported for other nanowire materials, including SiC[390], Si_3N_4[393] and Si[394]. In our case, the unique perfectly-square shape and high crystal quality of the SnO_2 nanotubes results in an increased maximum strain in the material that allows the large degree of bending seen in Figure 7.8.

7.4 Summary and Conclusions

Several additional studies have been completed which reveal more information about the properties of the SnO_2 nanotubes and their growth conditions. The flow rates of the carrier and dilution gases during Mist-CVD were optimised for the deposition of the SnO_2 nanotubes The optimal flow rates of the carrier and dilution gases was found to be 1.6 $Lmin^{-1}$ and 4.5 $Lmin^{-1}$, respectively. The effect of possible carbon contamination on the growth of SnO_2 nanospirals was also discussed, with contamination of the growth substrate and the precursor solution identified as potential

sources of nanospiral formation. Carbon was proposed to act as a catalyst for the growth of these nanostructures, which could be tested in the future by the intentional inclusion of carbon-containing compounds such as organic solvents to the precursor solution.

STM was used to characterise the surface morphology of the SnO_2 nanotubes, which revealed two distinct regions on the nanotube (110) surface. An atomically smooth region was identified at the top end of an individual nanotube, with a surface roughness of $\rho \approx 0.3$ nm. A lamellar-like terrace region was identified further down the nanotube, with a surface roughness of $\rho \approx 1.2$ nm. The formation of these two regions was explained in terms of material availability during their growth, with the lamellar-like terrace region receiving less material compared to the atomically smooth region. STM imaging of the lamellar-like terraces revealed a consistent terrace width of 25-30 nm and atomic steps along the surface of the terraces. Steps between terraces were larger, with a step of ~28.5 Å observed.

Finally, the observation of the bending of a nanotube following an unsuccessful lift-off process allowed some insight into the mechanical properties of the SnO_2 nanotubes. No cracking or fractures were produced by the large bending involved, despite a large calculated stress and strain of 9.5 GPa and 2.1%, respectively. Such a large strain is typically infeasible in bulk SnO_2 materials, however the SnO_2 nanotube could likely withstand the strain due to a combination of its unique morphology and high single-crystal material quality.

Chapter 8

Conclusions

8.1 Overview

Previous work performed by Dr Jonty Scott resulted in the discovery of a process to grow square, hollow SnO_2 nanotubes by a mist chemical vapour deposition technique. These square SnO_2 nanotubes have a unique geometry with abrupt 90° corners and edges, high crystal quality and large surface-to-volume ratios. Combined with other fundamental properties of SnO_2, such as a wide bandgap, controllable n-type electrical conductivity and an electronically sensitive surface, these remarkable structures are well-suited for applications such as UV photodetectors, gas sensors and neuromorphic devices.

In this thesis, undoped and 0.15% Sb-doped square SnO_2 nanotubes have for the first time been fabricated into resistors, back-gated field-effect transistors (FETs), Schottky diodes and ionic-gated transistors, all of which demonstrate excellent electrical properties. These different device structures allowed further investigations into the fundamental properties of these square nanostructures alongside the testing of nanotube devices for different applications, with a specific focus on UV photodetection and artificial synapses for neuromorphic applications. However, the electronically sensitive surfaces and strong downward band bending of the SnO_2 nanotubes that are created by electron-donating surface hydroxyl groups can have a negative impact on the operation and long-term stability of some devices. To further understand the removal and recovery of this surface hydroxyl layer on the SnO_2 nanotubes, synchrotron X-ray photoelectron spectroscopy (XPS) investigations

were performed on annealed samples of nanotubes under different hydroxyl recovery conditions. Building on previous work by Dr Liam Carroll, synchrotron XPS and near-edge X-ray absorption fine structure (NEXAFS) spectroscopy were used to investigate the effectiveness of sulfur treatments in reducing the downward band bending and passivating the surface of the SnO_2 nanotubes. Different effects were observed compared to the same treatments on ZnO and NEXAFS was performed on sulfur-treated ZnO and SnO_2 nanotube samples to investigate the reasons behind these differences. Finally, a range of additional studies on the growth and properties of the SnO_2 nanotubes revealed further information that may be useful in future work on these unusual nanostructures.

8.2 Electrical and UV Sensing Properties of SnO_2 Nanotube Resistors

Improving on preliminary work performed by Scott, single nanotube resistors consisting of robust Ti/Au contacts to individual undoped and 0.15% Sb-doped SnO_2 nanotubes were fabricated. The electrical characterisation of undoped devices showed asymmetrical IV curves and current hysteresis, which was attributed to partial Schottky barriers at each contact. Resistors fabricated from 0.15% Sb-doped nanotubes demonstrated higher quality ohmic contacts, with symmetrical IV curves and very low current hysteresis. The Sb-doped resistors also exhibited conductances 100 times larger than the undoped resistors, which demonstrated the effective doping of the SnO_2 nanotubes.

The UV photoresponse of the SnO_2 nanotube resistors was investigated as a potential application of these devices and to further understand the surface chemistry of the SnO_2 nanotubes. Despite large initial responses to ~200 μW cm^{-2} 285 nm UV light, the large surface-to-volume ratios of these nanostructures and the dominant role of atmospheric adsorbates on the photocurrent recovery led to the observation of significant persistent photoconductivity (PPC) effects following UV exposure. Both undoped and 0.15% Sb-doped SnO_2 nanotubes demonstrated significant PPC effects, exhibiting fitted PPC recovery time constant (τ) values of 542 s and 136 s respectively after the first UV exposure.

To further investigate the roles of atmospheric adsorbates on the UV photoresponse and PPC of the SnO_2 nanotubes, UV measurements were performed at a series of increased operating temperatures, up to 150 °C. Faster photoresponse rises and PPC recoveries were observed on both undoped and Sb-doped devices at increasing operating temperatures. These observations were attributed to the thermal desorption of surface O_2 and H_2O adsorbate species, which reduced the influence of the UV-induced O_2/H_2O desorption and readsorption process on the photodetector response and recovery. An Arrhenius equation was fitted to the temperature dependence of τ, which provided activation energies of 180 meV and 210 meV for the undoped and Sb-doped devices respectively, agreeing well with values in the literature.

UV measurements performed in different ambient gas environments revealed further insights into the effect of PPC on SnO_2-based photodetectors operating in air, which have not been previously reported. Despite most current literature attributing PPC to the UV-induced desorption and readsorption of superoxide ions (O_2^-) from the SnO_2 surface, UV measurements performed in a dry O_2 gas environment exhibited a slower photodetector recovery compared to air, for both undoped and Sb-doped nanotubes. Similarly, UV measurements performed in a humid N_2 environment also produced a slower recovery than in air, demonstrating that the sole presence of oxygen or water vapour did not effectively minimise the PPC response. A humid O_2 gas environment illustrated the co-operative role of both atmospheric water vapour and O_2 on the minimisation of the PPC recovery time, with dramatically smaller PPC recovery time constants of 17 s and 8.6 s for the undoped and Sb-doped devices respectively observed in this ambient. A mechanism was proposed which involved the formation of a strongly hydrogen-bonded network between adsorbed H_2O and O_2^- ions which inhibits the desorption of O_2^- from the SnO_2 surface under UV illumination.

8.3 Advanced Devices of SnO_2 Nanotubes and their Applications

Individual back-gated undoped SnO_2 nanotube FETs were successfully fabricated on p-Si/SiO_2 substrates, with the ~77 nm thick SiO_2 layer acting as the gate dielectric. These devices demonstrated good n-type FET characteristics with a maximum on/off ratio of ~4×10^6, average subthreshold slope of 187 mV/decade and a small average threshold voltage of -1.96 V, comparable to other results seen in the literature. A 2D electrostatic simulation yielded a calculated gate capacitance of 146 aF μm^{-1} for an individual nanotube device, allowing the further calculation of a field-effect mobility and modulated carrier concentration. The calculated average field-effect mobility of 6.13 $cm^2V^{-1}s^{-1}$ is similar to that found in other undoped SnO_2 FETs, and actually likely underestimates the true field-effect mobility of the device due to interface states at the device gate and the partial Schottky barriers formed at the Ti/Au ohmic contacts to undoped nanotubes. The carrier concentration of the undoped SnO_2 nanotube was estimated to be ~5×10^{16} cm^{-3}. Large amounts of gate hysteresis was observed under extended gate voltage biases (V_G = ±20 V), which was explained by charge trapping most likely caused by water vapour present between the gate dielectric and the nanotube channel. FETs created from 0.15% Sb-doped SnO_2 nanotubes demonstrated current enhancement and depletion, however the carrier concentration was too high to pinch off these devices, even under large gate bias voltages.

Individual undoped and 0.15% Sb-doped SnO_2 nanotube Schottky diodes were fabricated using an IrO_x/Ir Schottky contact. Devices measured immediately following 400 °C annealing demonstrated excellent electrical characteristics that outperformed all existing SnO_2 nanostructure-based devices in the literature. Large average barrier heights of 0.82 eV and 0.67 eV were found for the undoped and Sb-doped devices respectively, with the difference in barrier heights due to the image-force lowering effect which increases with carrier concentration. Large current rectification ratios greater than 10^7 and low average ideality factors of 1.24 and 1.09 were also observed for the undoped and Sb-doped devices respectively. Despite the excellent initial electrical properties, a severe ageing effect was observed after leaving the

fabricated devices in air for six days, which resulted in a lowering of the Schottky barrier and a decrease in the forward current. Reannealing the devices restored their excellent electrical properties, which suggested that atmospheric adsorbates play an important role in this effect. The UV sensing performance of a recently-annealed Sb-doped SnO_2 nanotube Schottky diode was tested by back-illuminating the device through the sapphire substrate with ~360 μW cm^{-2} 275 nm UV light. A large sensor response of ~10^7 was observed under negative bias voltages, and was accompanied by the shortest observed PPC recovery with a fitted τ value of 3.4 s. Despite this significantly faster photodetector recovery, PPC effects were still observed to some degree in the reverse bias leakage current.

To demonstrate further applications of the SnO_2 nanotubes and to address the low yields and labour-intensive processes associated with individual-nanotube devices, multi-nanotube ionic gated SnO_2 nanotube transistors were fabricated using an interdigitated finger contact arrangement. This contact arrangement produced electrical contacts to nanotubes that were still of a reasonable quality, and with a significantly higher device yield that removed the need for scanning electron microscopy imaging to identify functional devices. $LiClO_4$ and polyethylene oxide were used as the electrolyte and ion-conducting medium respectively in the ionic-gated transistors. Fabricated devices exhibited good n-type transistor characteristics with on/off ratios up to ~2.2×10^4 and an average subthreshold slope of 173 mV/decade. Neuromorphic testing of the ionic-gated transistors demonstrated several synaptic behaviours such as excitatory postsynaptic current, paired-pulse facilitation, spike-rate-dependent-plasticity and adaptation. This demonstrated the potential use of these devices as artificial synapses in spiking neuromorphic systems, although more work is required to reduce the noise within these devices during operation.

8.4 Band Bending Studies of the SnO_2 Nanotubes

To further investigate the roles of surface adsorbates on the band bending and surface chemistry of the SnO_2 nanotubes — particularly following high temperature annealing — synchrotron XPS was used to study the removal and recovery of surface hydroxyls and the formation and repair of oxygen vacancies from undoped and

0.15% Sb-doped SnO_2 nanotubes. On undoped nanotubes, 600 °C heating removed much of the electron donating surface hydroxyl layer, although this did not result in a weakening of the downward band bending due to the formation of in-plane oxygen vacancies, which are also electron donors and compensate the loss of surface hydroxyls. Combined or separate O_2/H_2O dosing at low pressures of approximately 10^{-5} mBar demonstrated little to no recovery of the surface hydroxyl coverage, with only a slight weakening of the downward band bending due to oxygen adsorption. Venting with H_2O produced a small increase in the surface hydroxyls and downward band bending, likely limited by the removal of bridging oxygen required for water dissociation on the SnO_2 surface. Venting to atmosphere produced a recovery of the surface hydroxyls and the repair of oxygen vacancies, returning the downward band bending to its strong initial condition. A comparison of venting to atmosphere for 20 minutes and 10 hours demonstrated how the process of complete hydroxyl recovery likely occurs over the time scale of several hours.

The Sb-doped SnO_2 nanotubes showed similar trends to the undoped nanotubes in the same experiments, although fewer oxygen vacancies were formed on the Sb-doped sample, likely due to a higher defect formation energy. This resulted in minor differences to the surface band bending and surface hydroxyl recoveries, such as a slight upward shift in band bending following 600 °C heating and a stronger hydroxyl recovery following venting with H_2O compared to the undoped sample.

Undoped and 0.15% Sb-doped square SnO_2 nanotubes underwent sulfur treatments in $(NH_4)_2S$ and Na_2S solutions in an attempt to produce an upward shift in the surface band bending, and the effects of these treatments were investigated by SEM, XPS and NEXAFS. A 1 hour $(NH_4)_2S$ treatment produced no significant changes to the surface band bending, with only a weak sulfur environment present on treated samples. A treatment in an Na_2S solution for 20 minutes demonstrated small shifts of the valence band edge when observed with low X-ray doses. Larger upward band bending shifts were observed with increasing X-ray irradiation, with the chemical reduction of oxidised sulfur species correlated with these shifts.

NEXAFS studies on Na_2S treated ZnO revealed that the surface of ZnO is converted to ZnS during the treatment, which produces the large valence band shift

observed in previous studies due to the band offset of the two materials. A similar process does not occur in SnO_2 due to the material's higher chemical stability compared to ZnO.

8.5 Additional Studies

Several additional studies of the SnO_2 nanotubes and related structures revealed more information about their growth and properties. The carrier and dilution gas flow rates during Mist-CVD growth were identified as important parameters for the growth of square SnO_2 nanotubes, with optimised flow rates of 1.6 $Lmin^{-1}$ and 4.5 $Lmin^{-1}$ respectively. Carbon-based contaminants are capable of producing square nanospirals of SnO_2, through either the contamination of the growth substrate or the precursor solution. Future investigations could be performed to attempt to grow these structures intentionally.

The different surface morphologies of the sides of an SnO_2 nanotube were investigated by STM. Two different regions were identified: an atomically smooth region near the top of the nanotube and a lamellar-like terraced region further down the nanotube. These two different regions likely form due to differences in material availability during Mist-CVD growth.

The observation of a nanotube experiencing significant bending without cracking or fracturing revealed insights into the mechanical properties of these structures. The nanotube was able to withstand a maximum strain of 2.1%, which is significantly larger than the yield strain of 0.07-0.1% seen in other bulk metal oxides similar to SnO_2.

8.6 Final Remarks

The work performed in this thesis has demonstrated that high-quality electronic devices are able to be fabricated that incorporate the square SnO_2 nanotubes, some of which exhibit electrical properties that are among the best compared to other devices in the literature that utilise SnO_2 nanostructures. The testing of these devices has revealed several new findings that are related to the SnO_2 nanotubes and are

strongly relevant for devices that incorporate SnO_2 nanostructures, such as effective Sb-doping and highlighting the critical roles of atmospheric adsorbates on the SnO_2 surface. In particular, the co-operative role of oxygen and water vapour in minimising the effect of persistent photoconductivity in SnO_2 UV photodetectors is a finding previously unseen in the literature, and is a strong contribution to the understanding of this effect that will likely benefit other researchers in this area.

The strong downward band bending and surface sensitivity of the SnO_2 nanotubes continues to be an issue in some applications. While sulfur teatments using Na_2S solutions are capable of almost completely removing this downward band bending after soft X-ray irradiation, more work is required to produce these effects without large X-ray doses. If this can be achieved, then the incorporation of passivated SnO_2 nanotubes into electronic devices will need to be performed to assess the environmental sensitivity of these devices. This thesis serves as a strong foundation for these potential future investigations.

Finally, neuromorphic systems are receiving a huge amount of research due to the increasing role of AI technologies in everyday life and the need to minimise the power used in our modern computational systems. The development of low-power neuromorphic devices is a crucial step to achieve these goals, and future endeavours should be undertaken in this area. The ionic-gated transistors developed in this thesis are only a starting point for future investigations, and could be further improved to reduce the noise experienced during operation. Alternative device structures could also be employed to create artificial synapses that utilise the SnO_2 nanotubes, and further computational or sensing tasks should be identified for these neuromorphic devices.

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